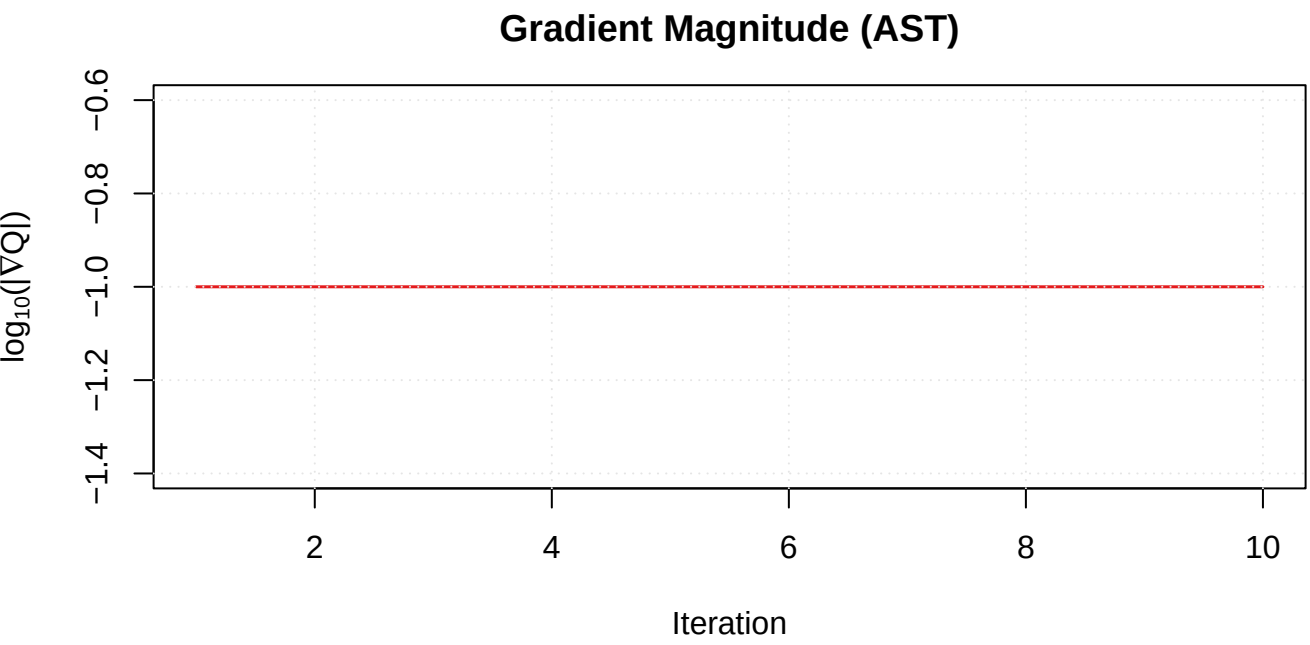
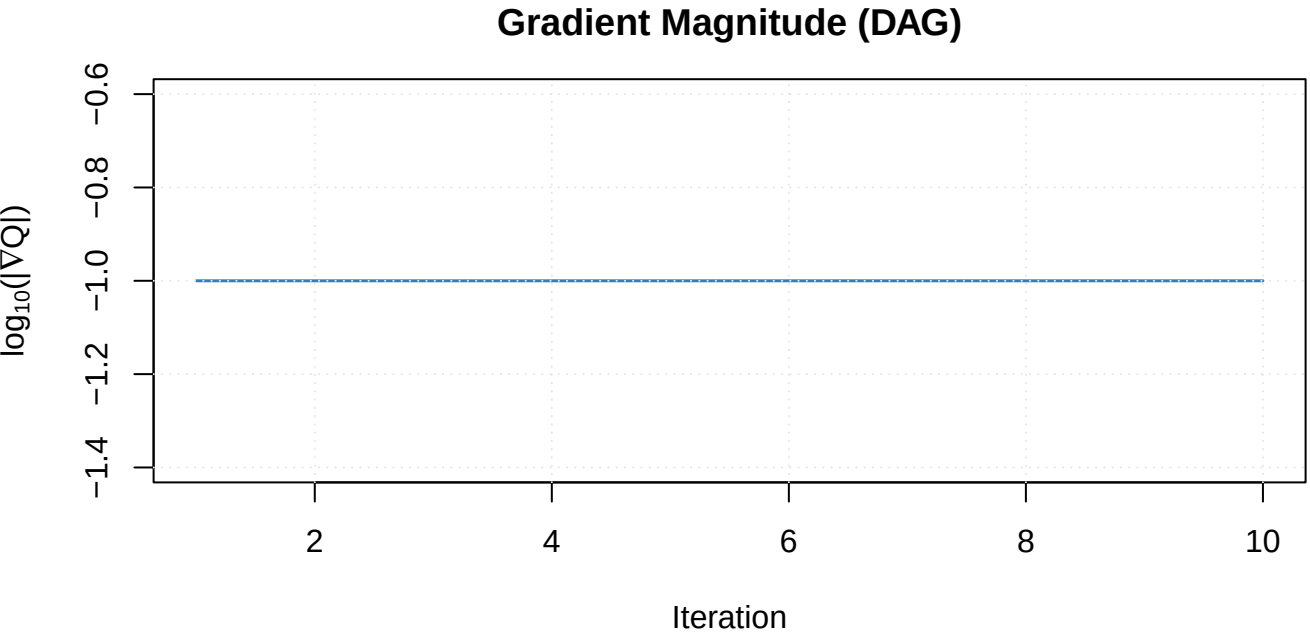
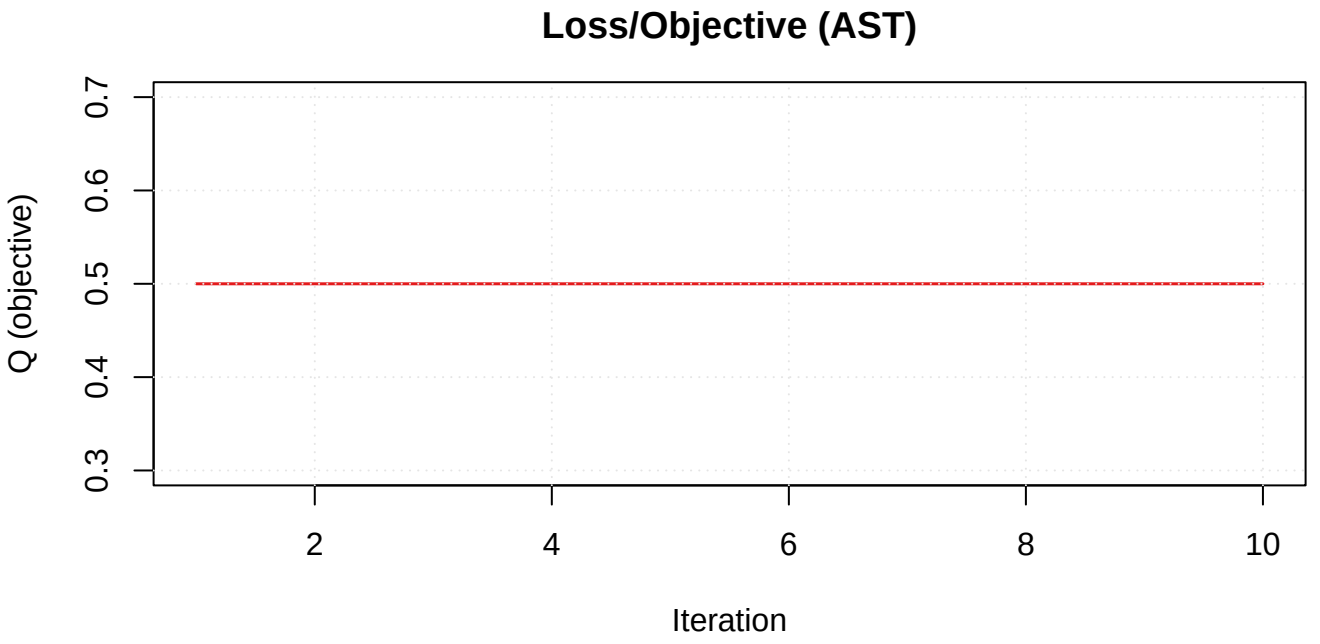
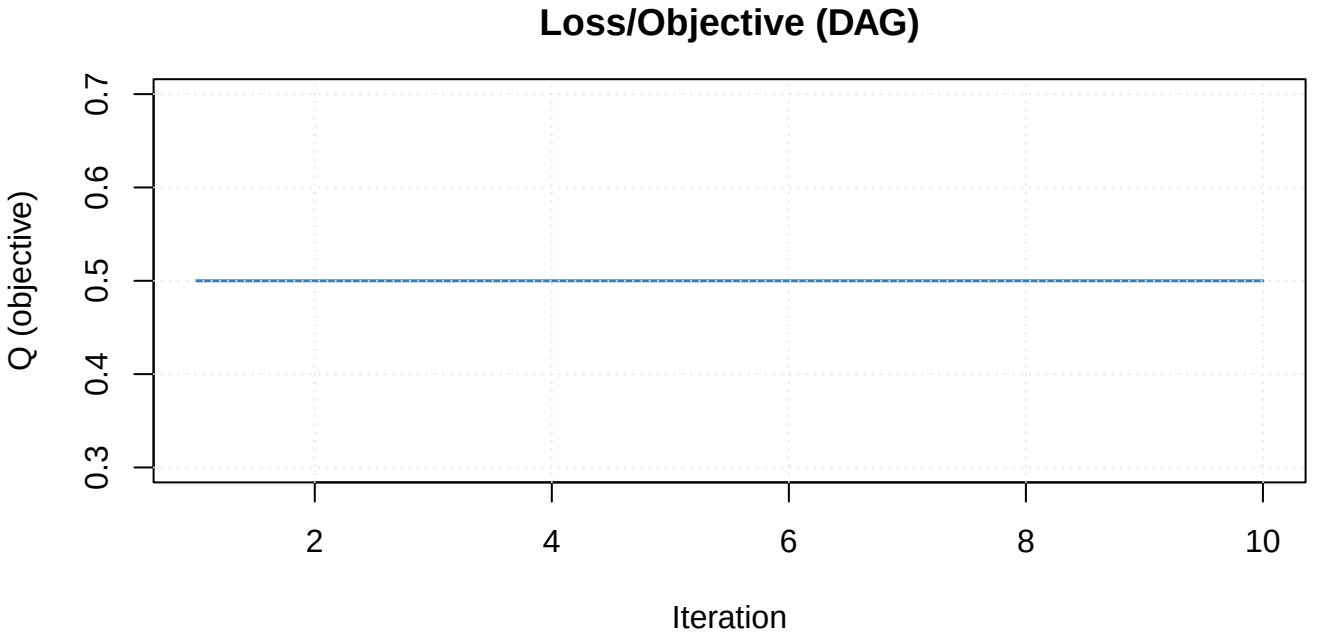
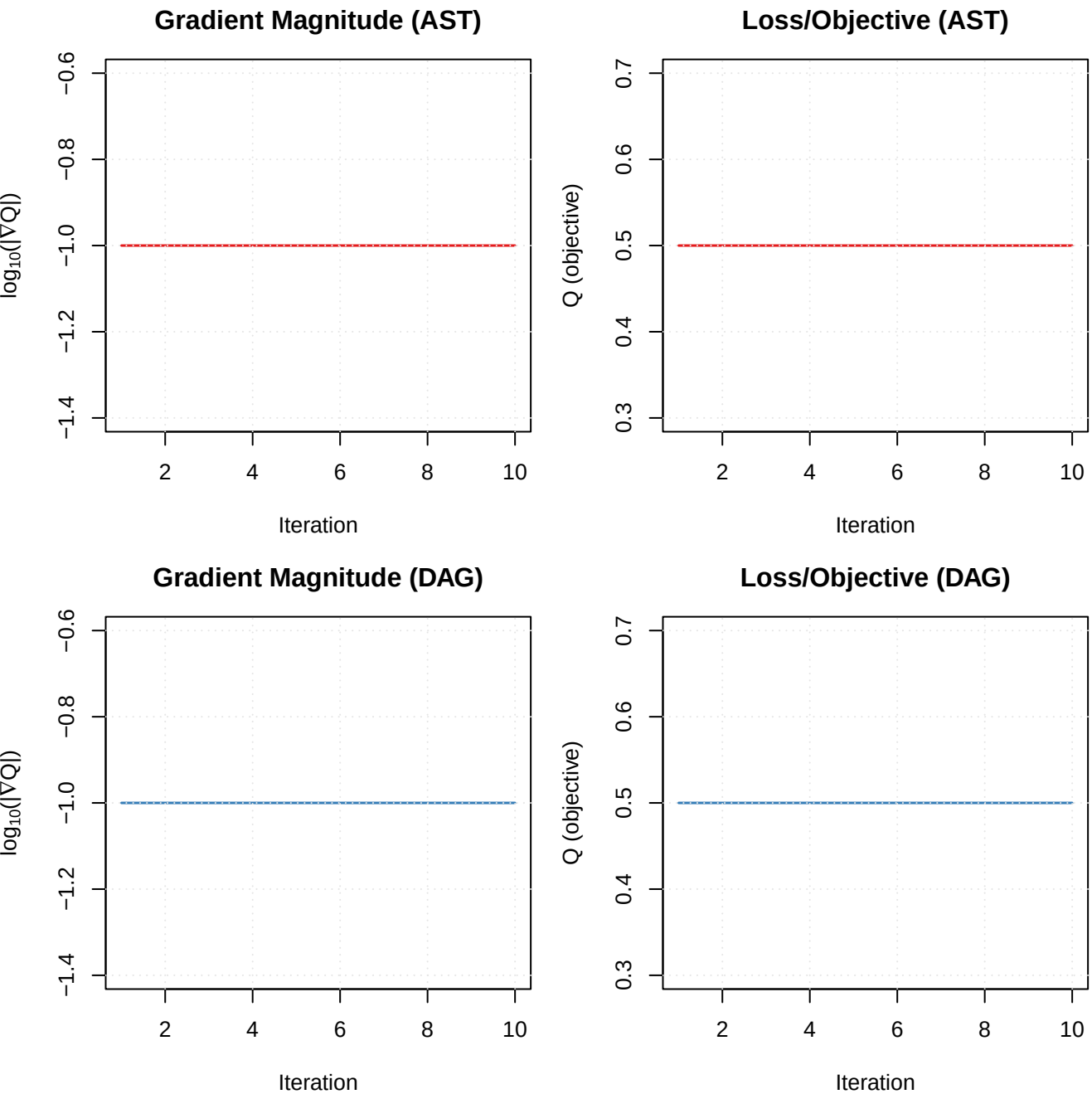
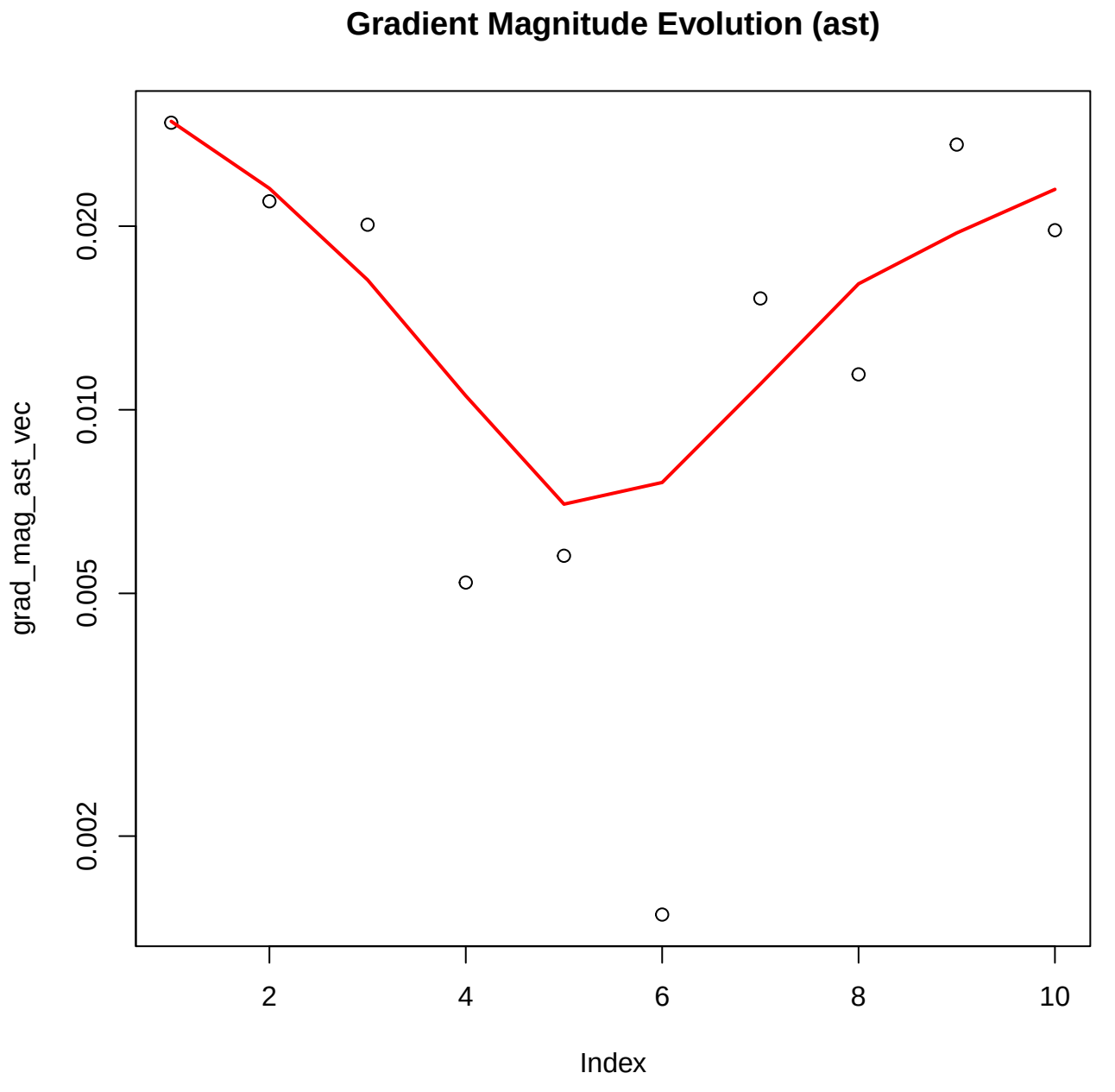

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

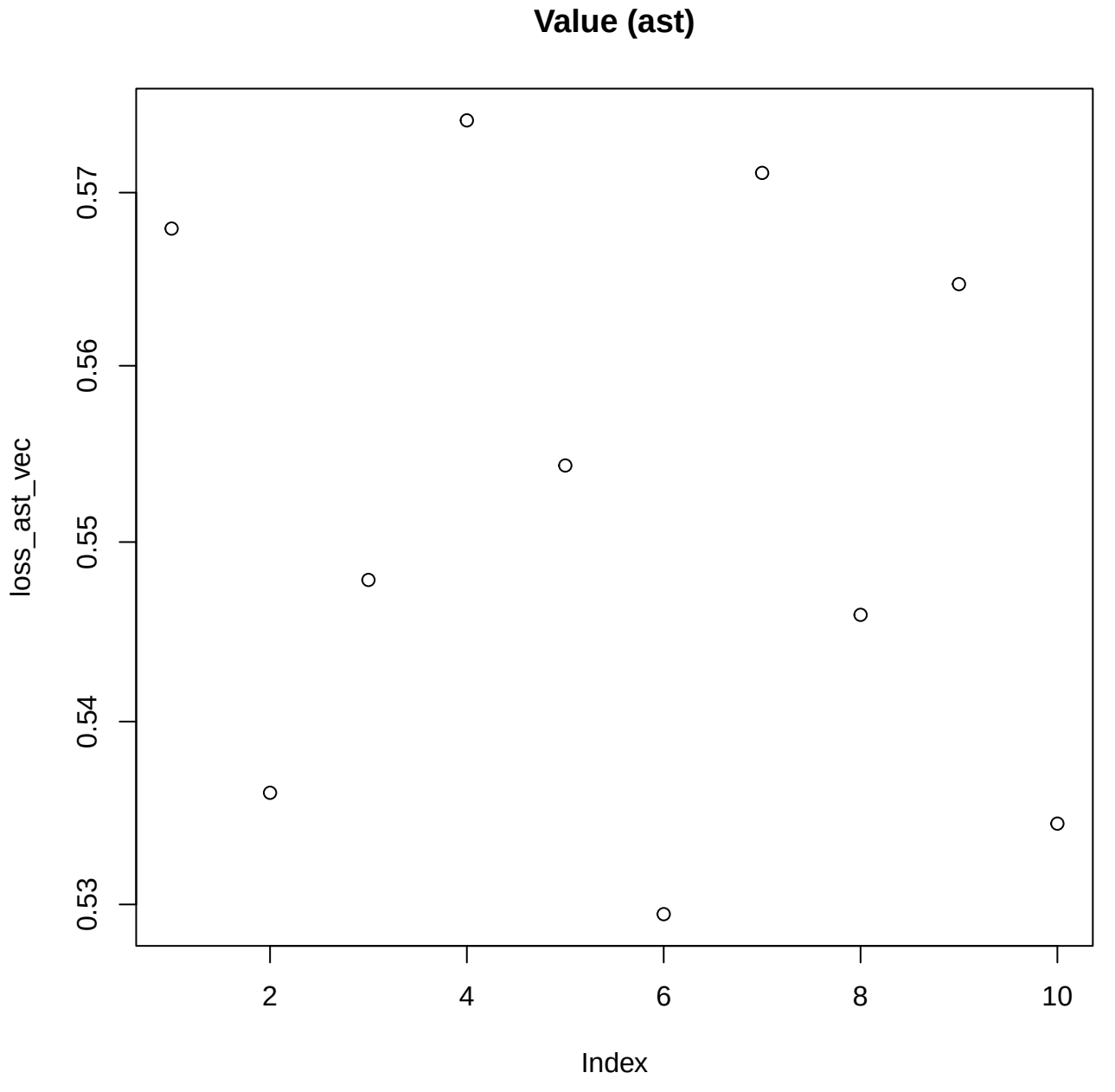

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

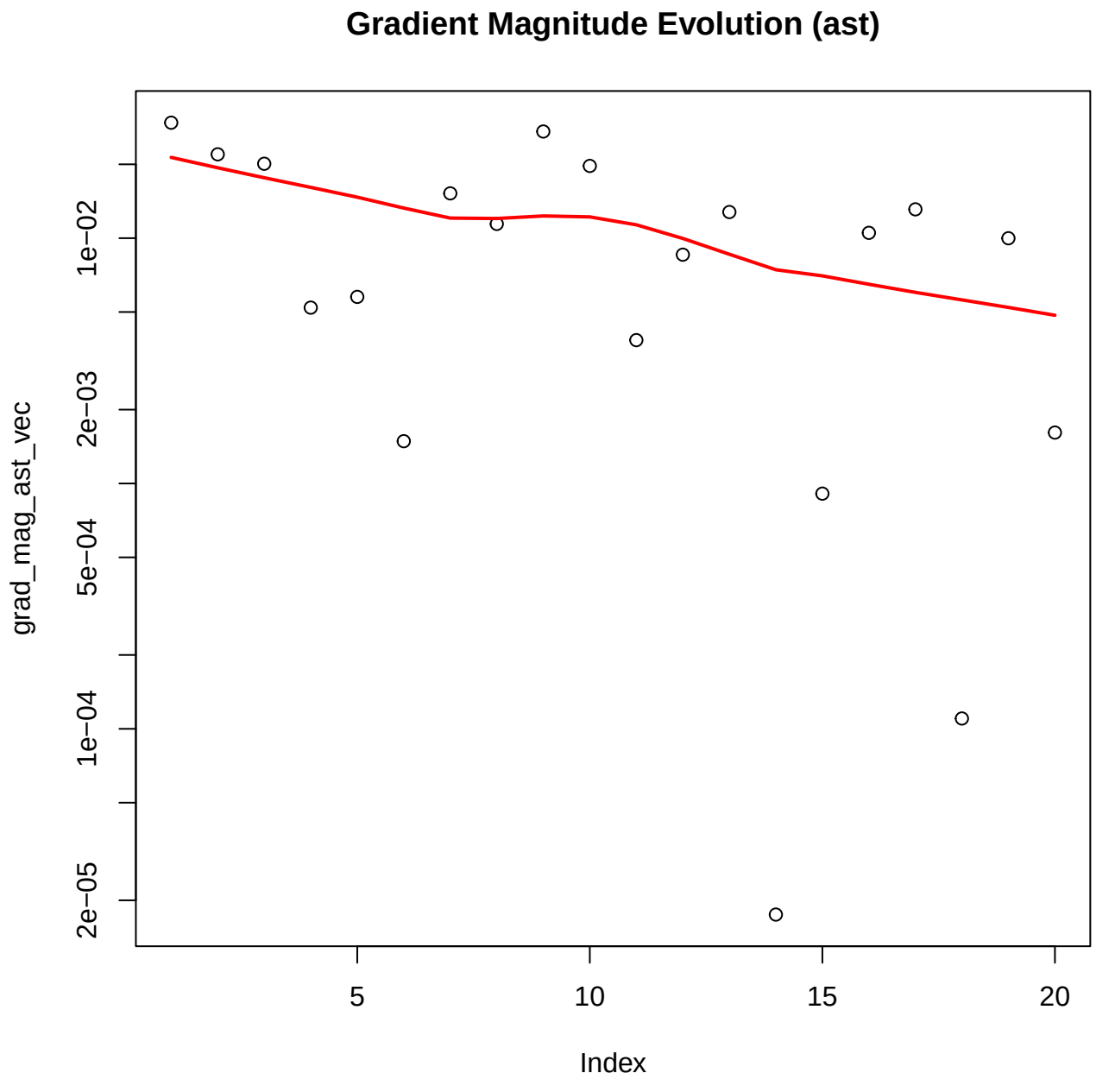

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

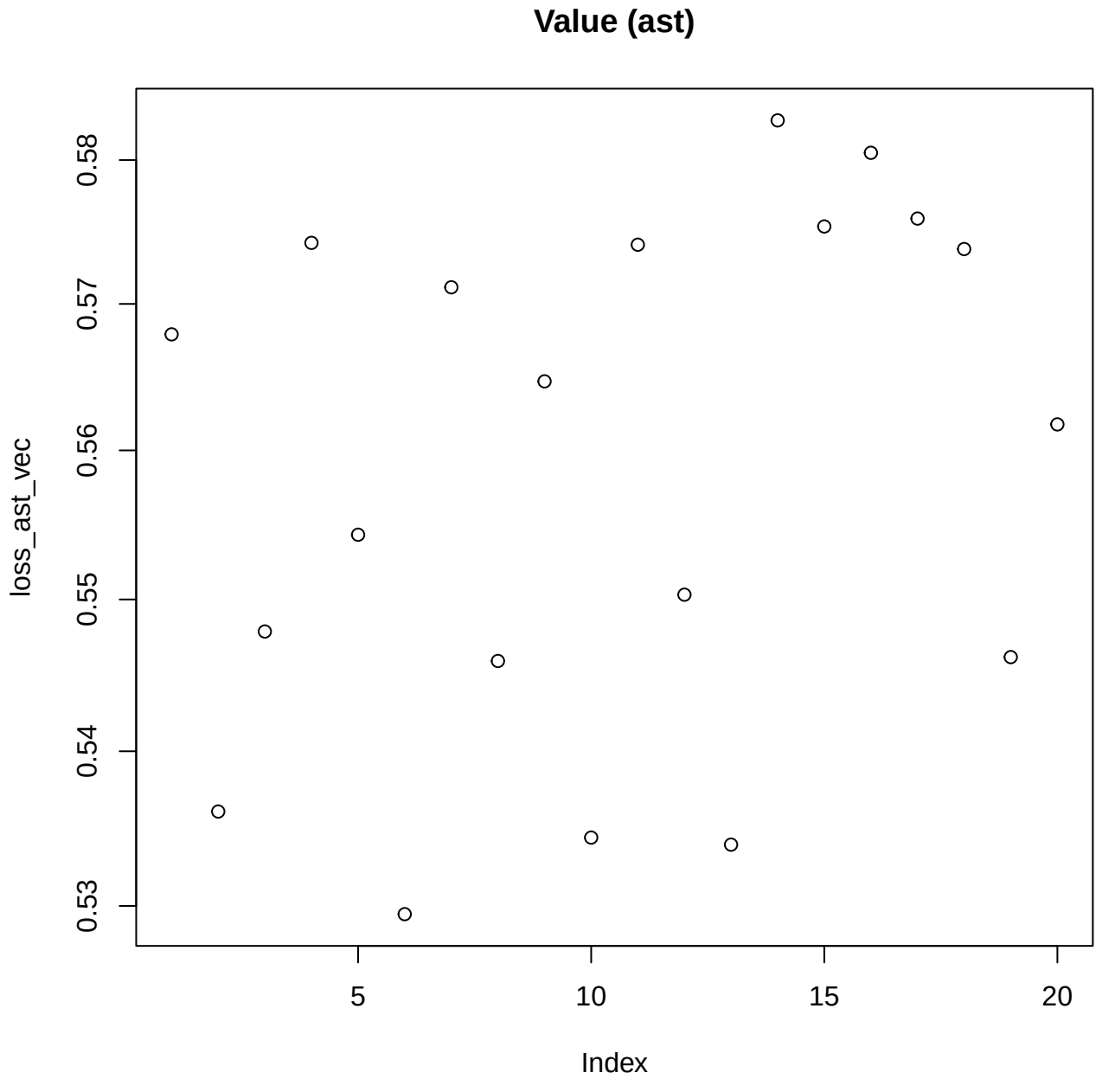

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

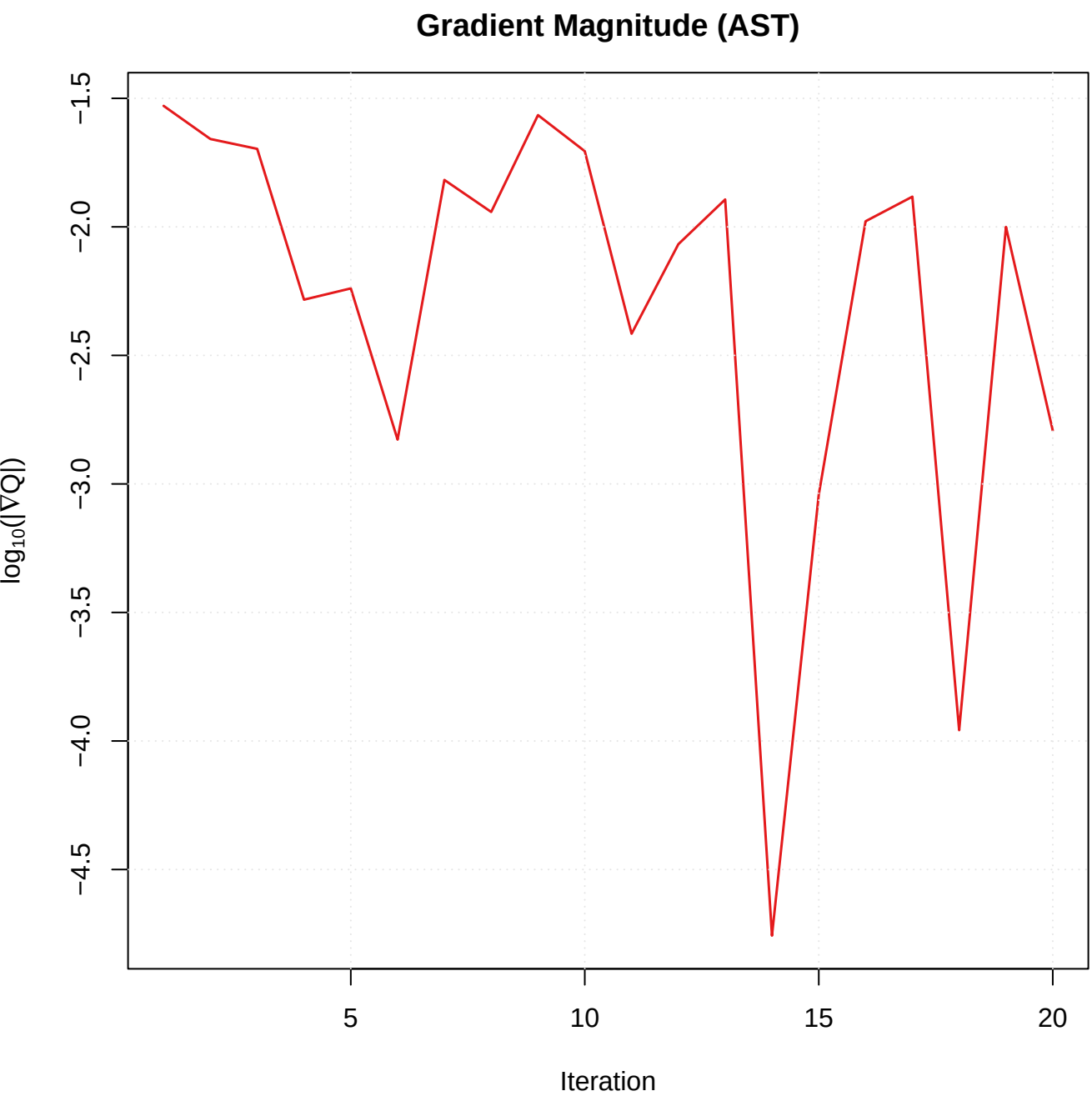

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

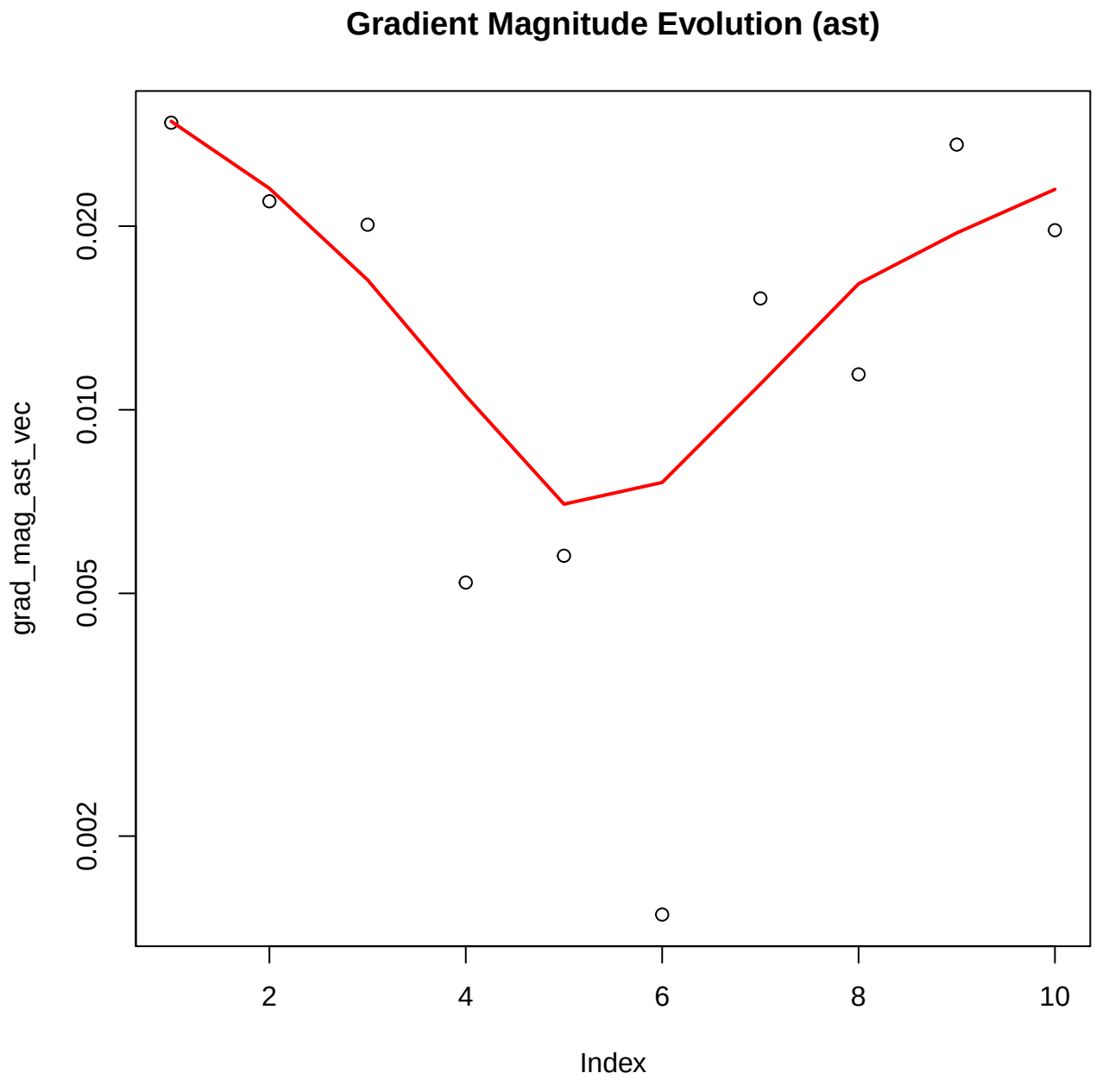

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.002
0.005
0.010
0.020
2
4
6
8
10
Index

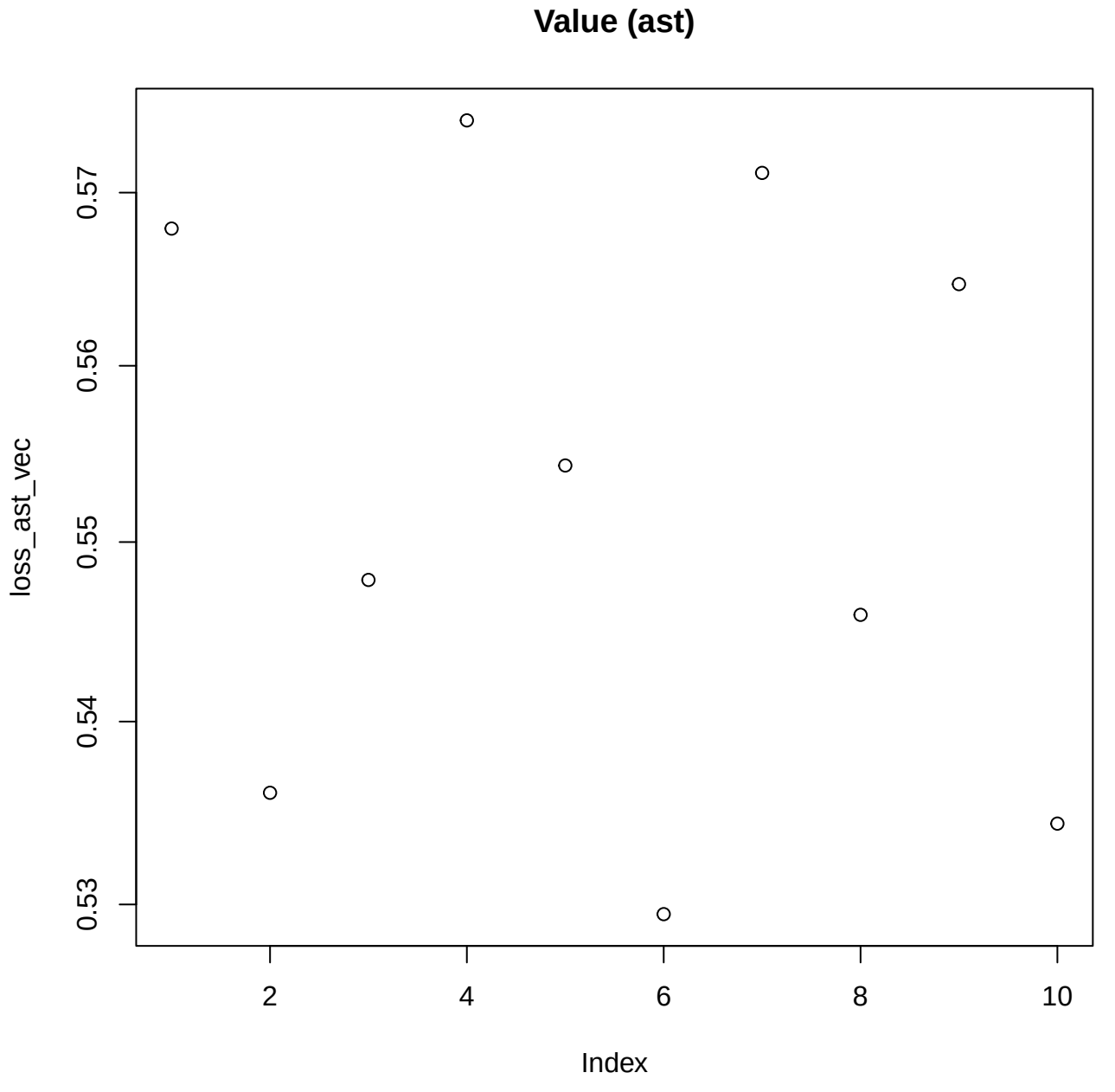

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
2
4
6
8
10
Index

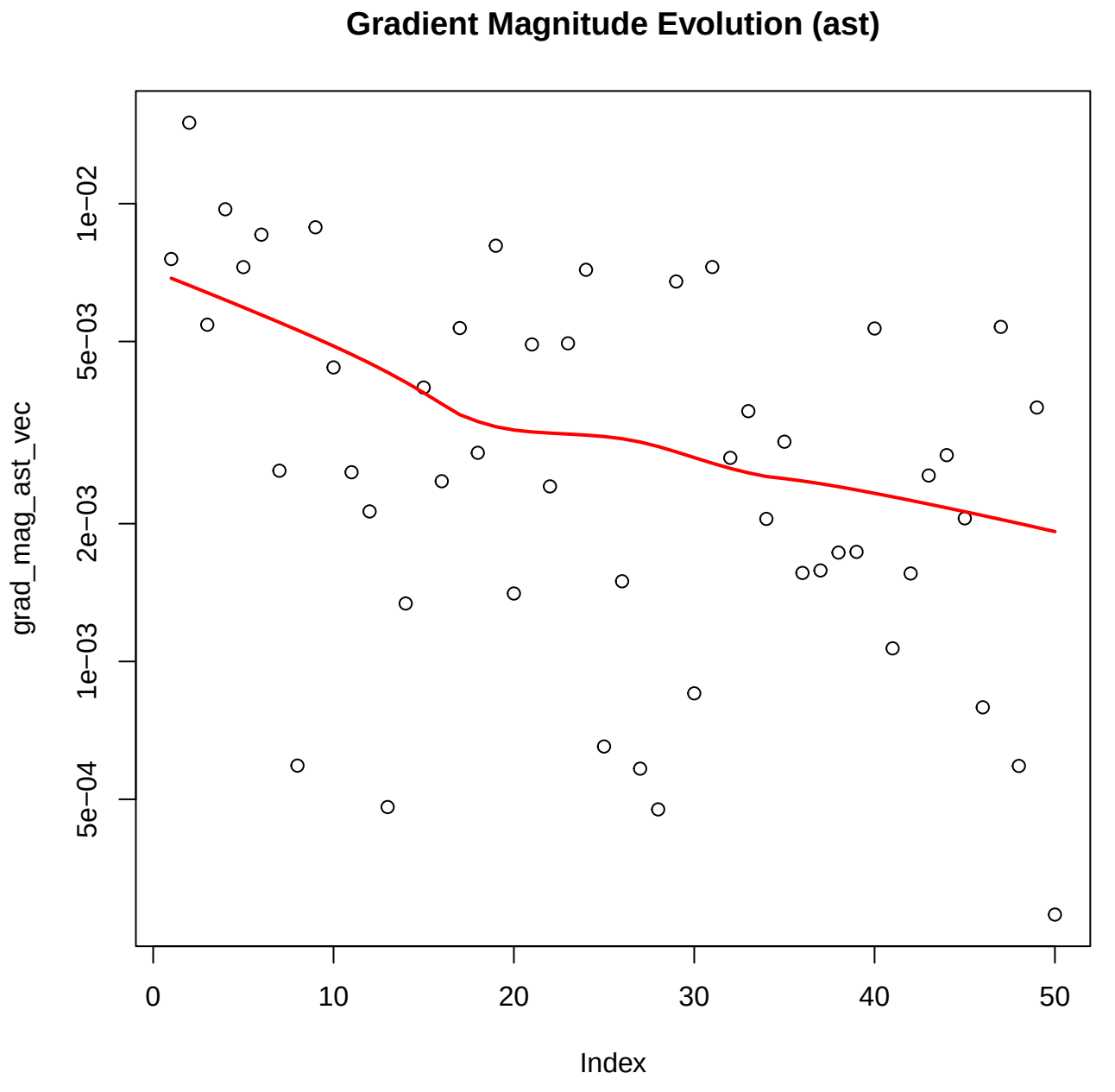

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
2e−03
5e−04
1e−04
2e−05
5
10
15
20
Index

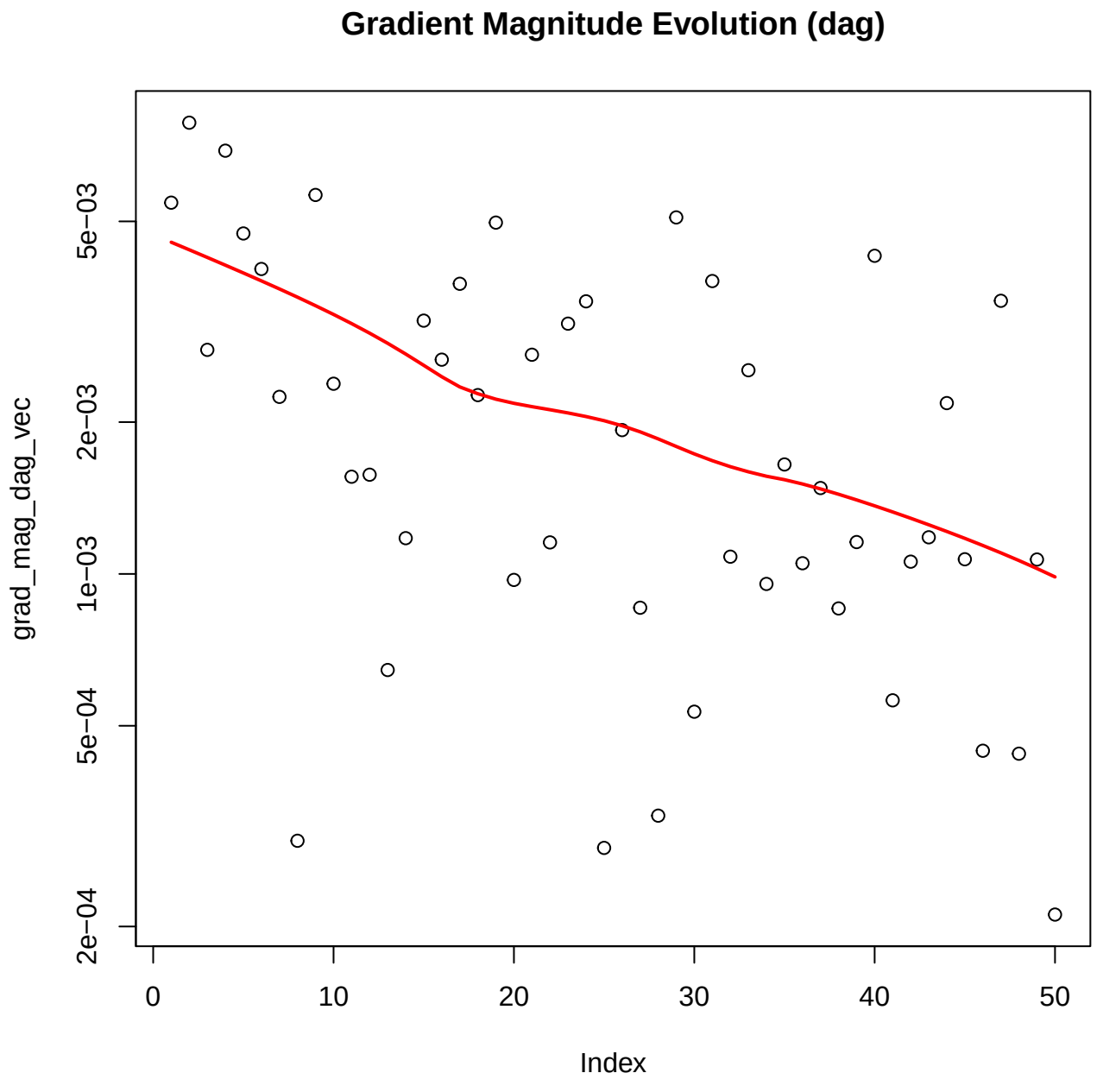

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
0.58
5
10
15
20
Index

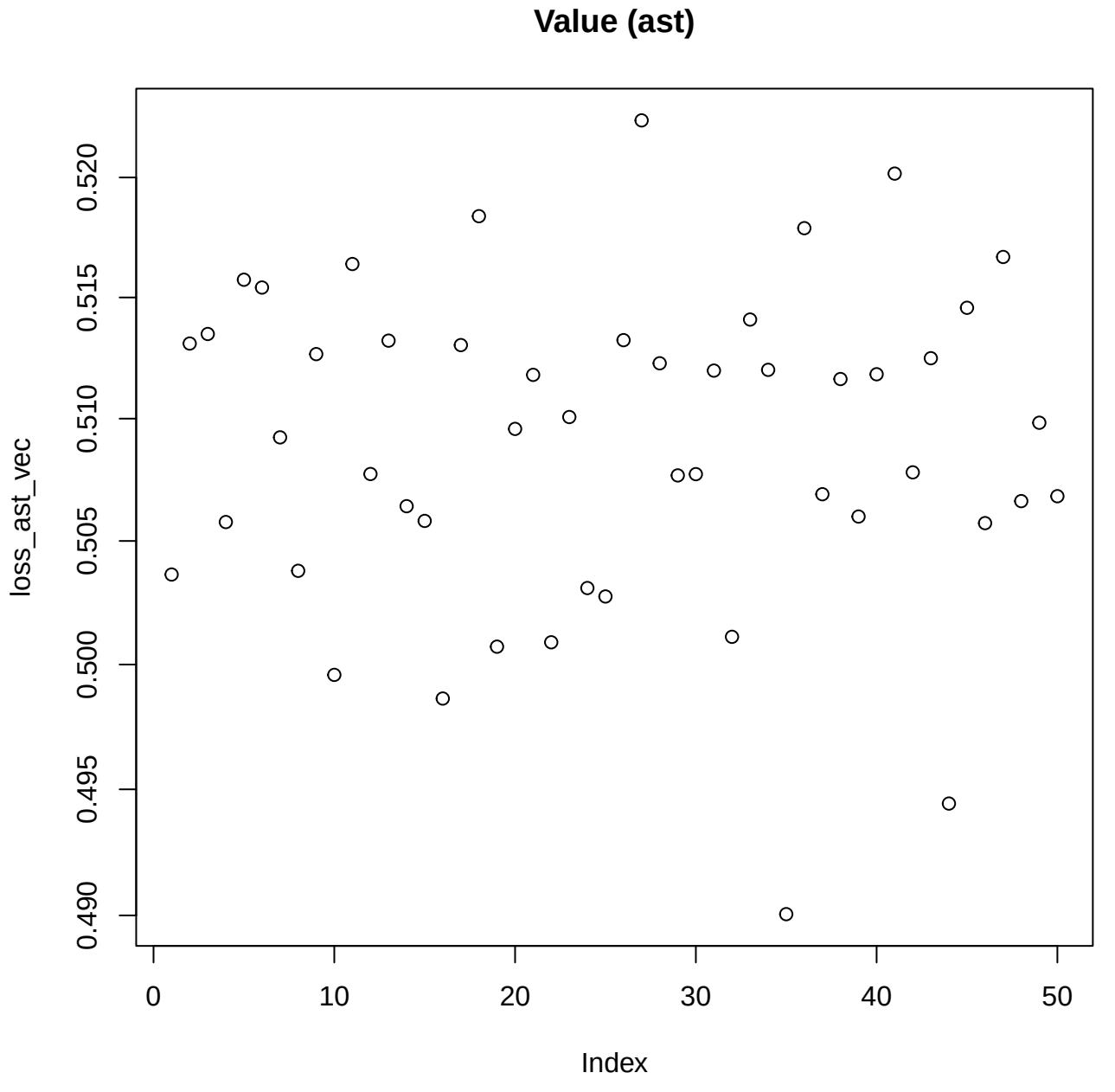

Gradient Magnitude (AST)
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
log10(|∇Q|)
5
10
15
20
Iteration

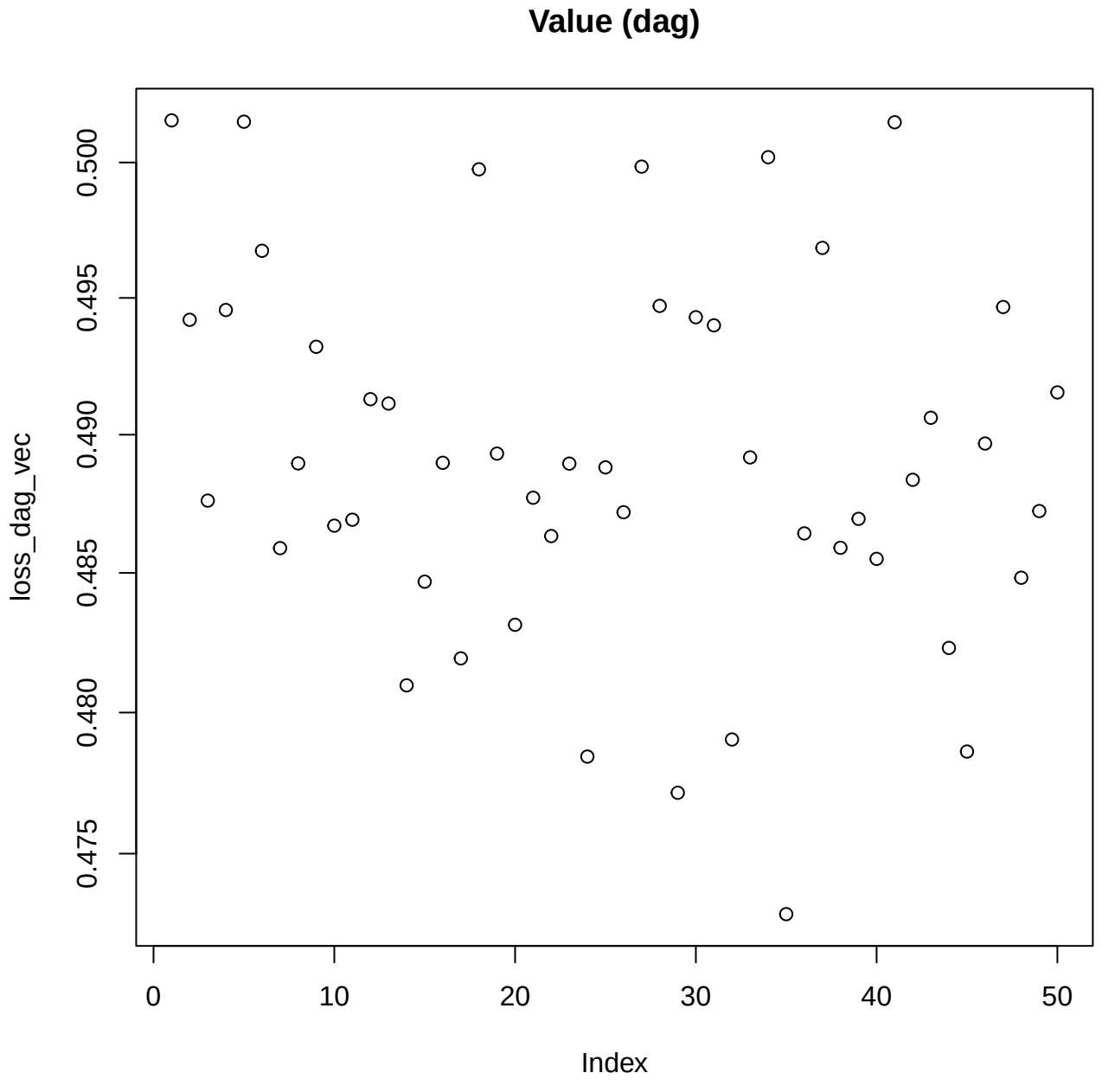

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.002
0.005
0.010
0.020
2
4
6
8
10
Index

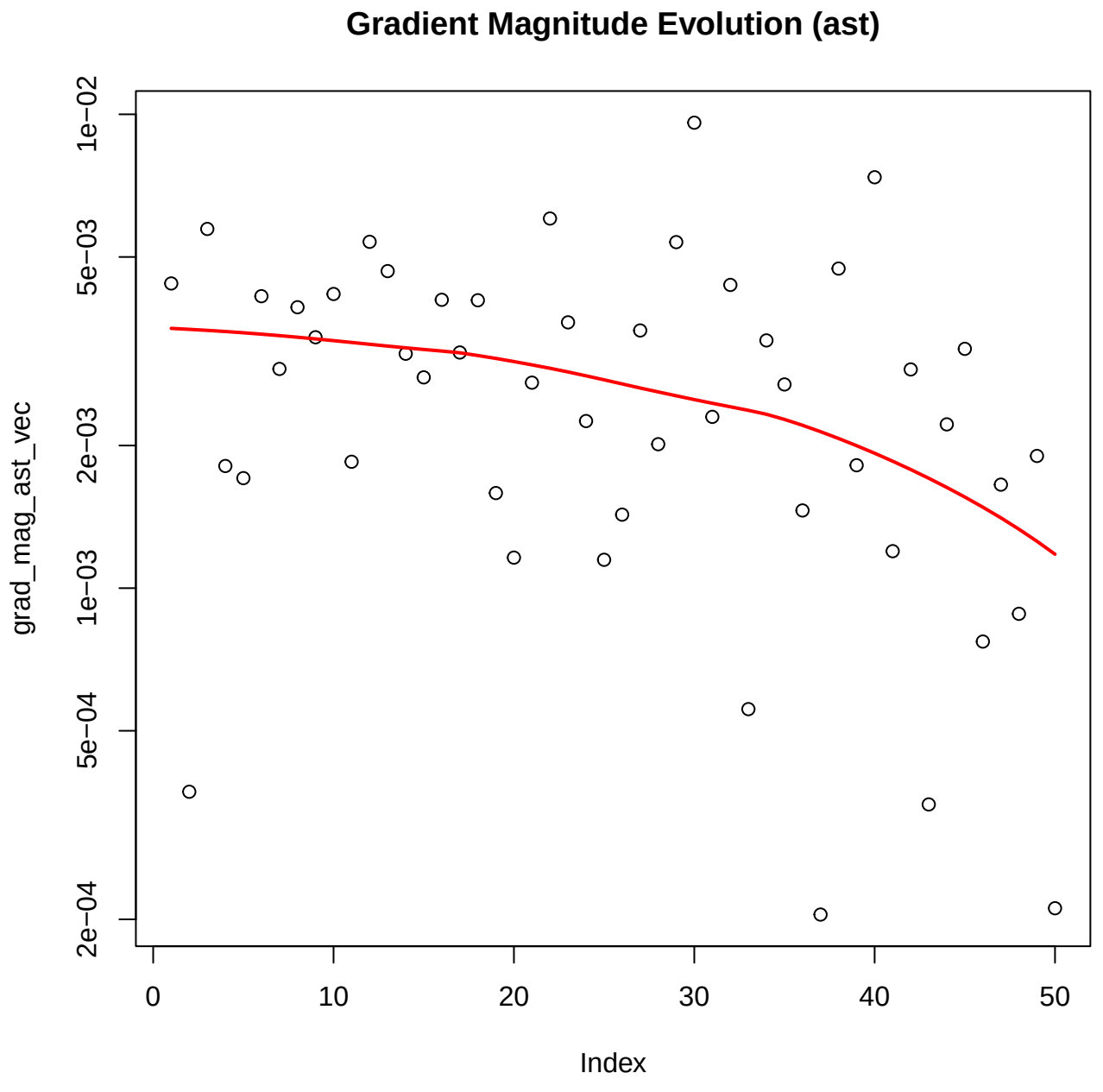

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
2
4
6
8
10
Index

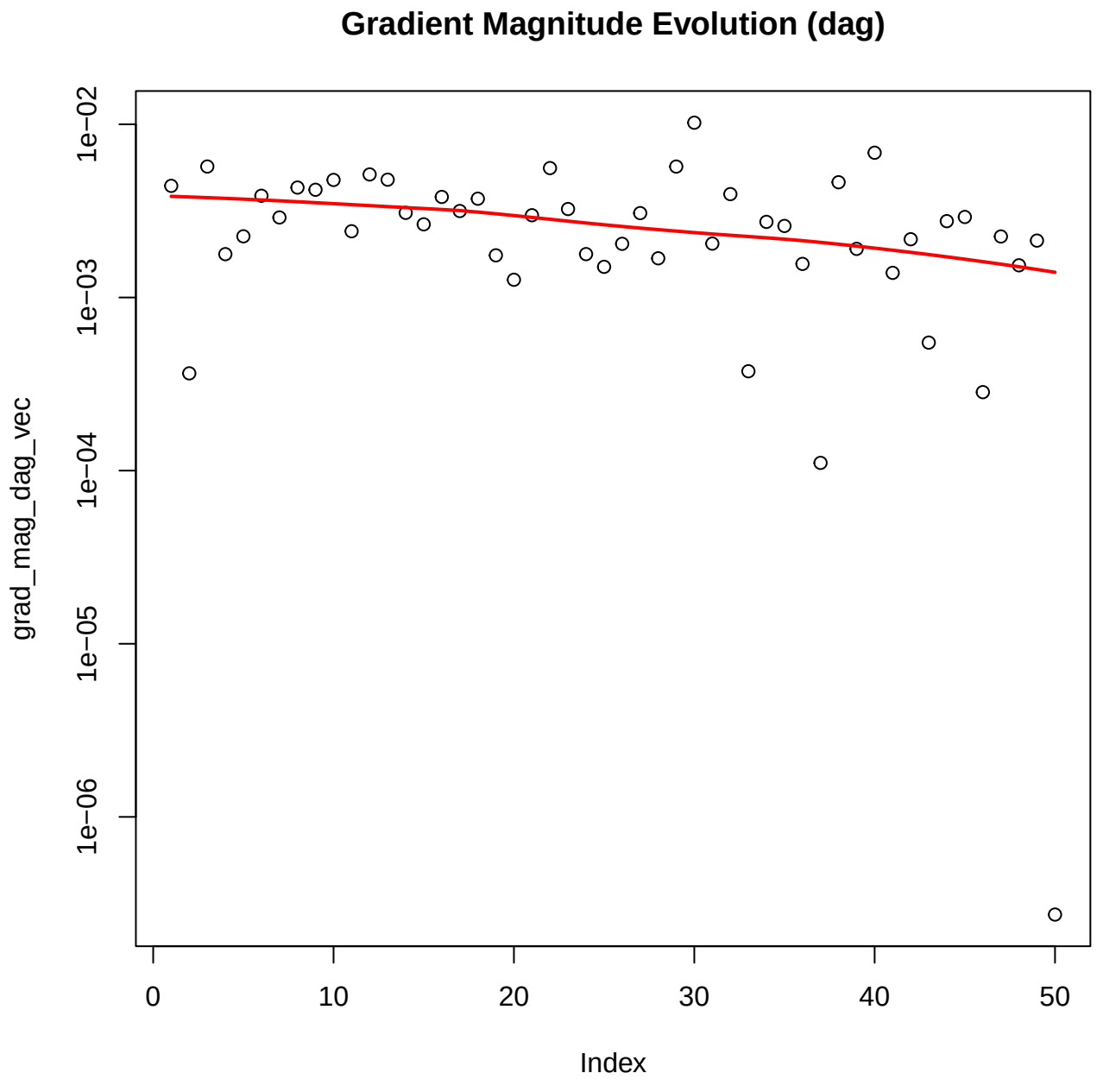

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
5e−03
2e−03
1e−03
5e−04
0
10
20
30
40
50
Index

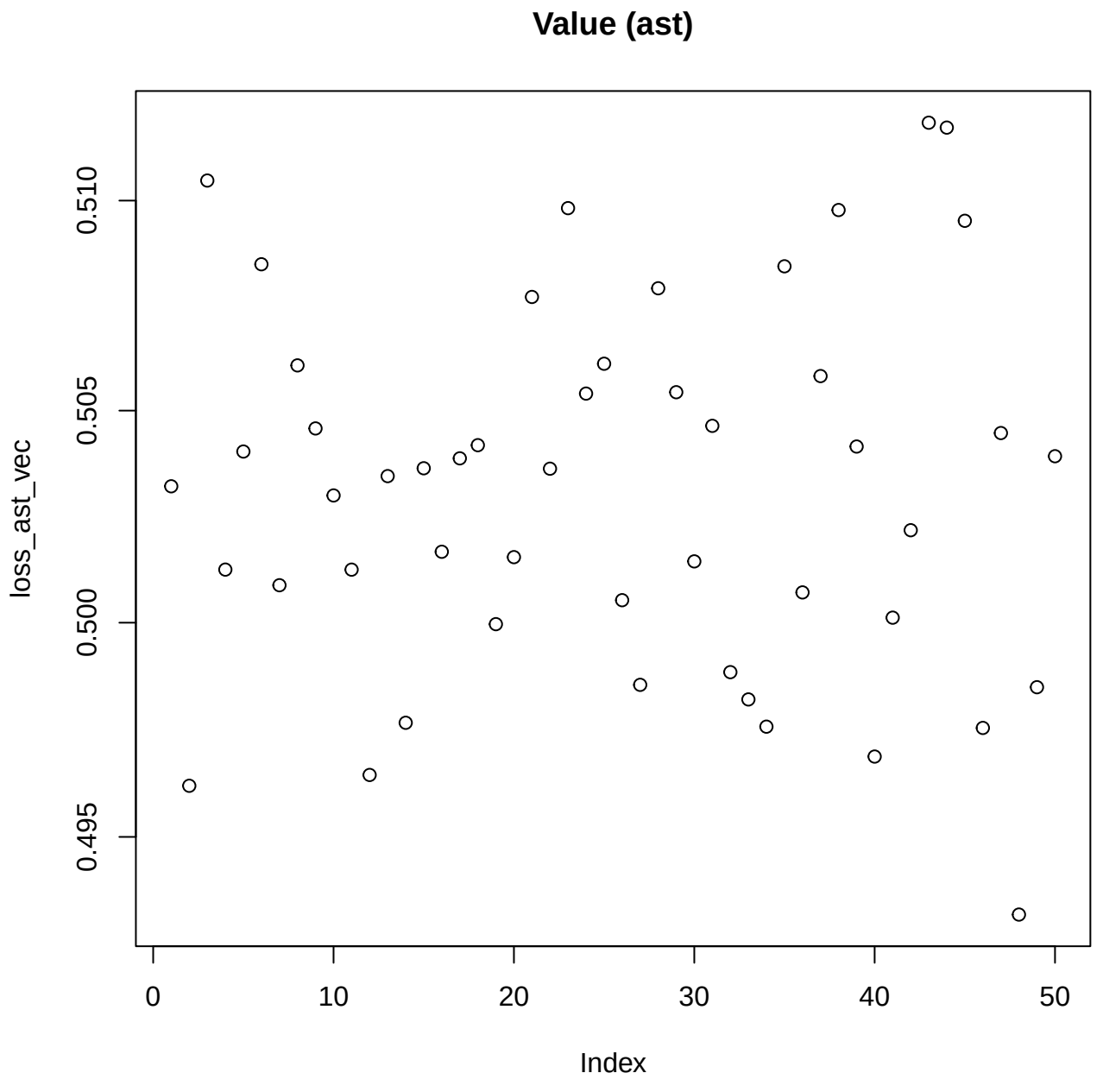

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2e−04
0
10
20
30
40
50
Index

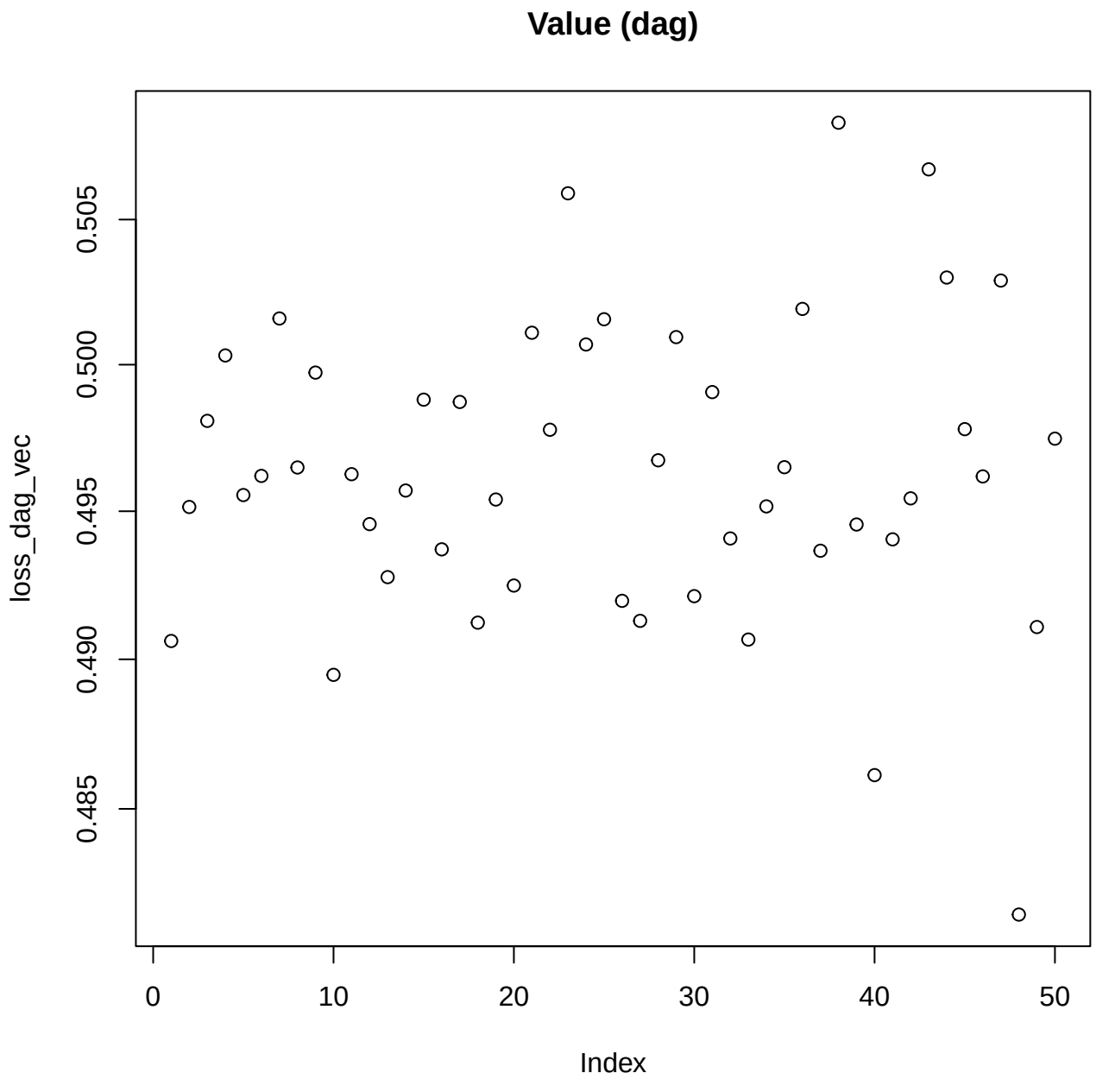

Value (ast)
loss_ast_vec
0.490
0.495
0.500
0.505
0.510
0.515
0.520
0
10
20
30
40
50
Index

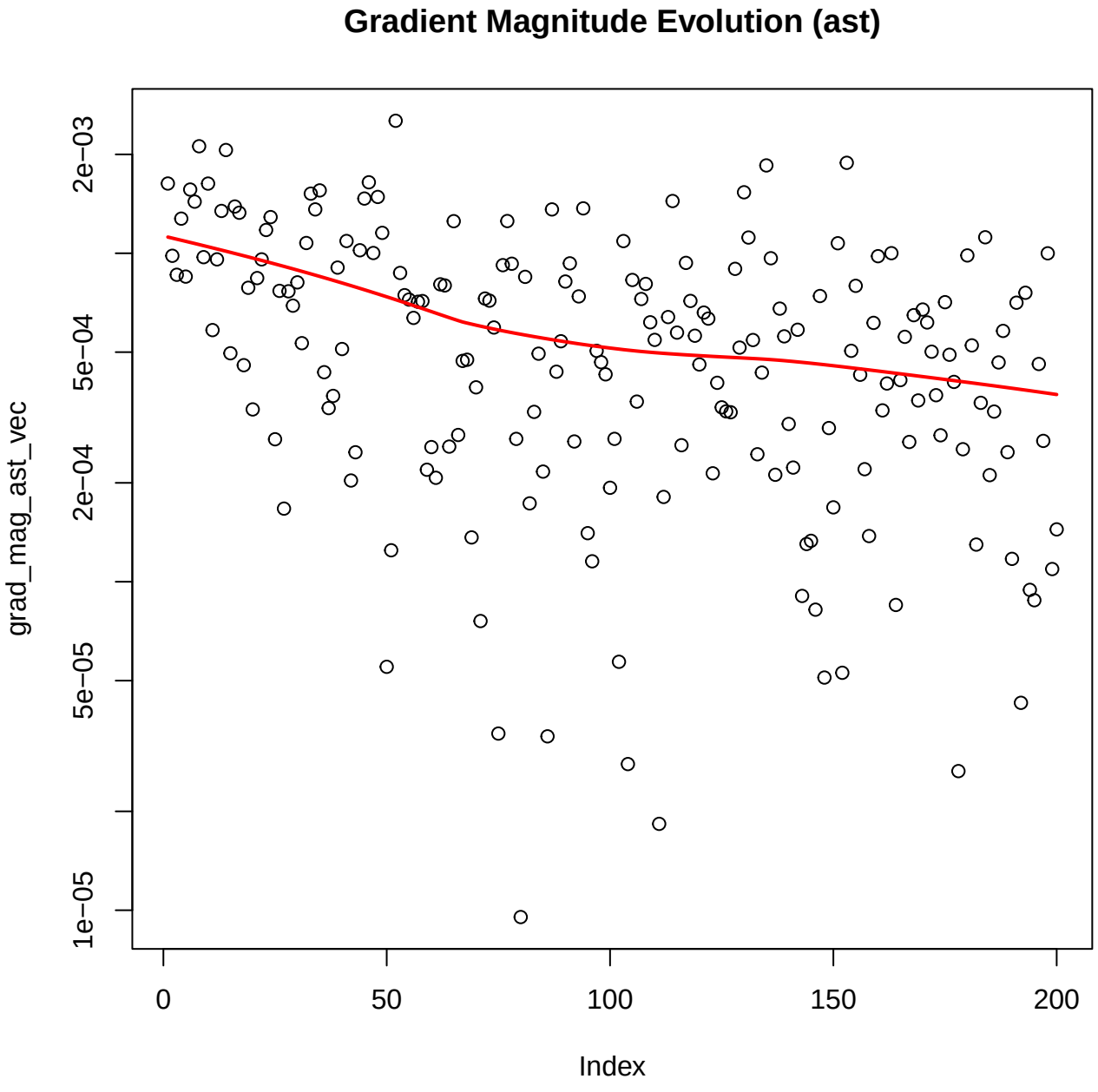

Value (dag)
loss_dag_vec
0.475
0.480
0.485
0.490
0.495
0.500
0
10
20
30
40
50
Index

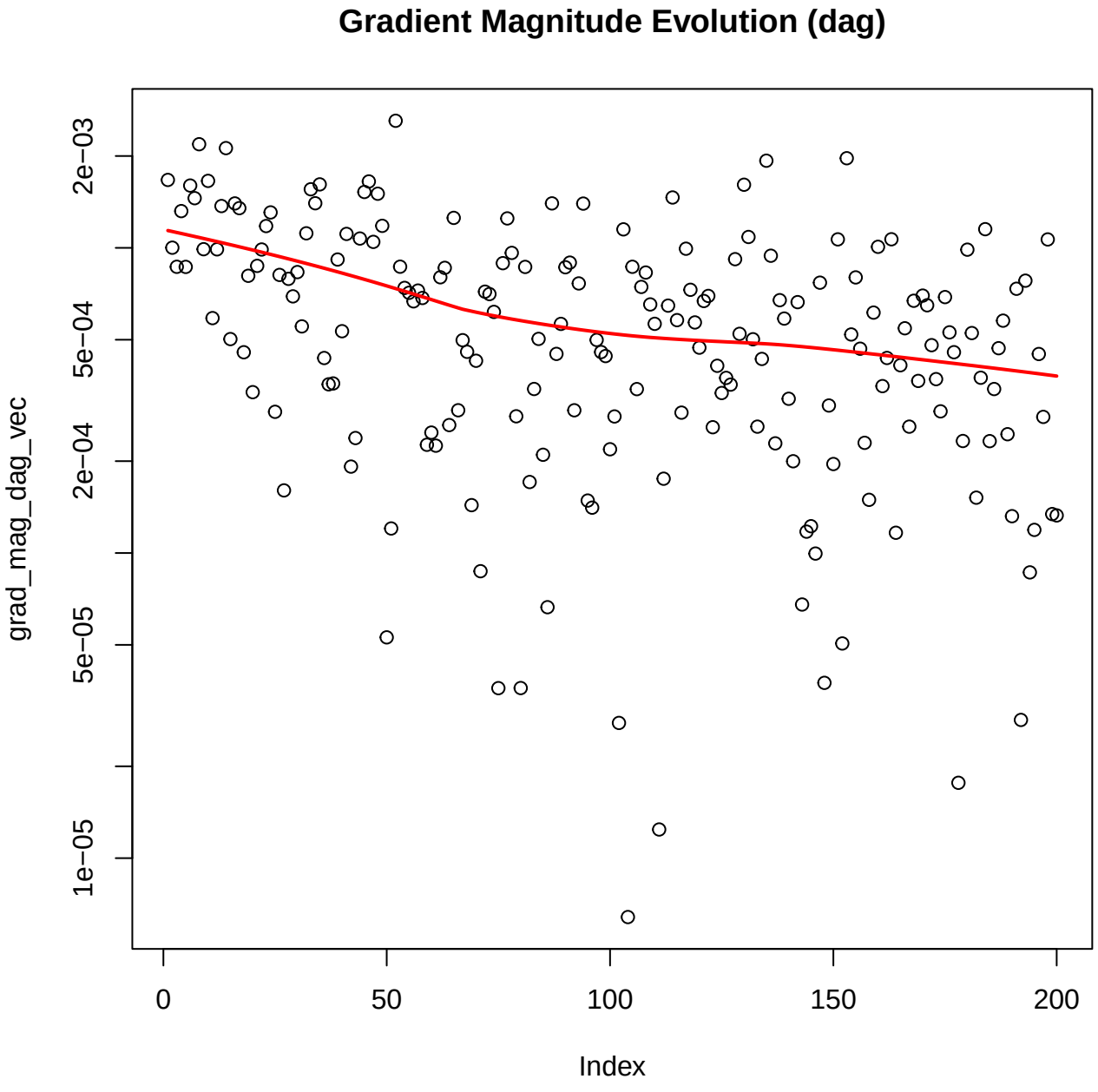

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
5e−03
2e−03
1e−03
5e−04
2e−04
0
10
20
30
40
50
Index

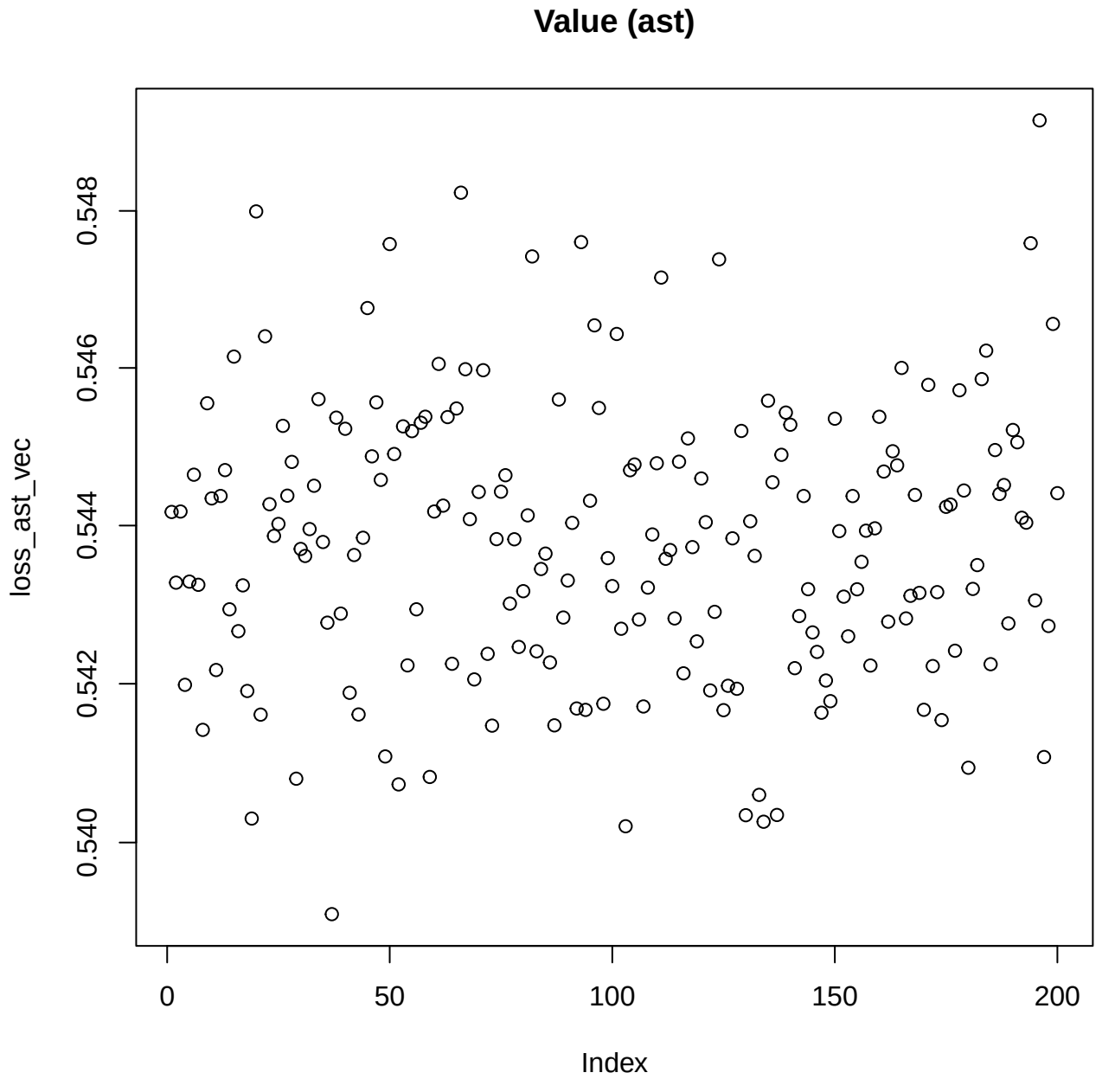

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−02
1e−03
1e−04
1e−05
1e−06
0
10
20
30
40
50
Index

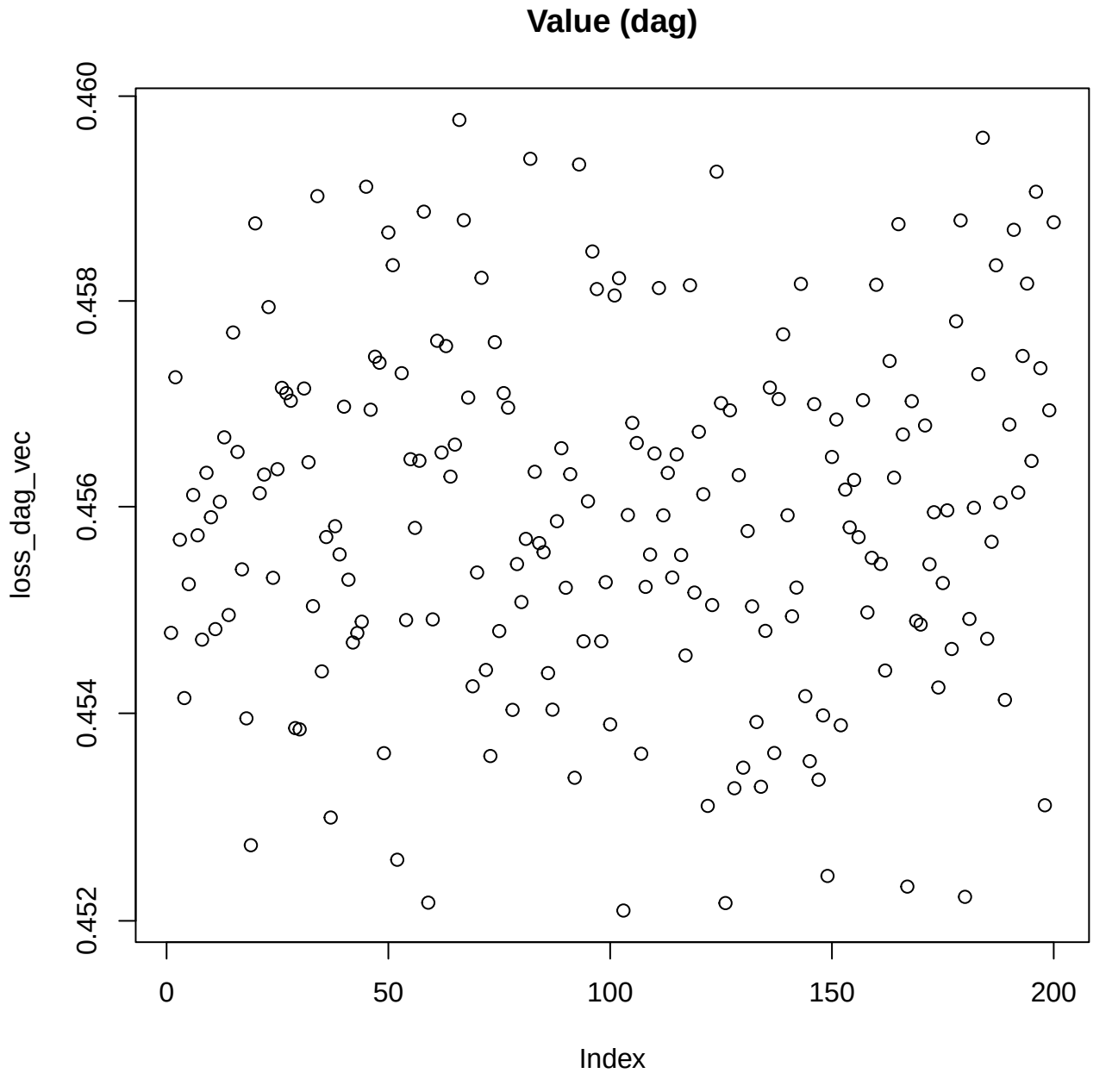

Value (ast)
loss_ast_vec
0.495
0.500
0.505
0.510
0
10
20
30
40
50
Index

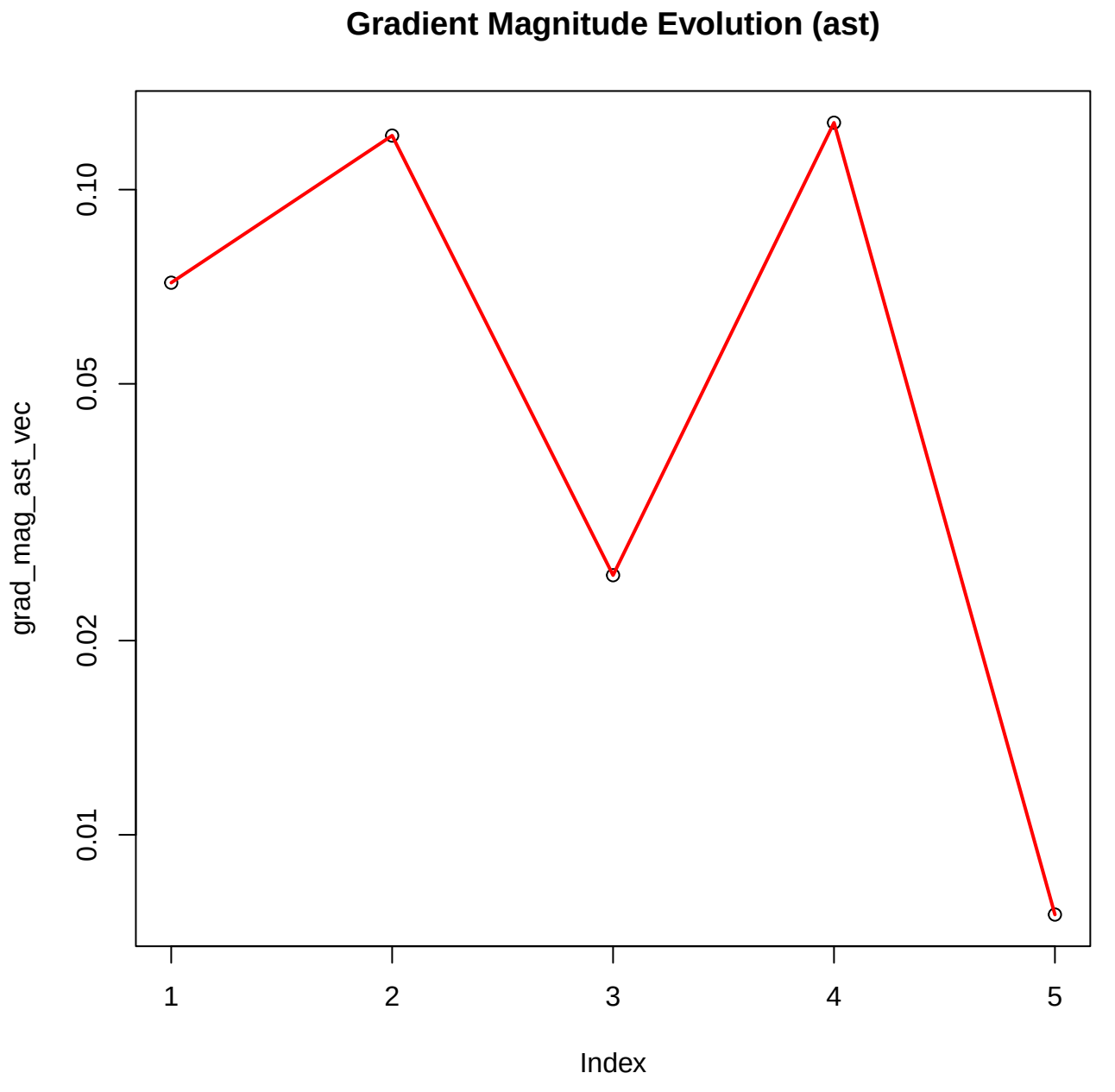

Value (dag)
loss_dag_vec
0.485
0.490
0.495
0.500
0.505
0
10
20
30
40
50
Index

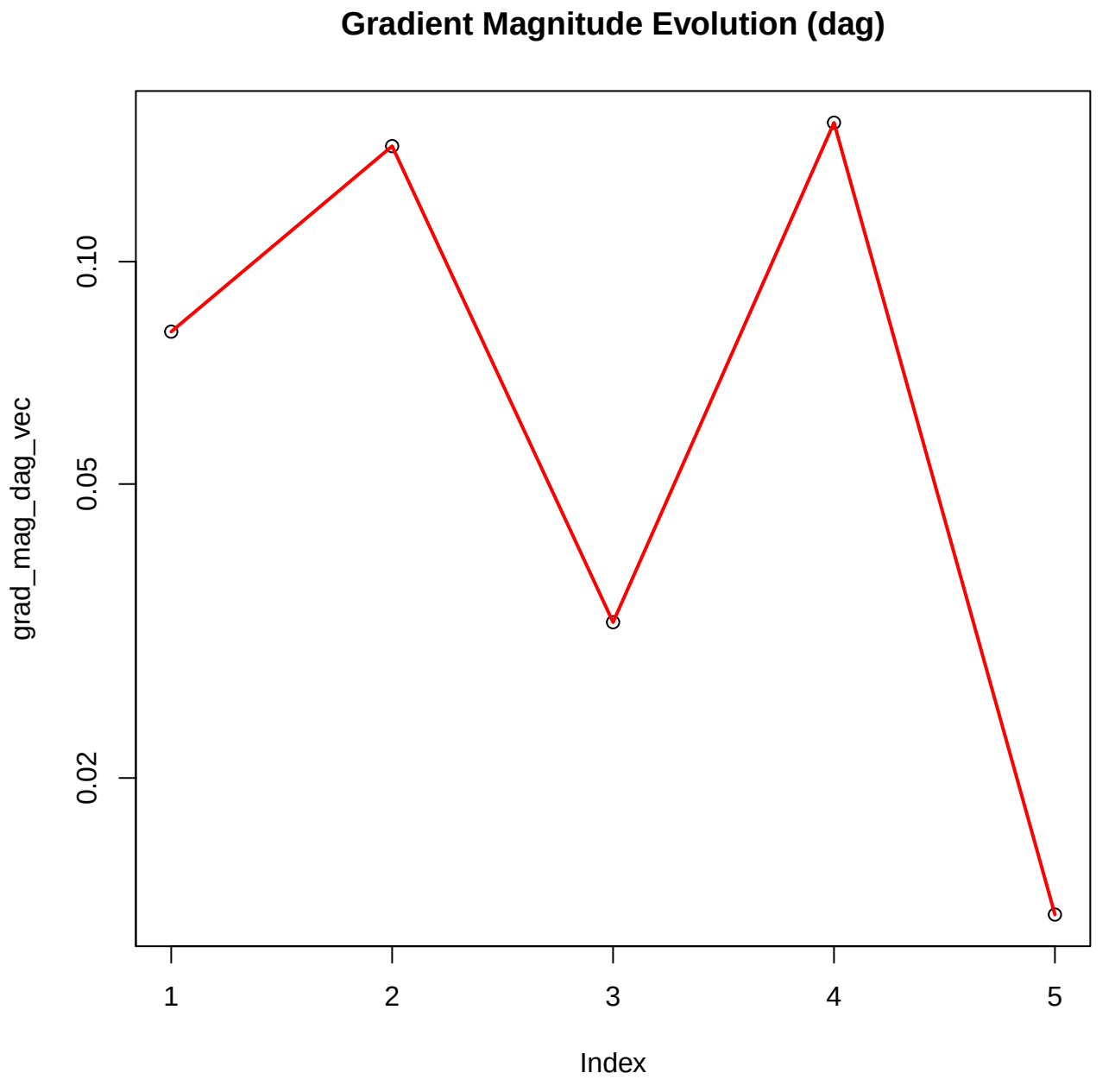

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

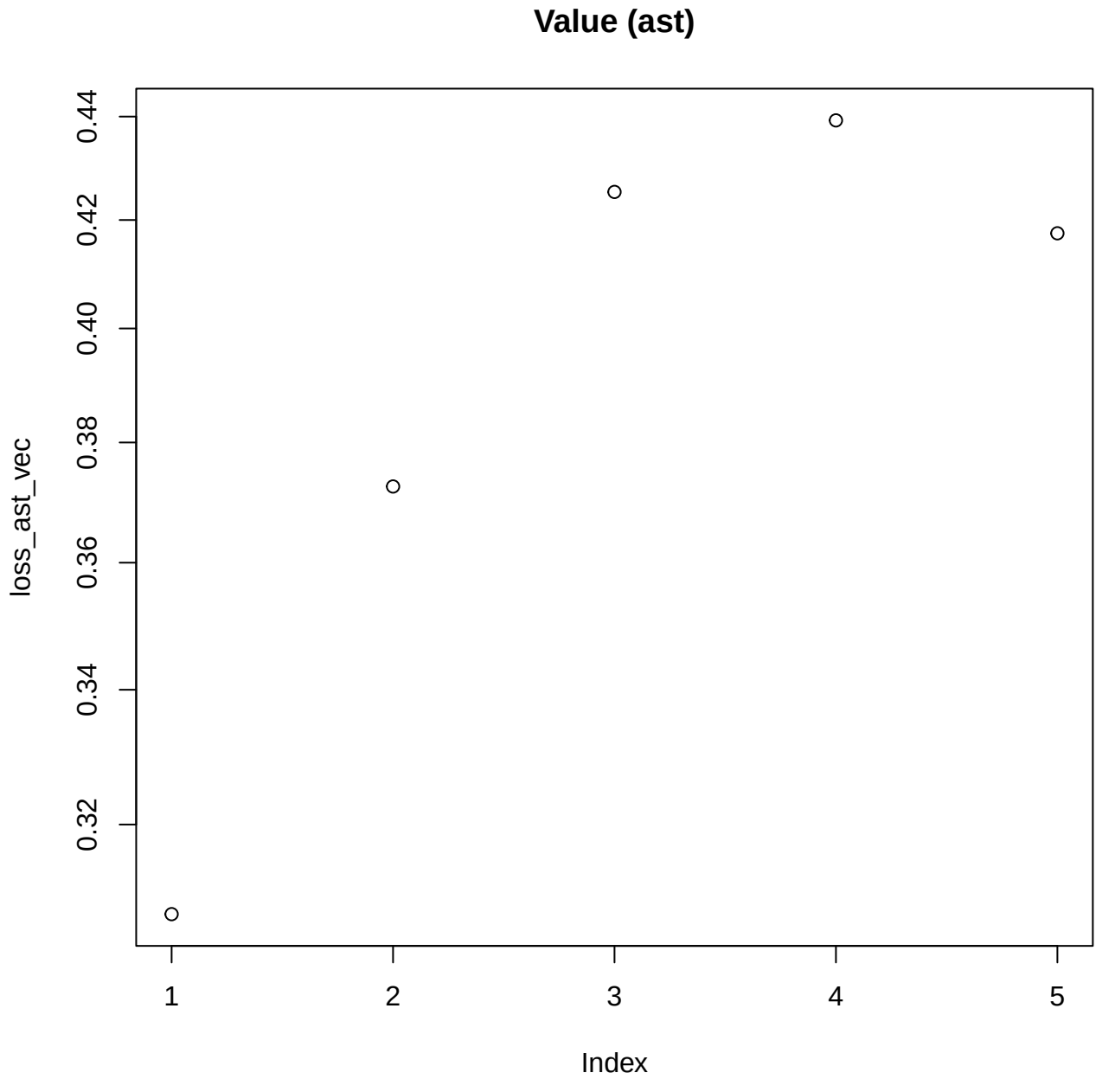

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

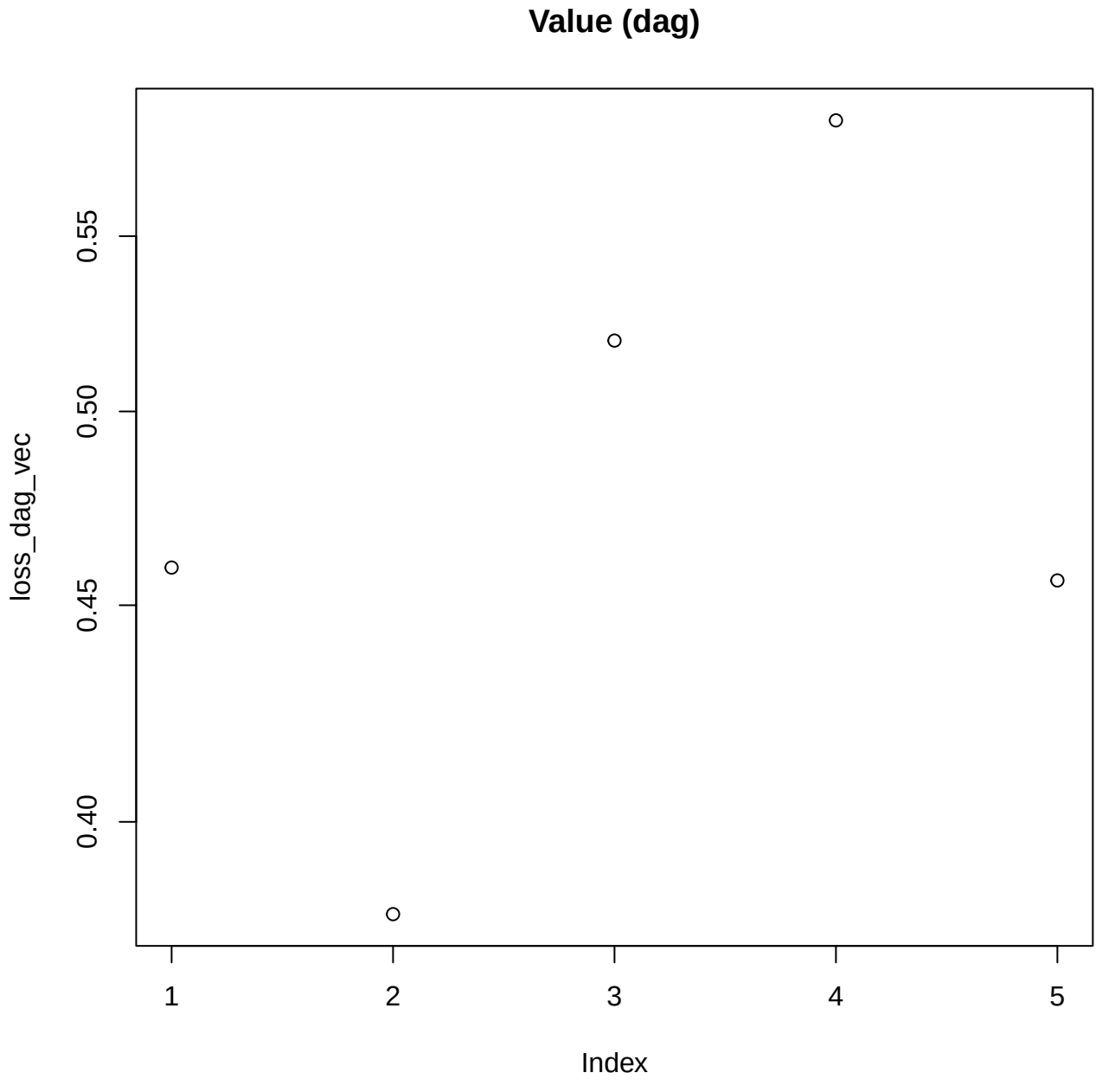

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

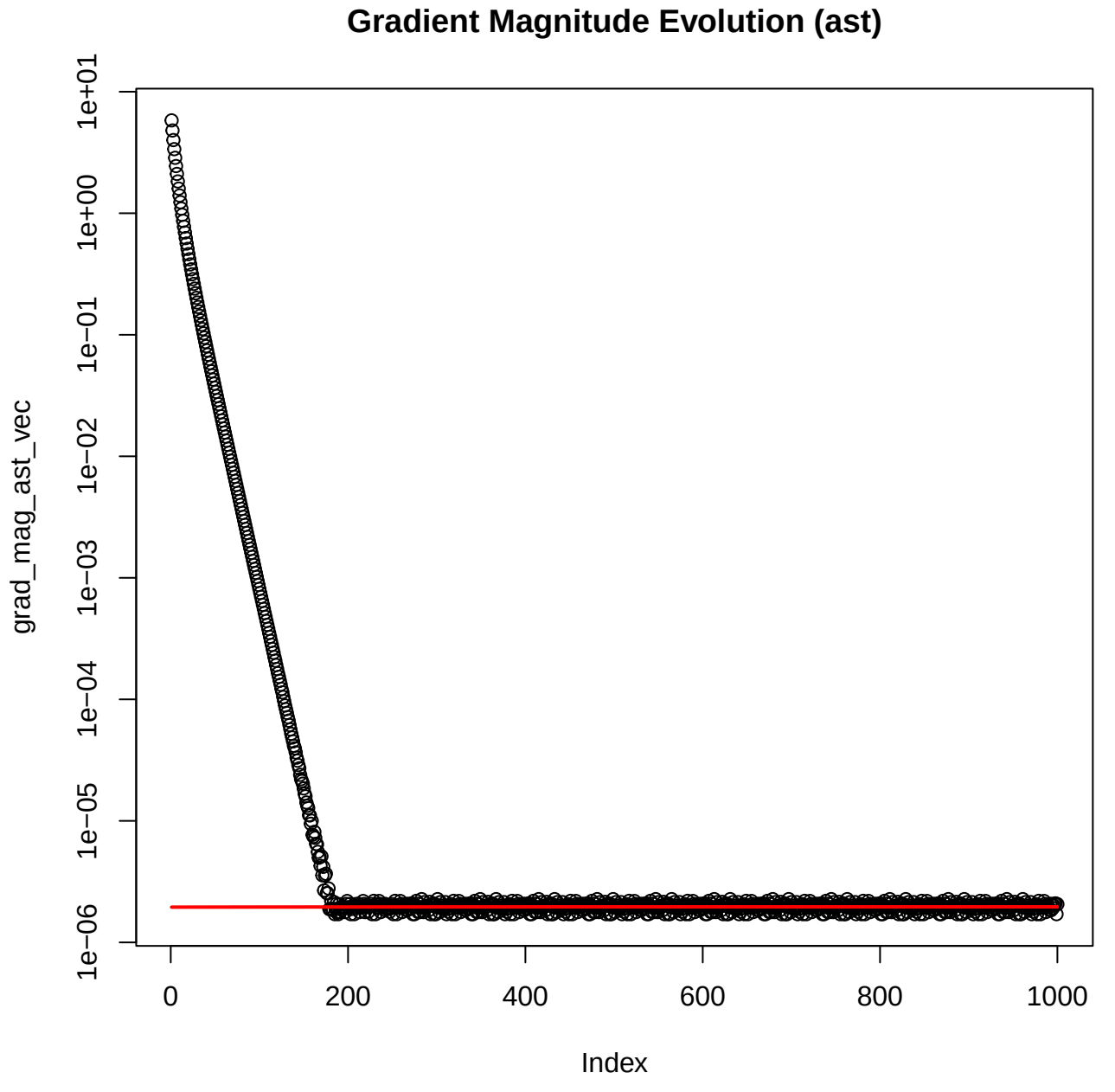

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

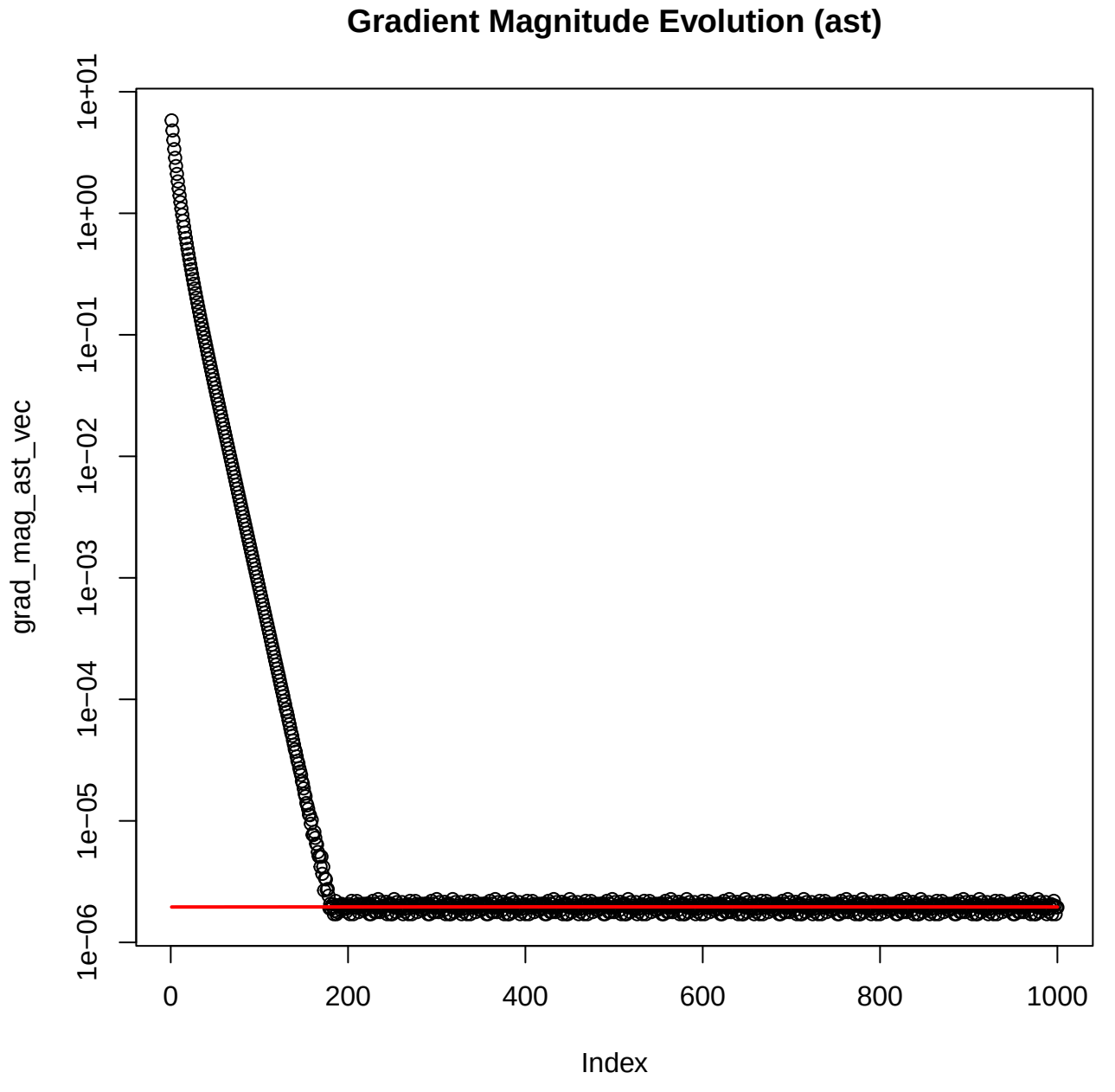

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

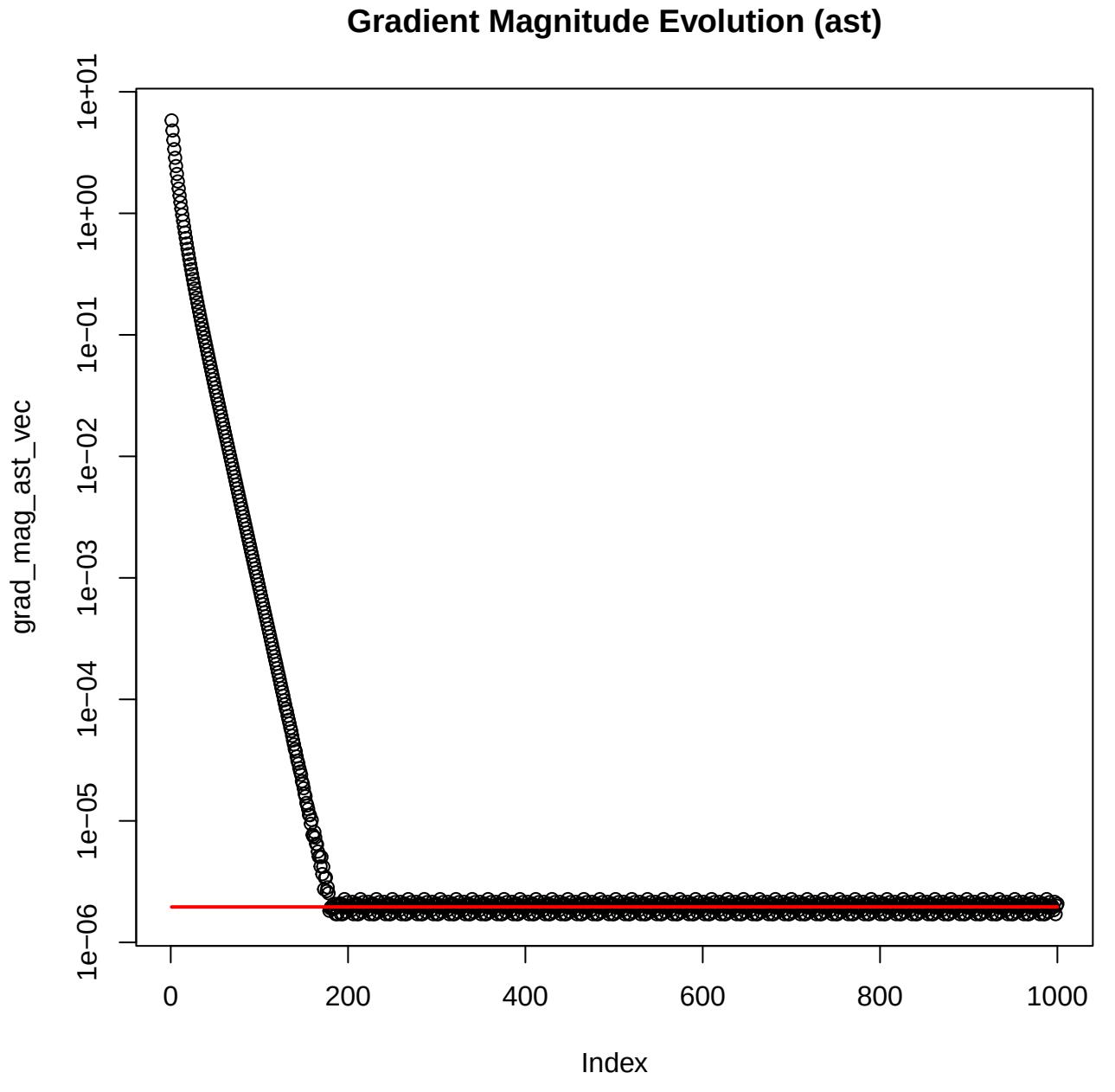

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

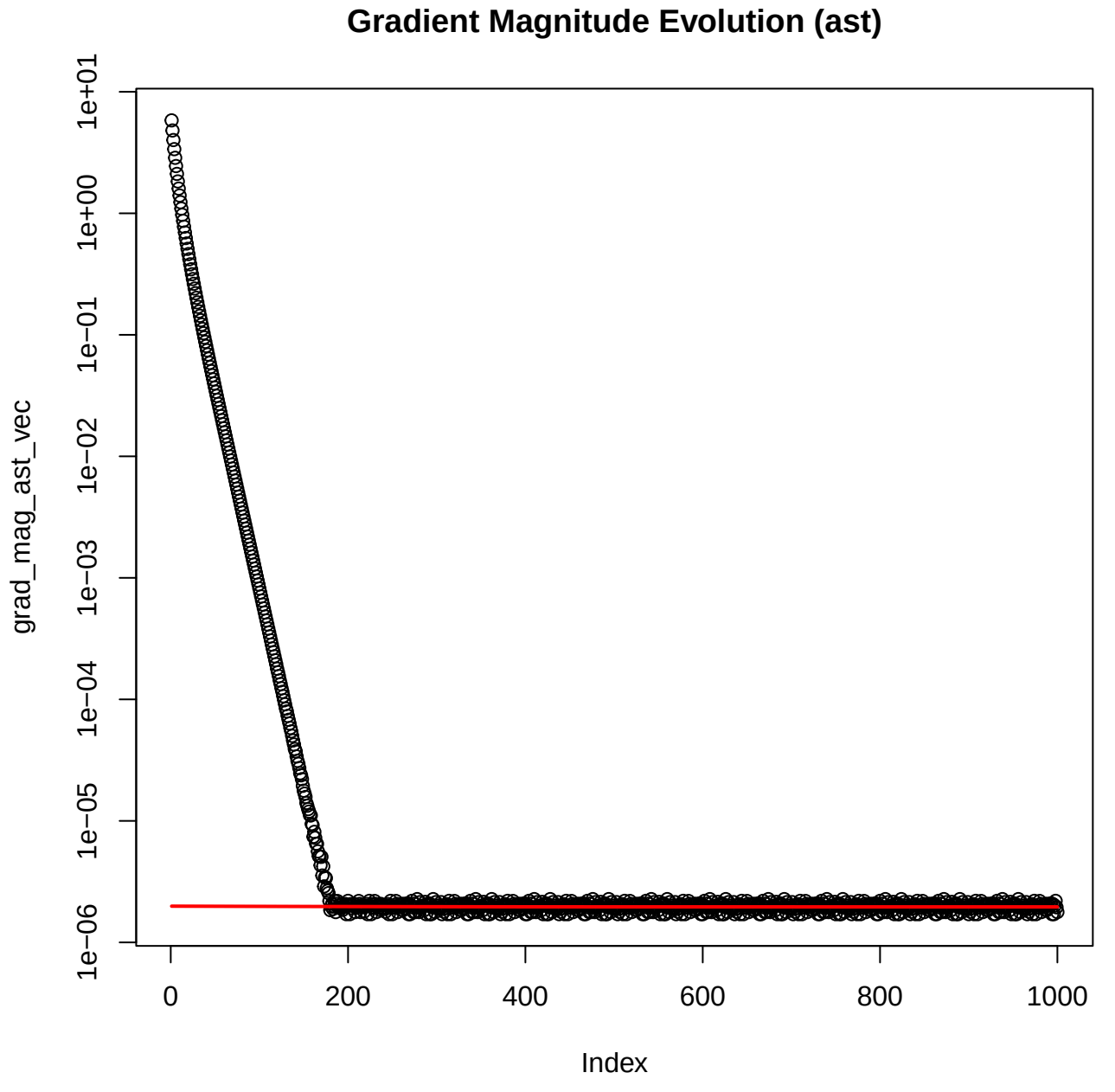

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

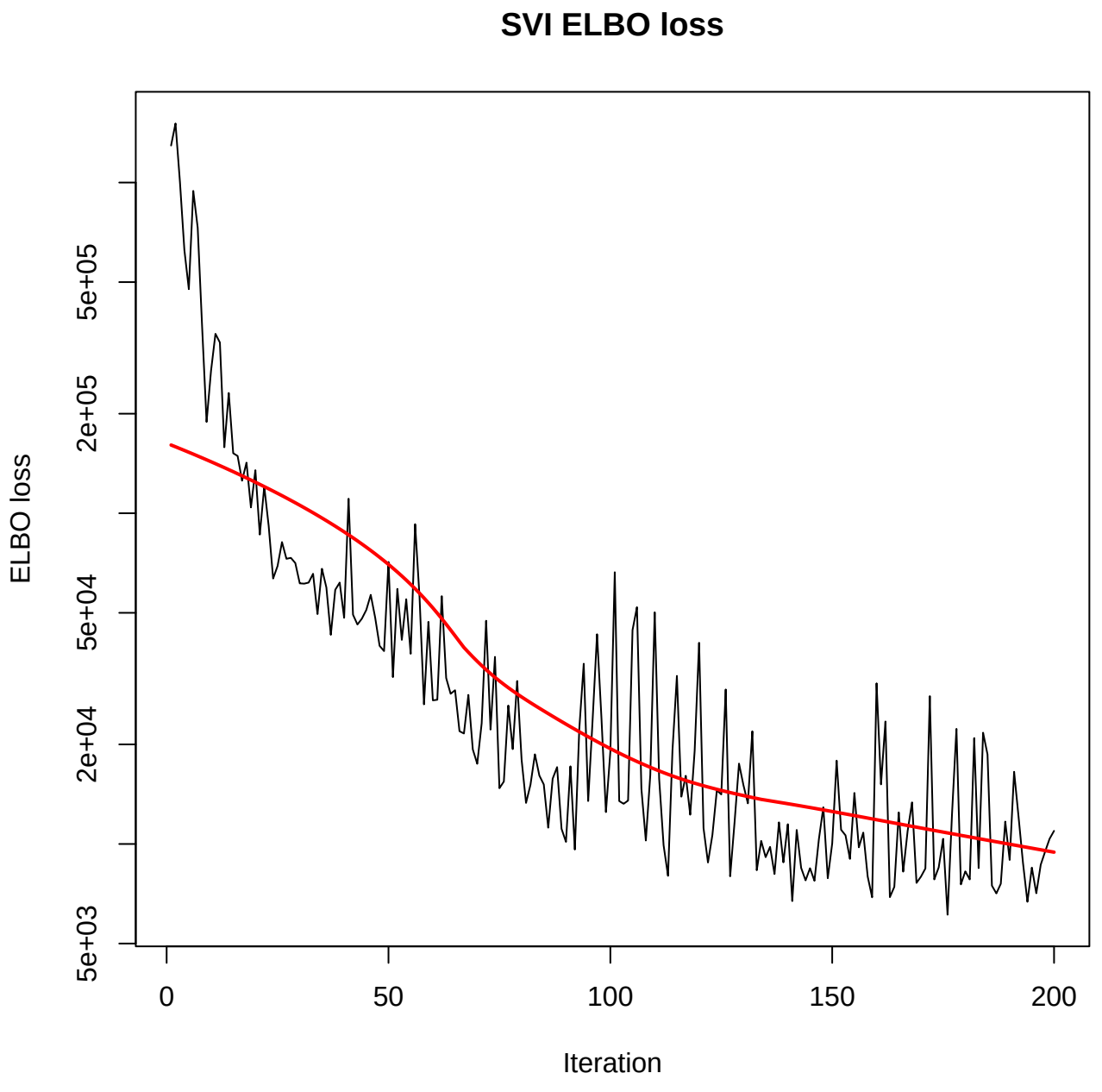

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

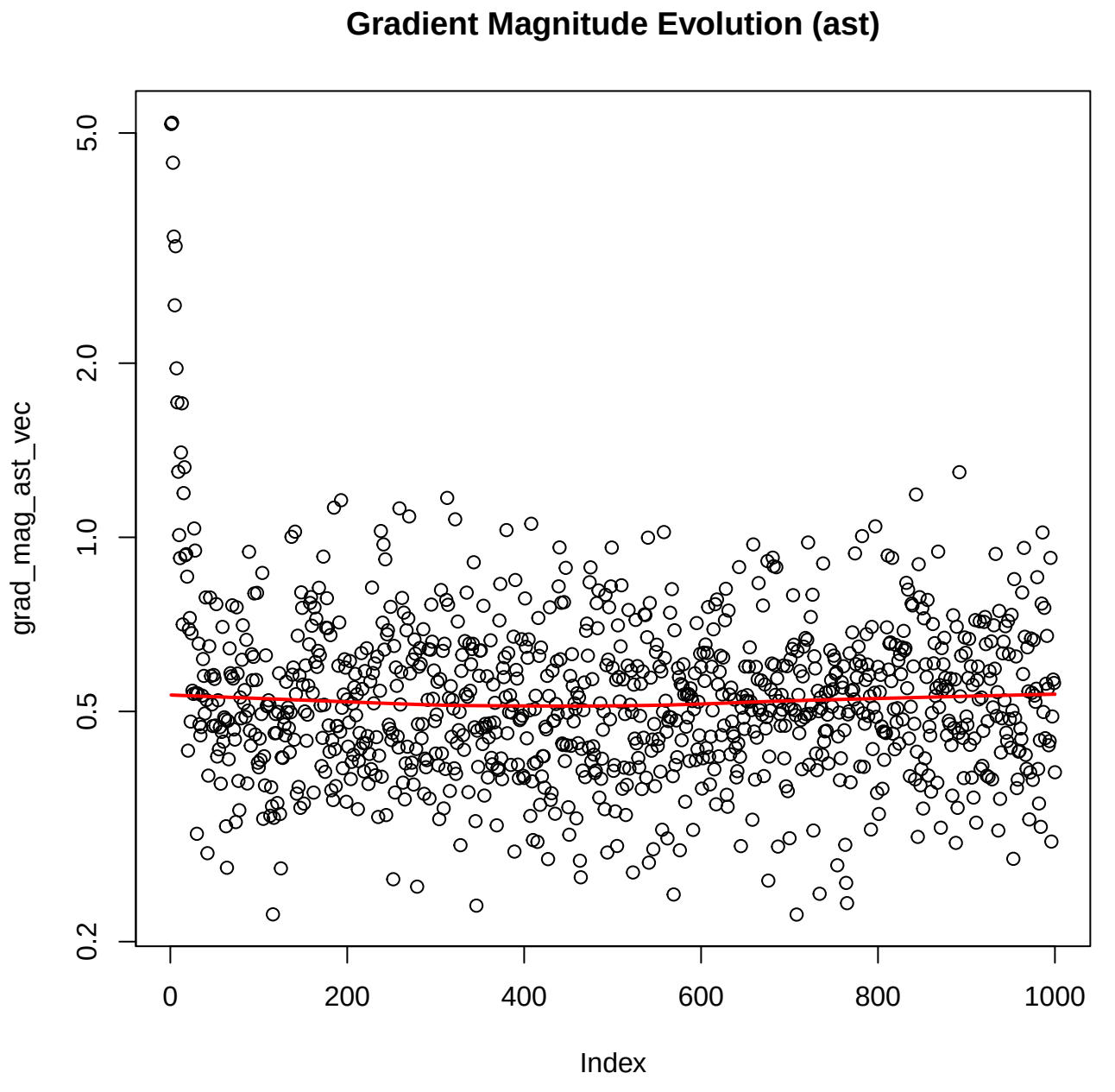

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

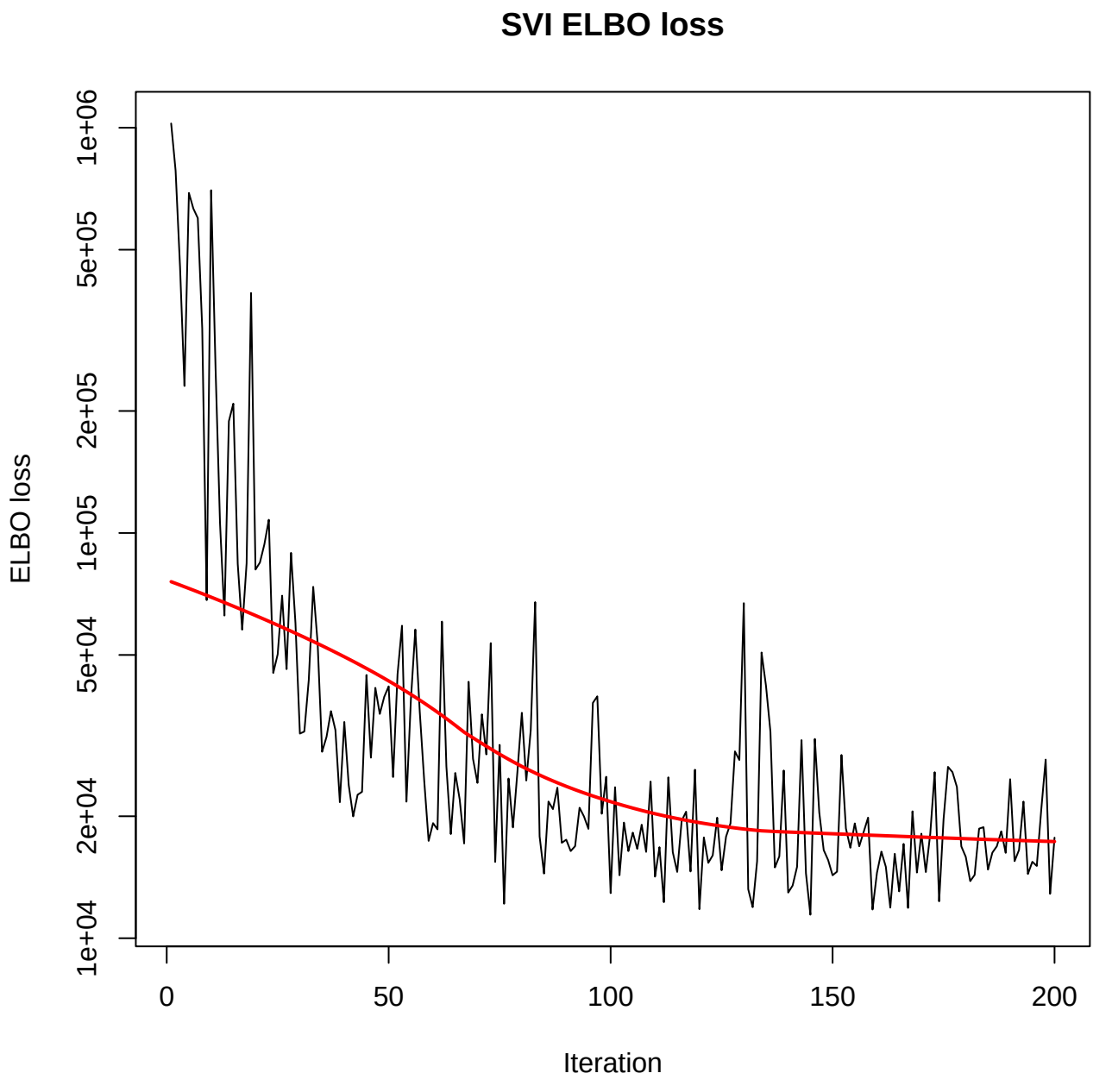

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

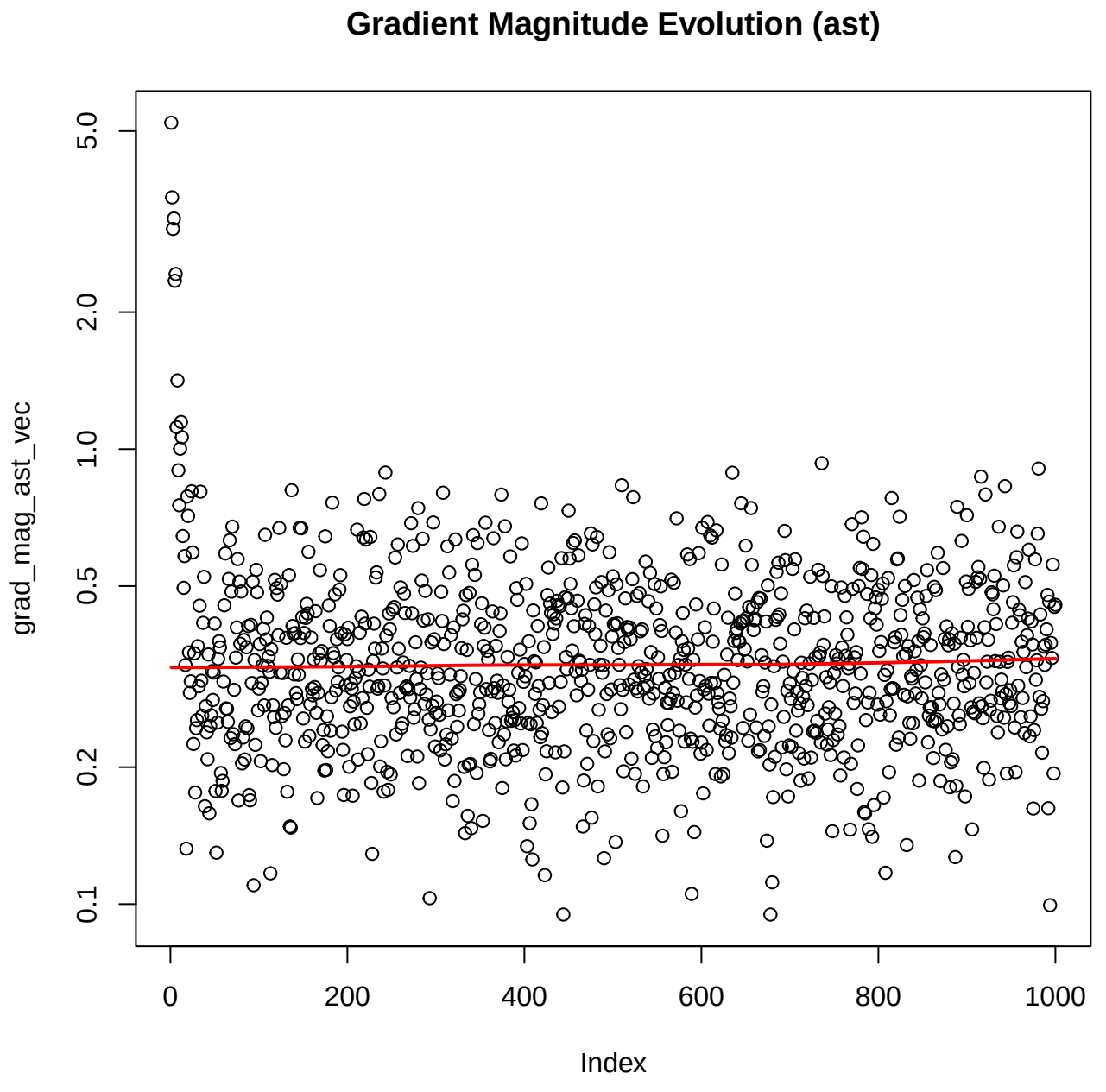

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

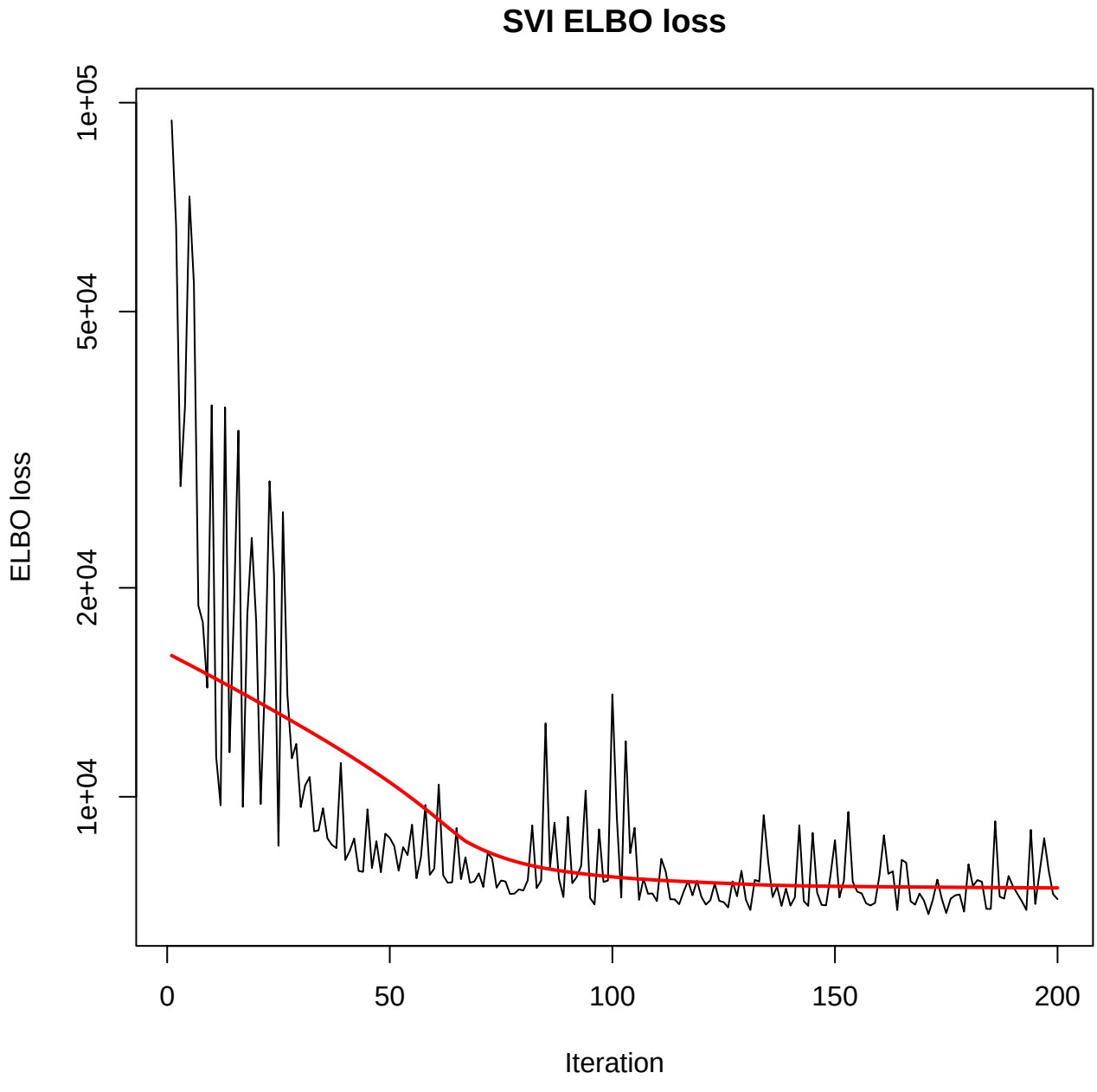

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

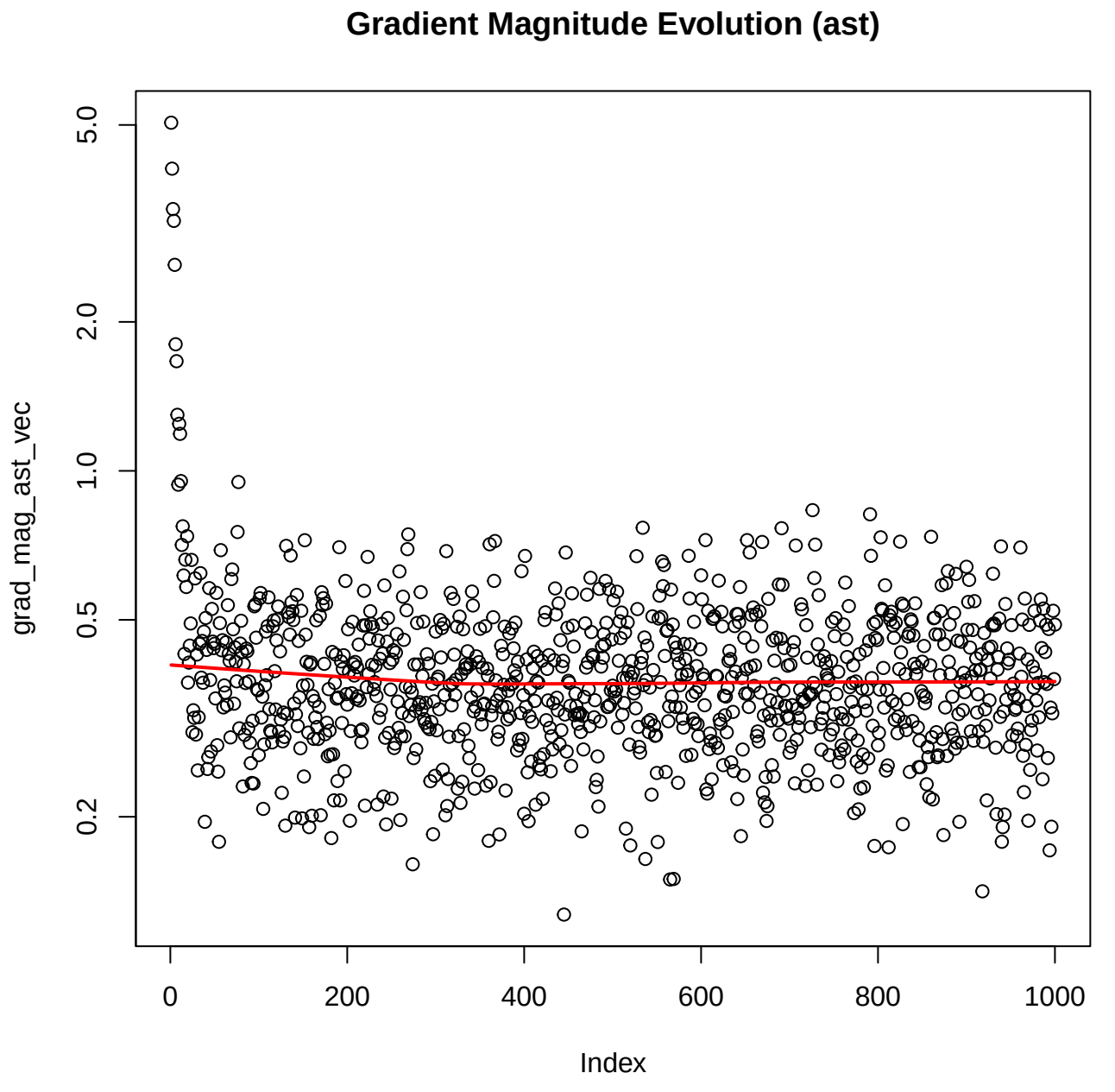

SVI ELBO loss
ELBO loss
5e+05
2e+05
5e+04
2e+04
5e+03
0
50
100
150
200
Iteration

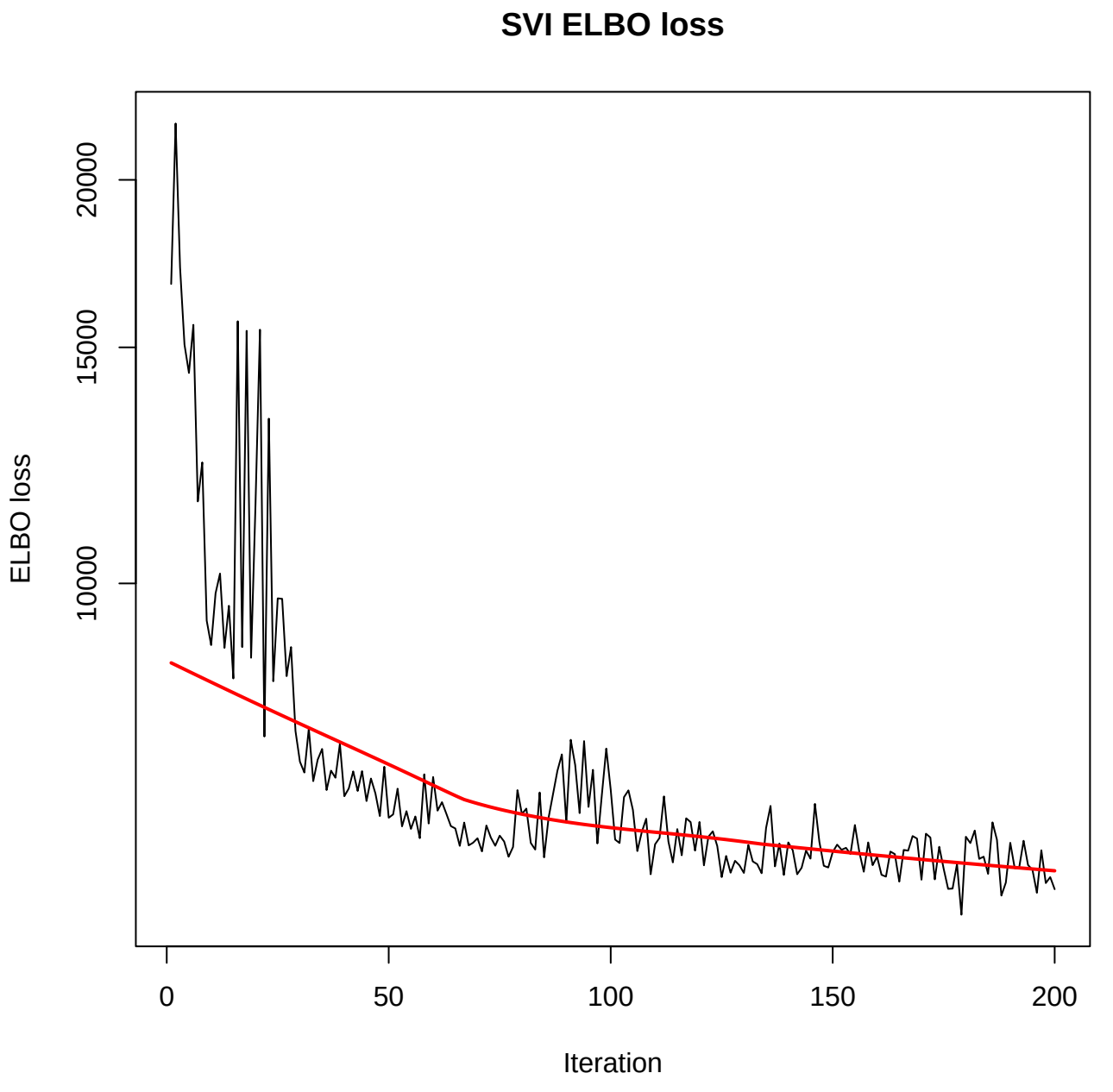

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
1e+06
5e+05
2e+05
1e+05
5e+04
2e+04
1e+04
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0.1
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
1e+05
5e+04
2e+04
1e+04
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0
200
400
600
800
1000
Index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ELBO loss
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10000
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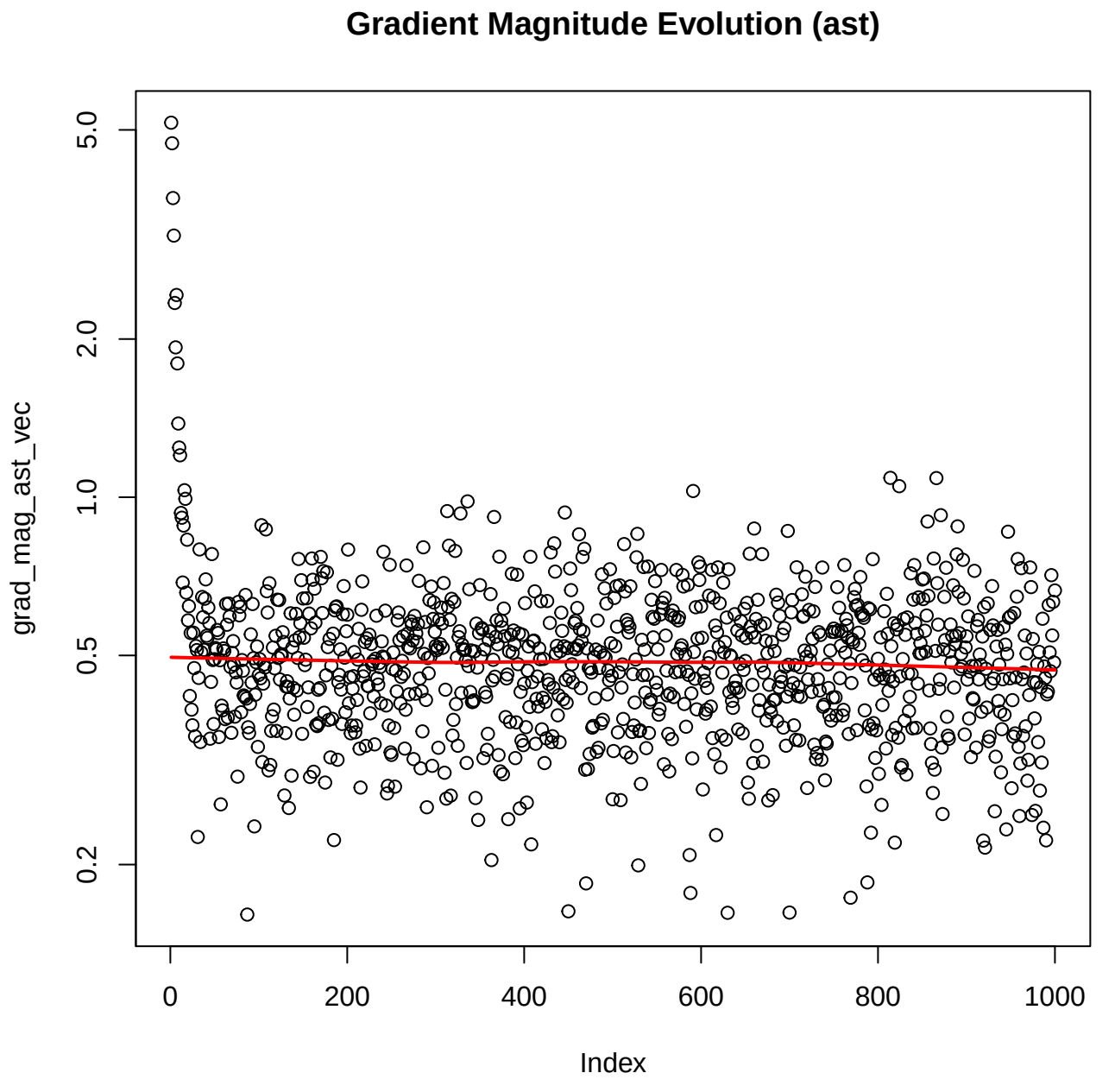
200
Iteration


Gradient Magnitude Evolution (ast)
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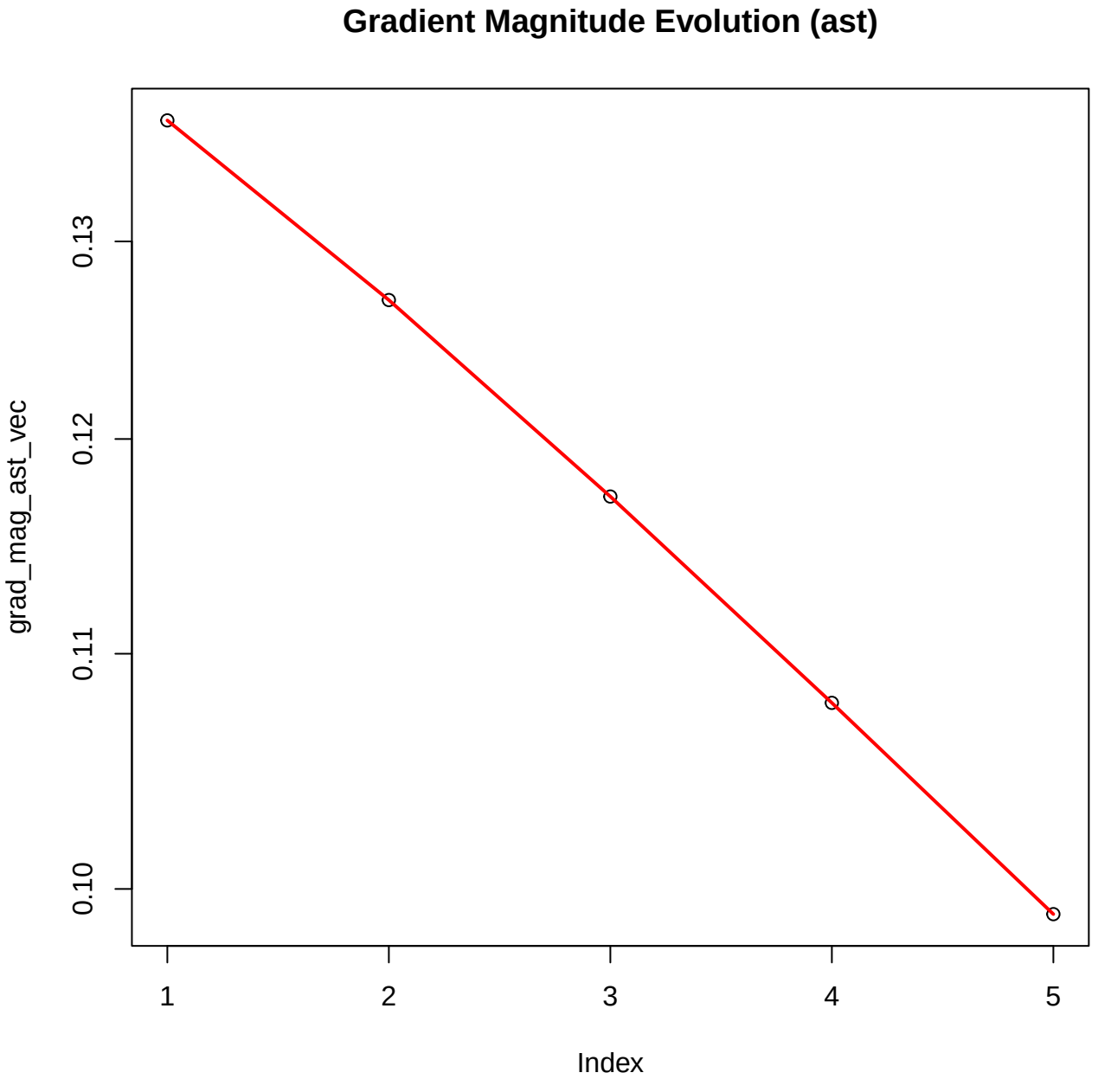
Gradient Magnitude Evolution (ast)
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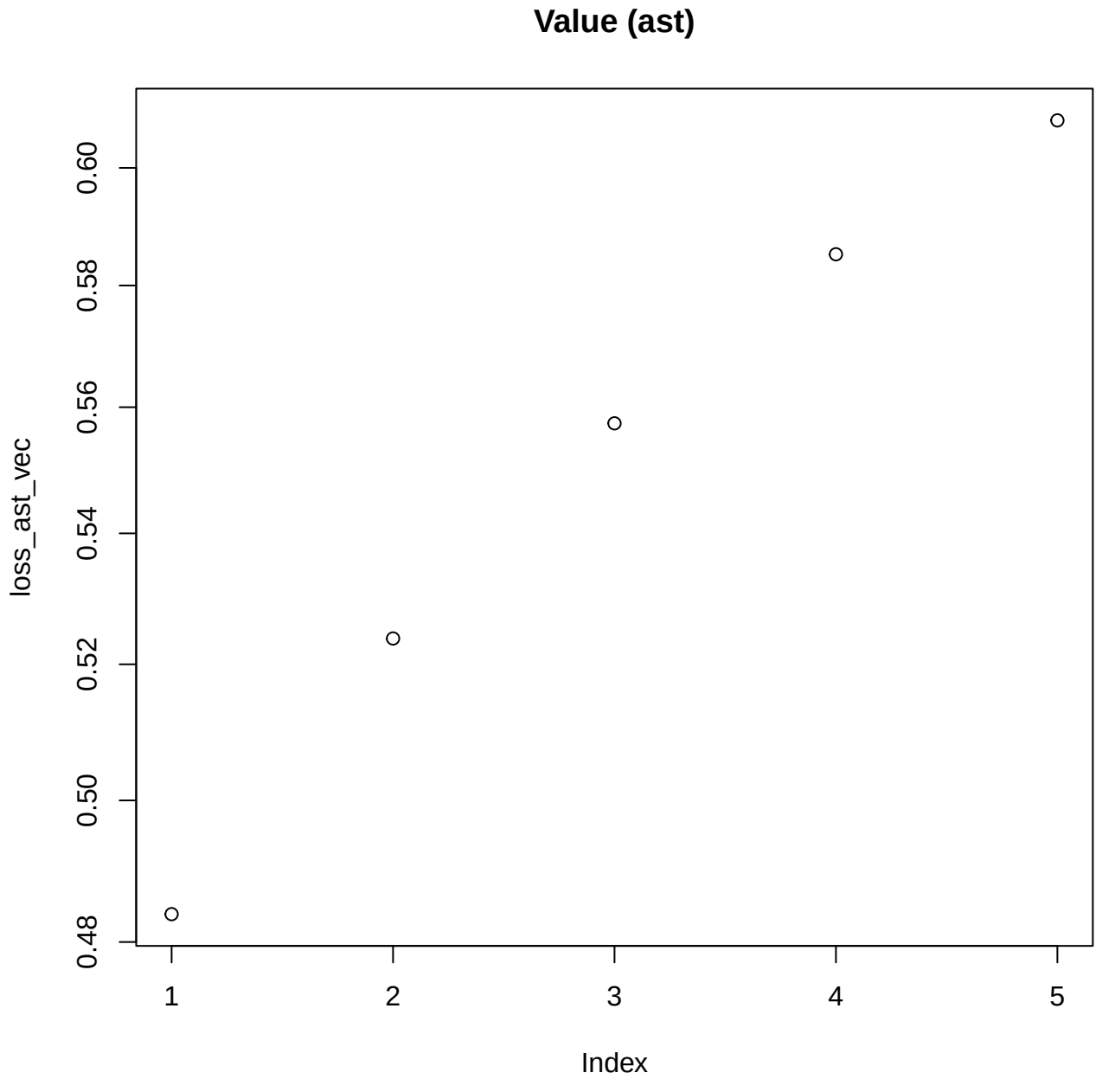
Value (ast)
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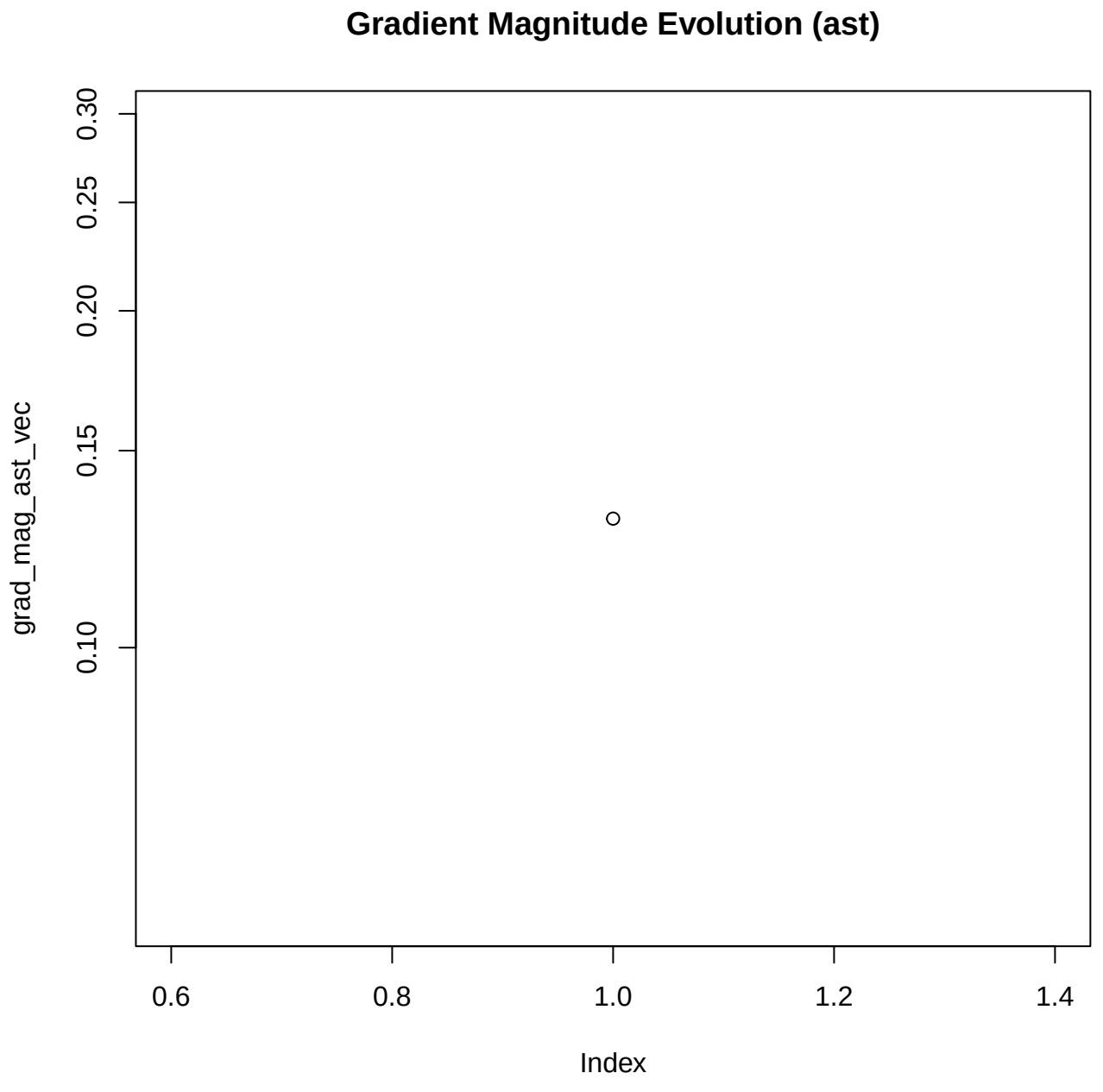
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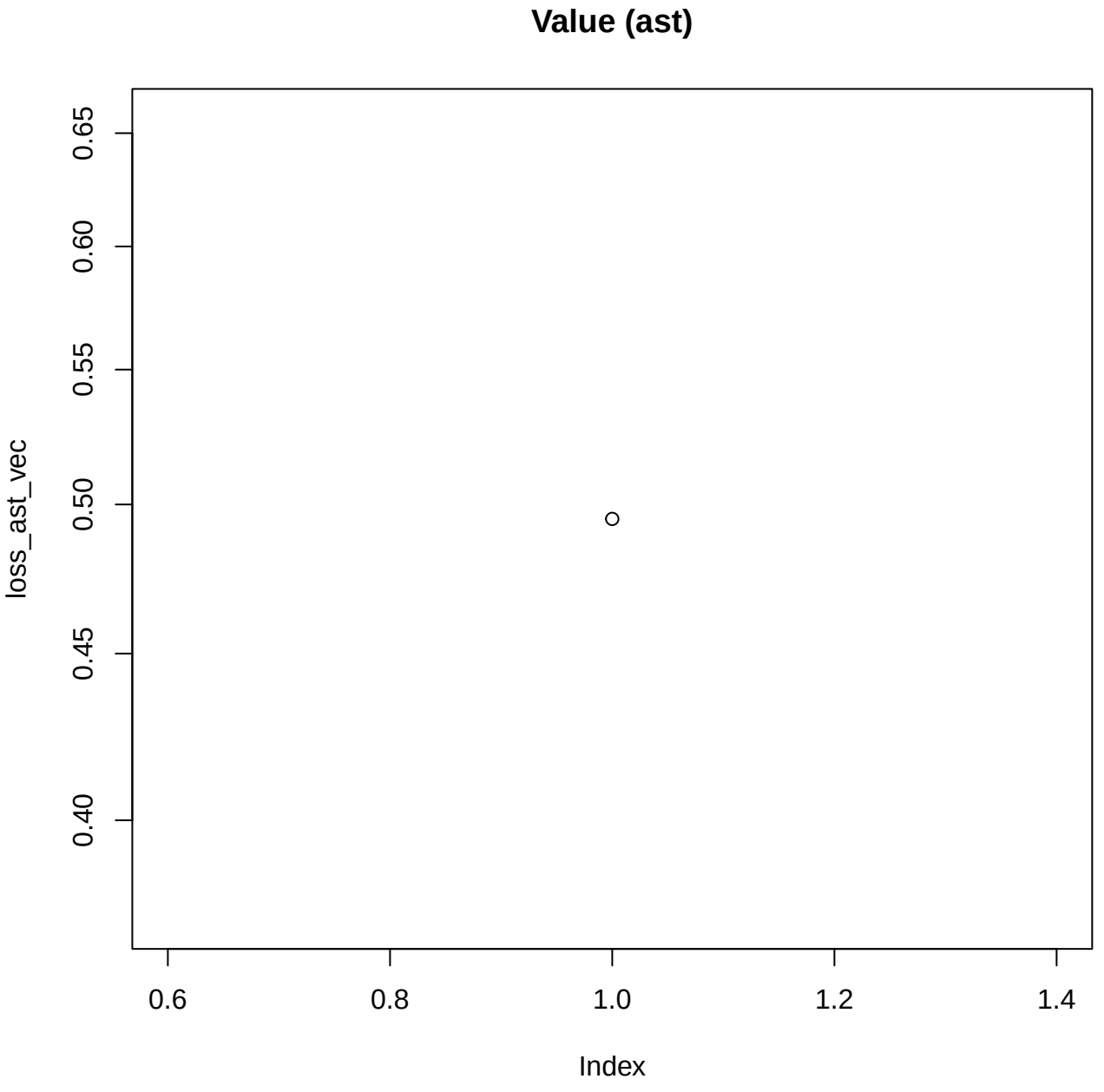
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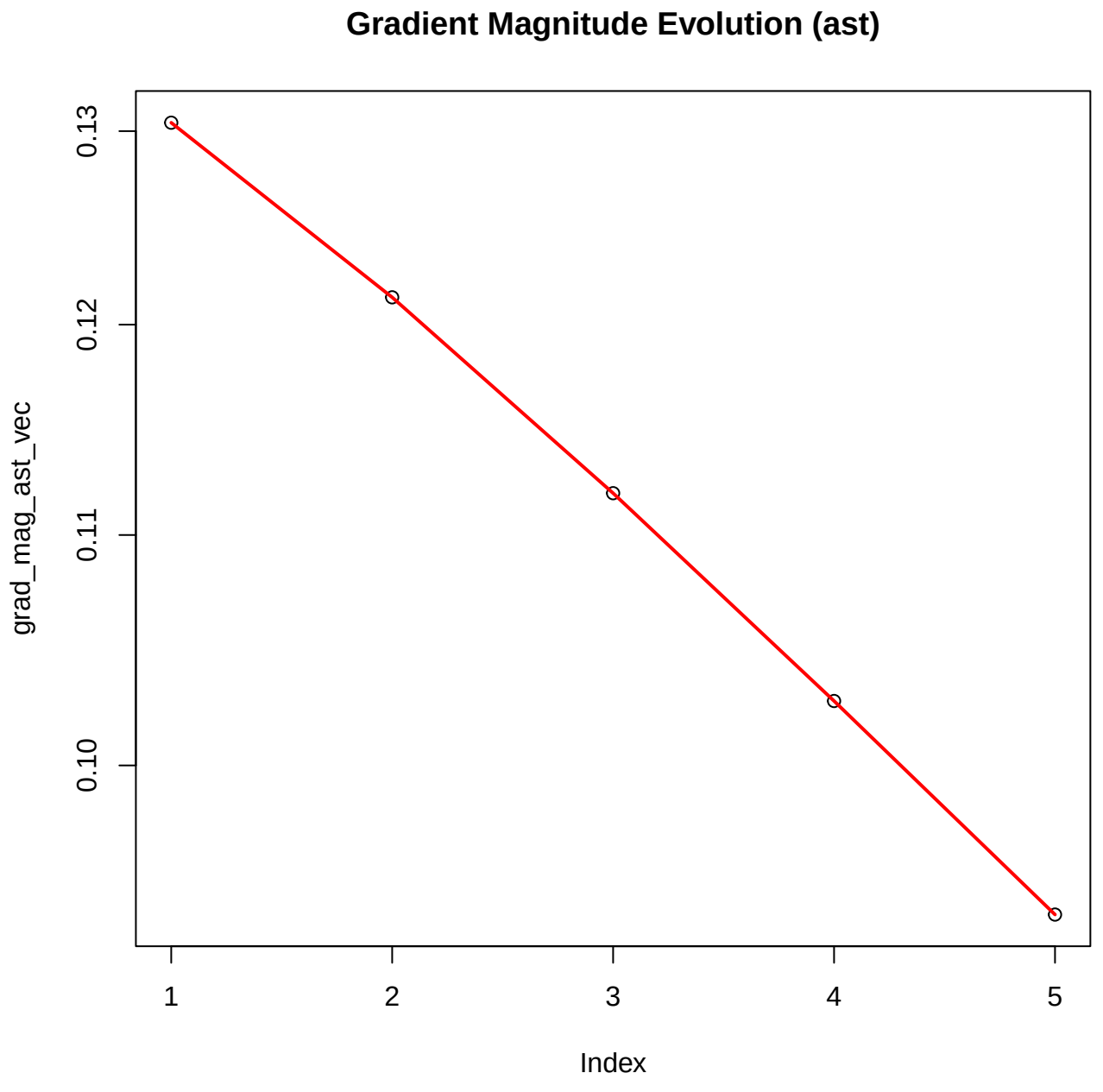
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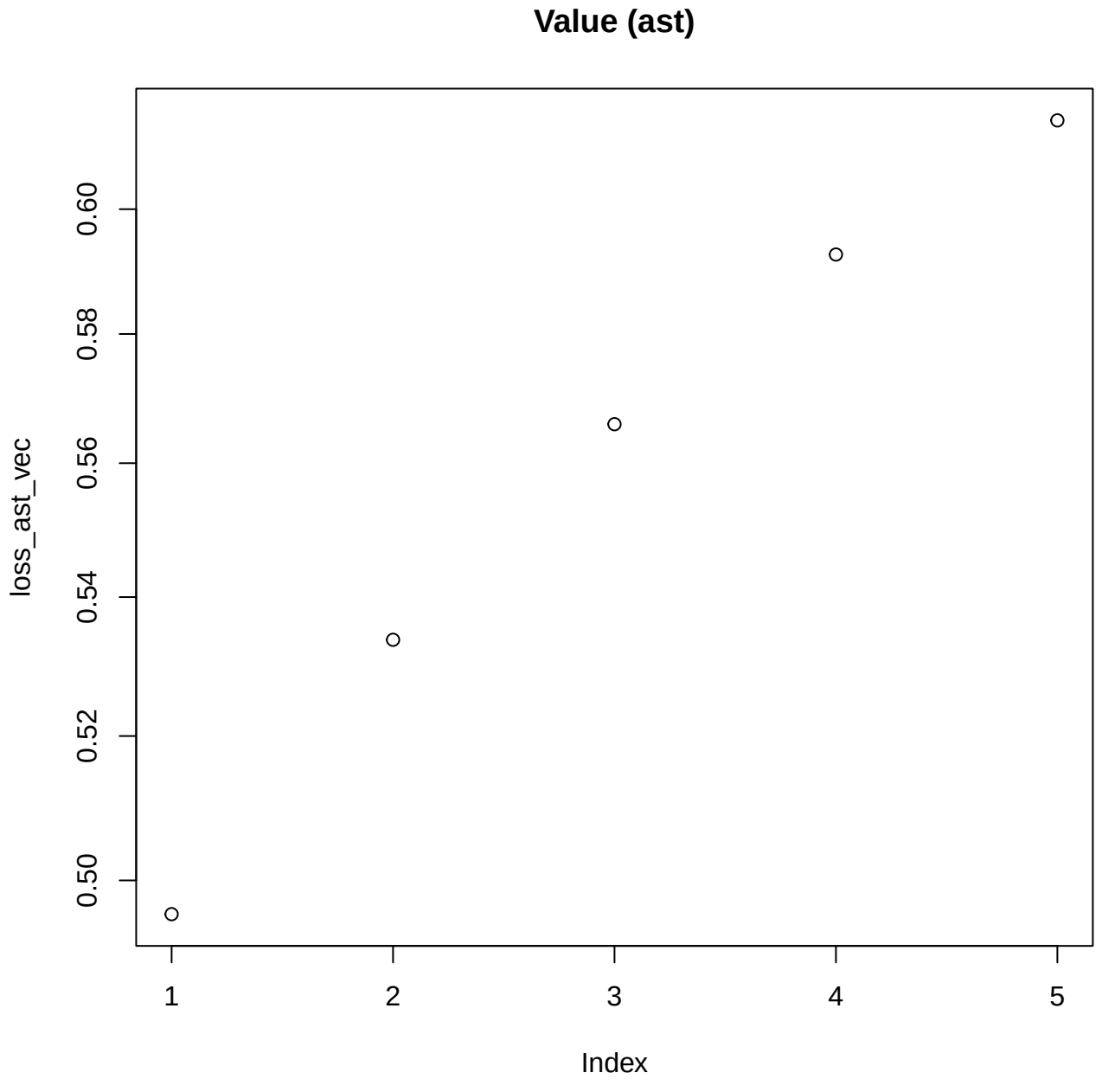
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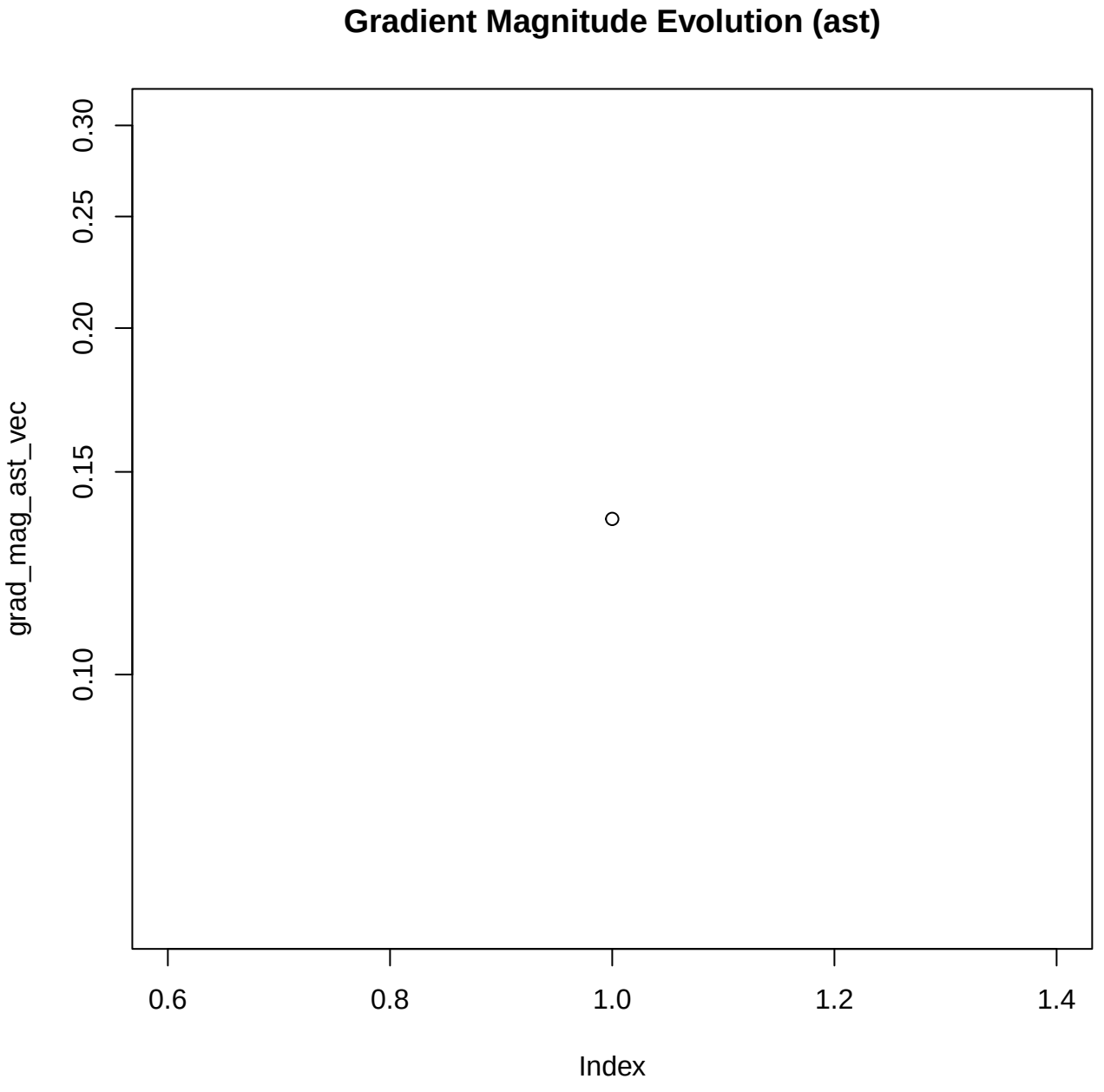
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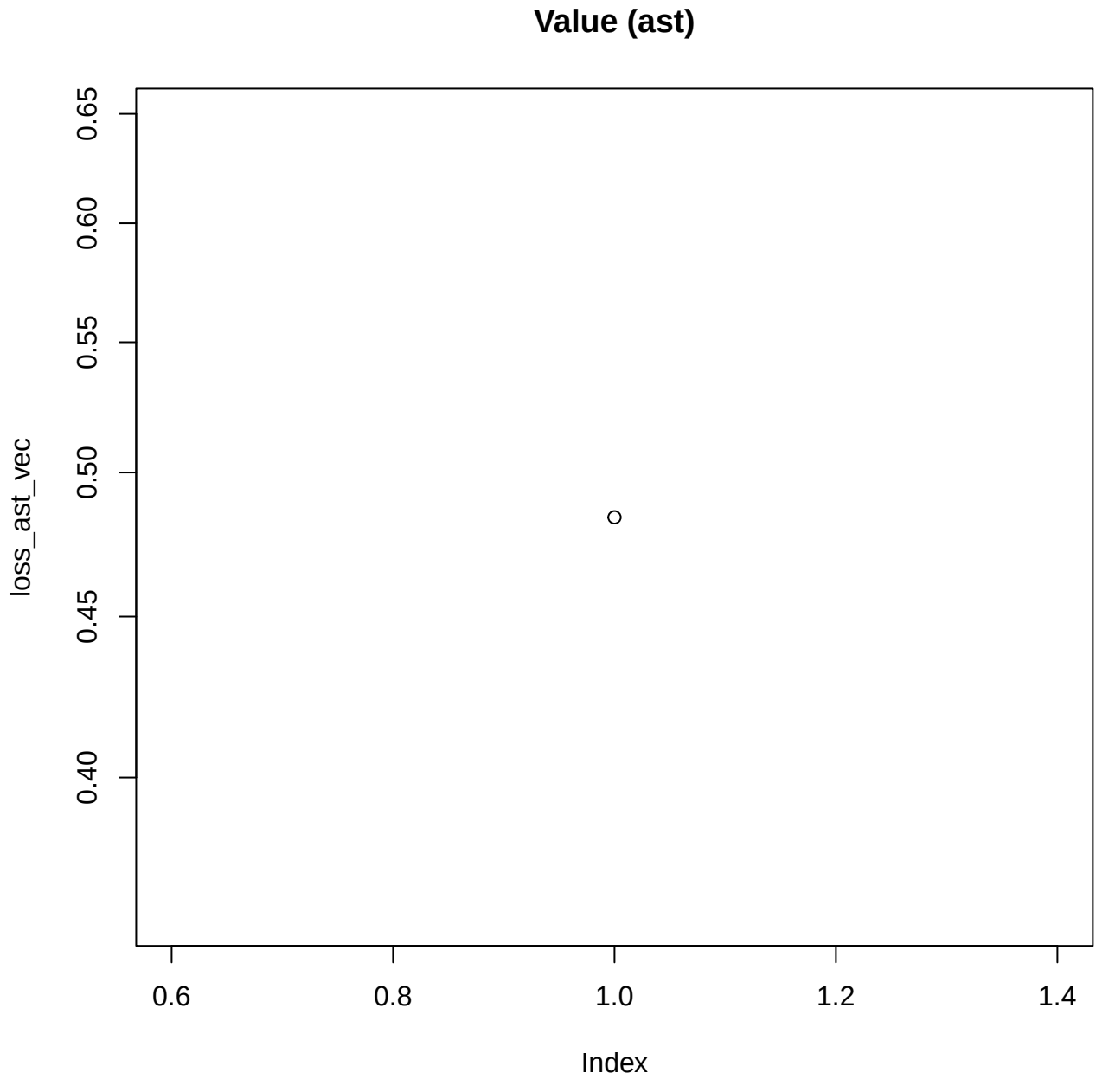
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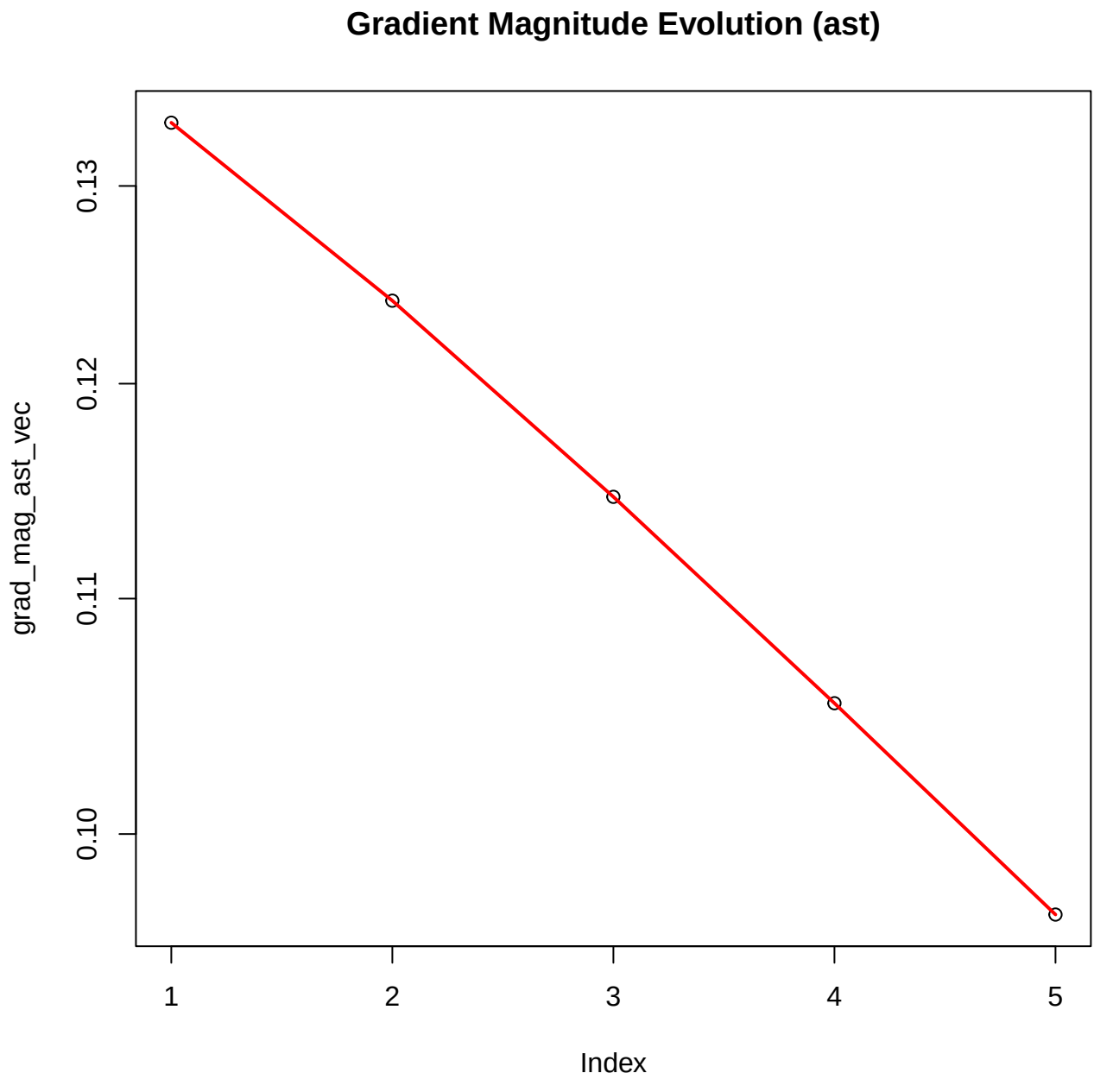
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grad_mag_ast_vec
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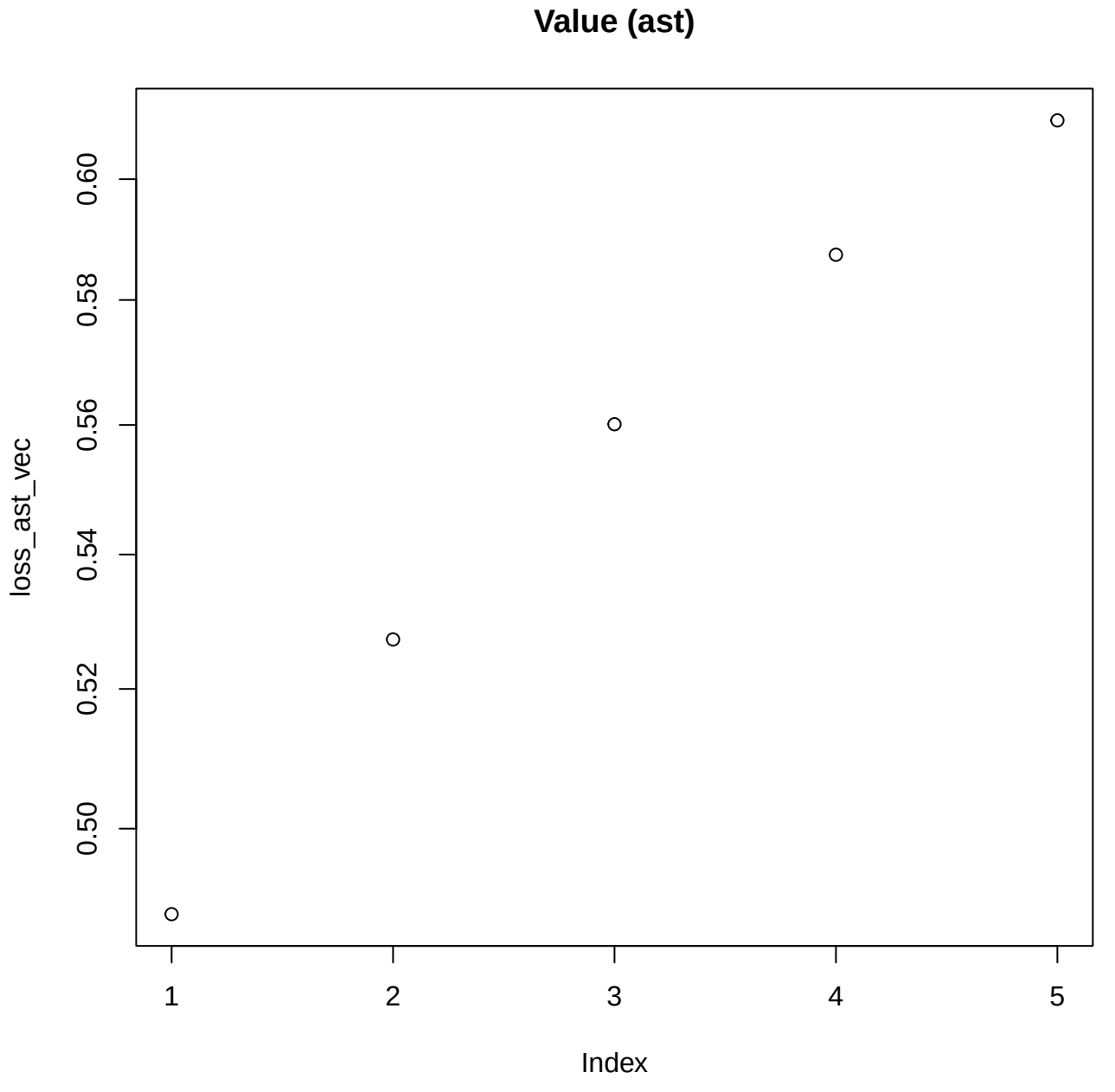
Value (ast)
loss_ast_vec
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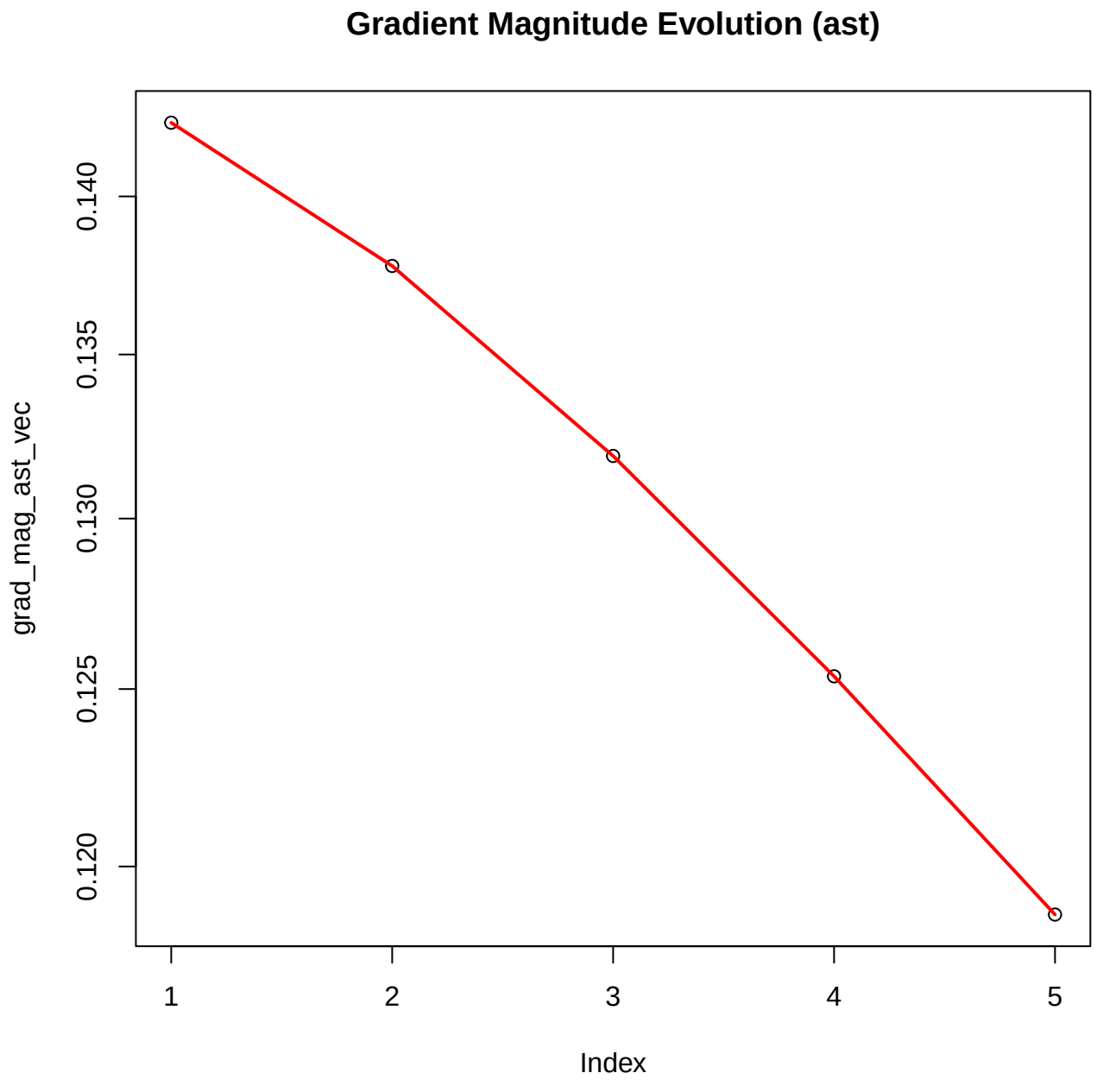
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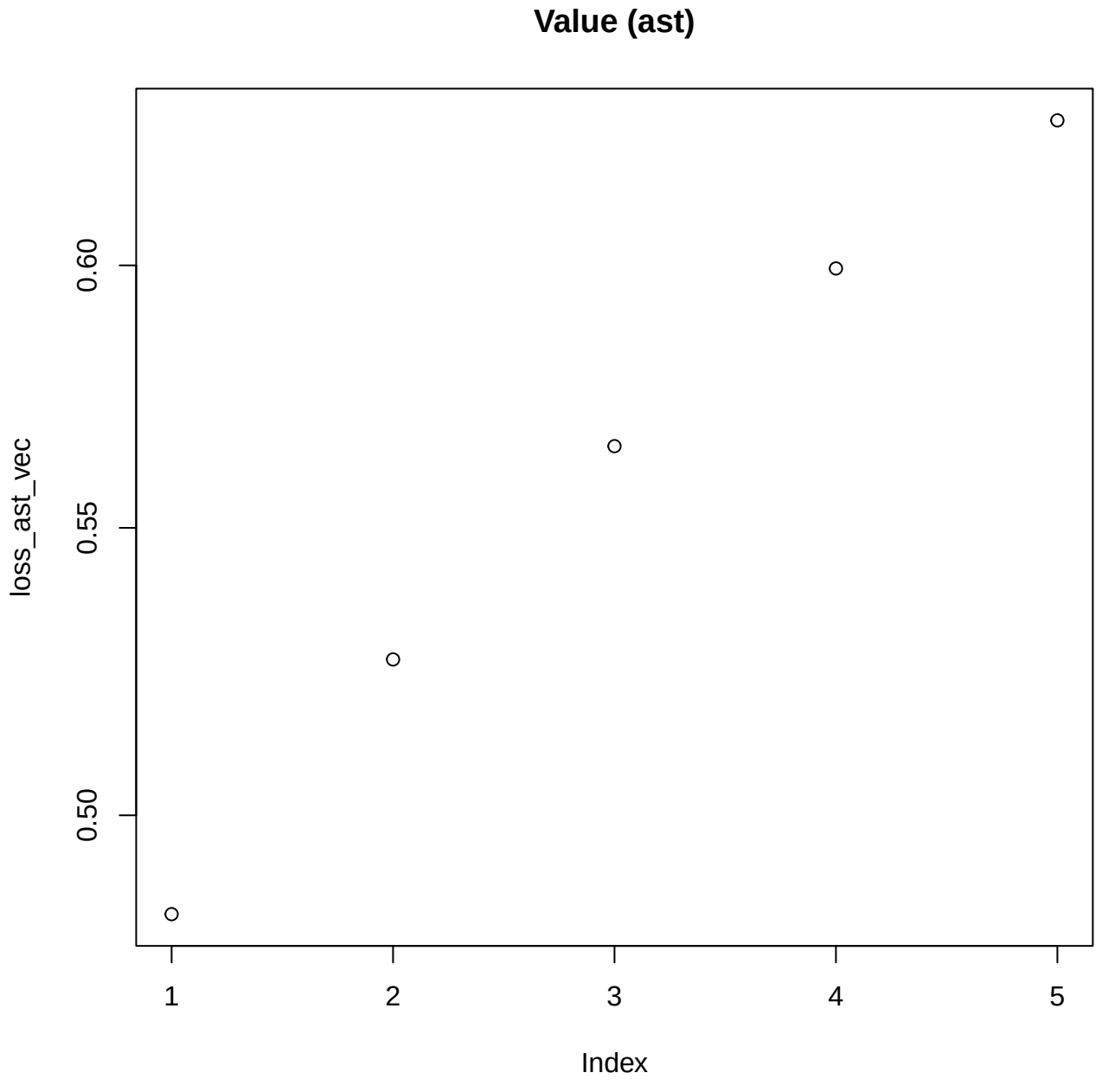
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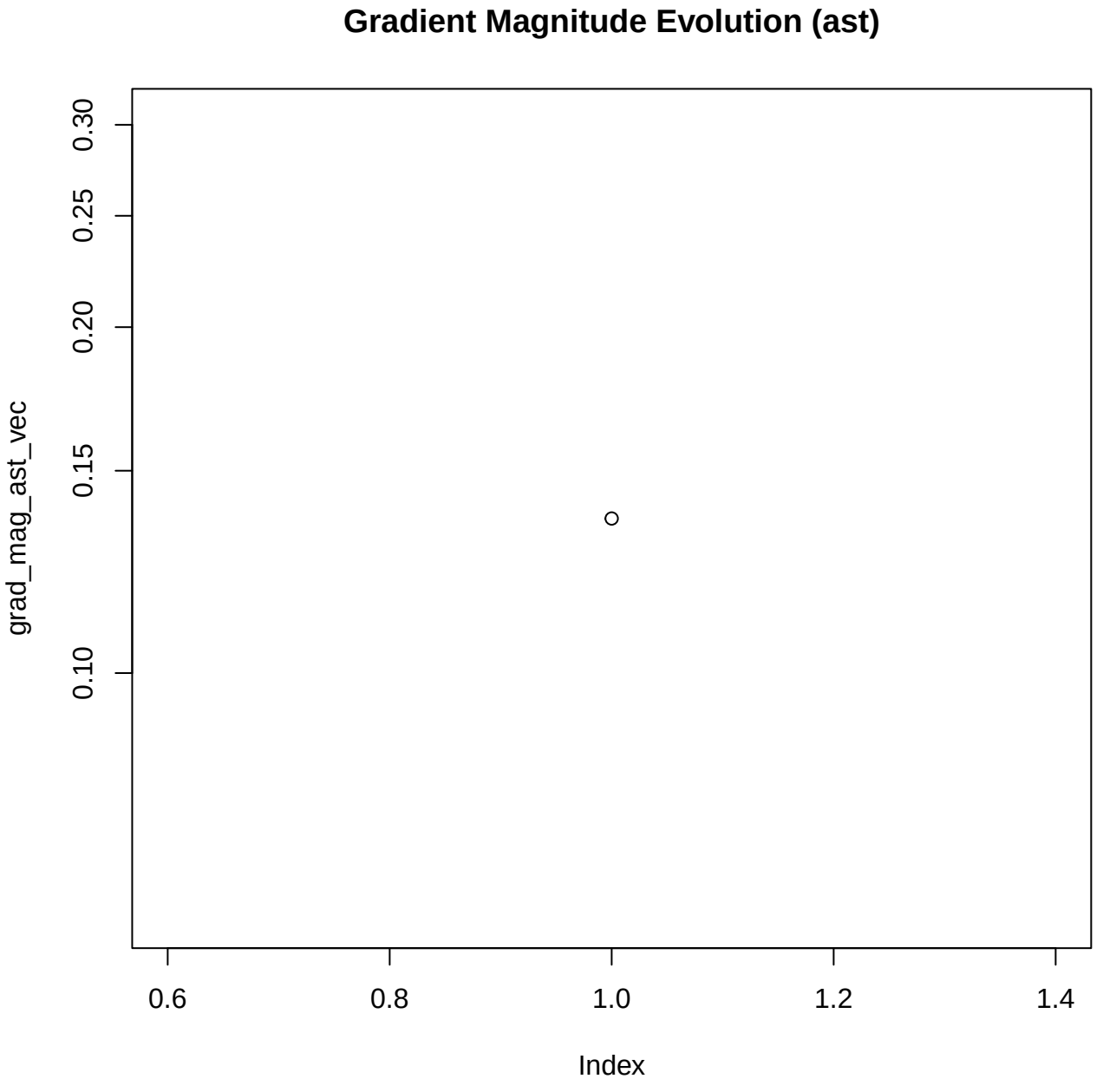
Gradient Magnitude Evolution (ast)
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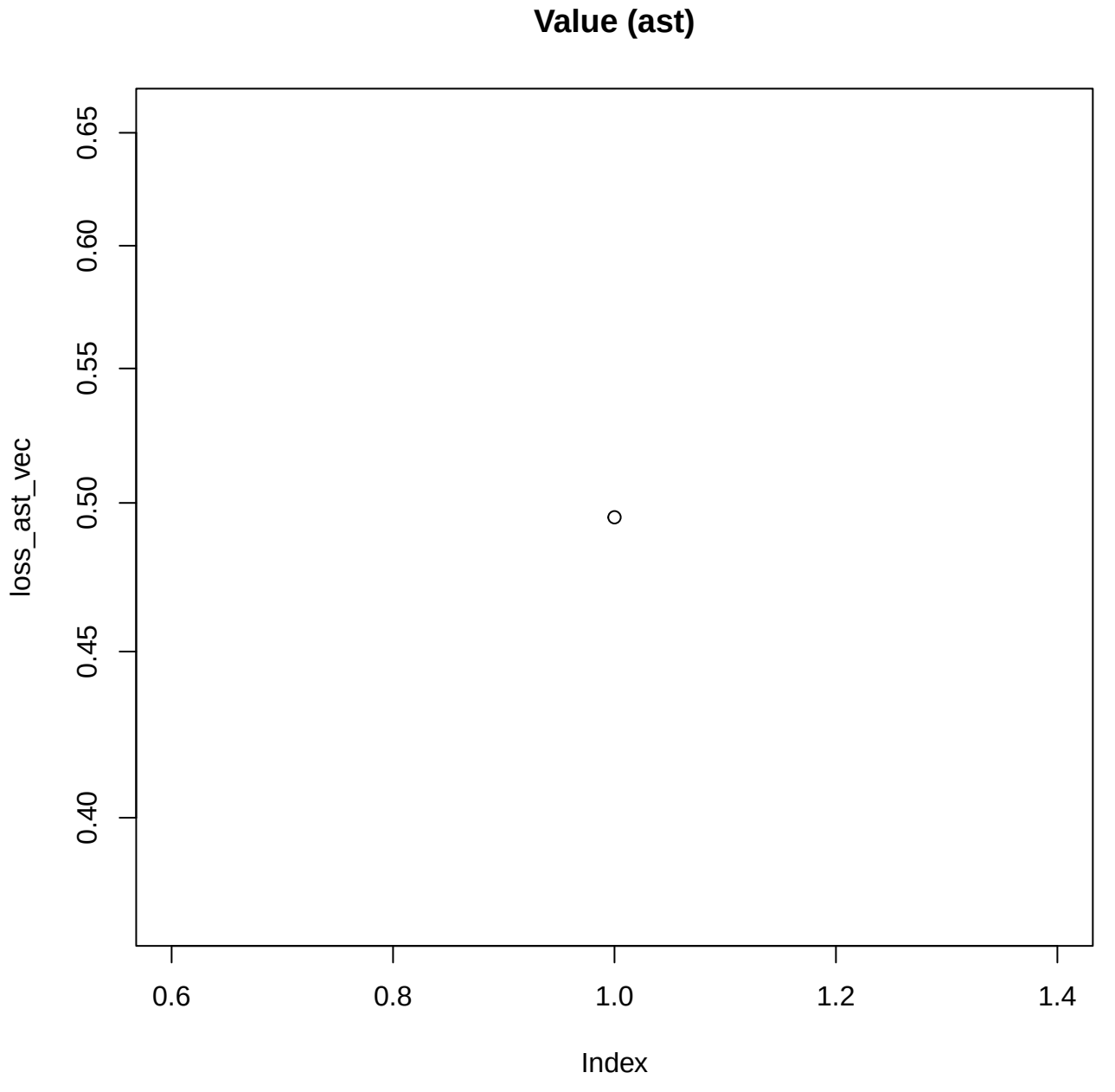
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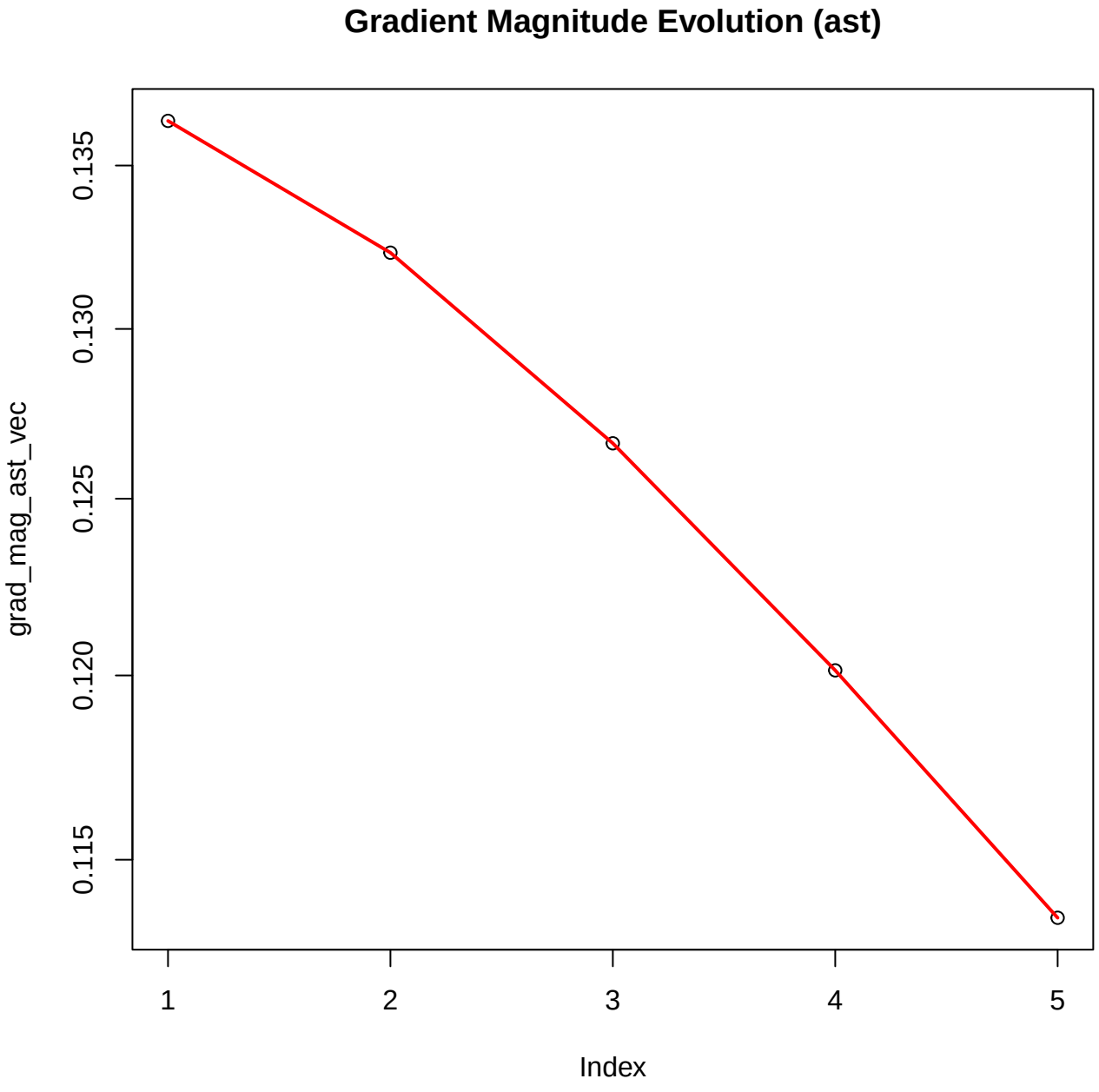
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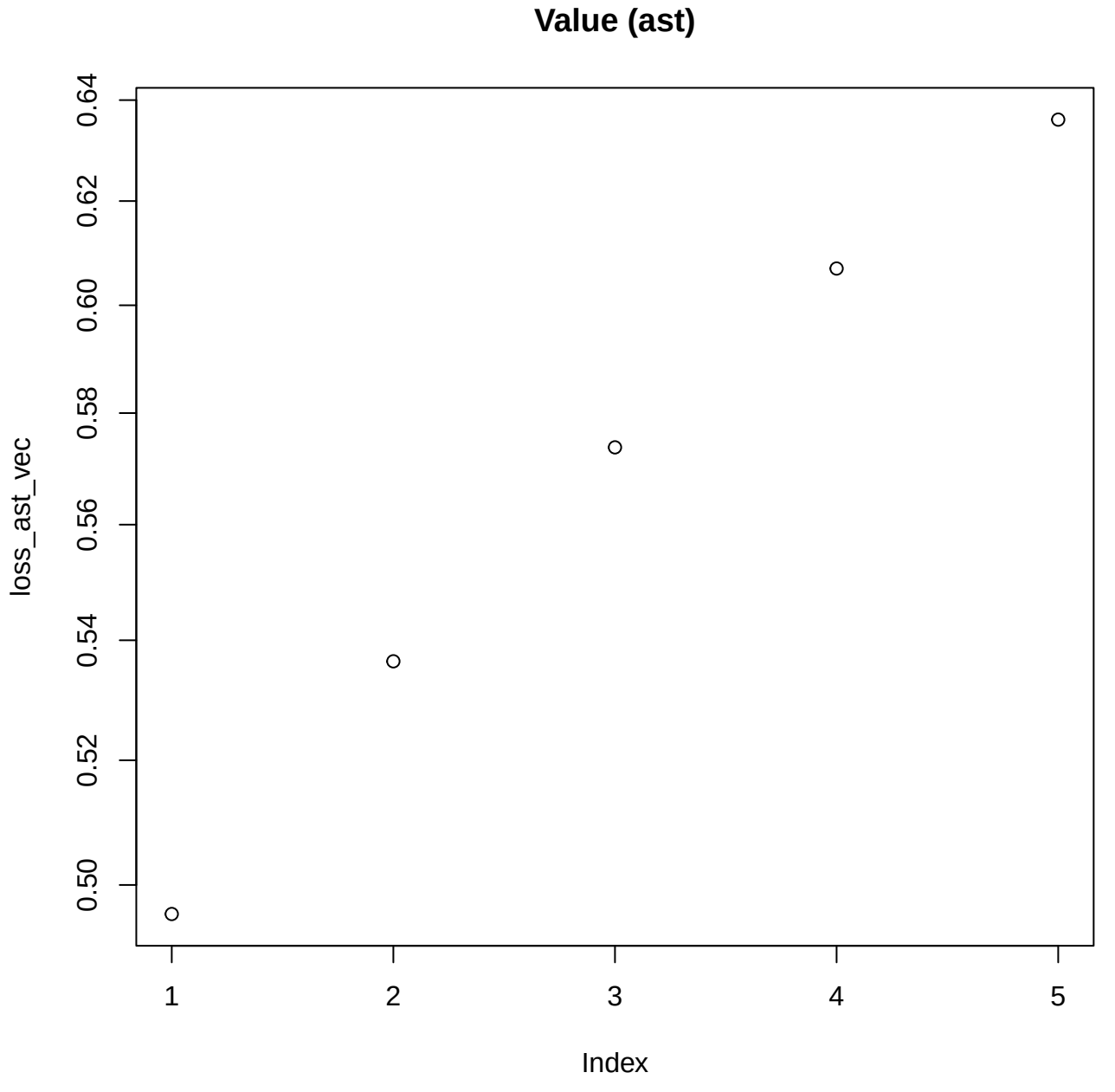
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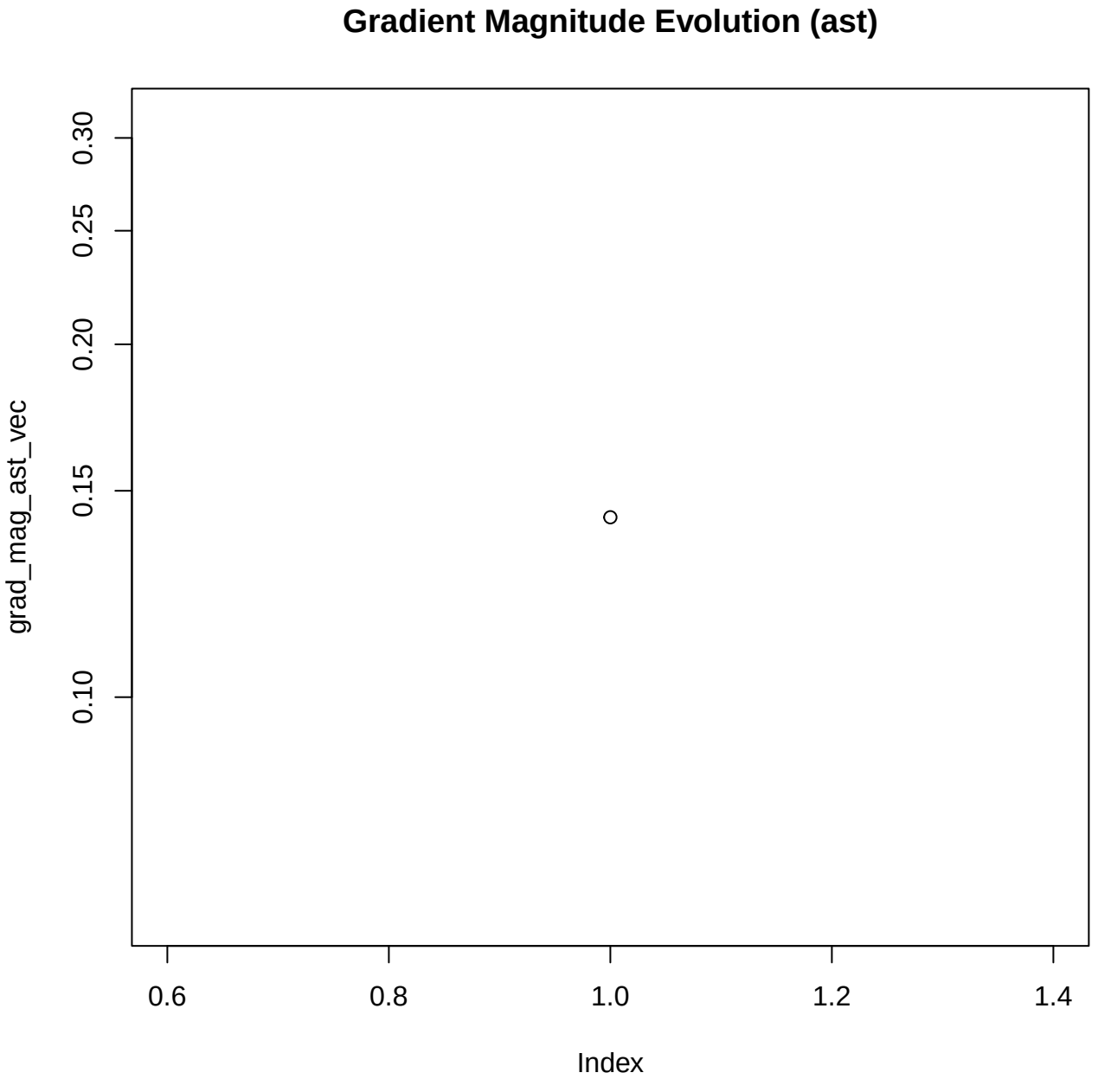
Gradient Magnitude Evolution (ast)
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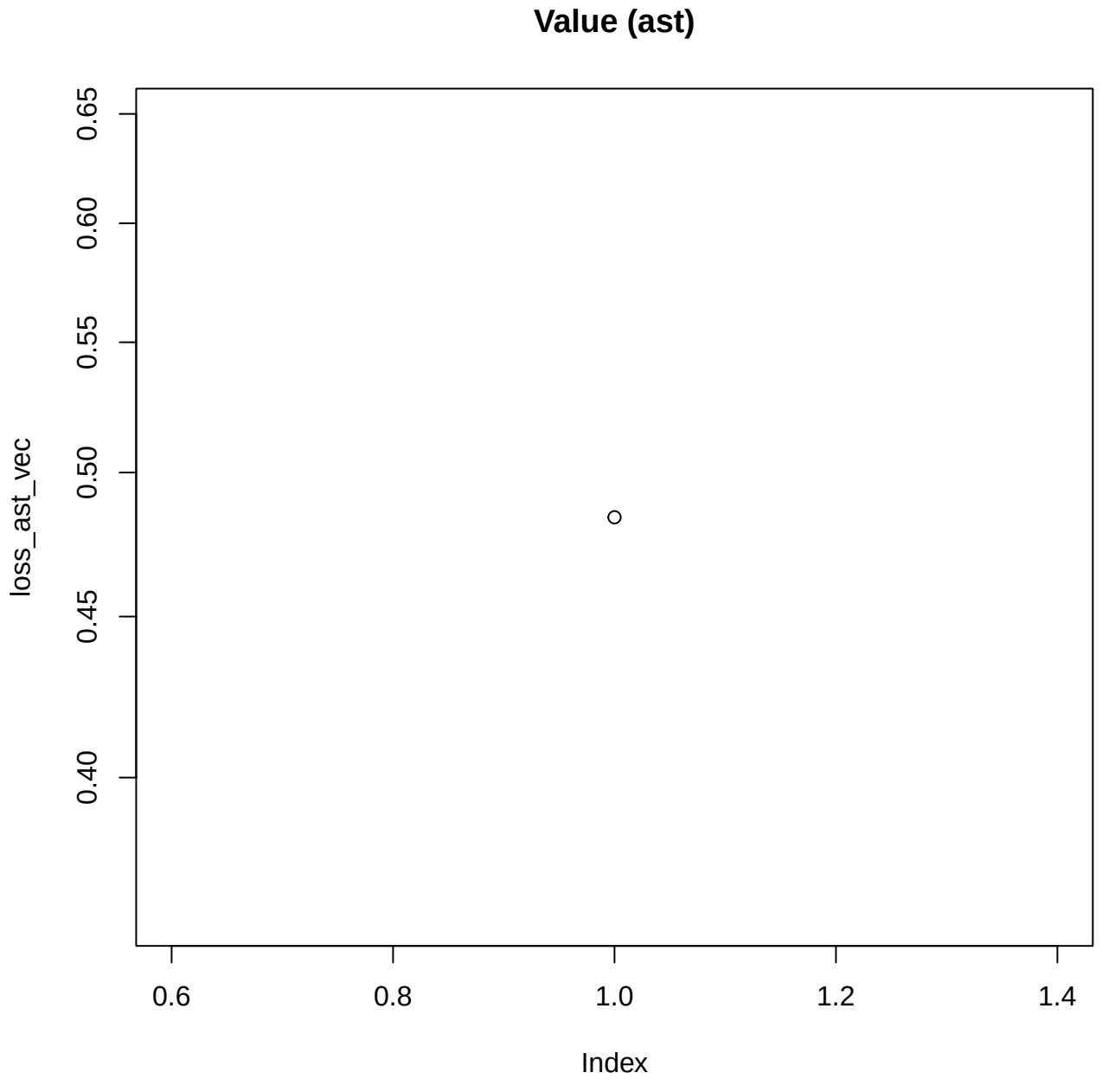
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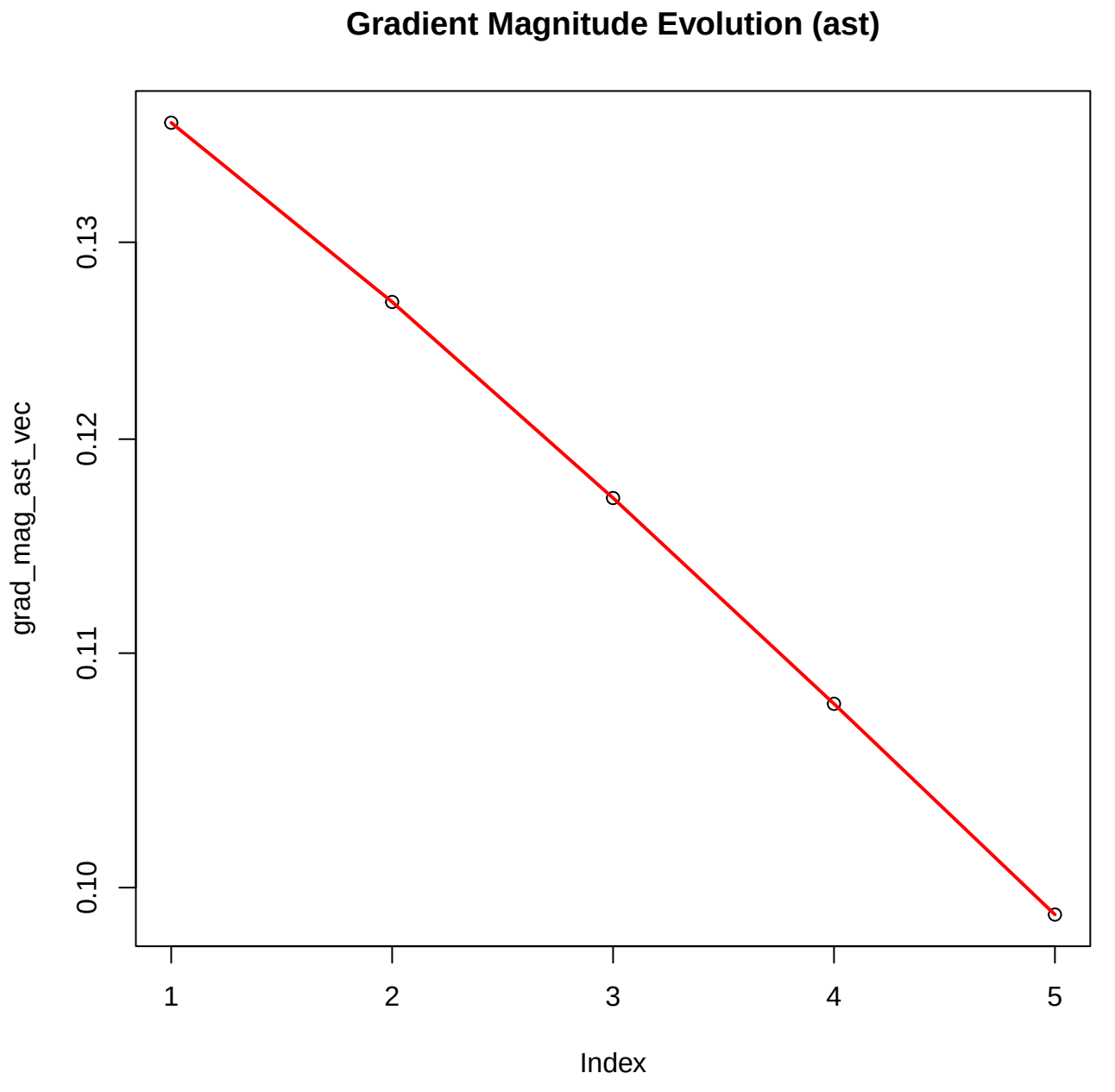
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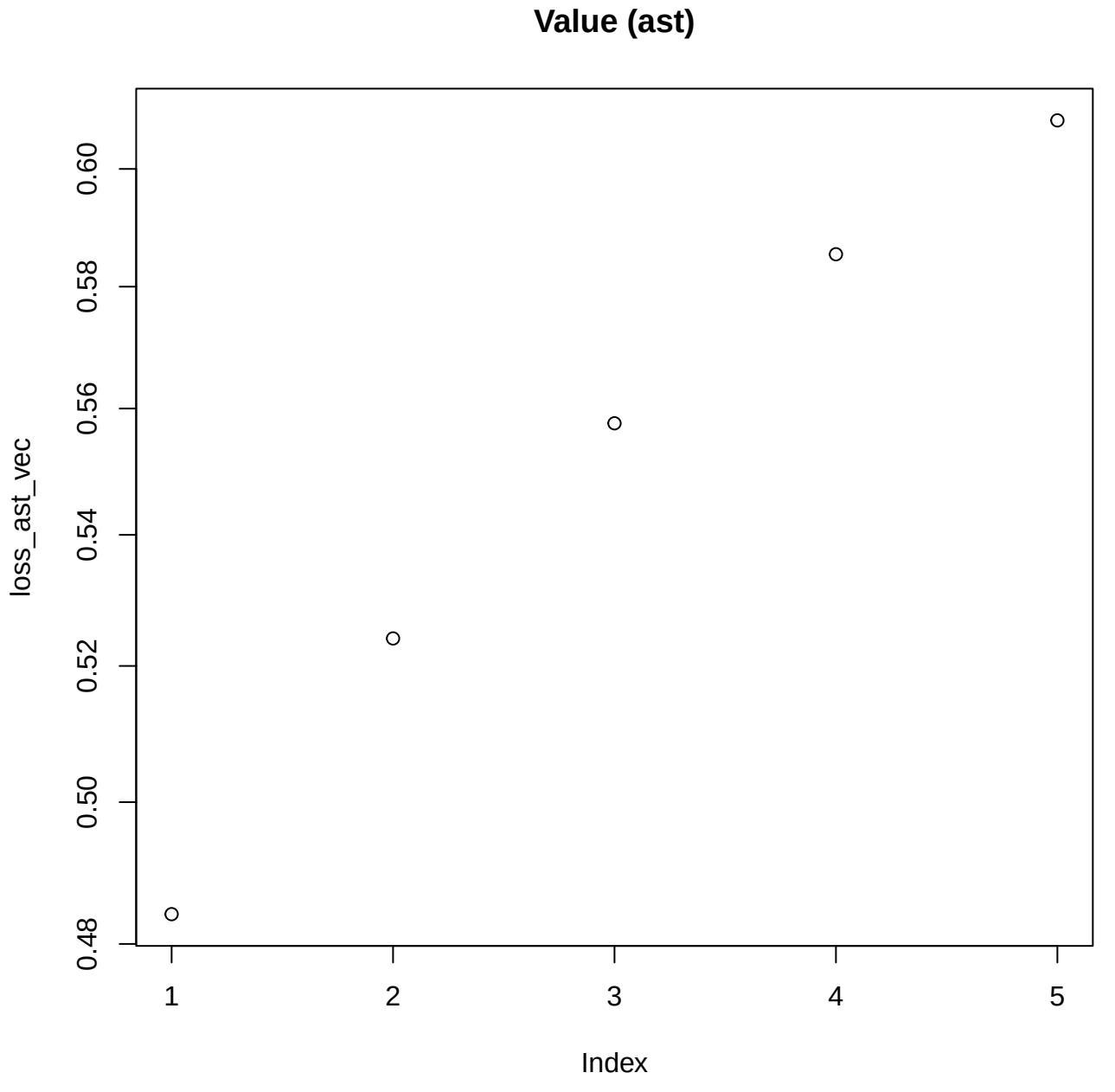
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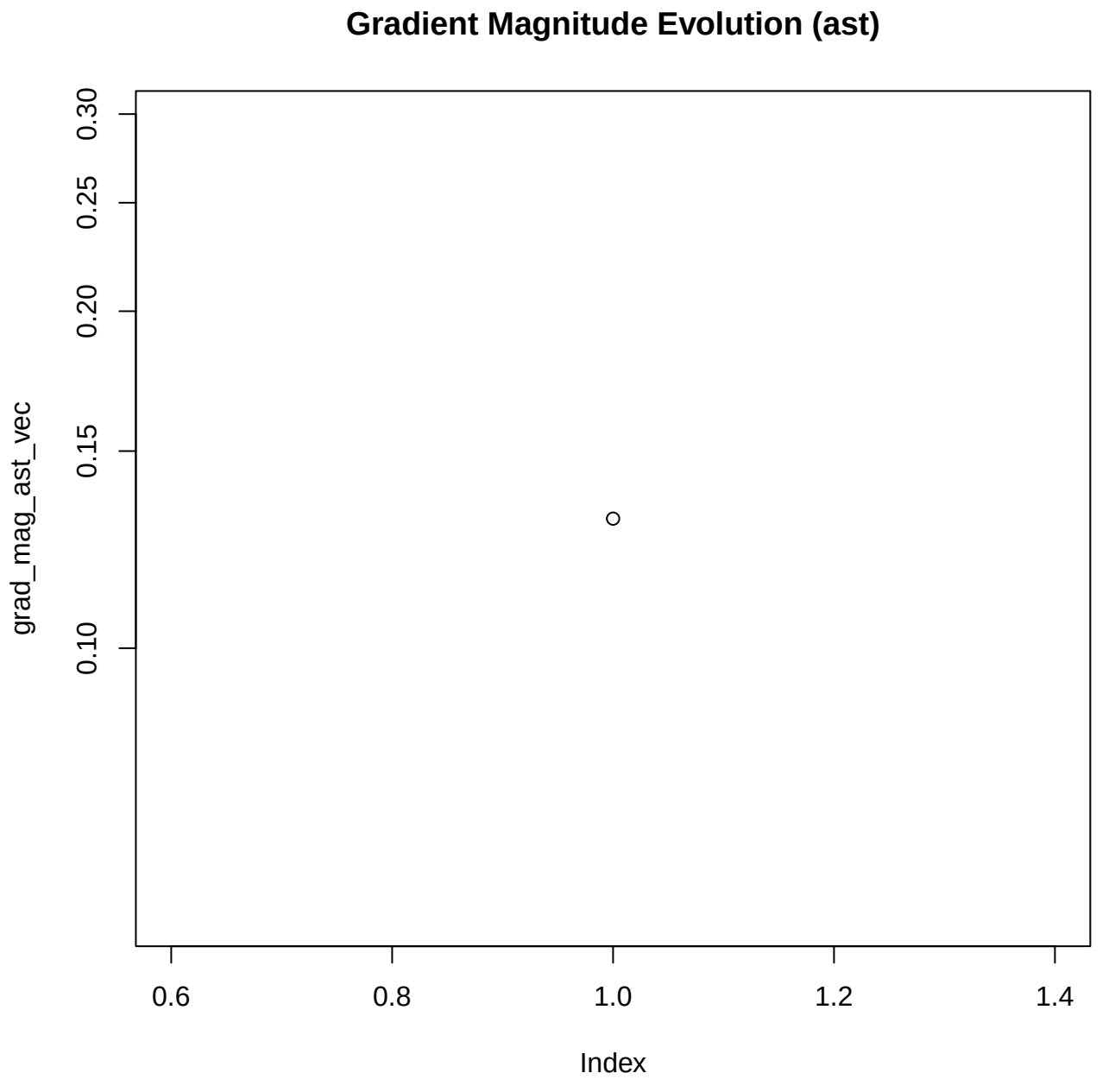
Gradient Magnitude Evolution (ast)
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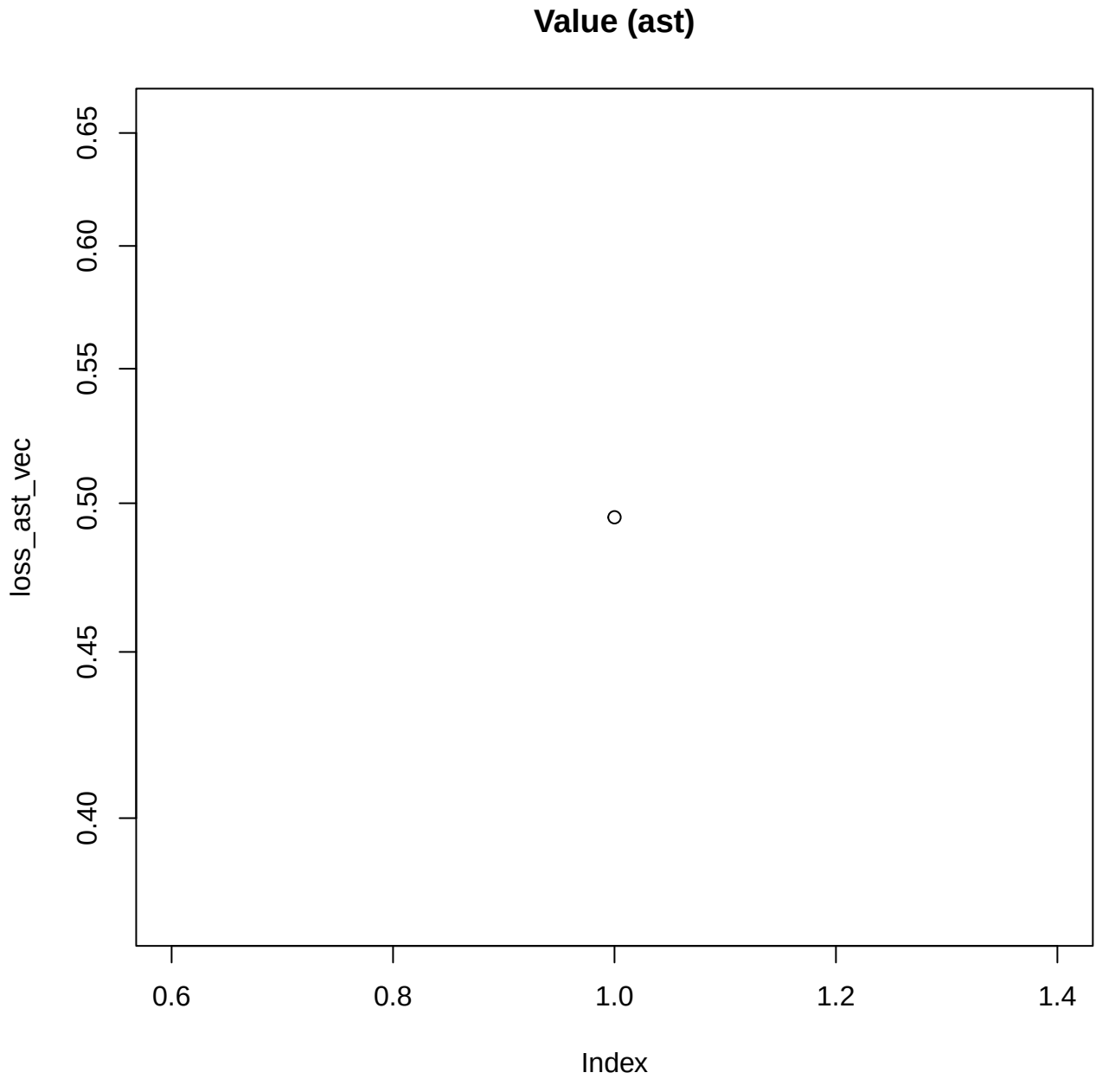
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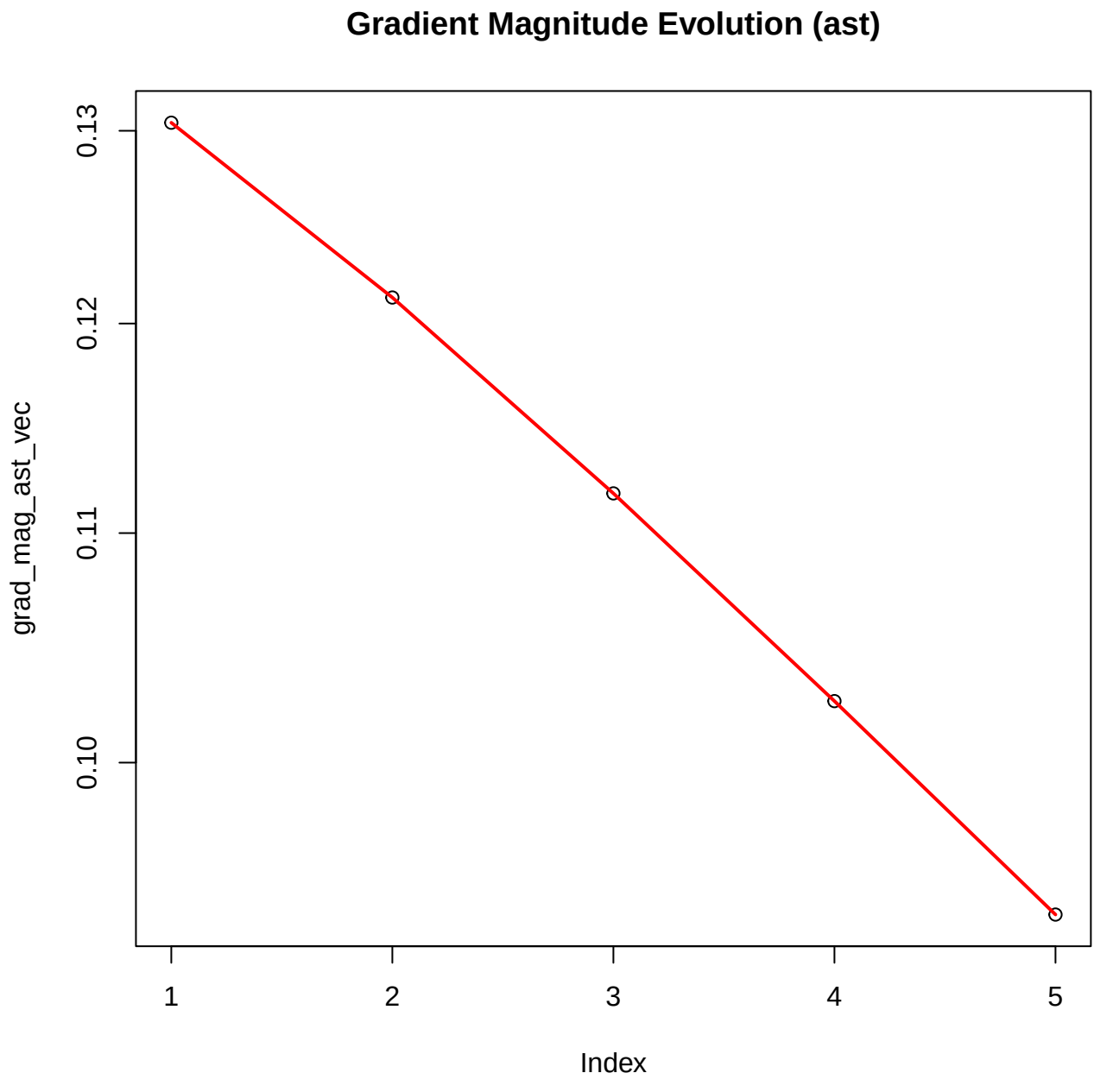
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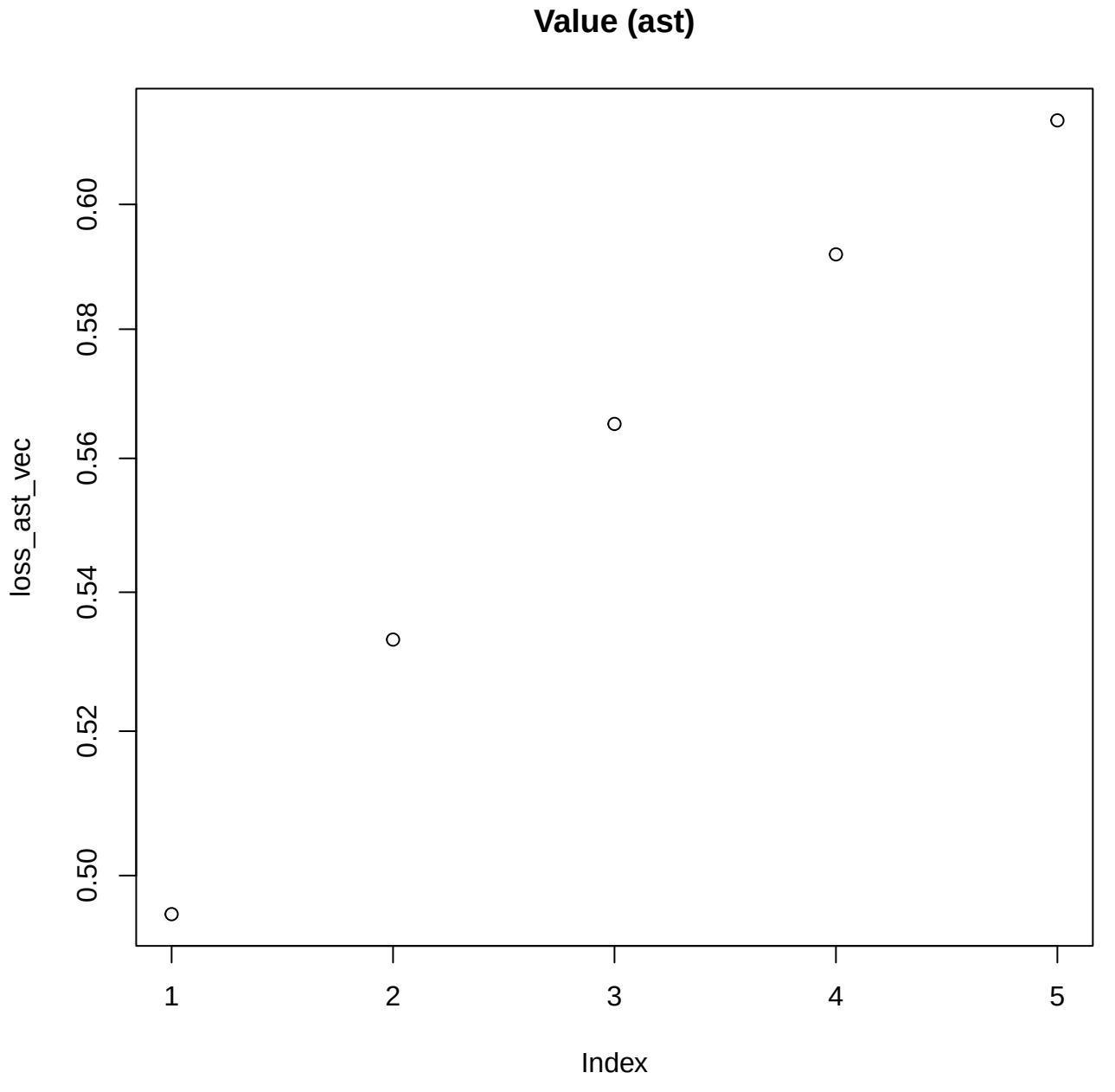
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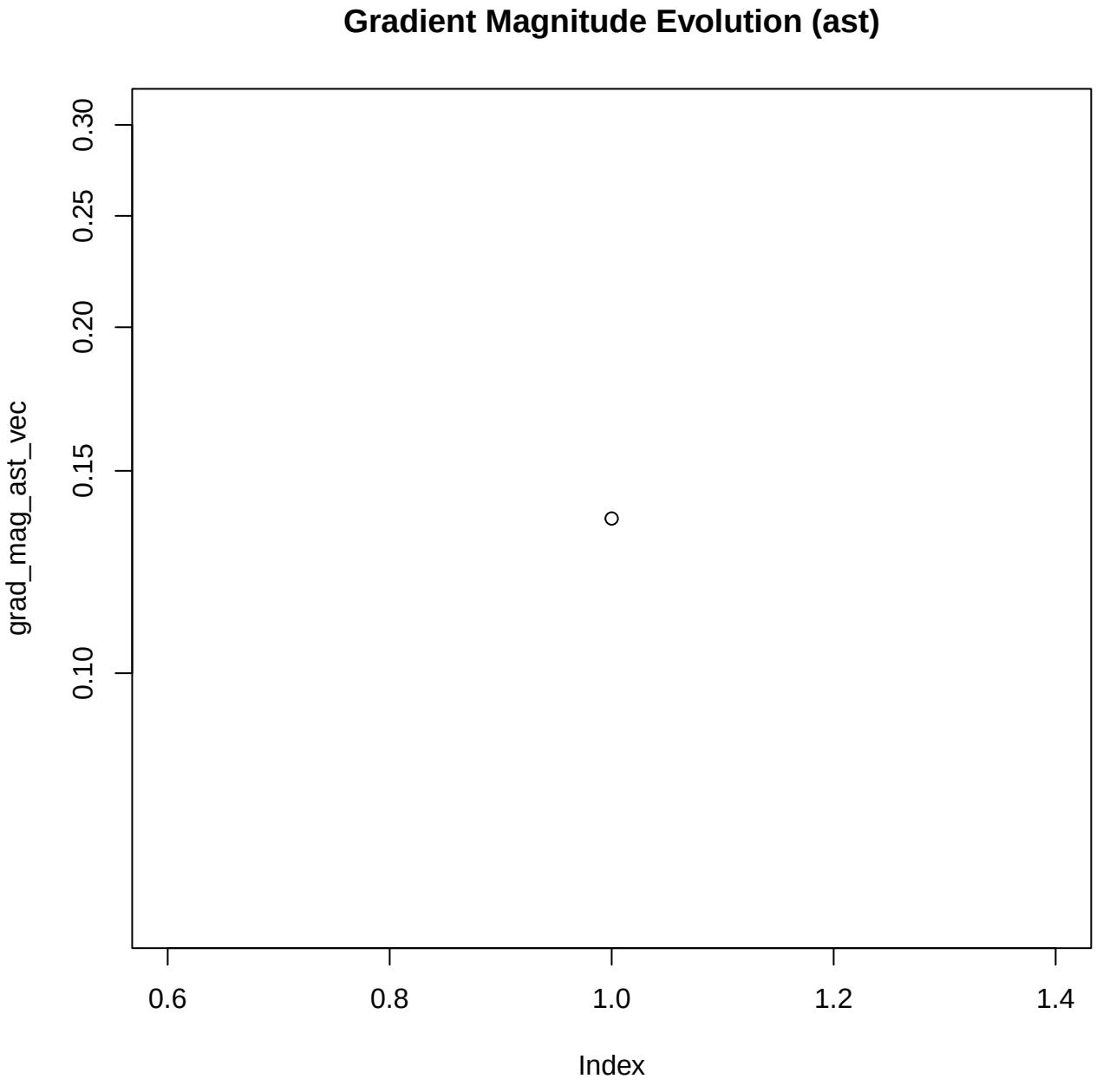
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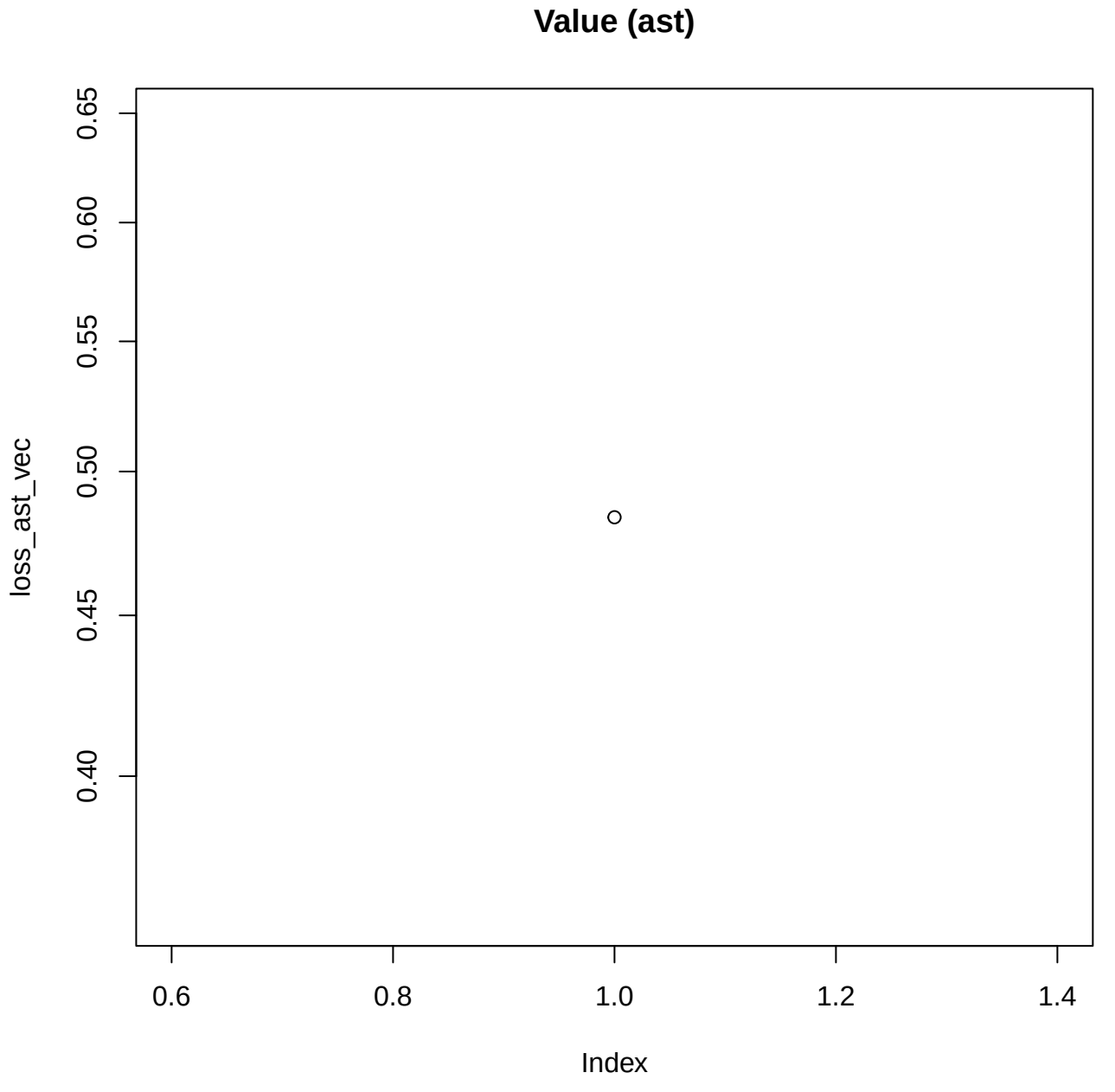
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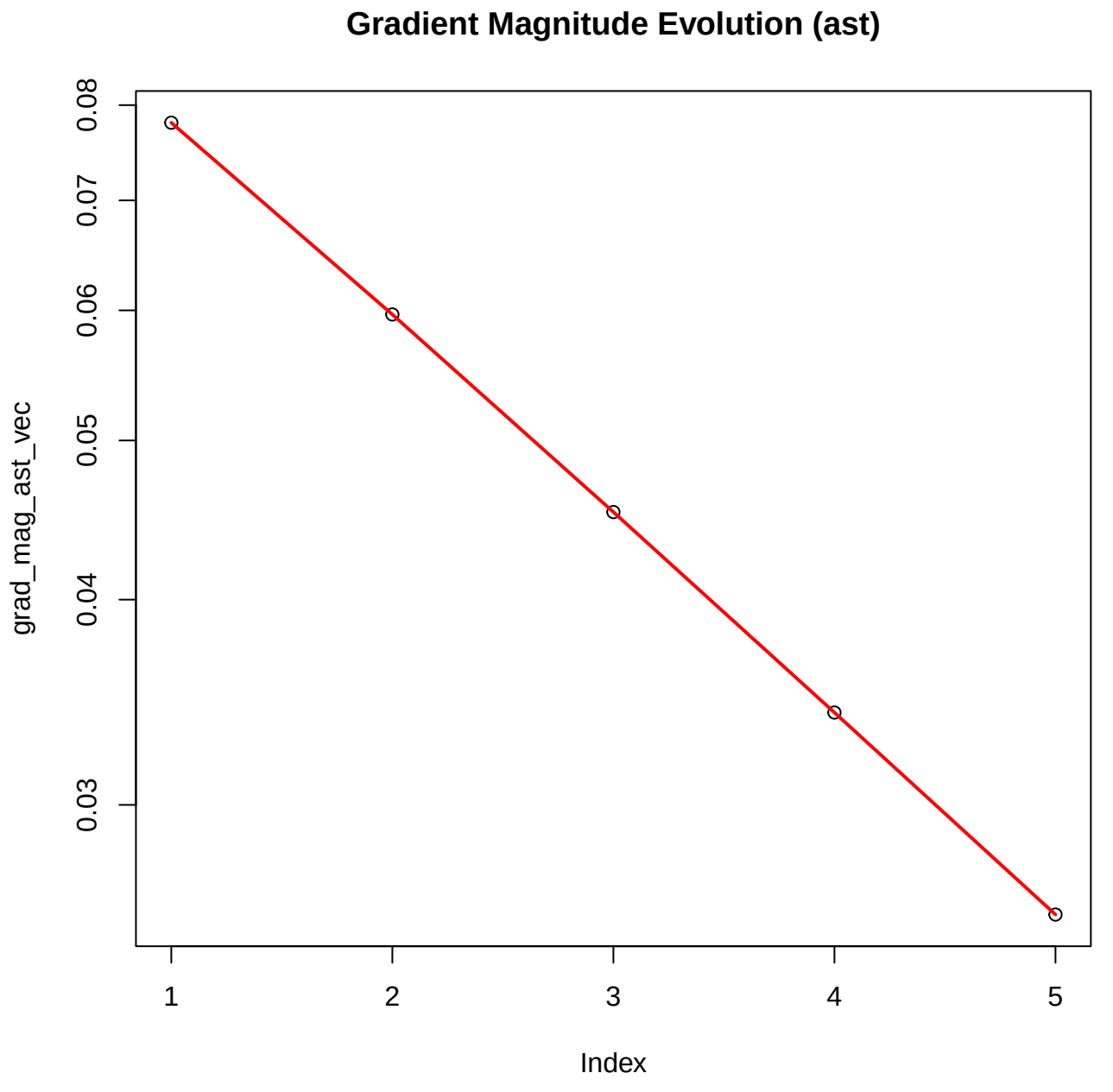
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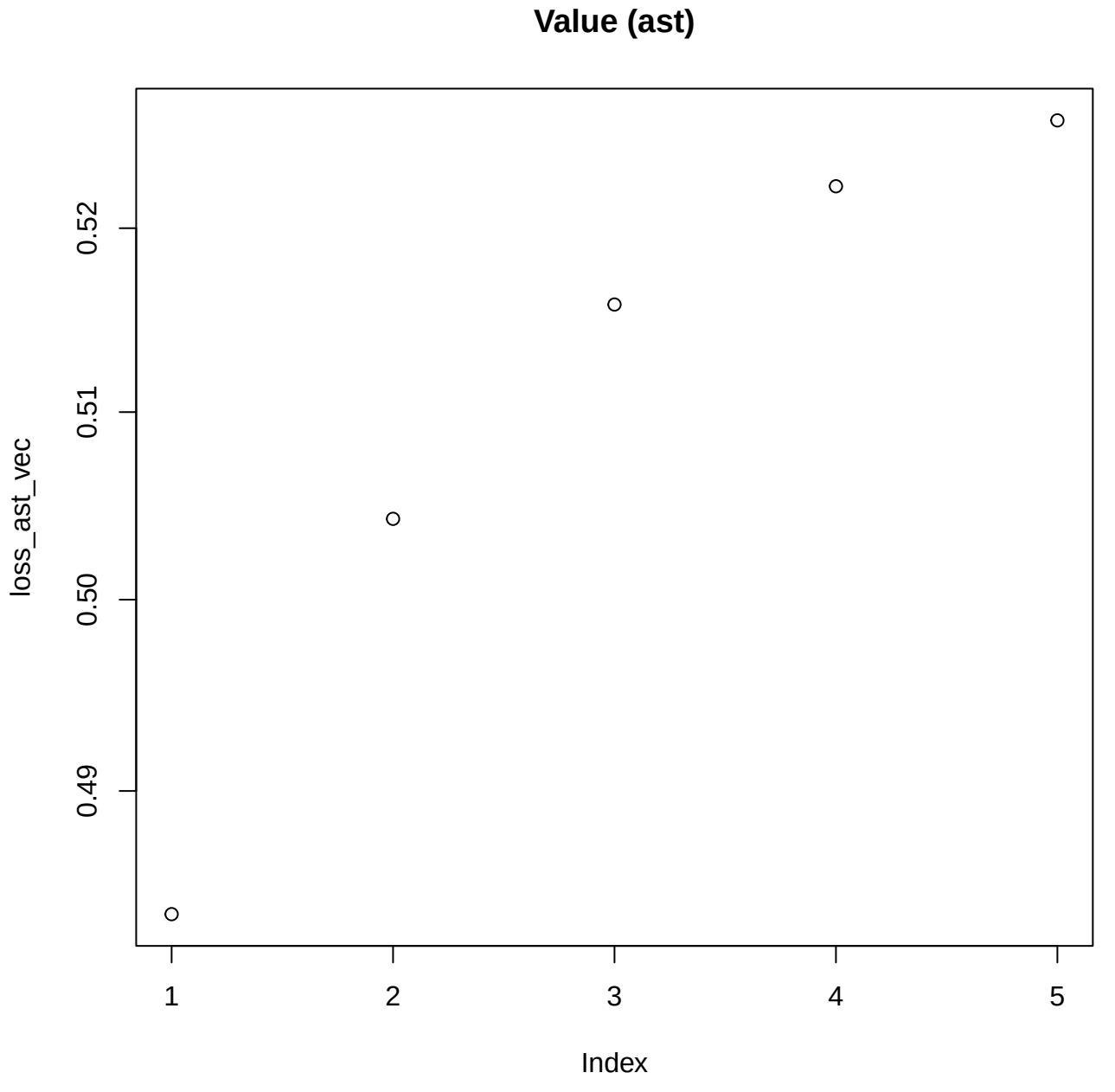
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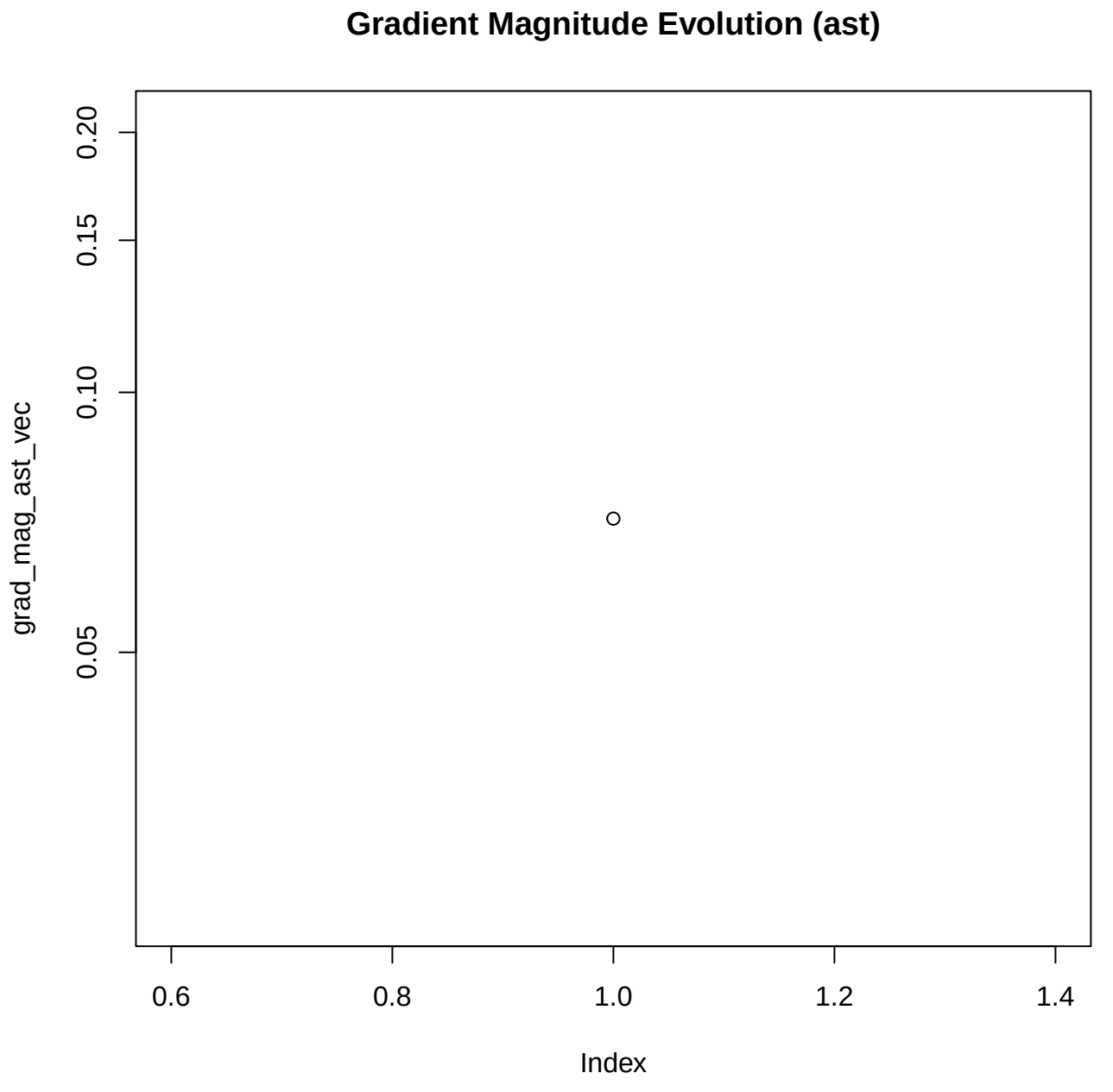
Gradient Magnitude Evolution (ast)
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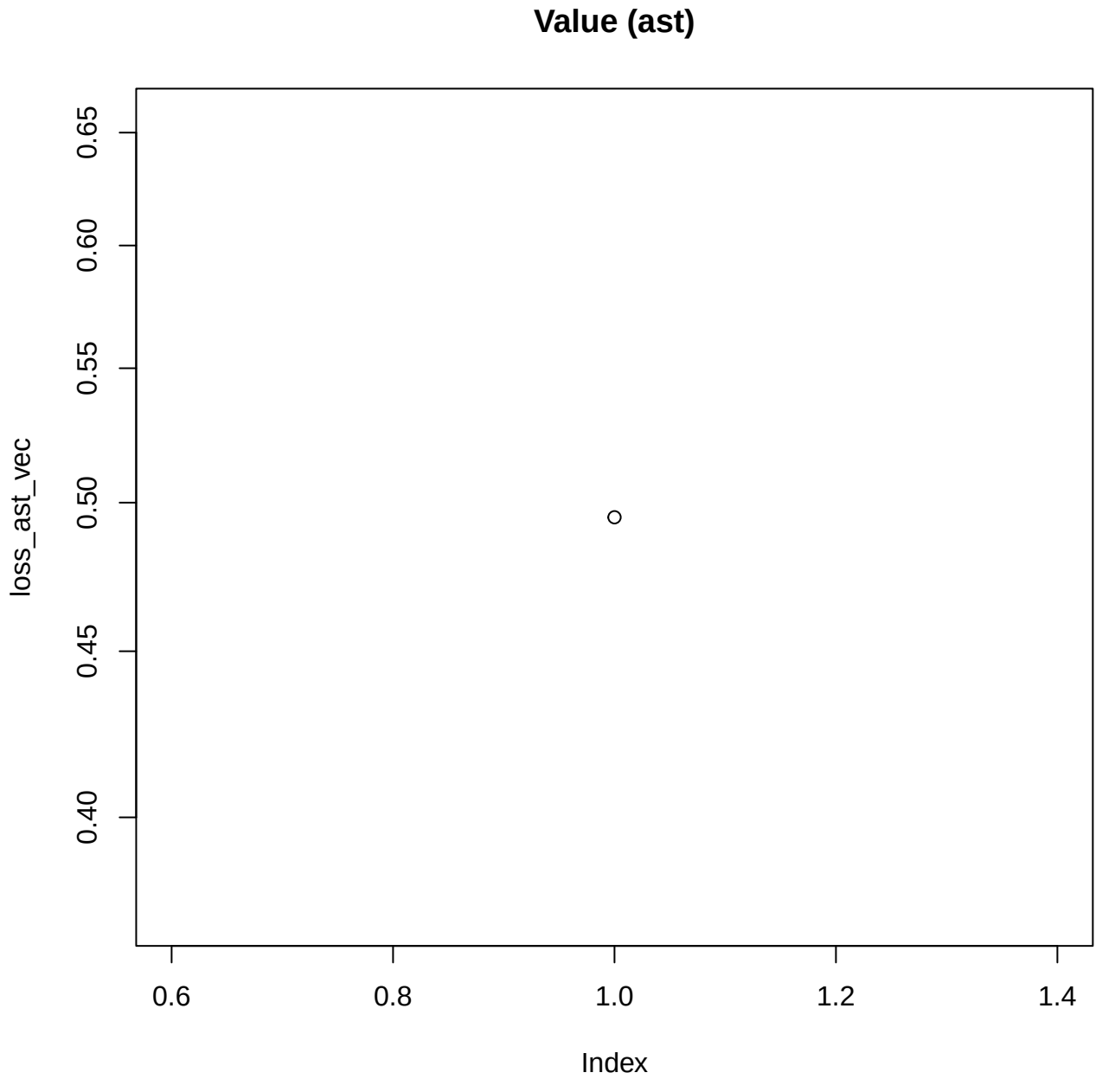
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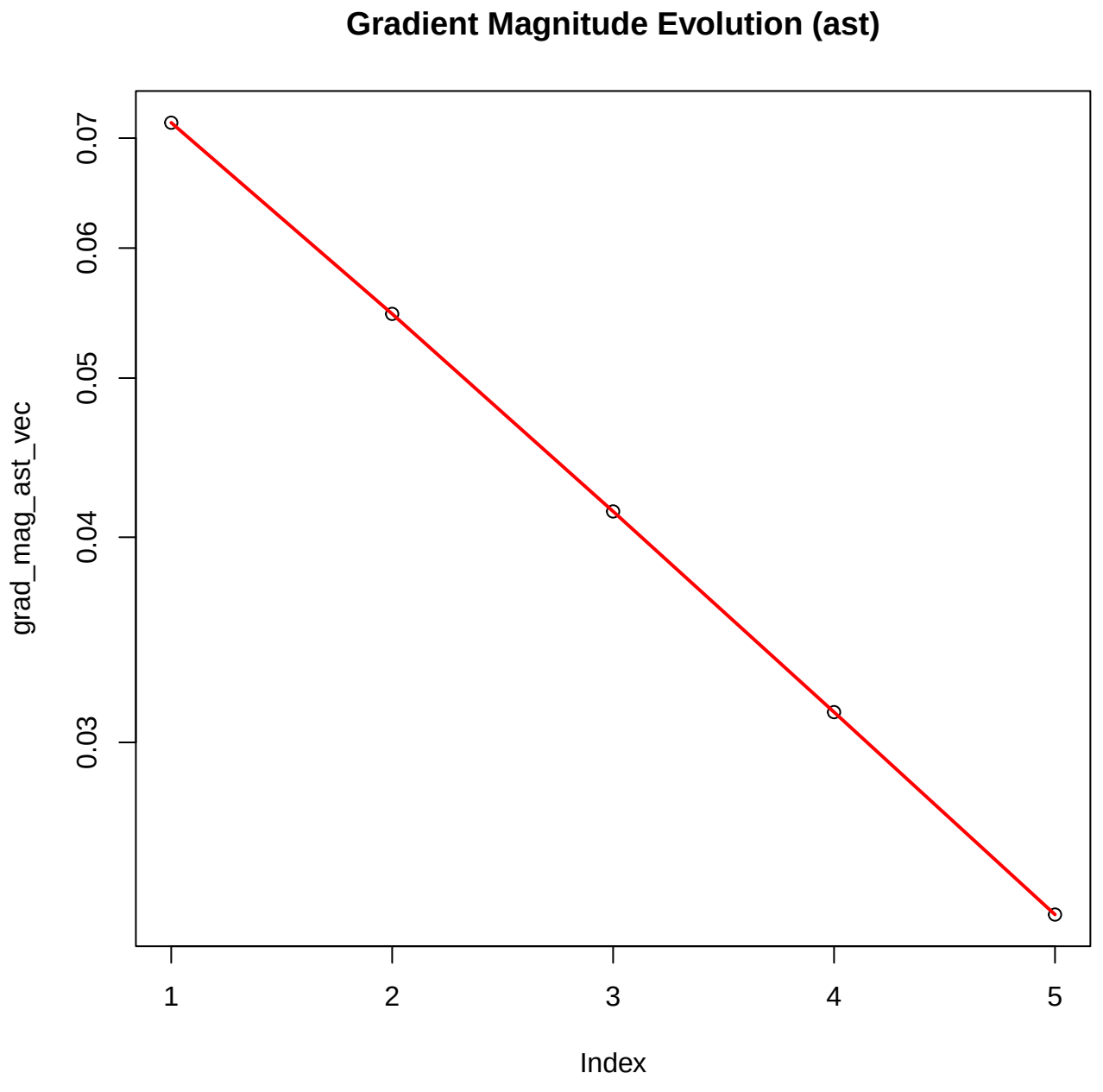
Gradient Magnitude Evolution (ast)
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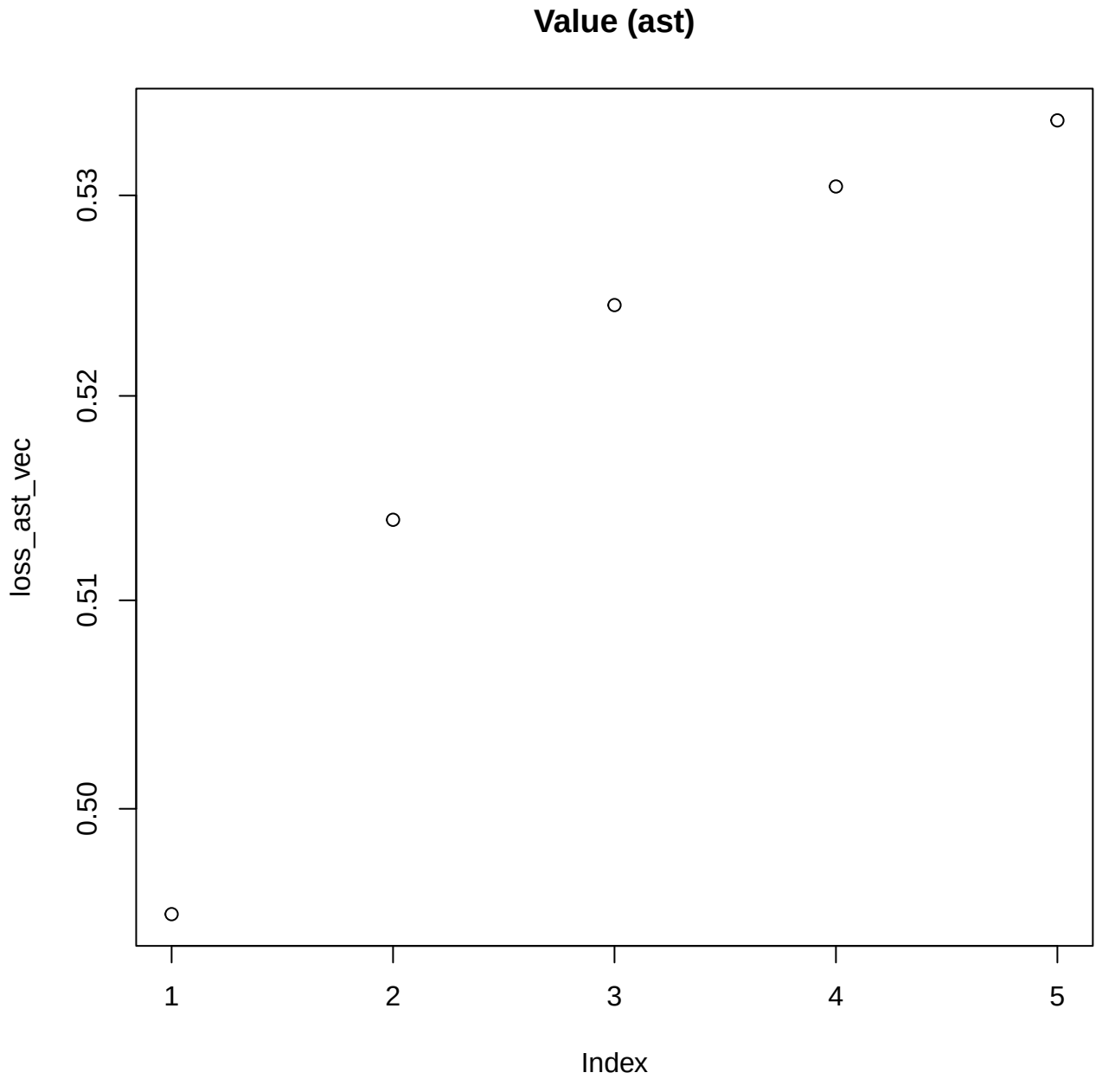
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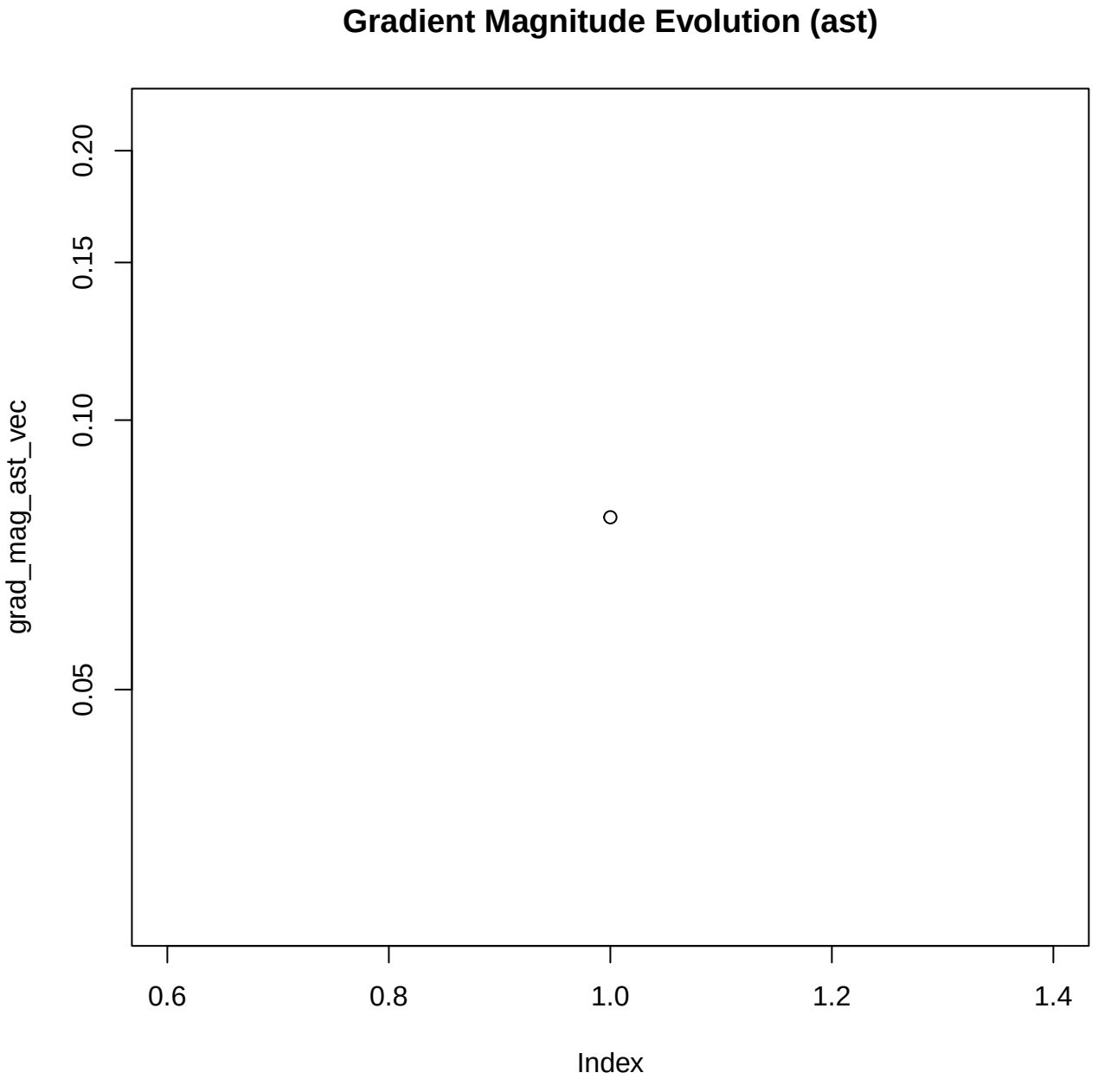
Gradient Magnitude Evolution (ast)
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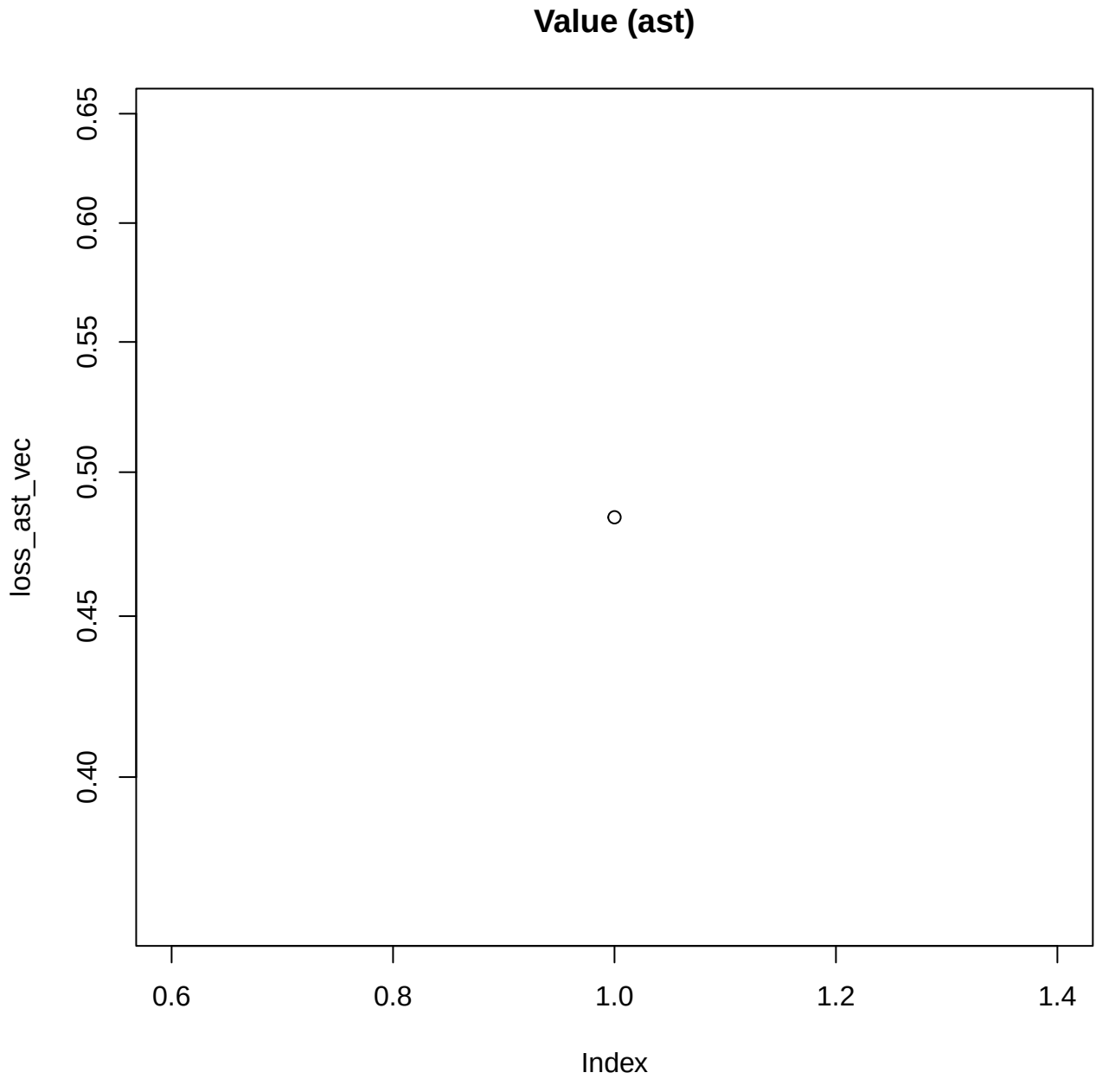
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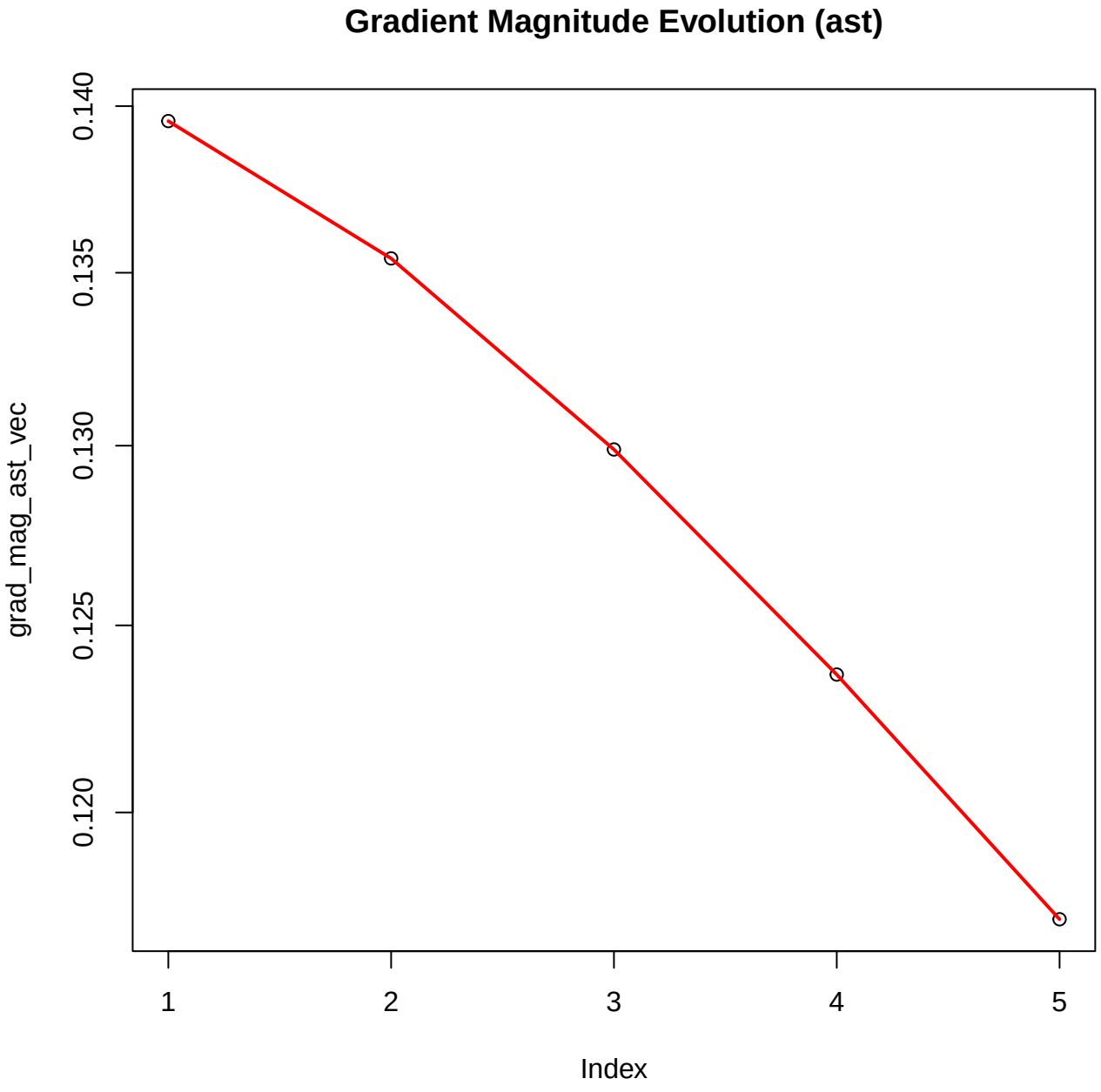
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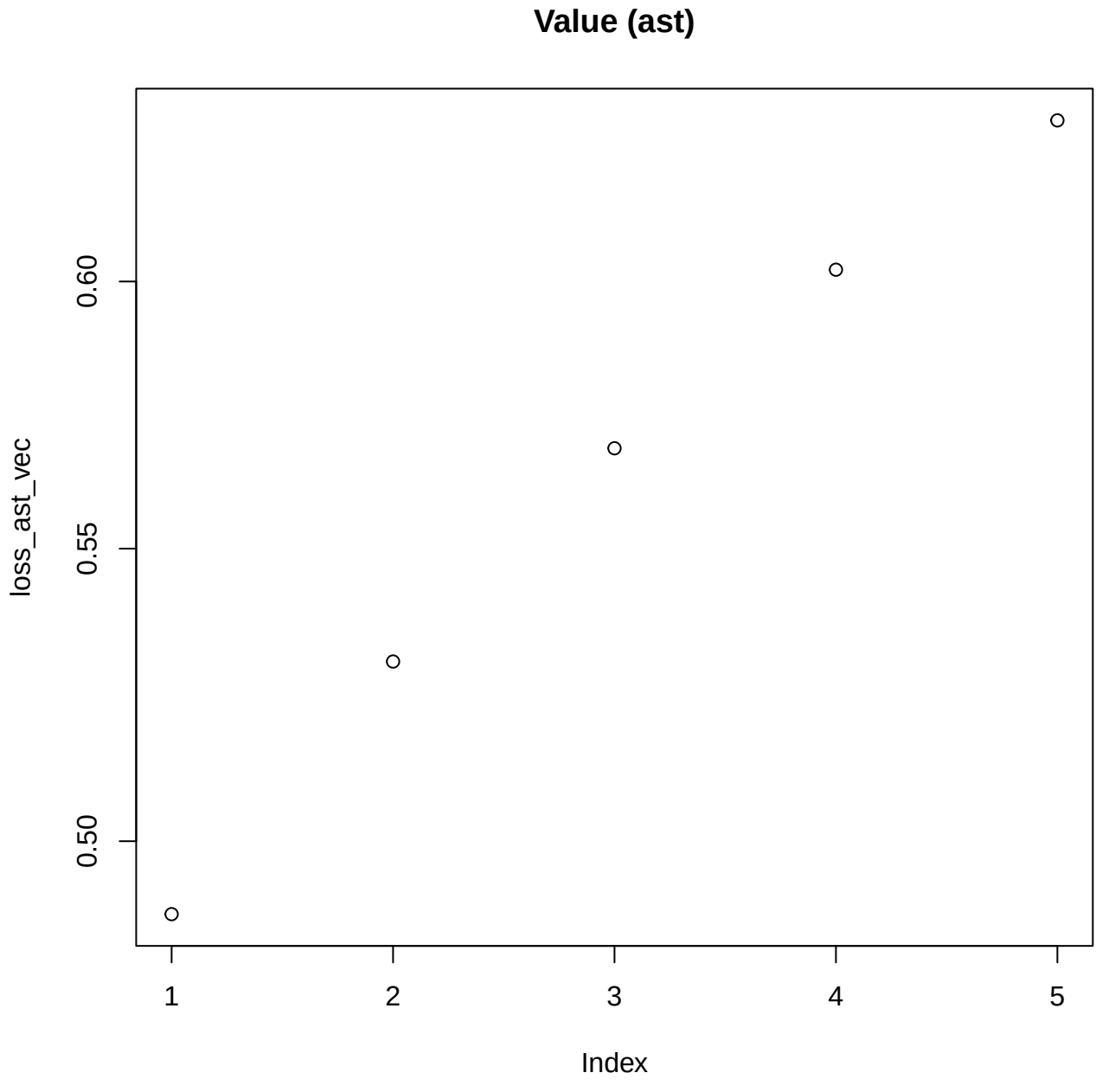
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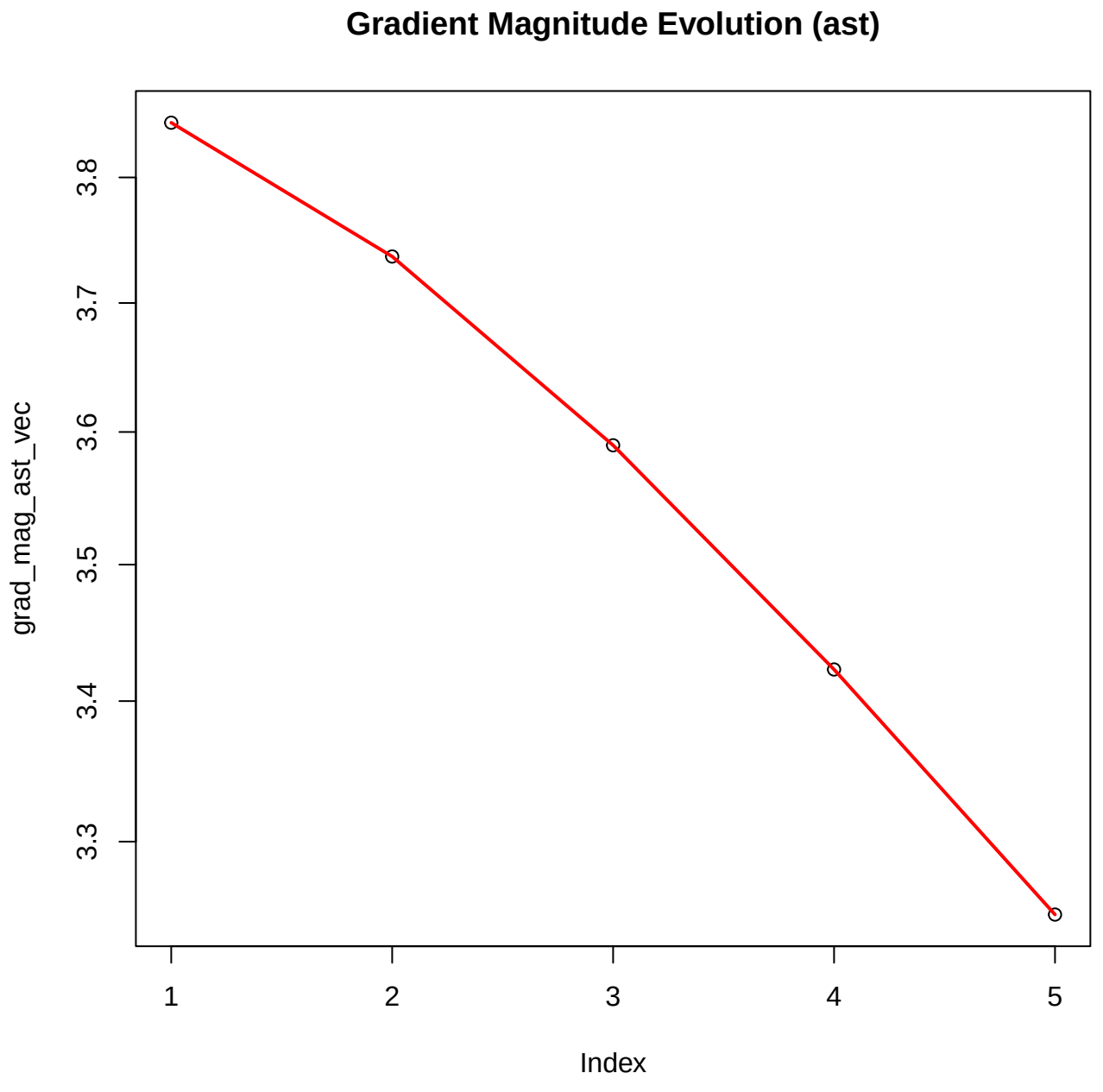
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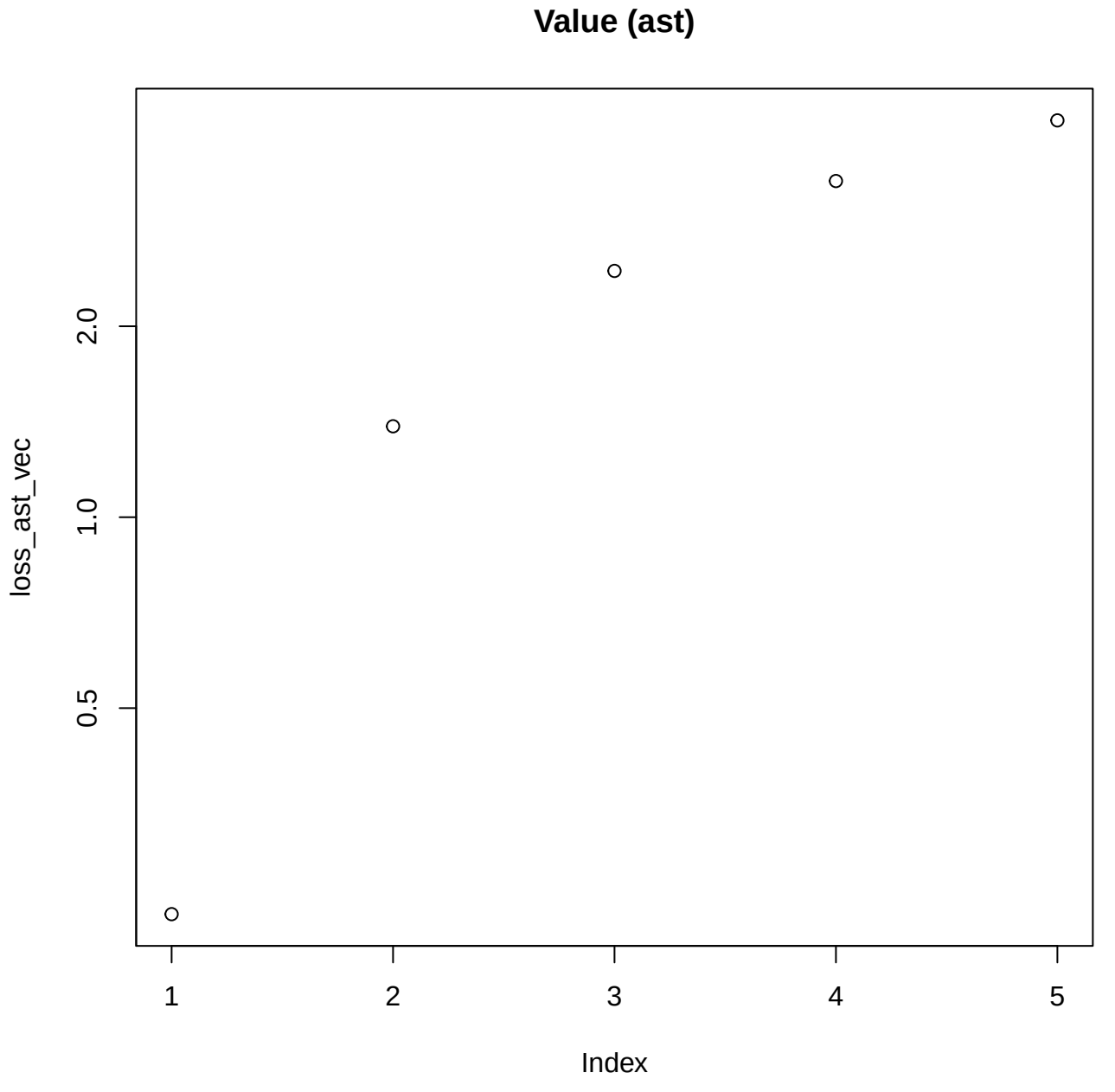
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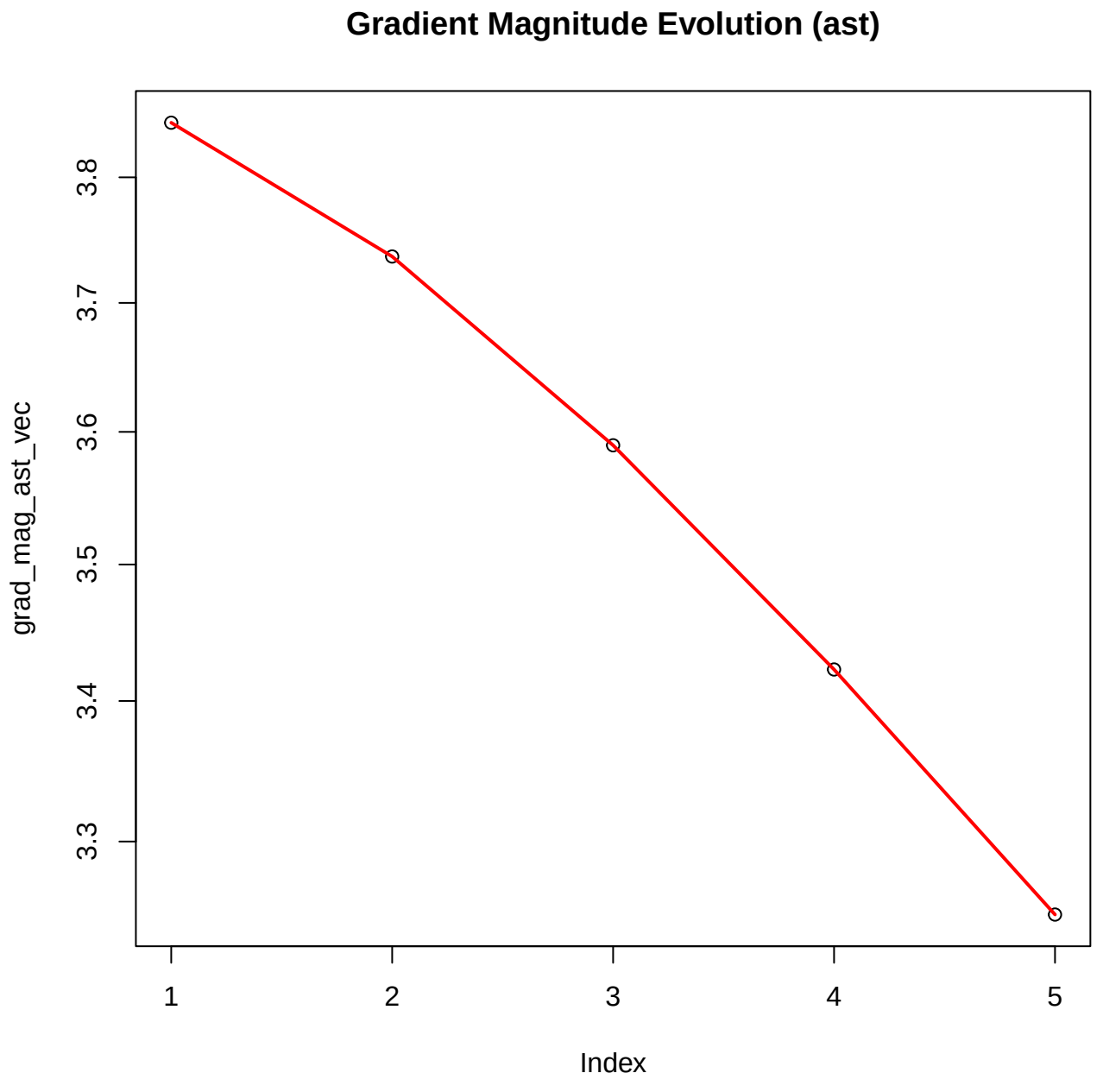
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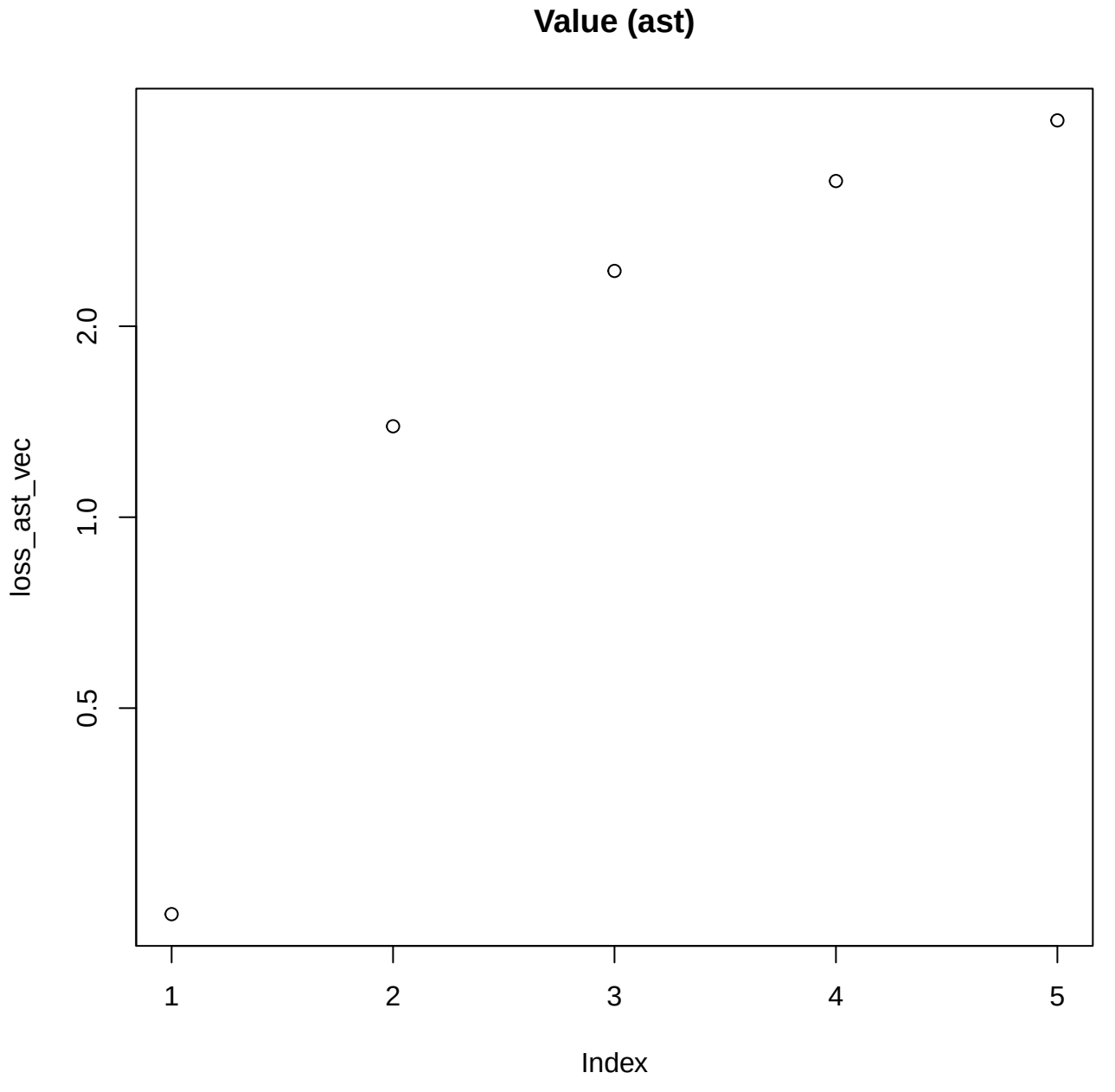
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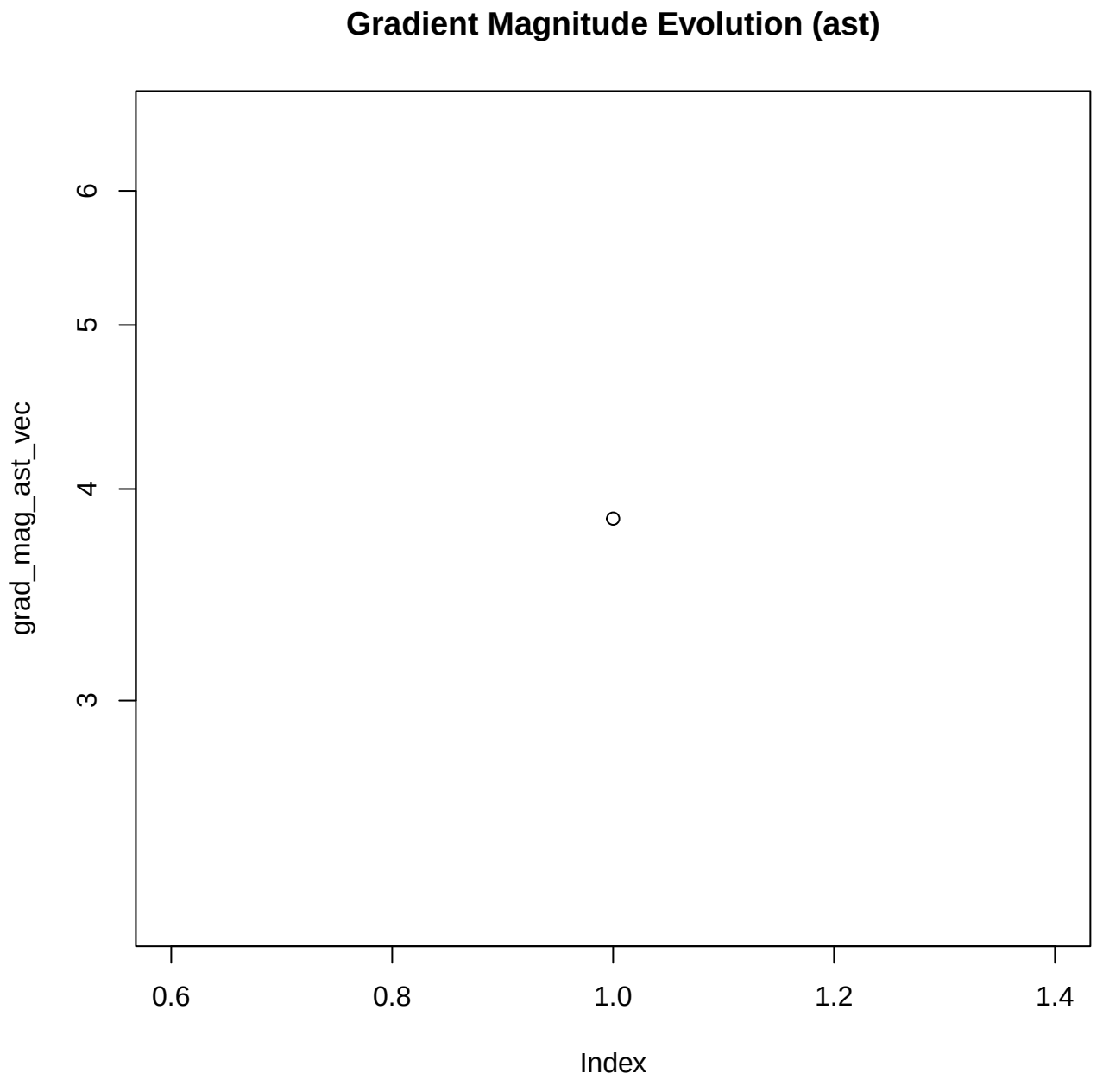
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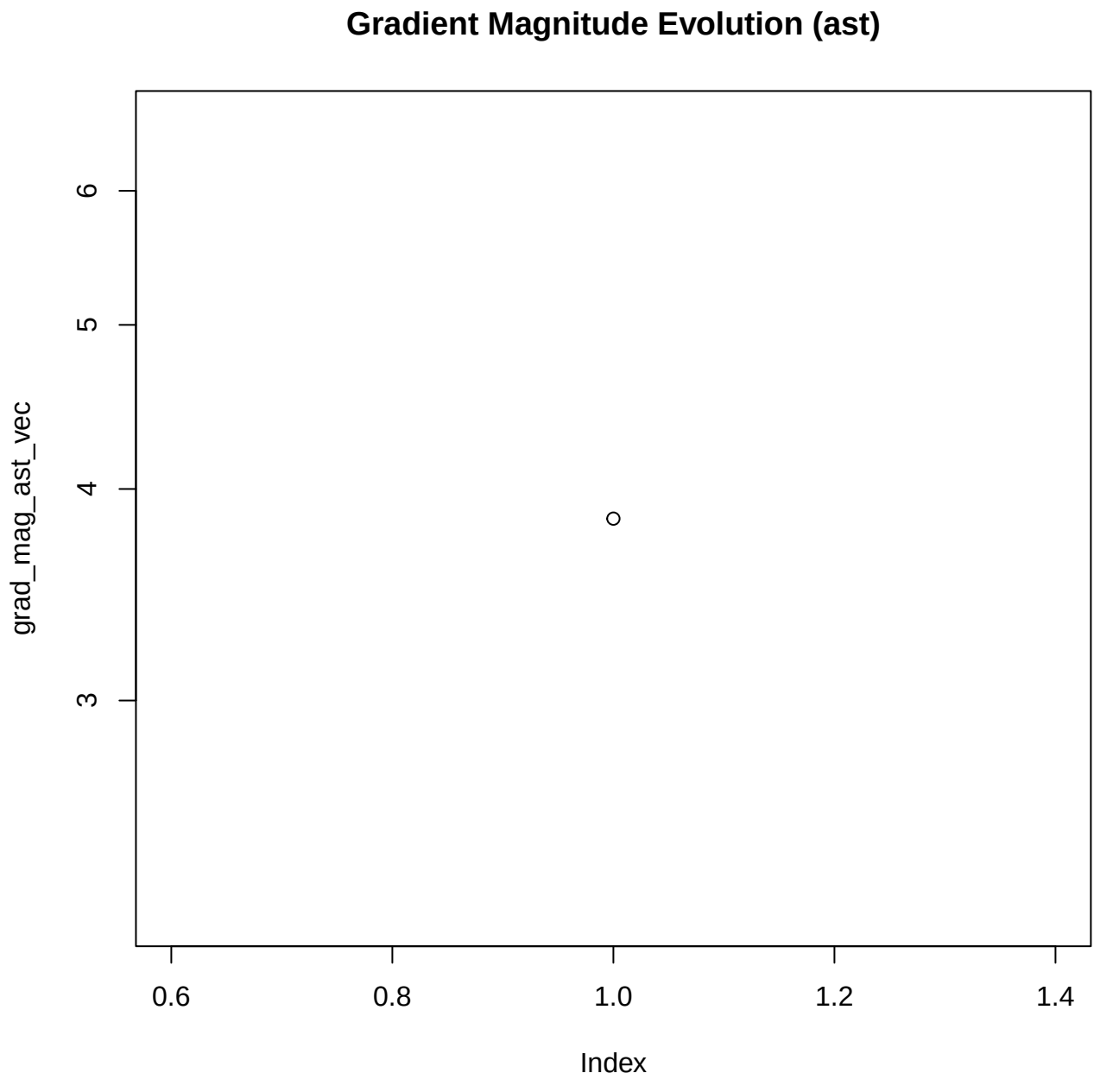
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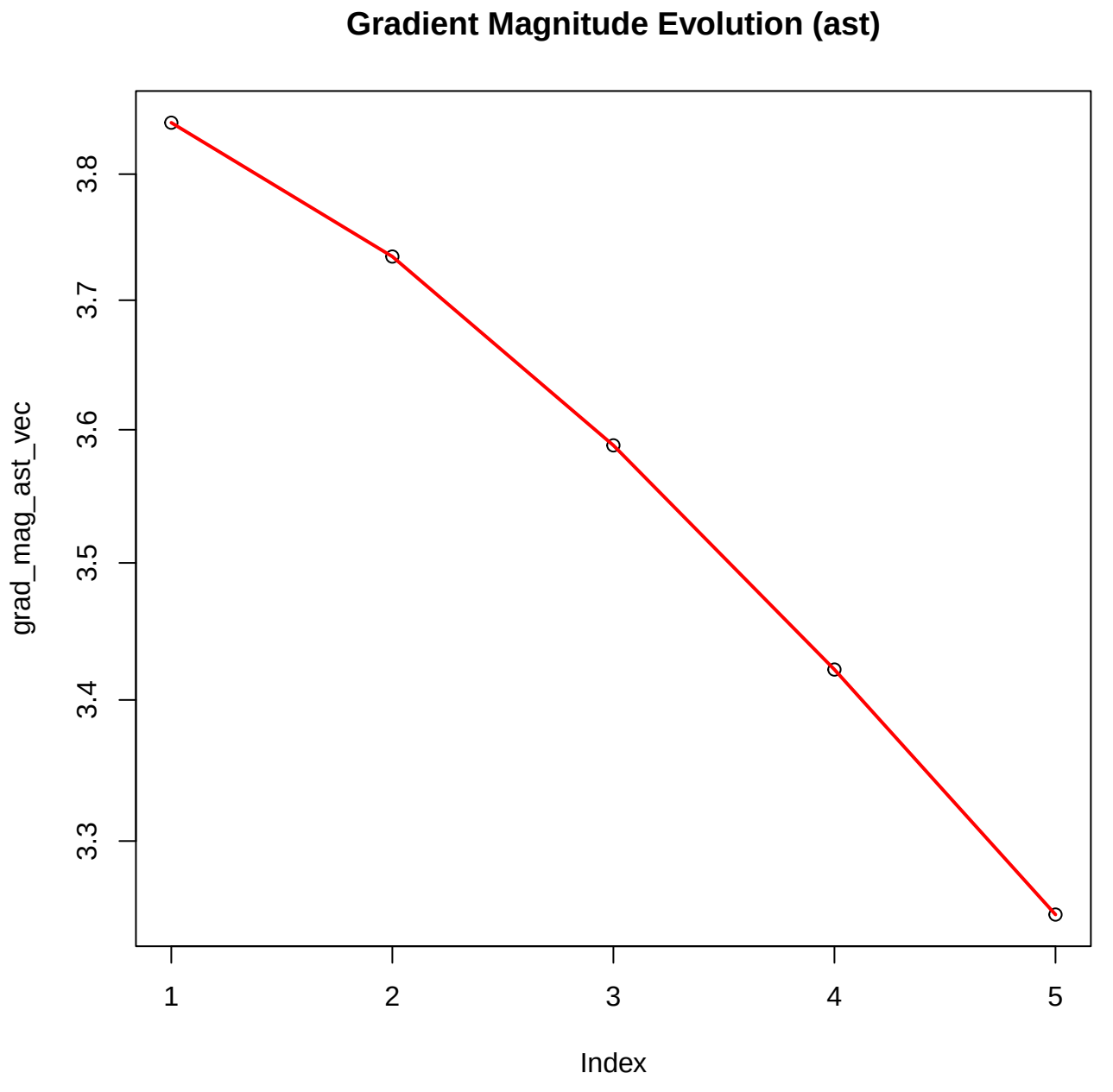
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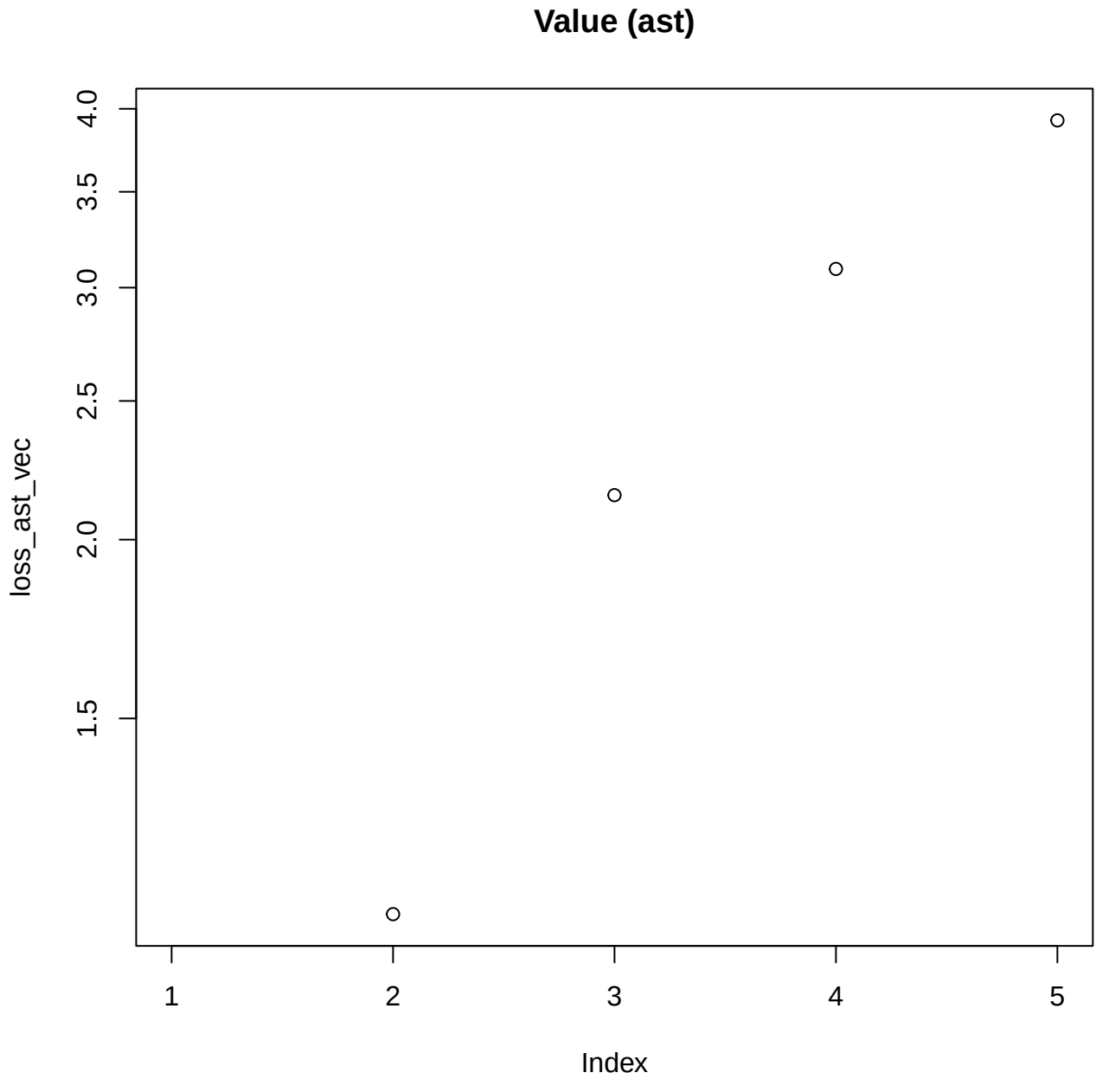
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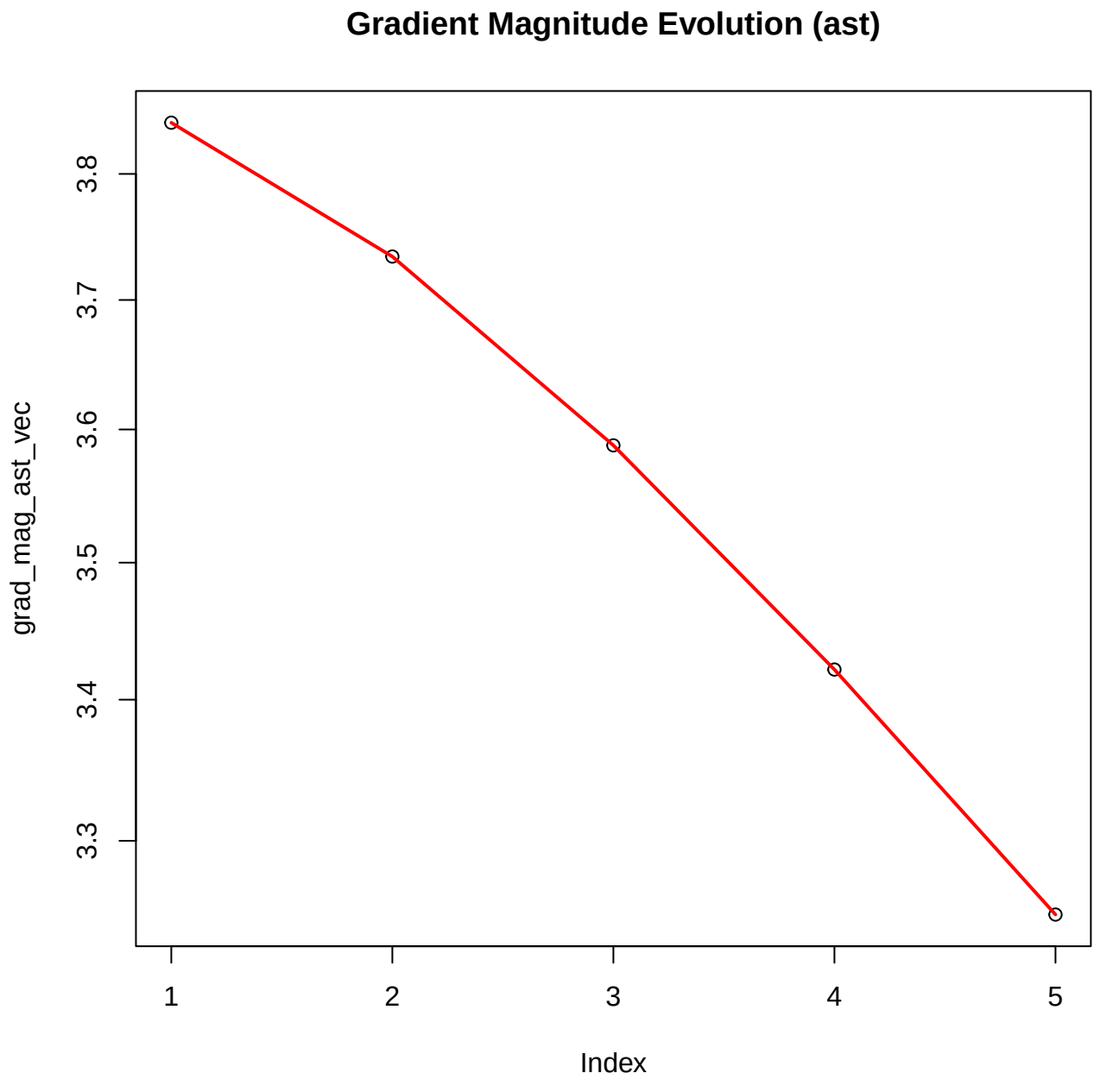
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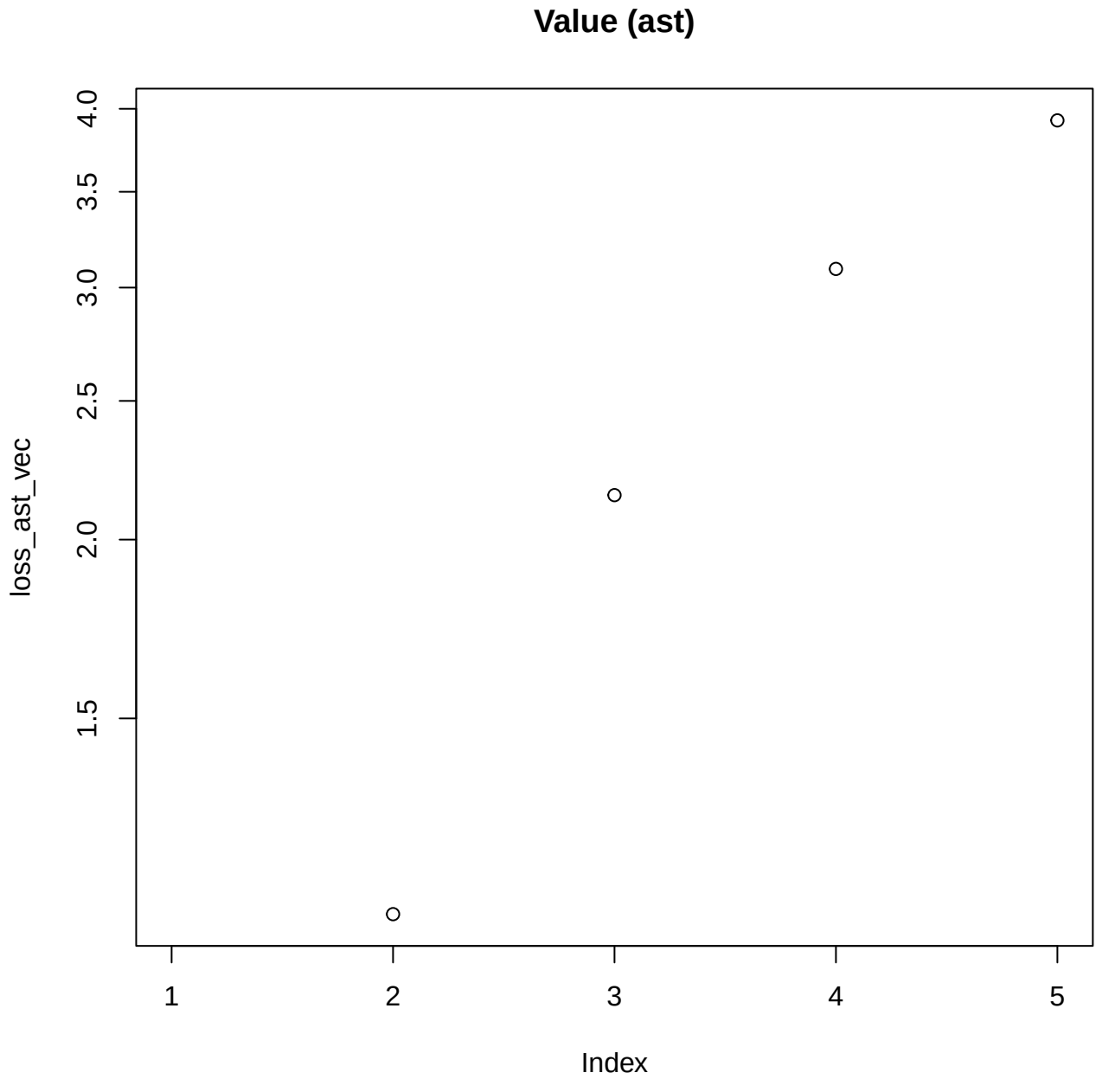
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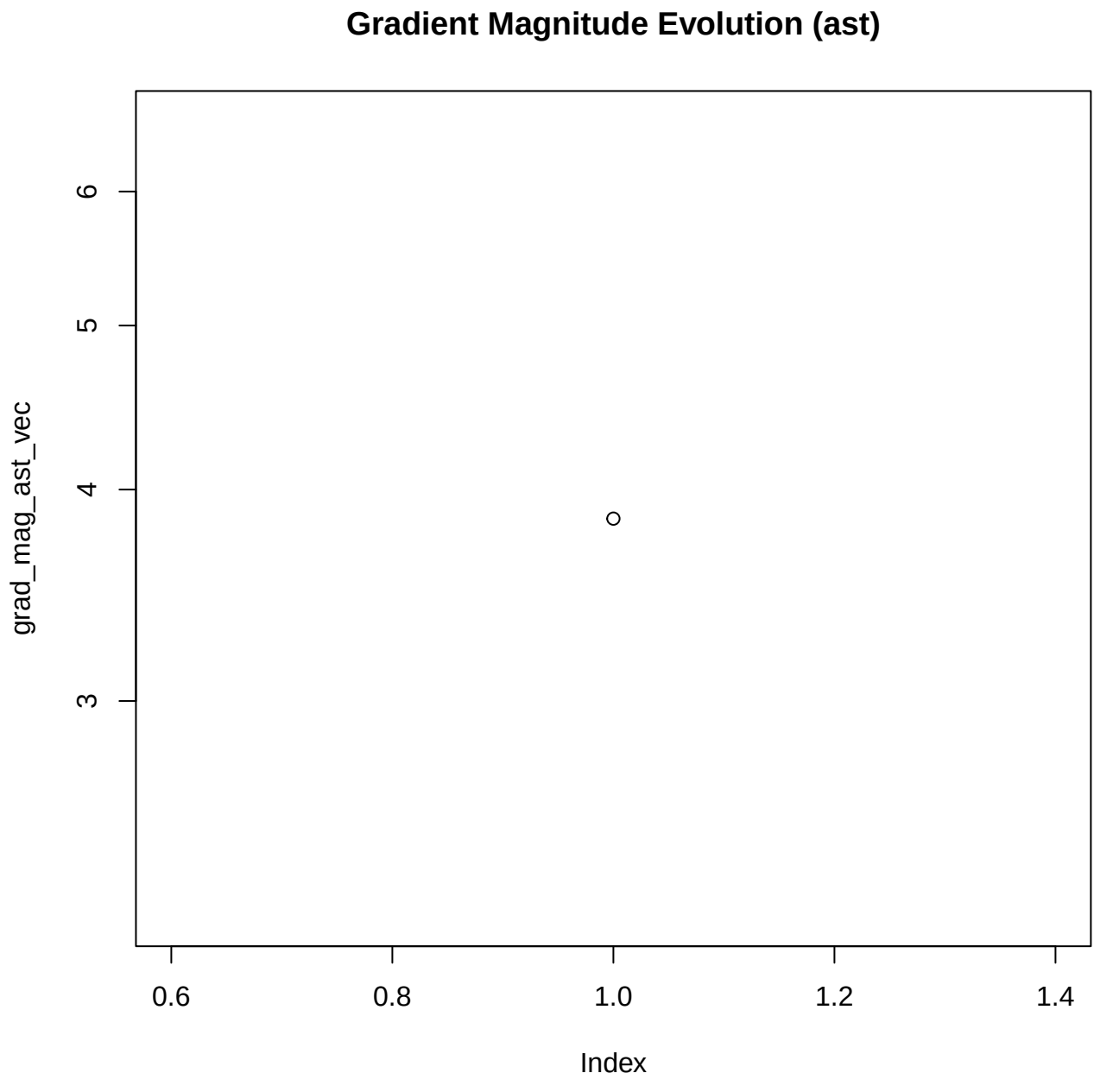
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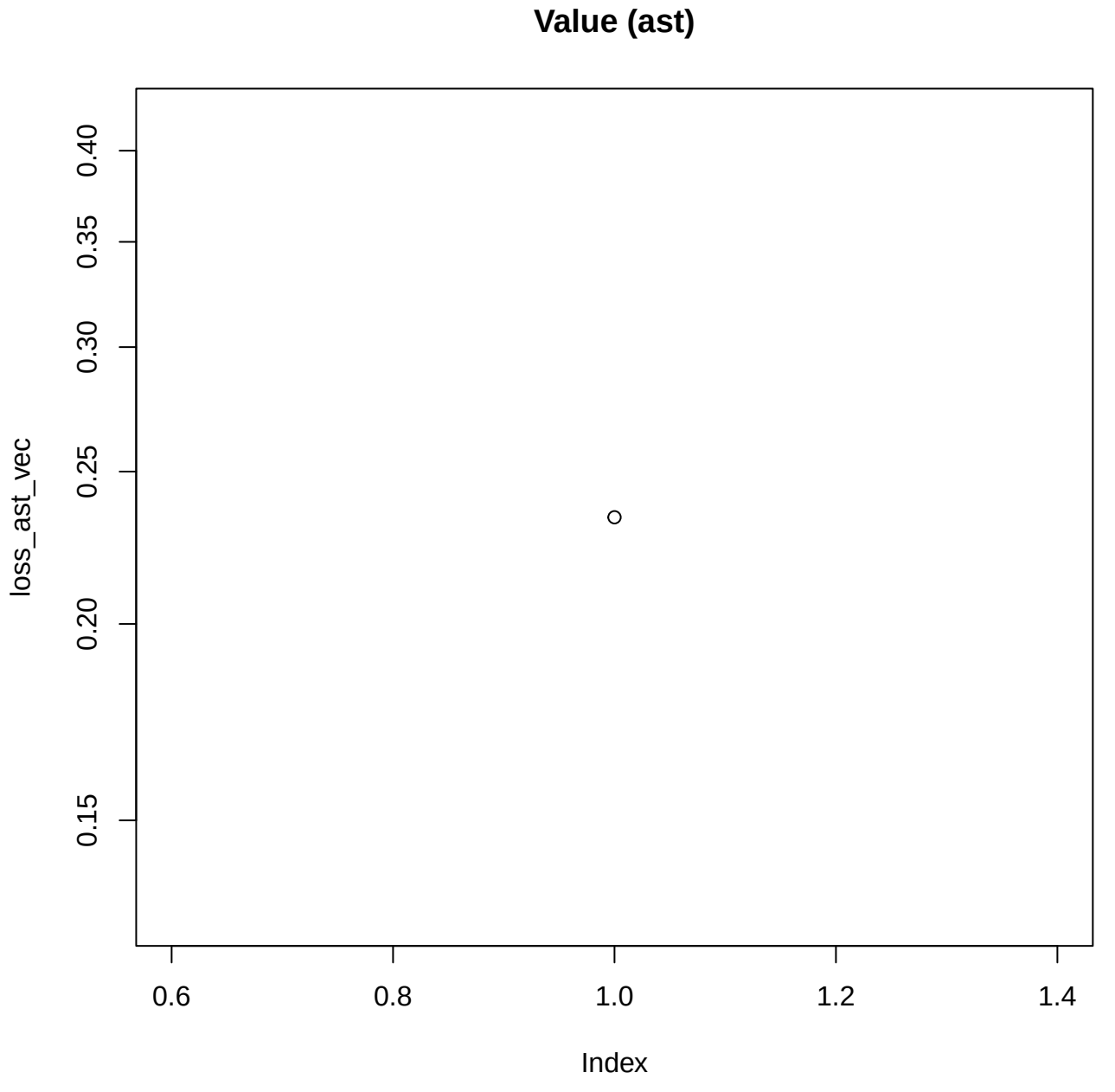
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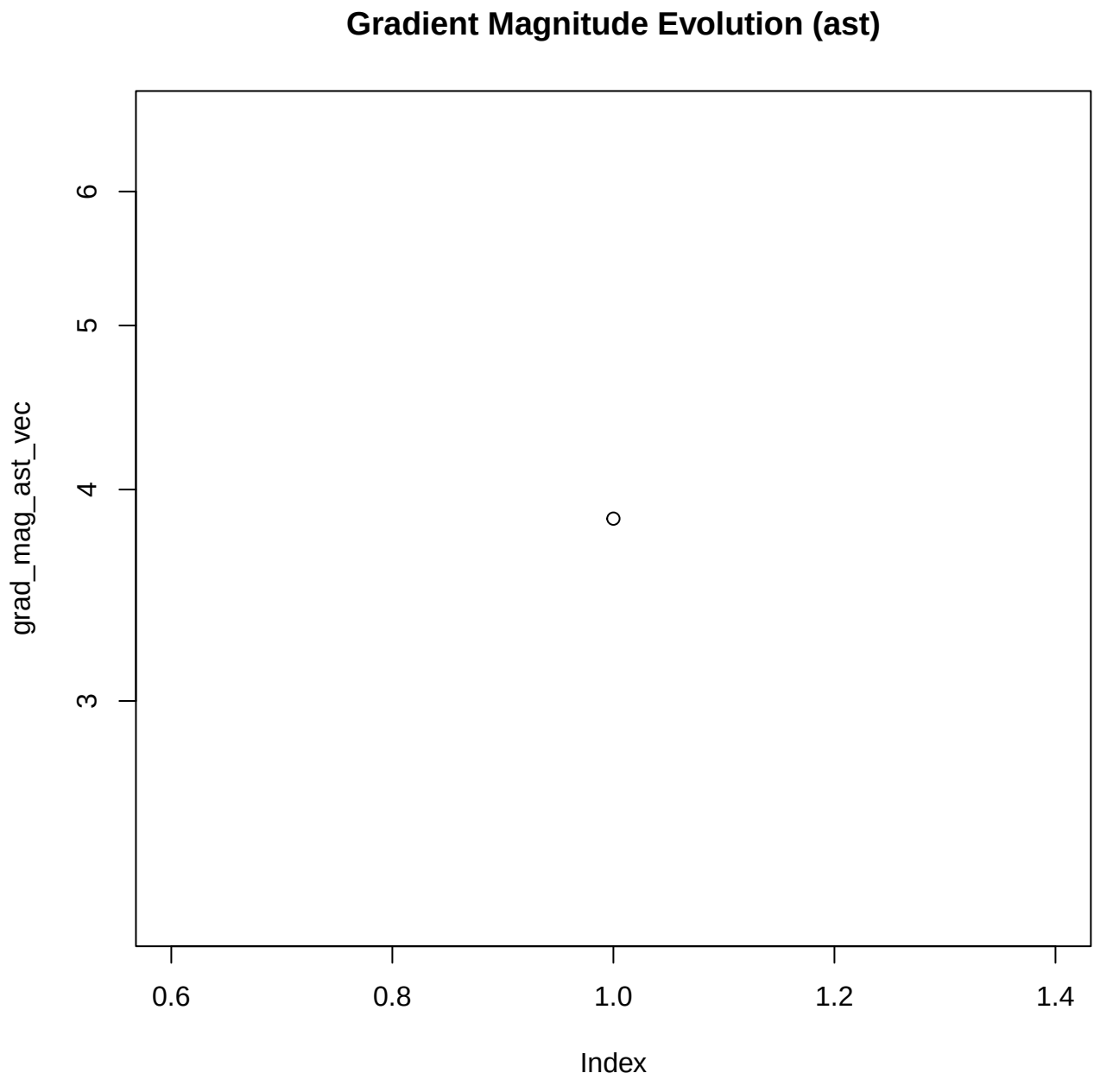
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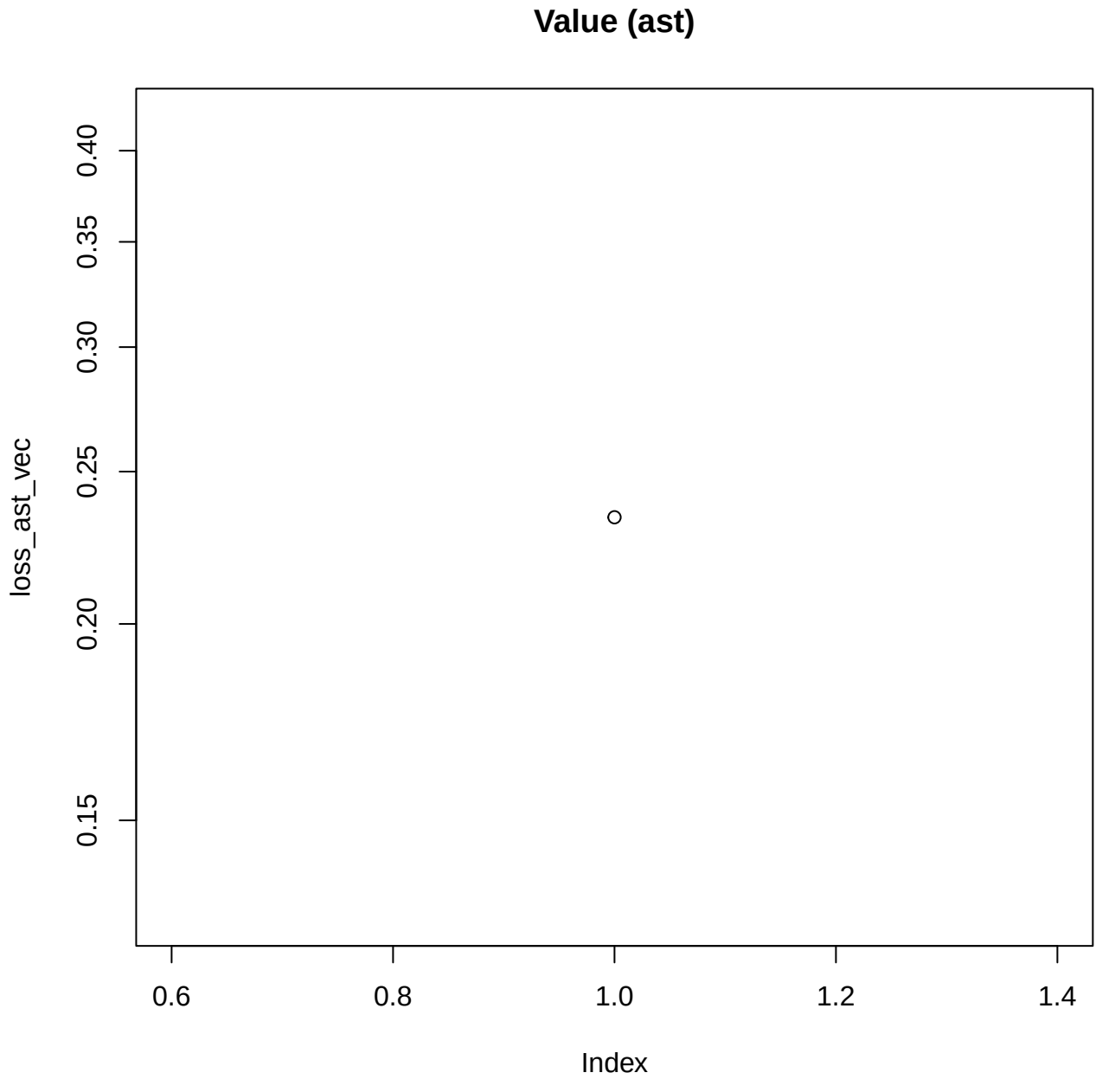
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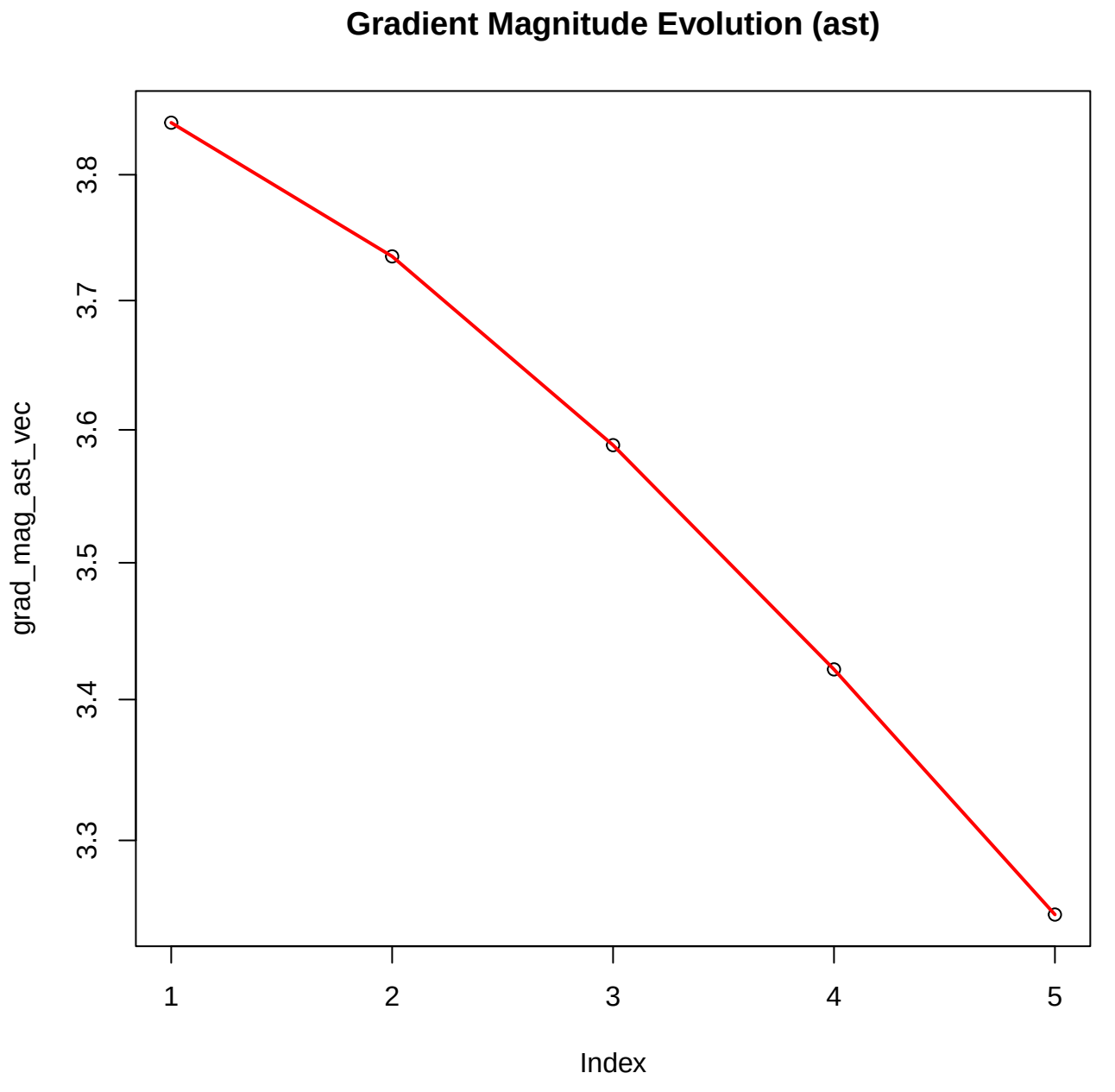
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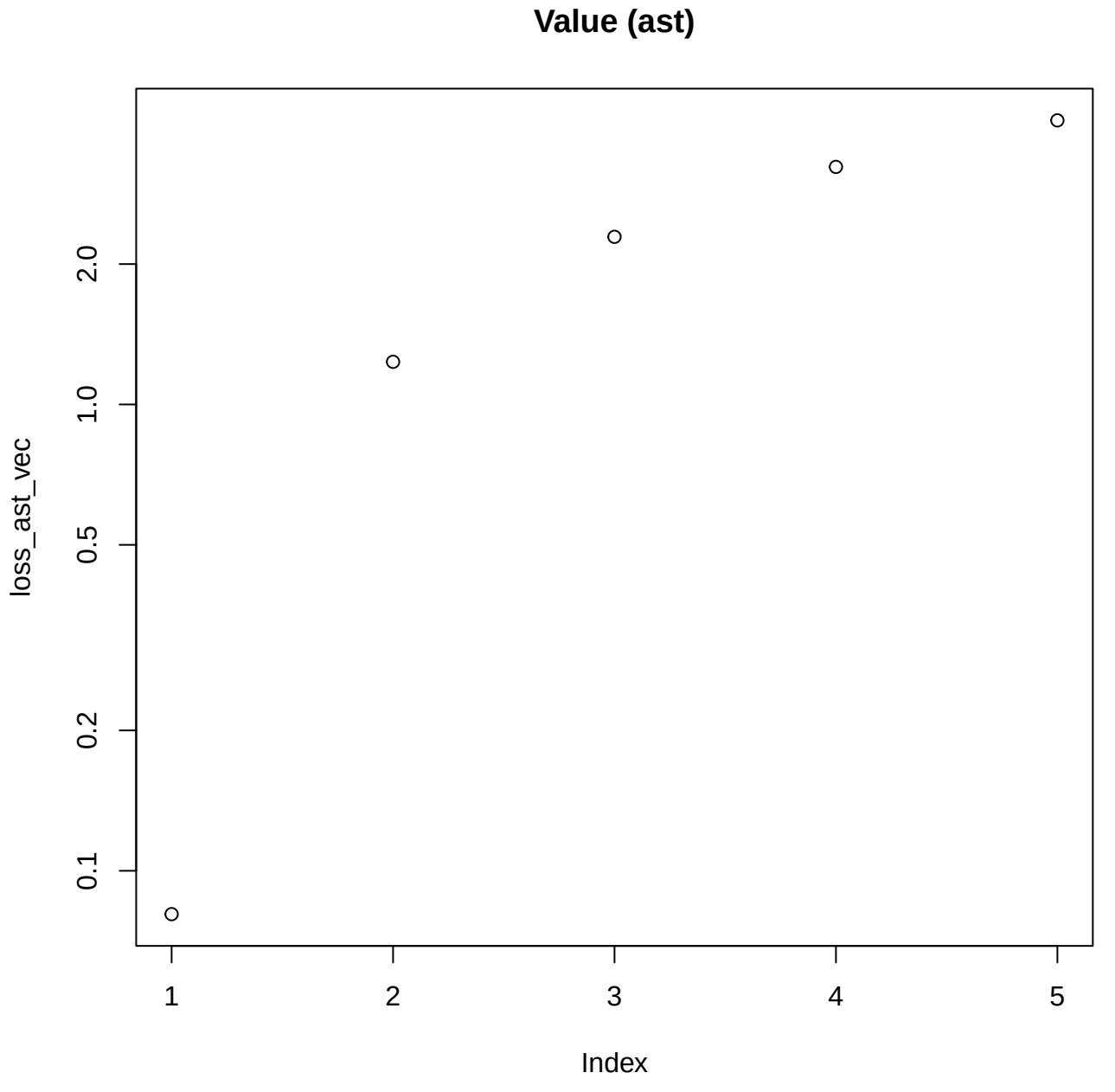
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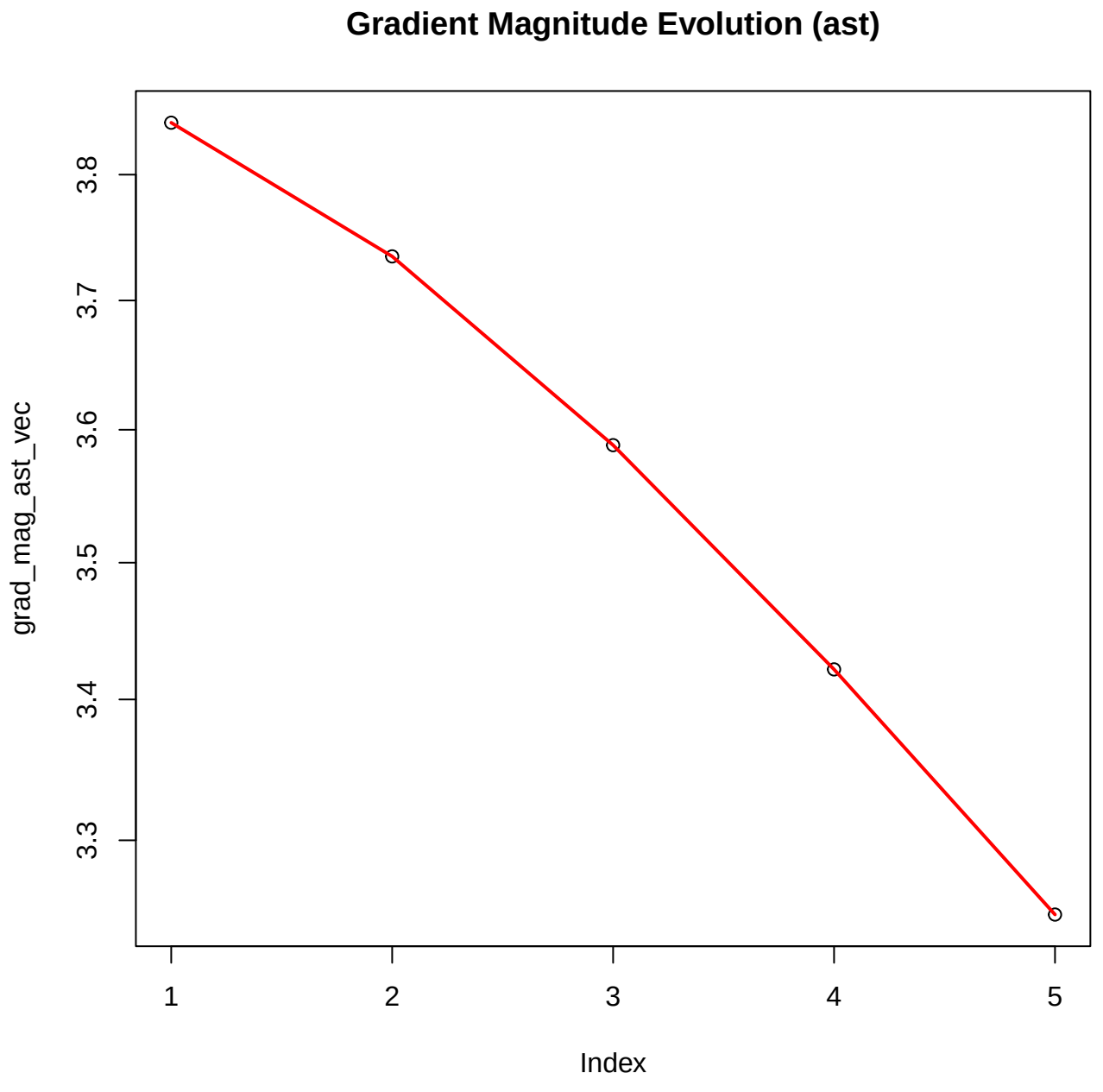
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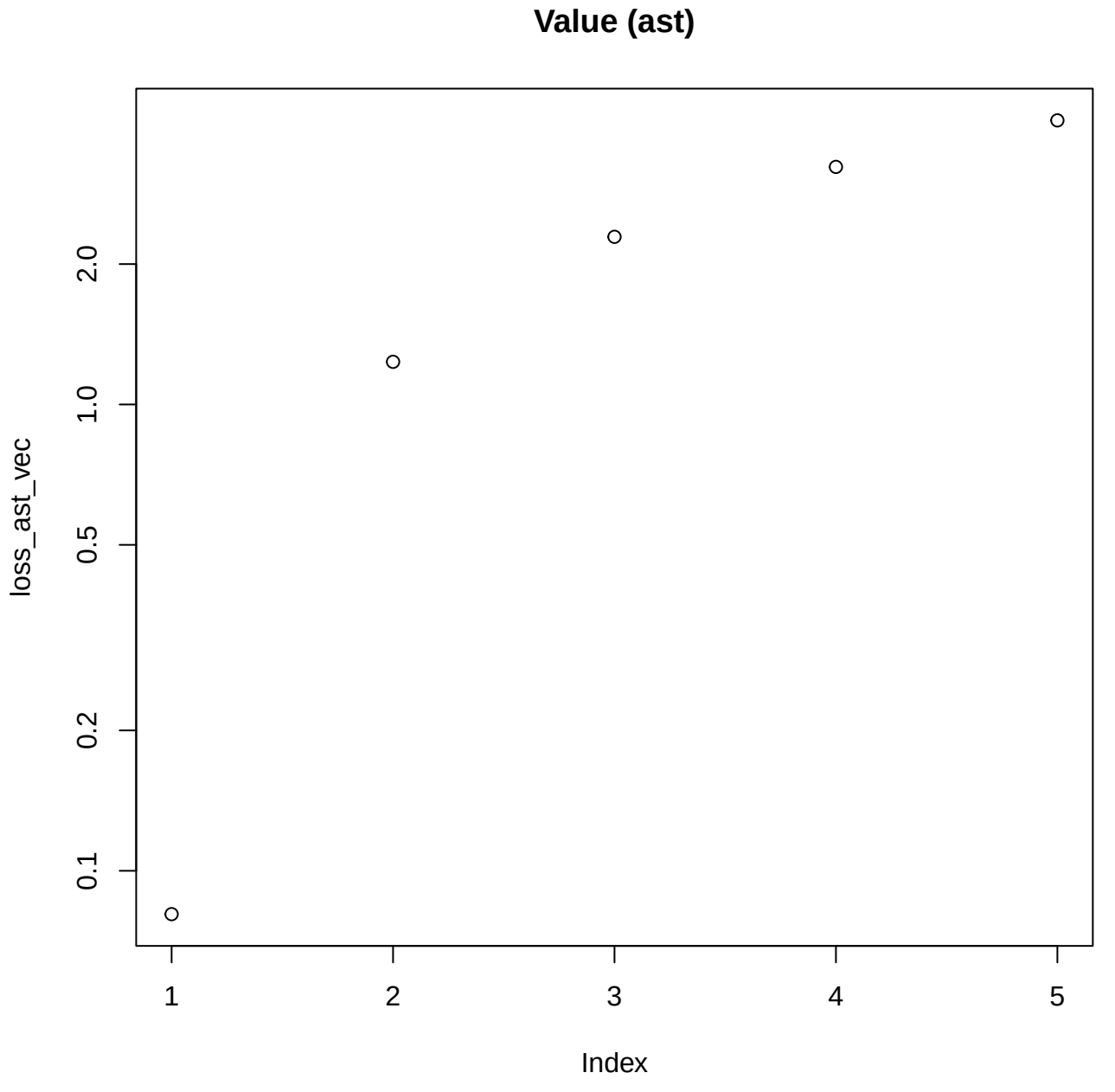
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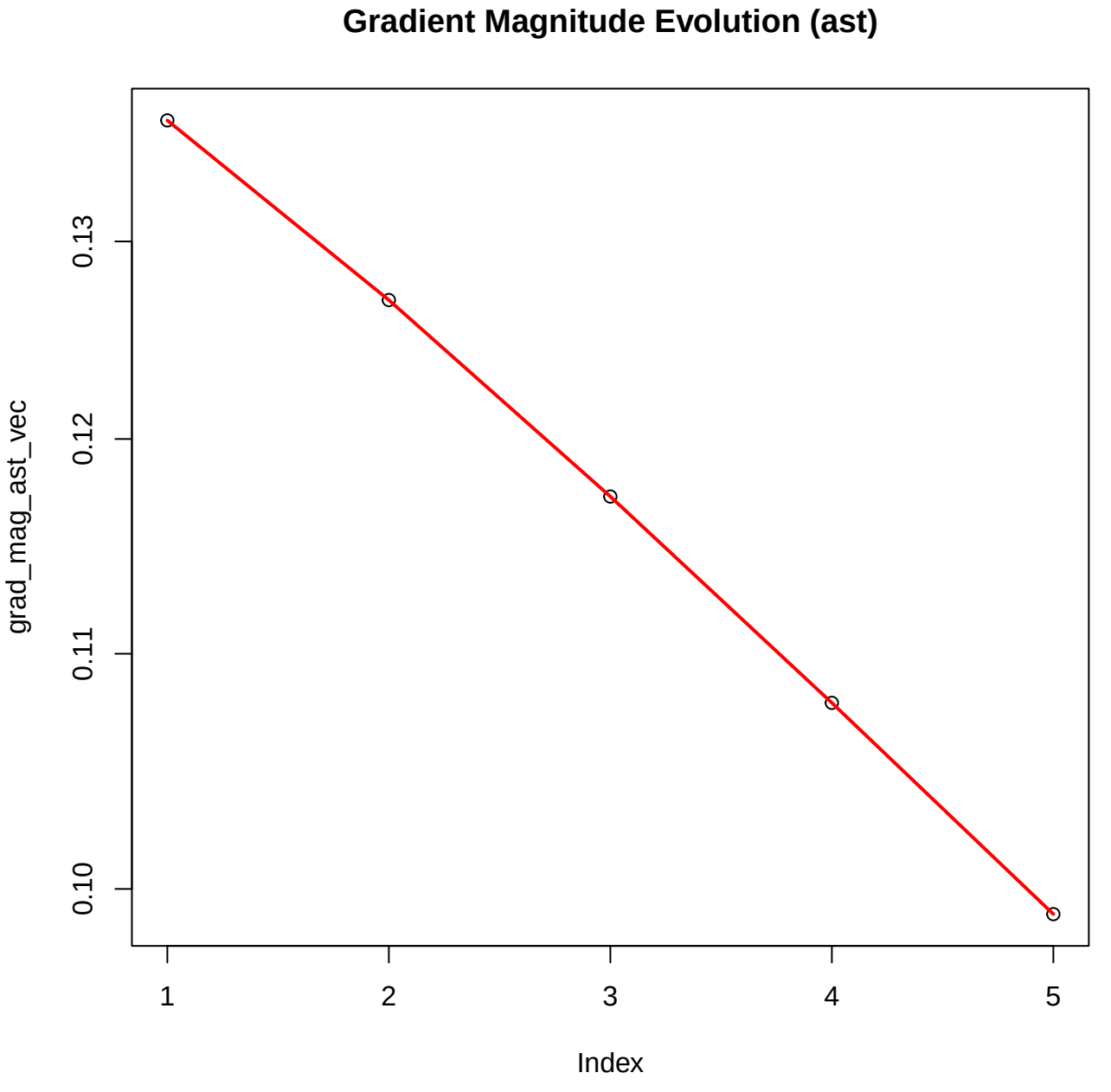
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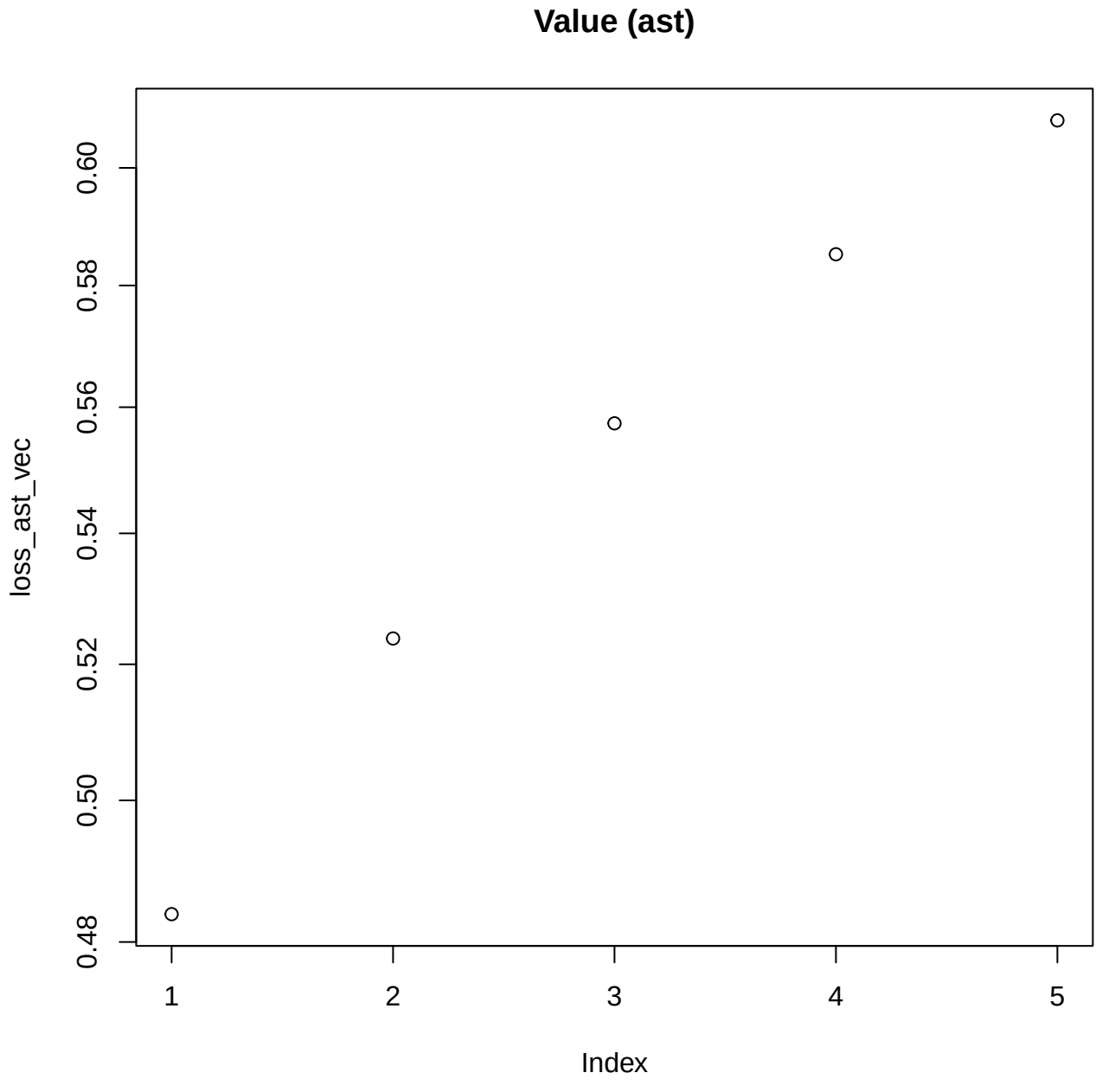
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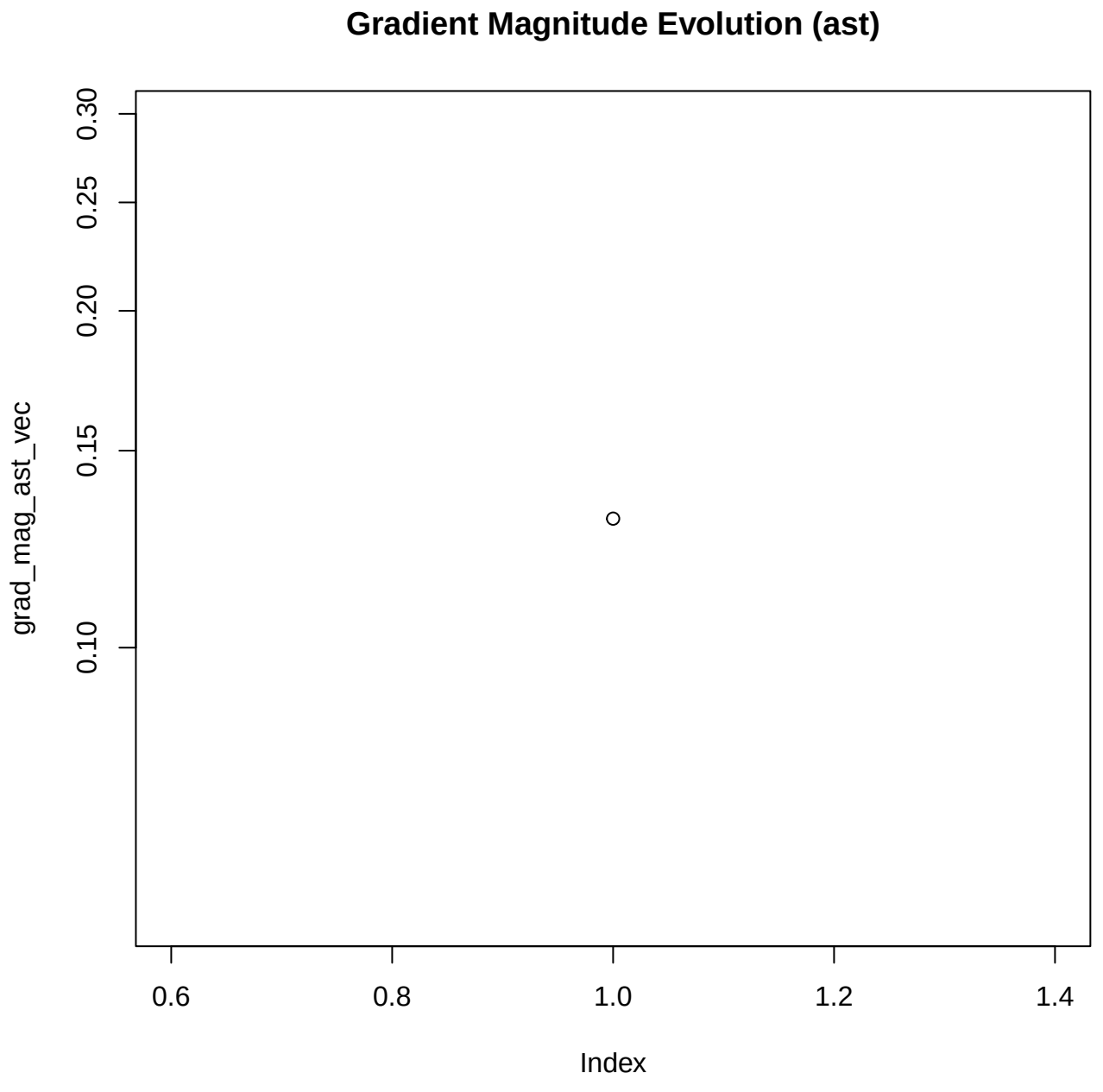
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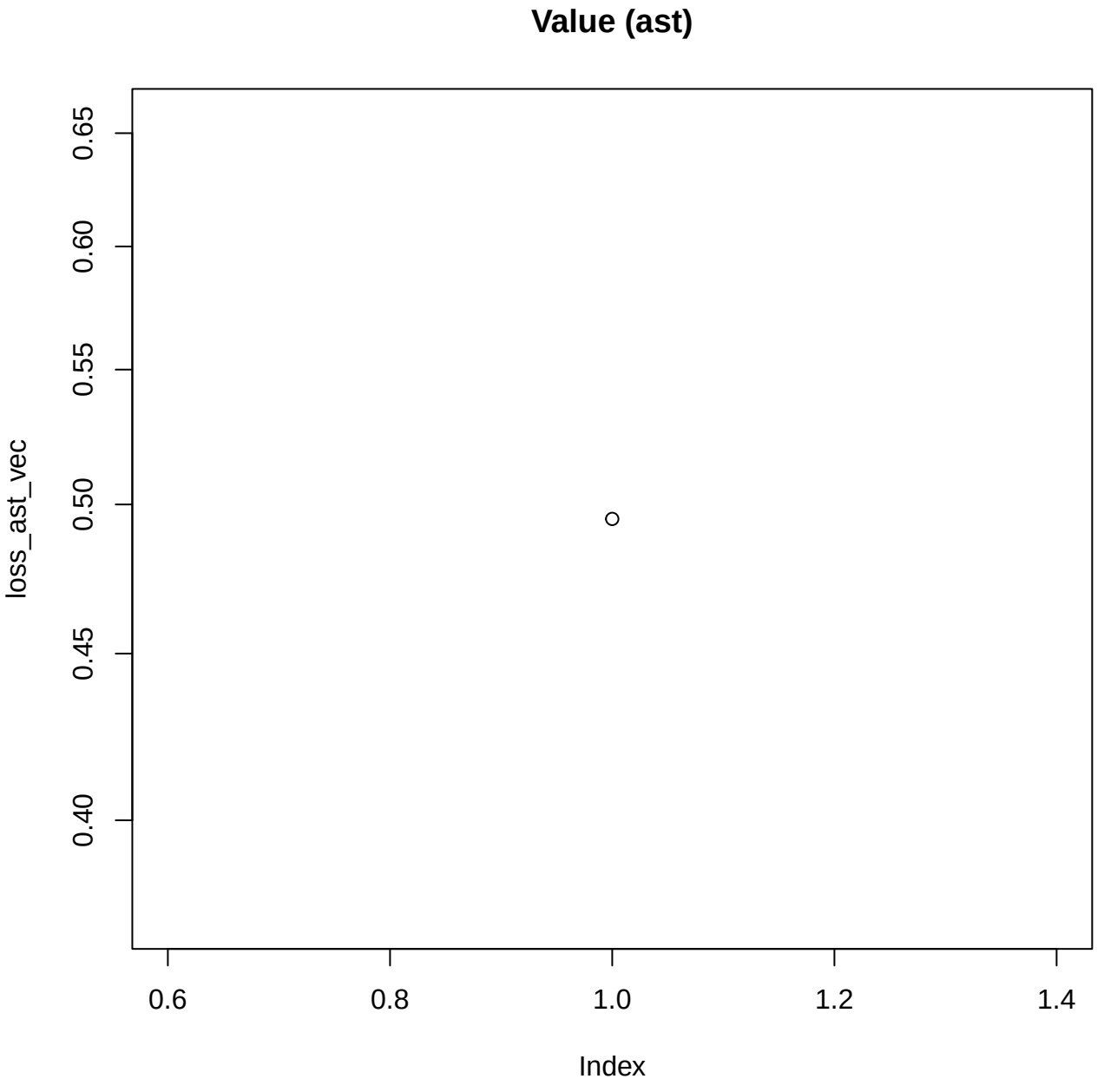
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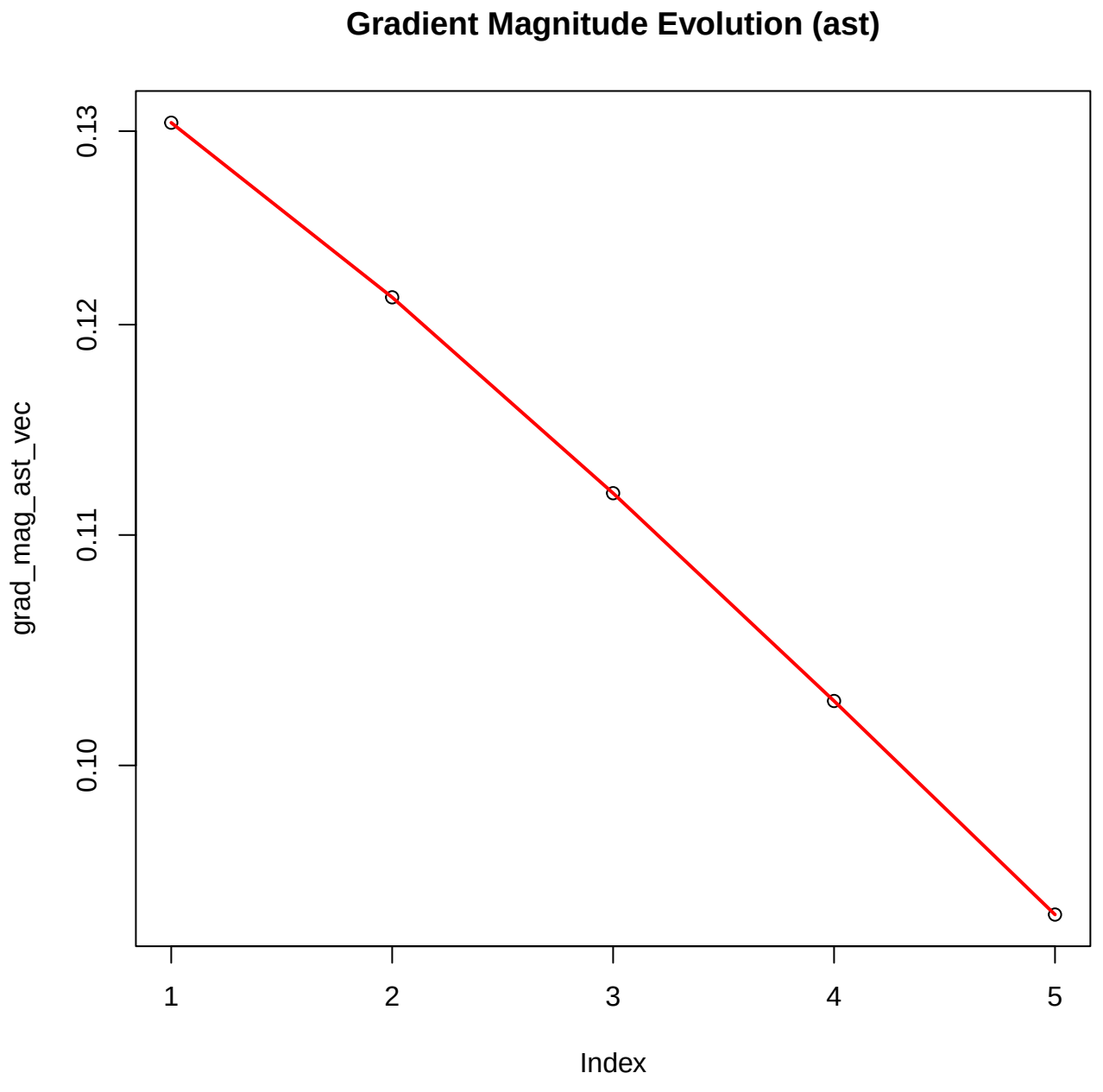
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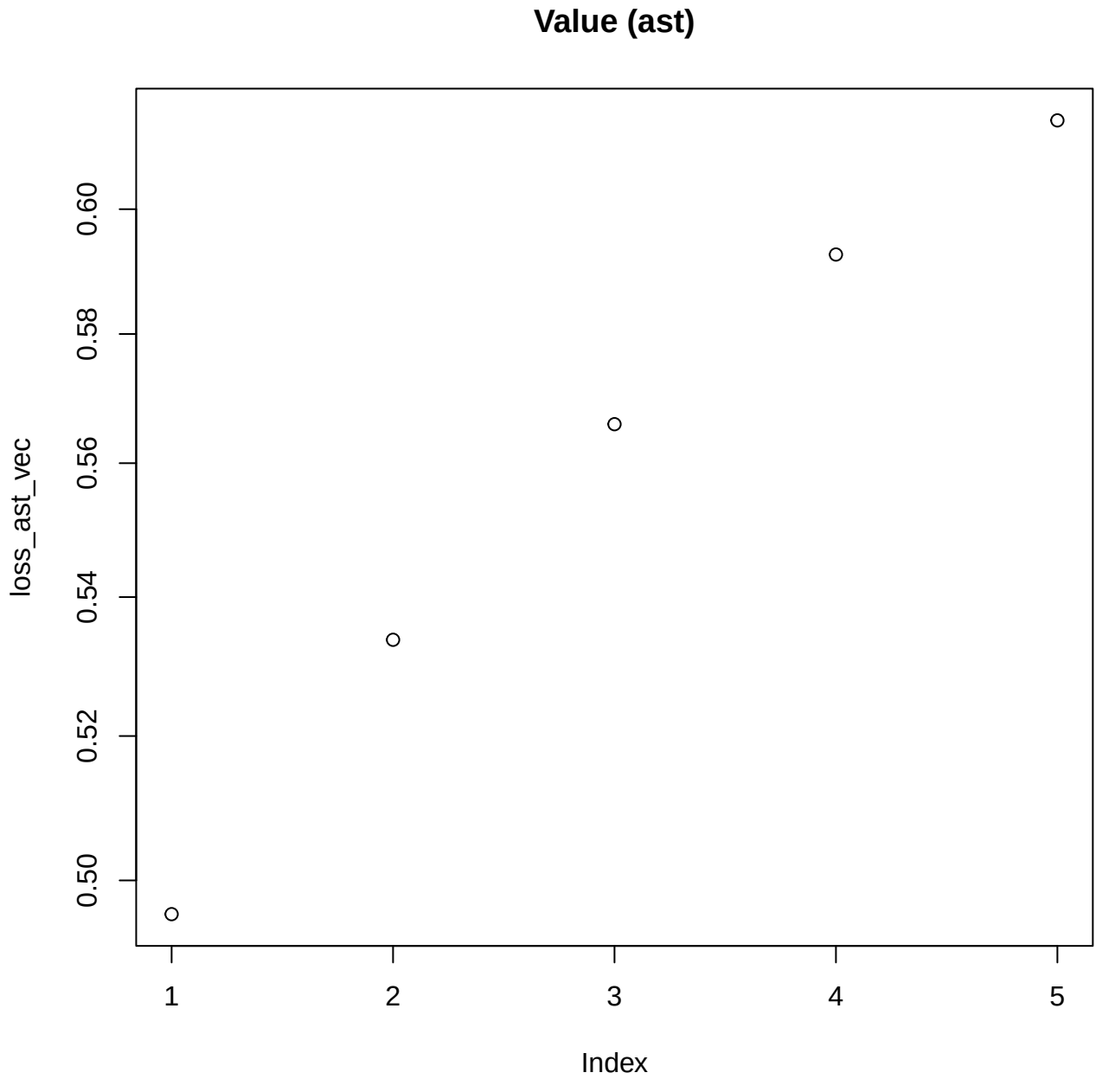
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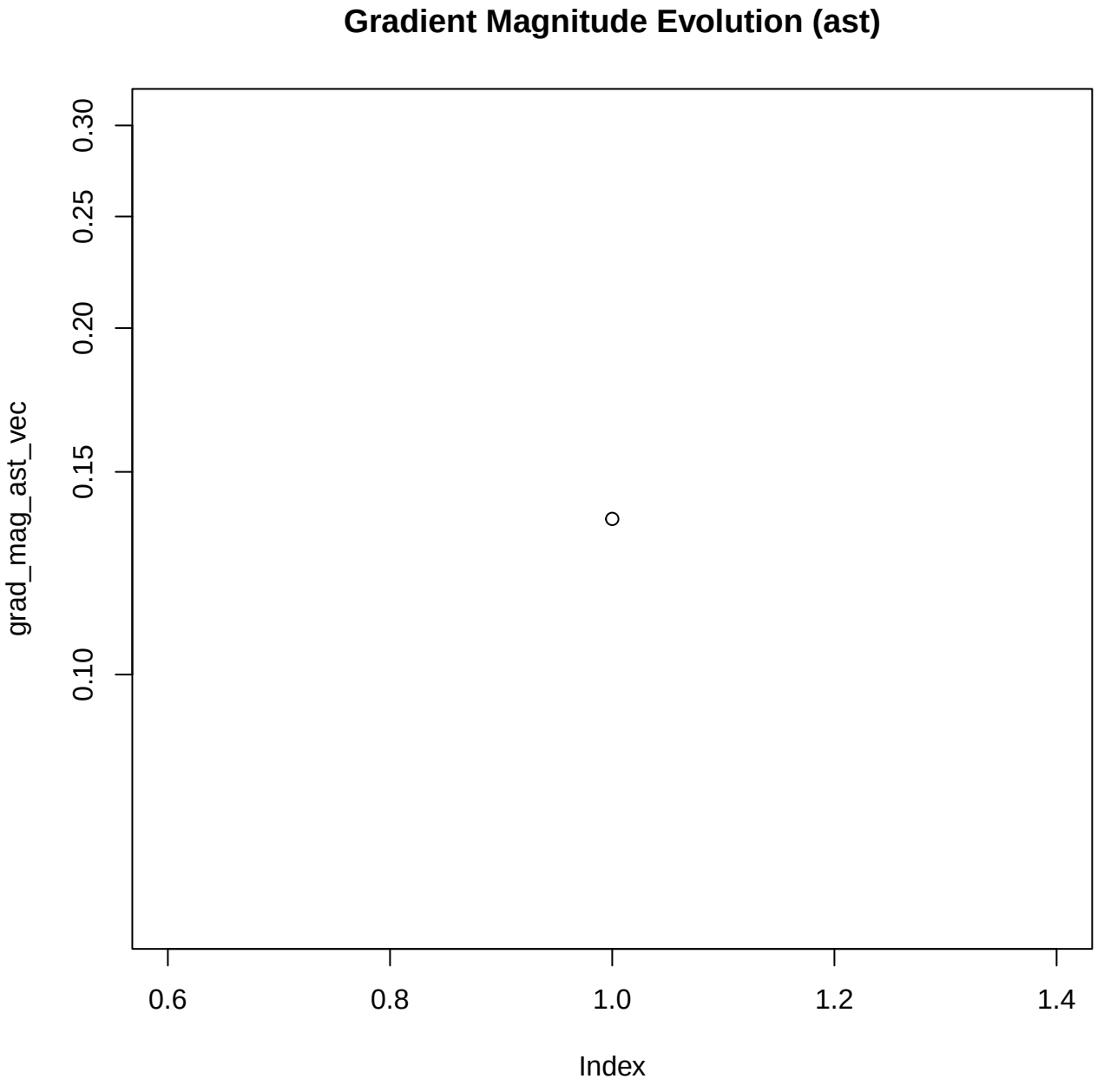
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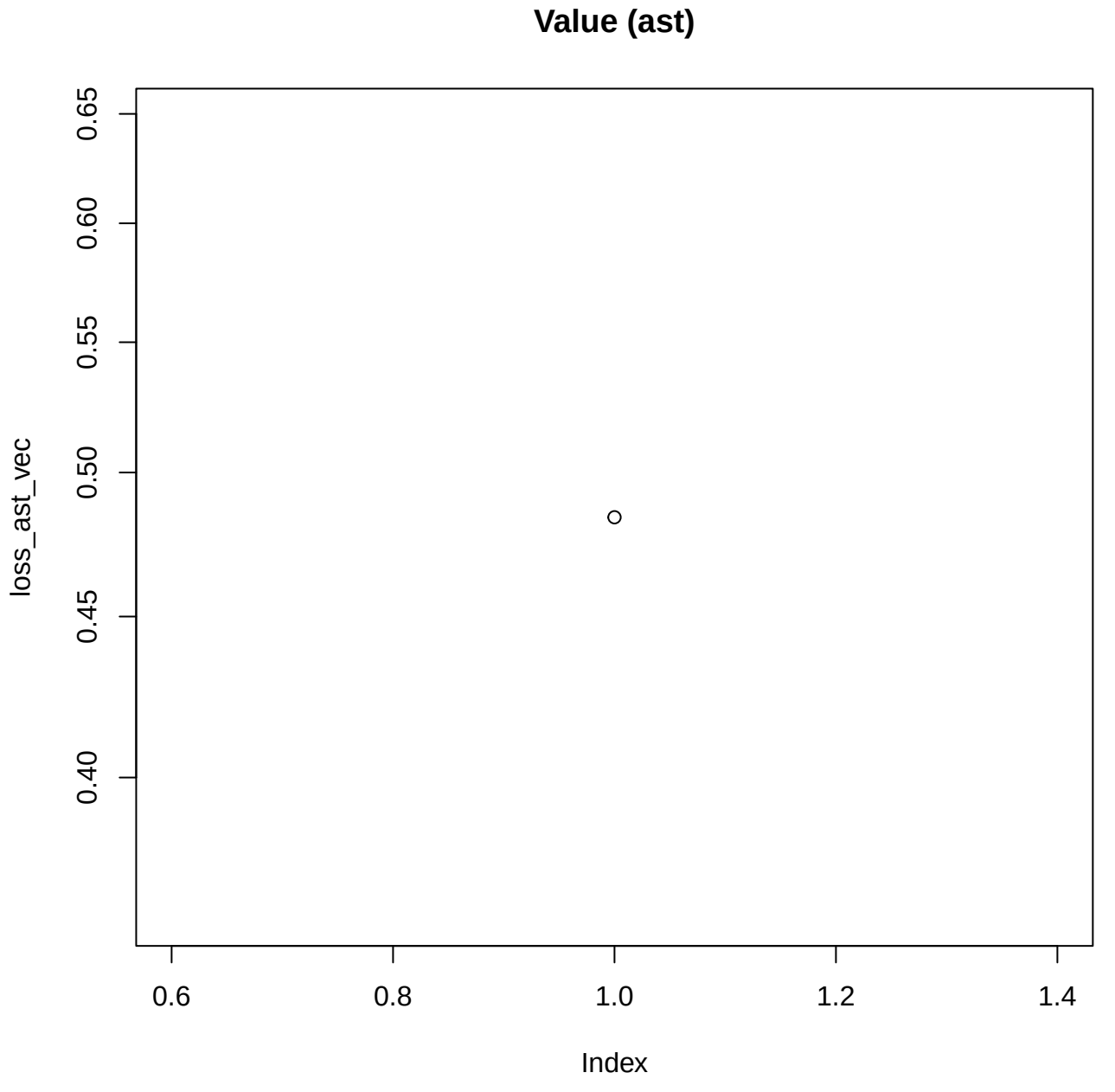
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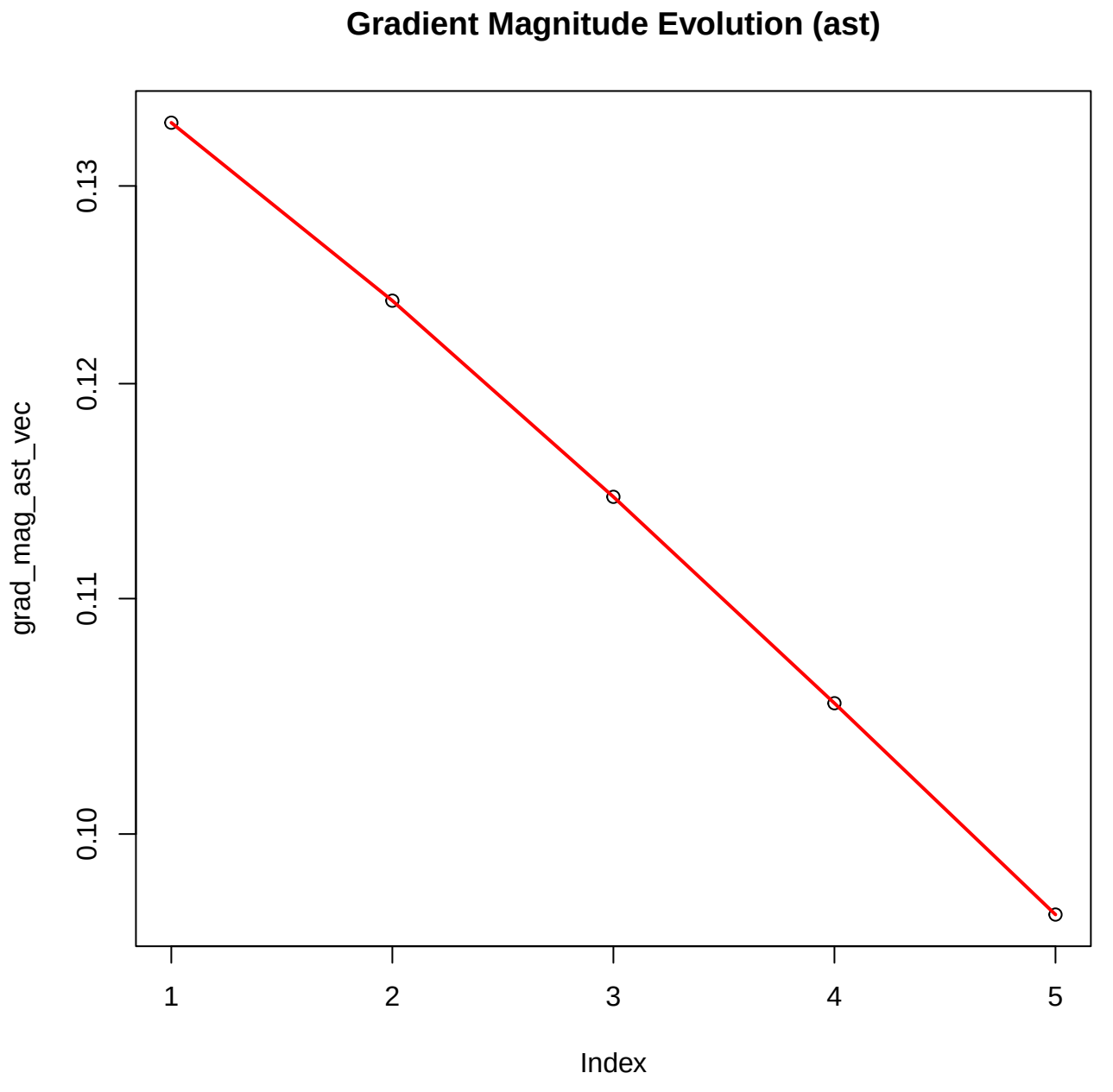
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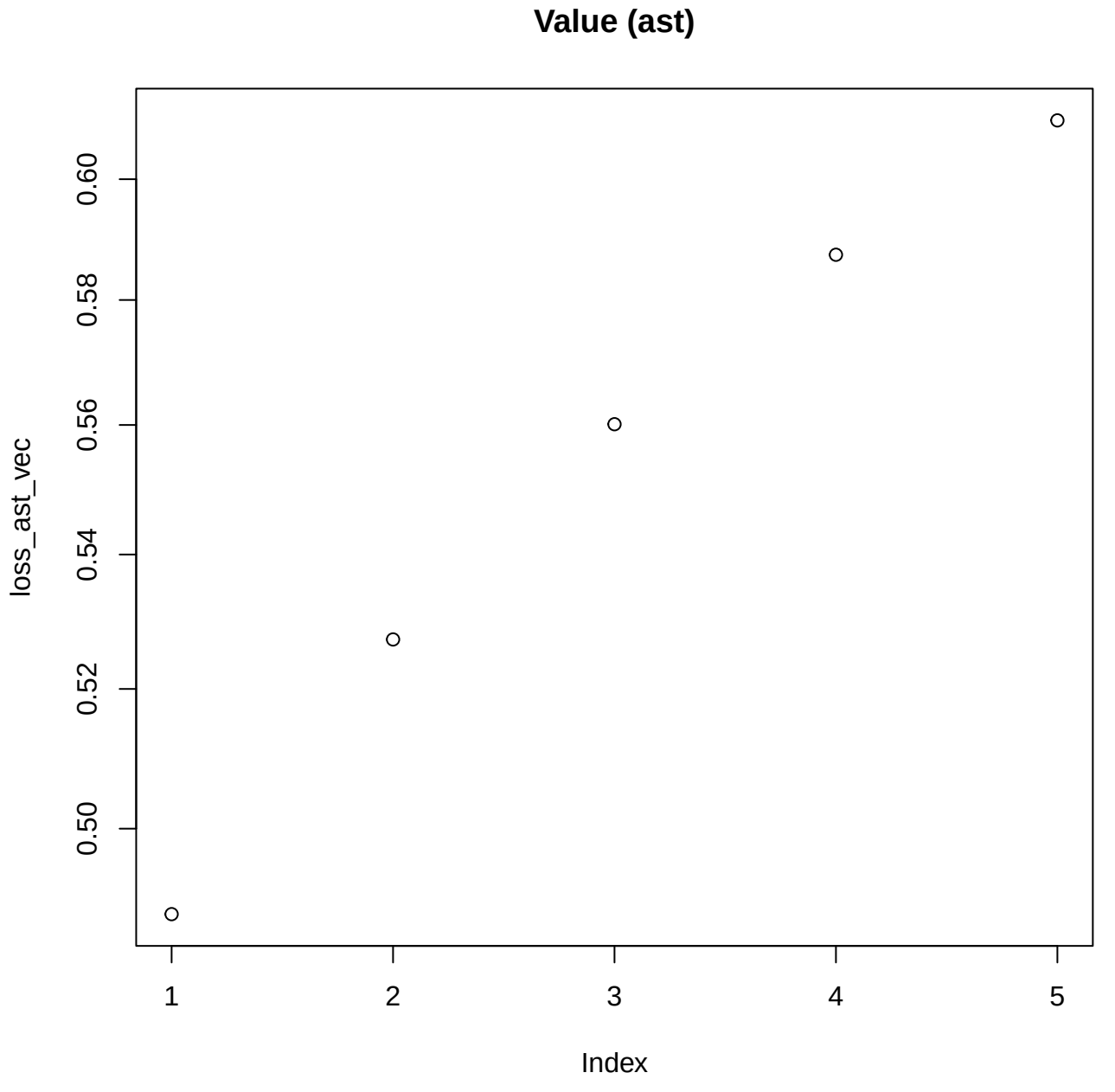
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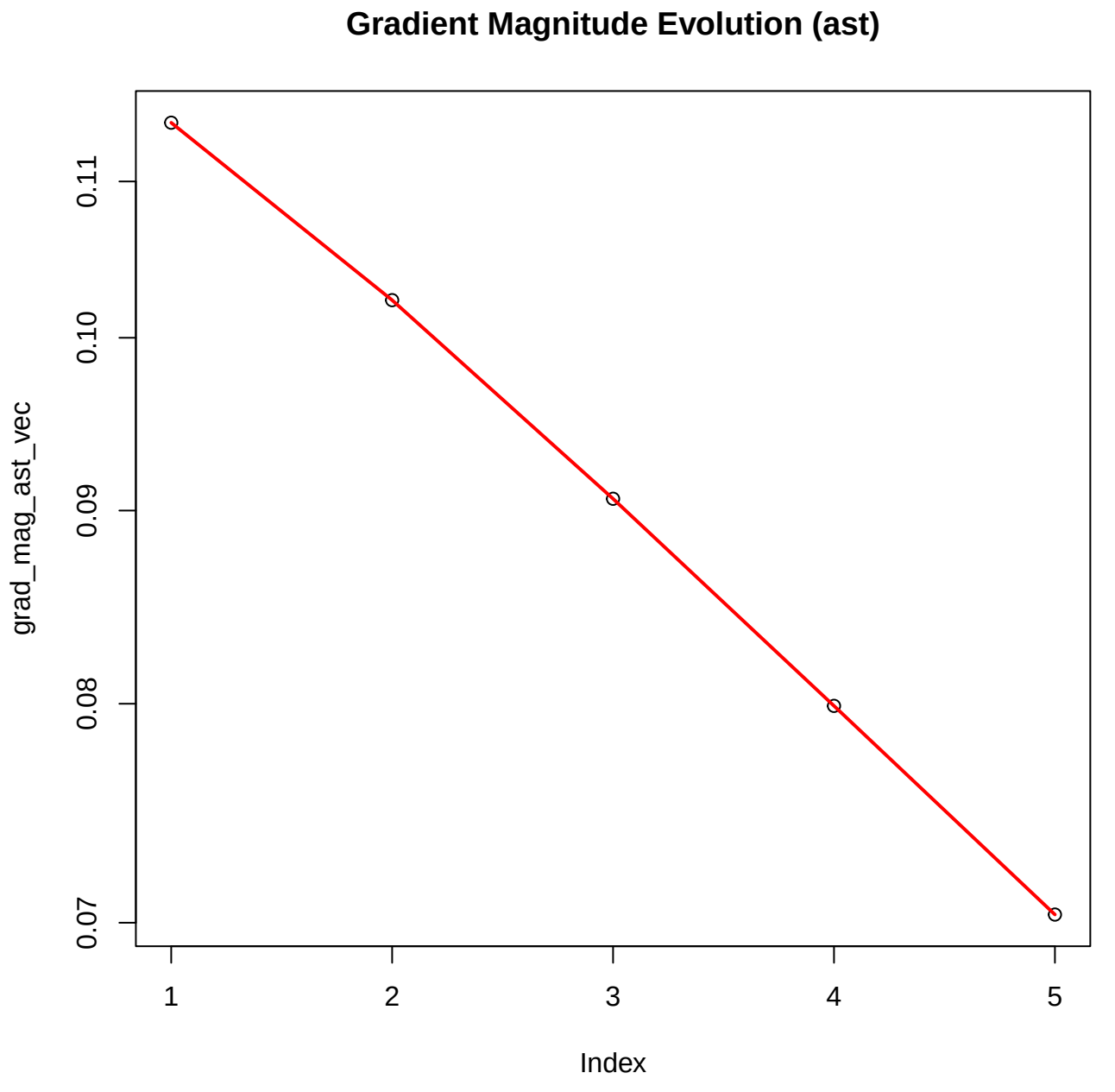
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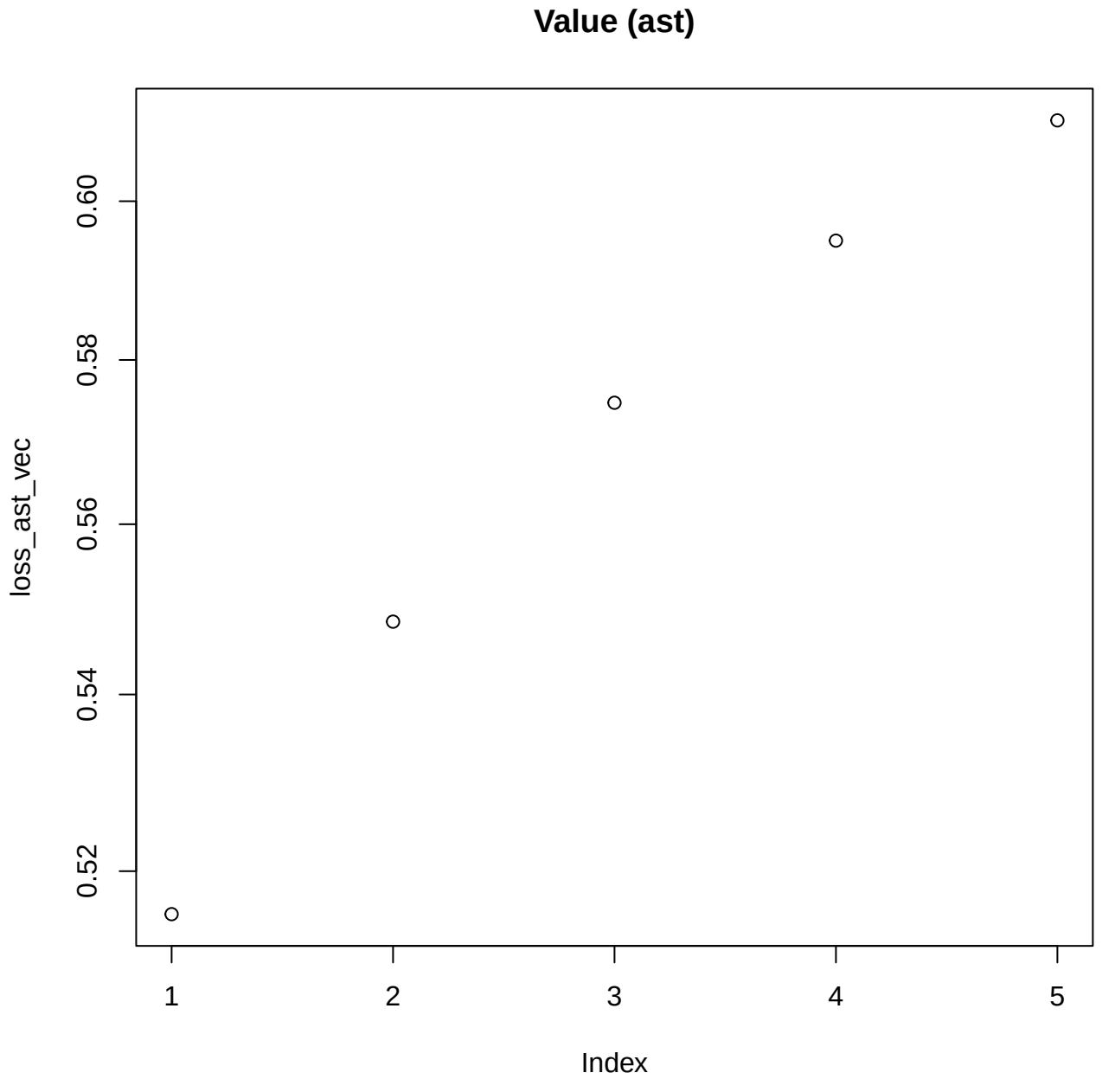
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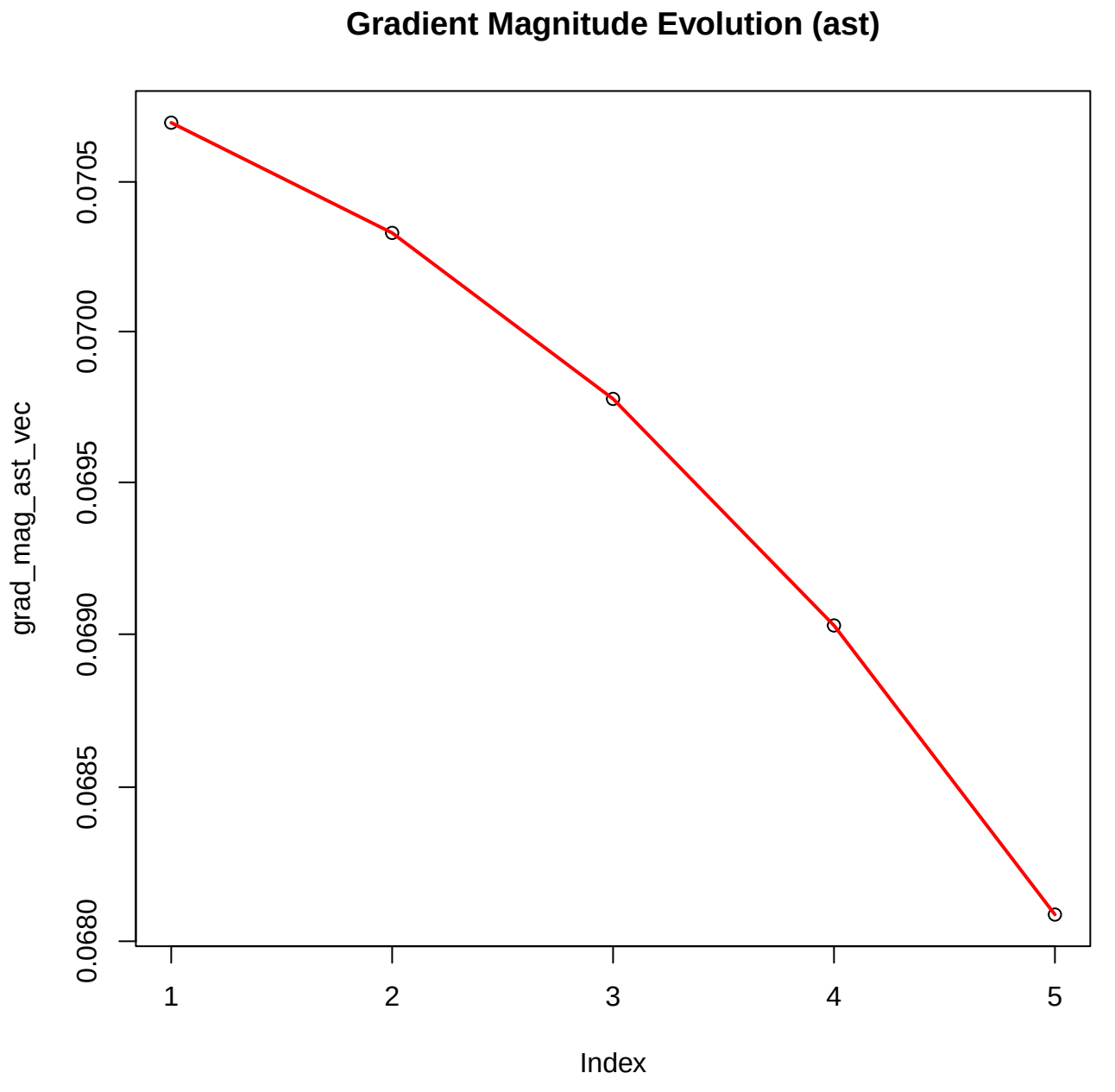
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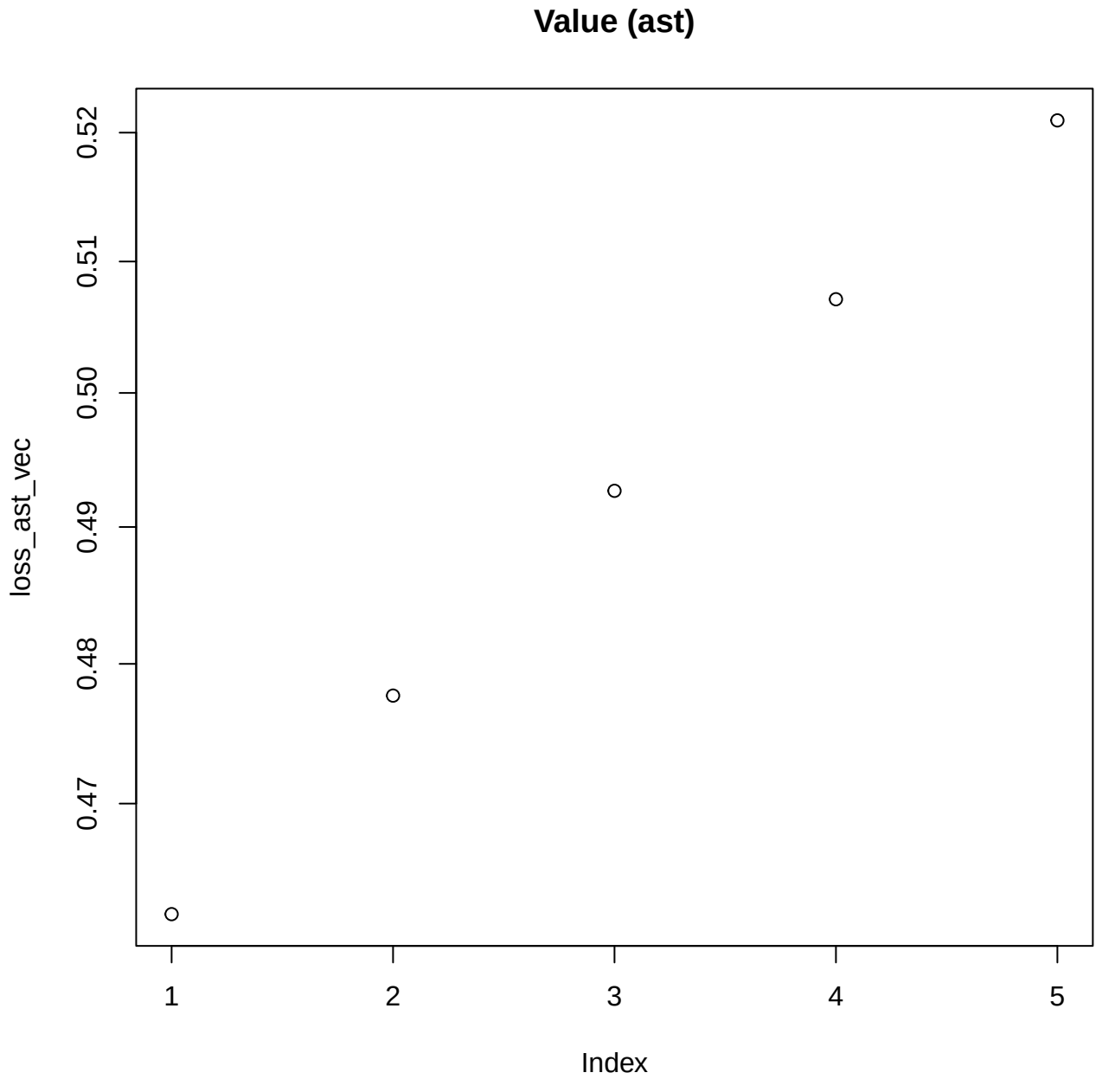
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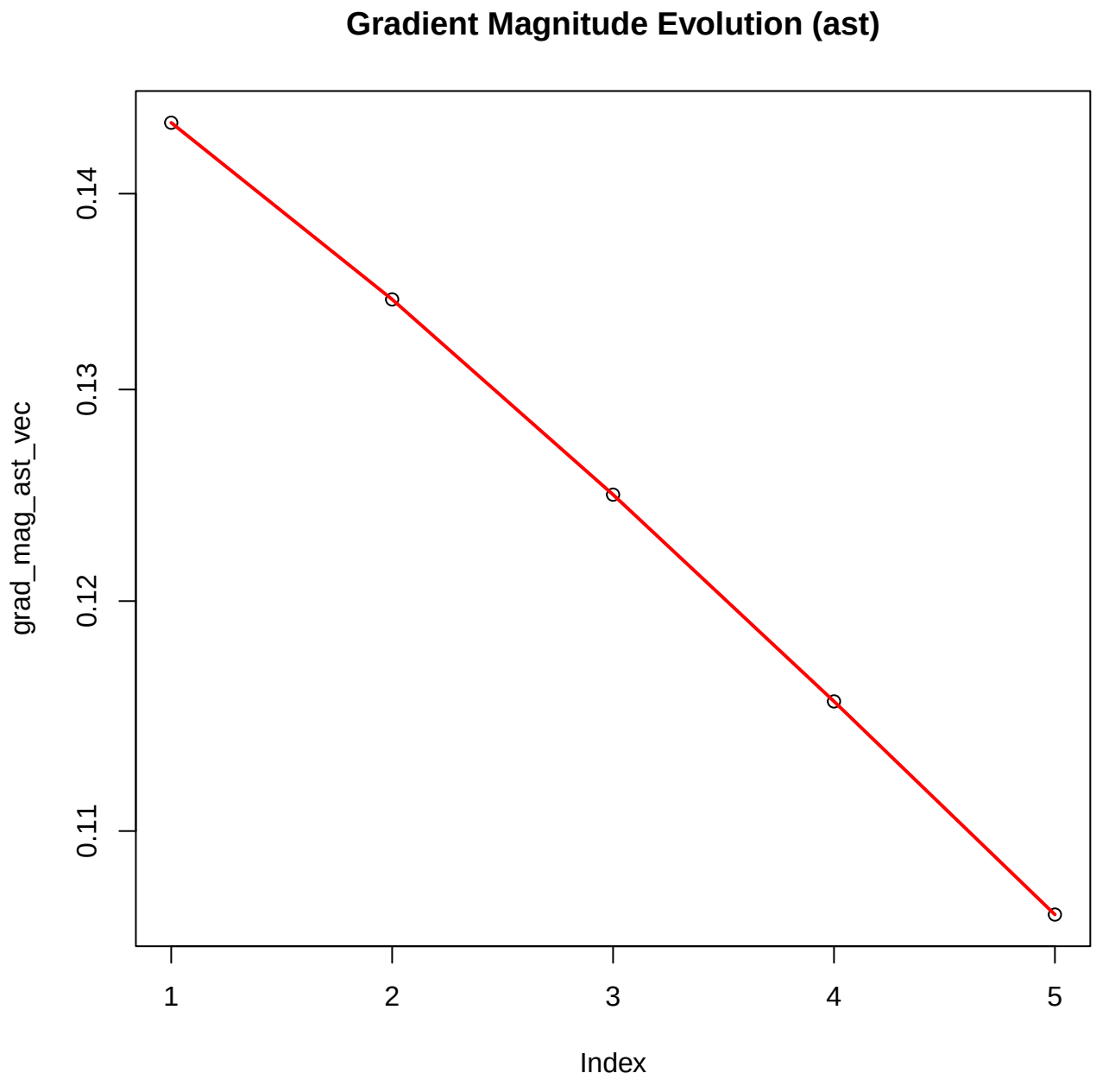
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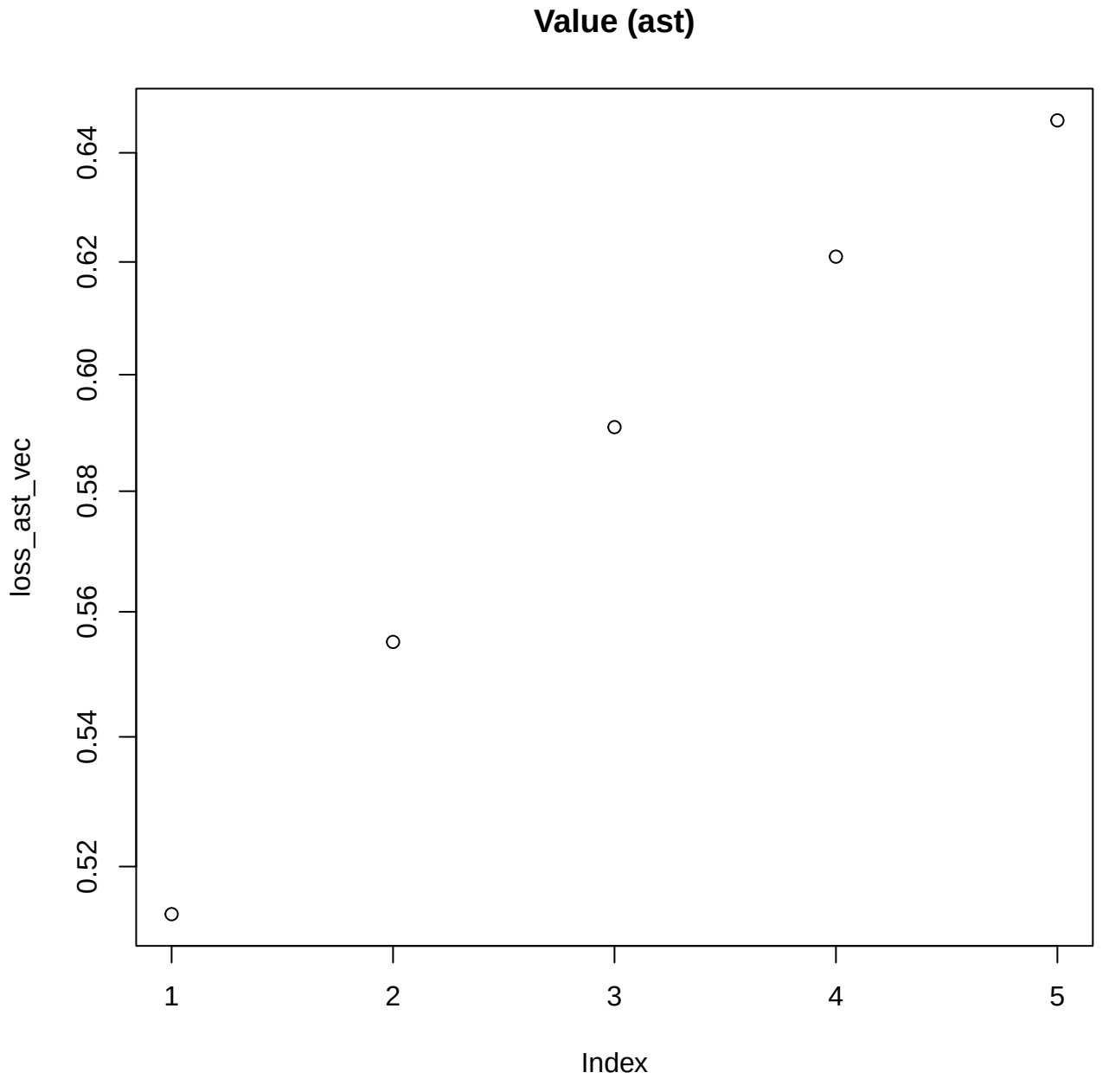
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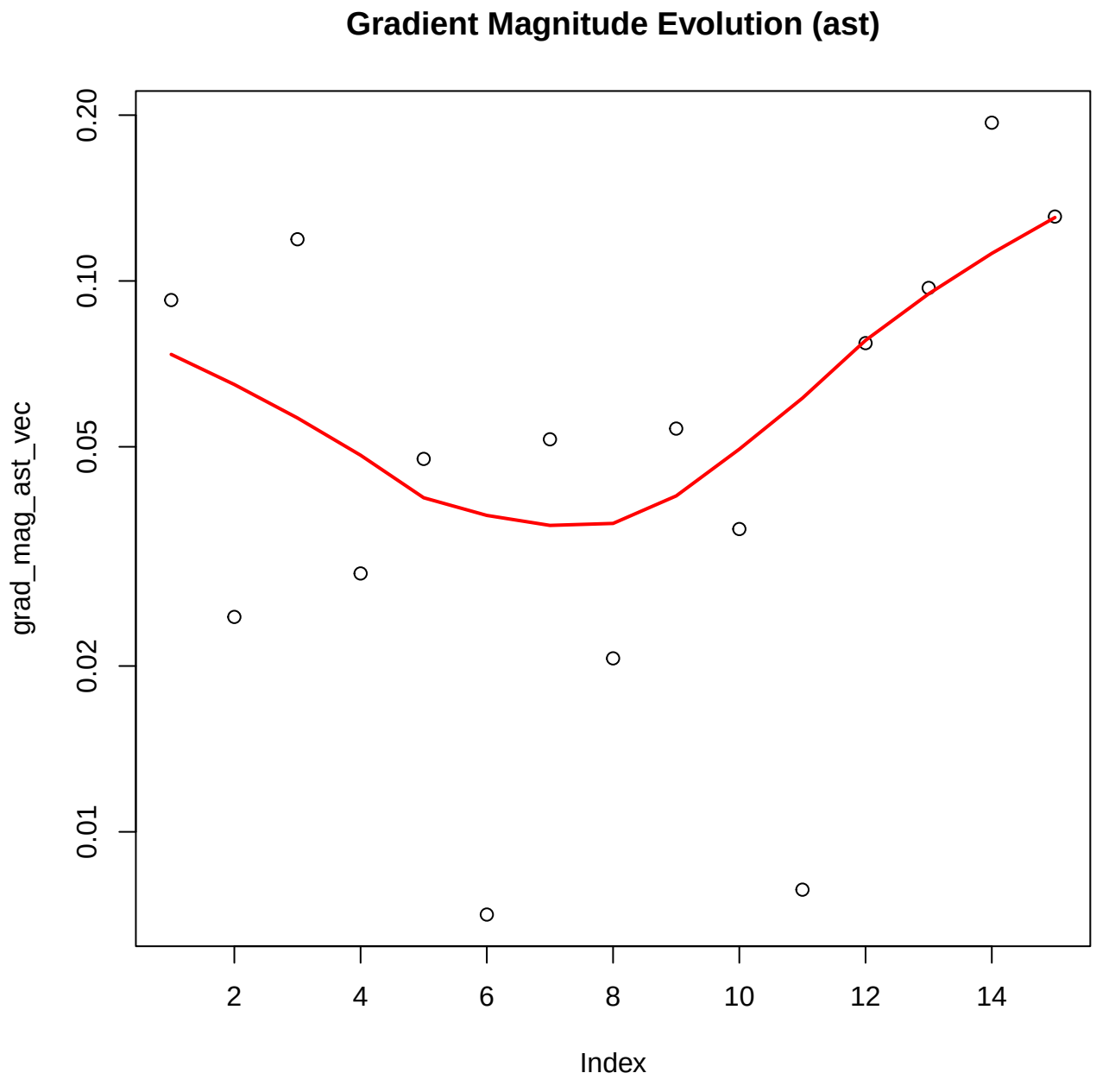
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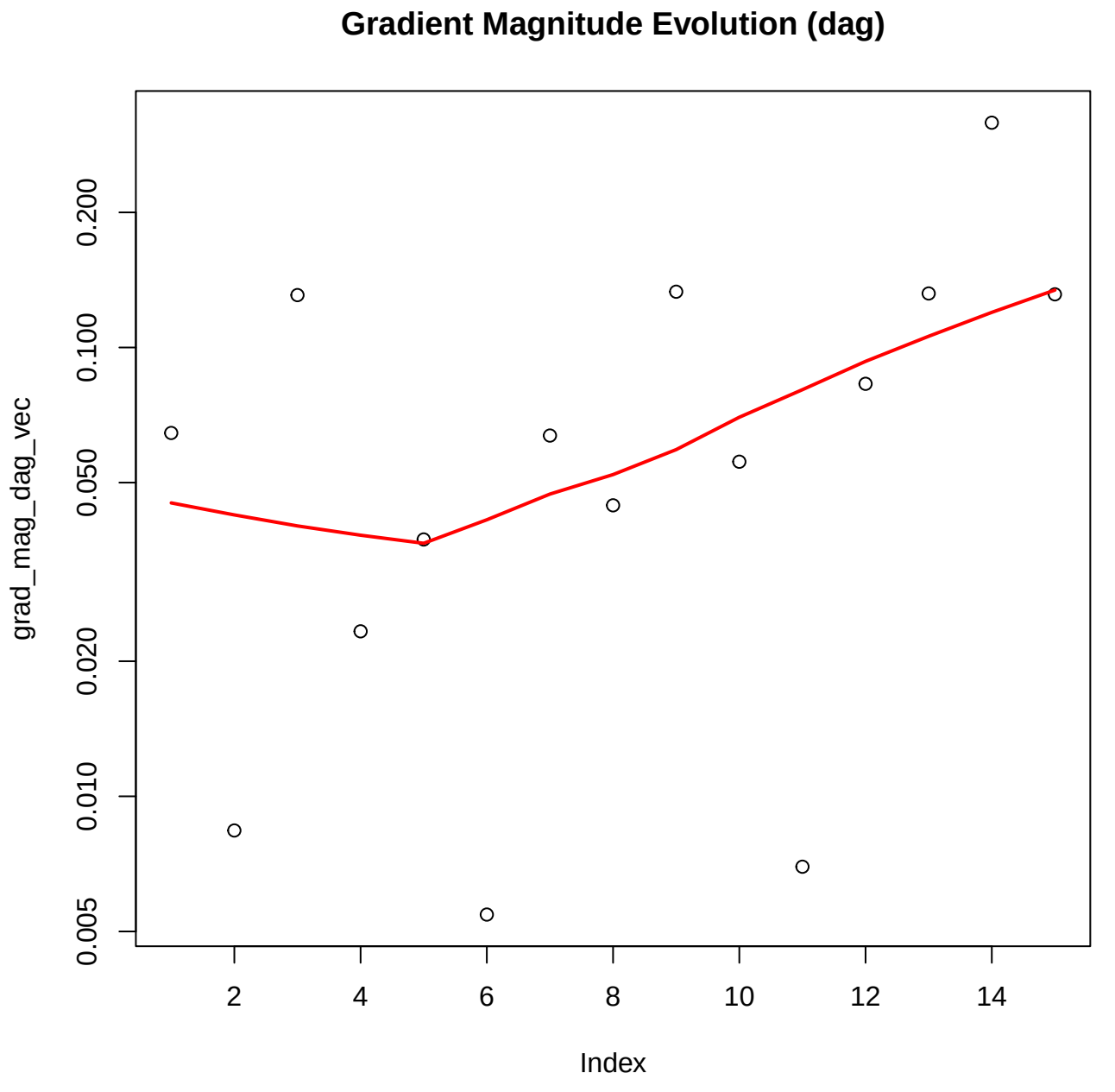
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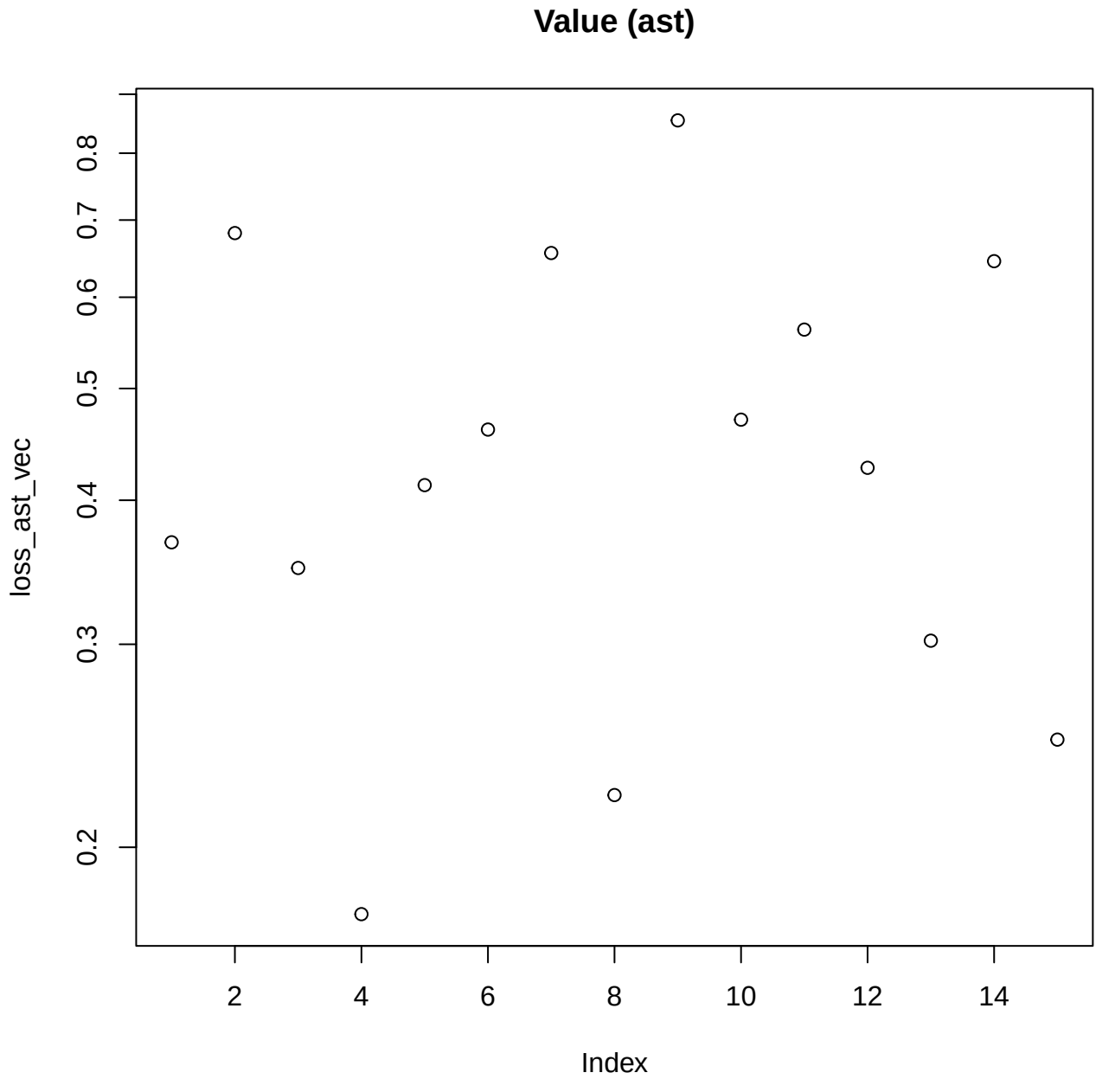
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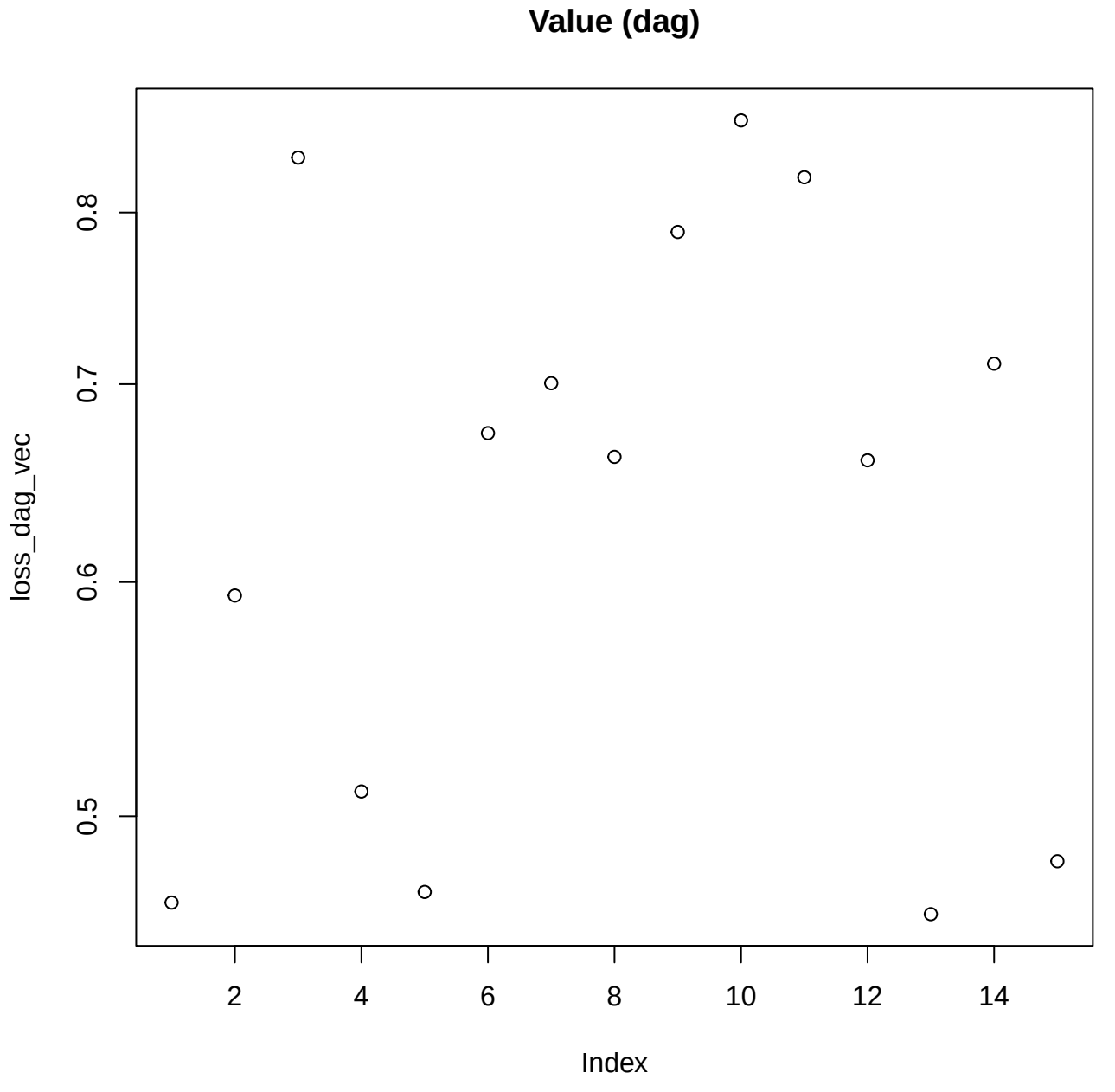
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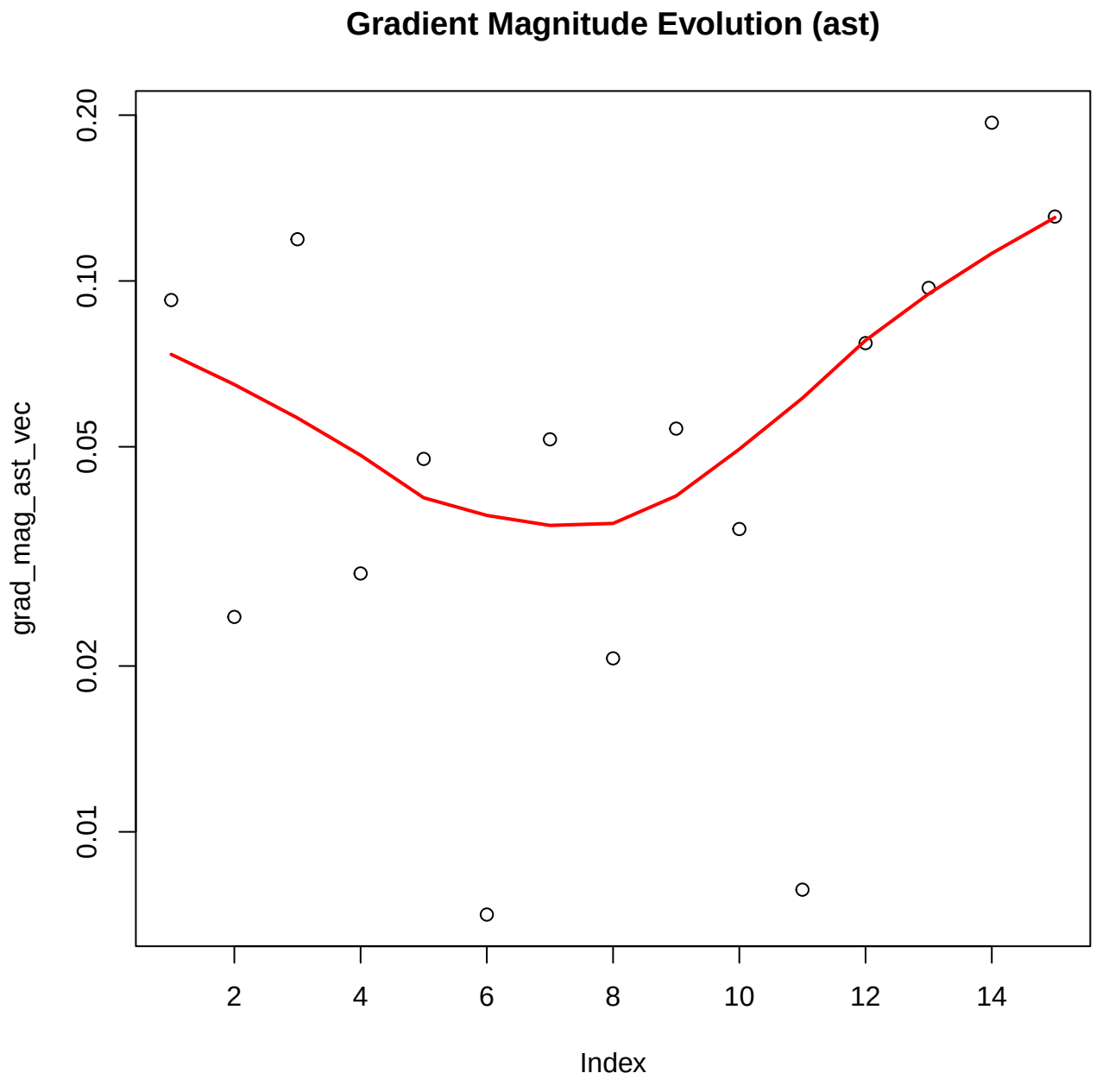
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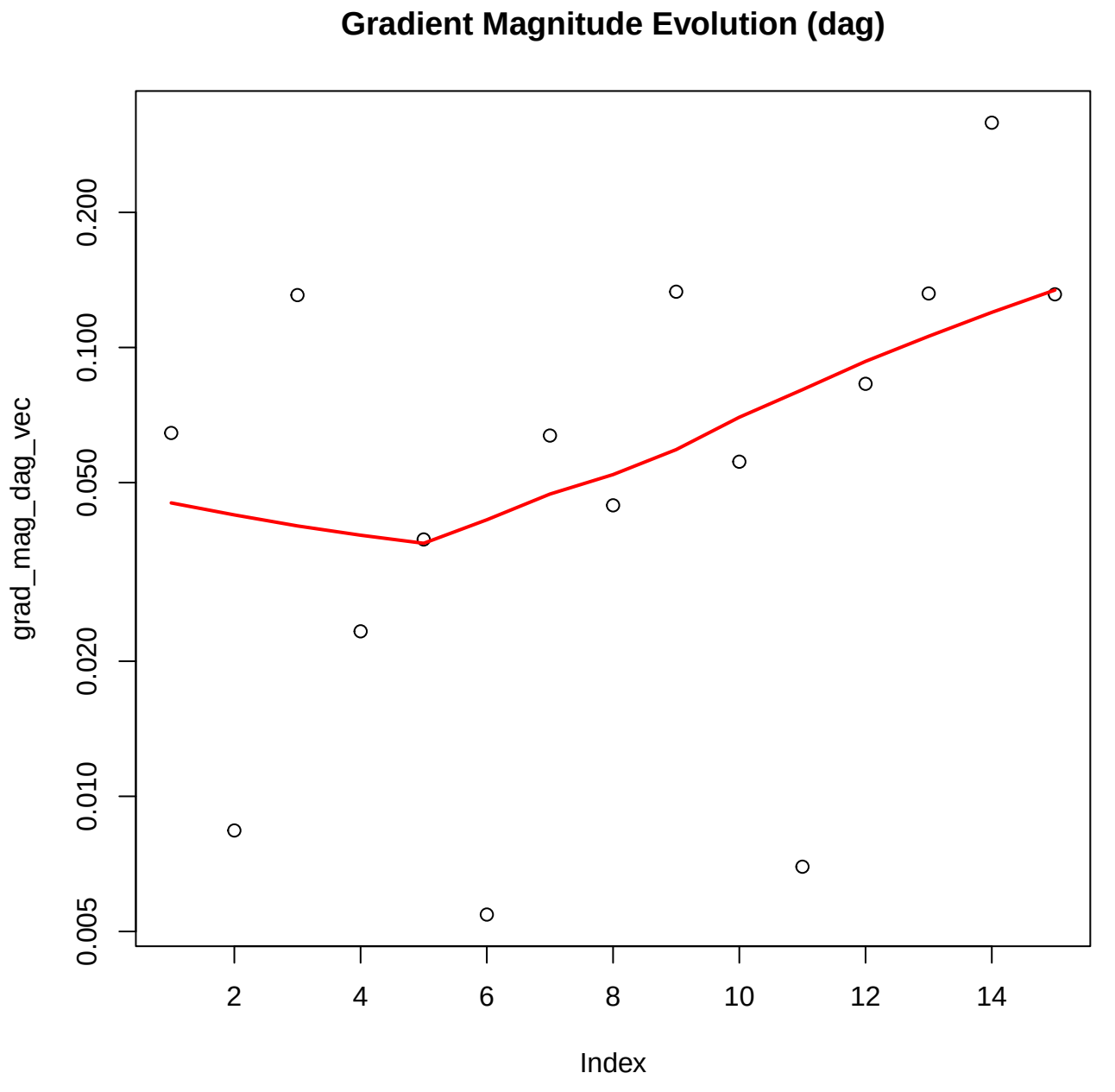
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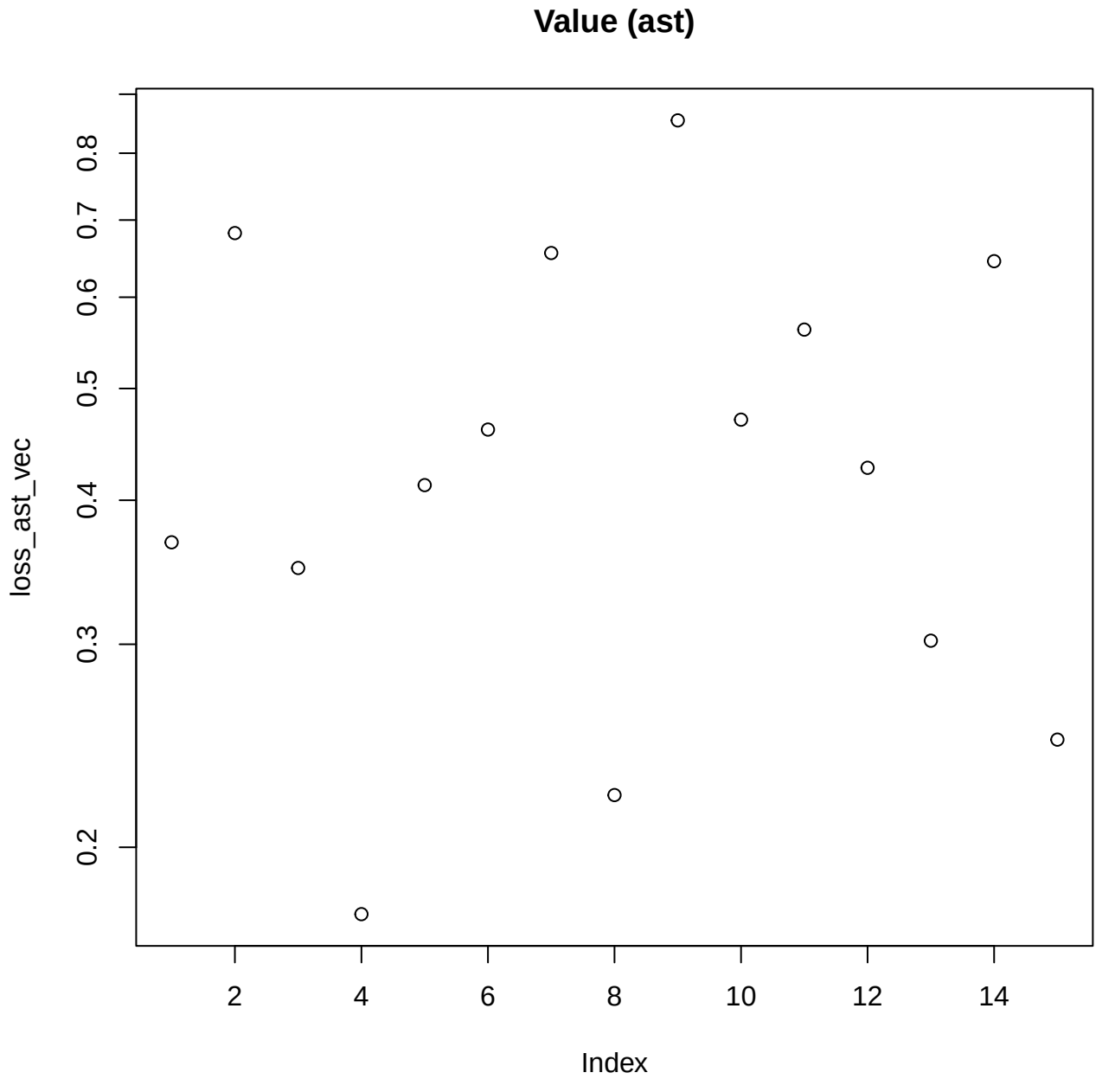
Value (ast)
loss_ast_vec
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2
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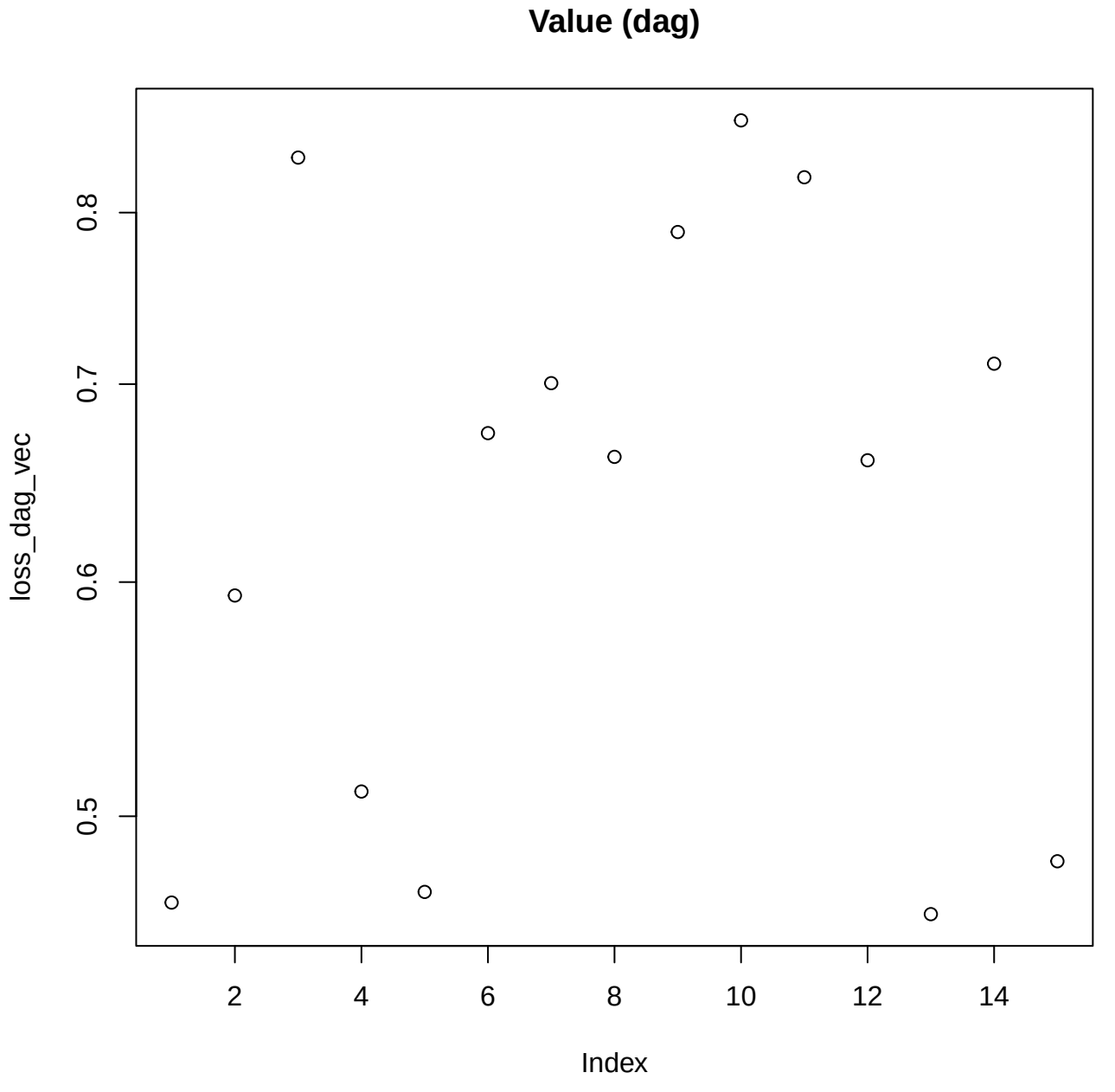
10
12
14
Index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14
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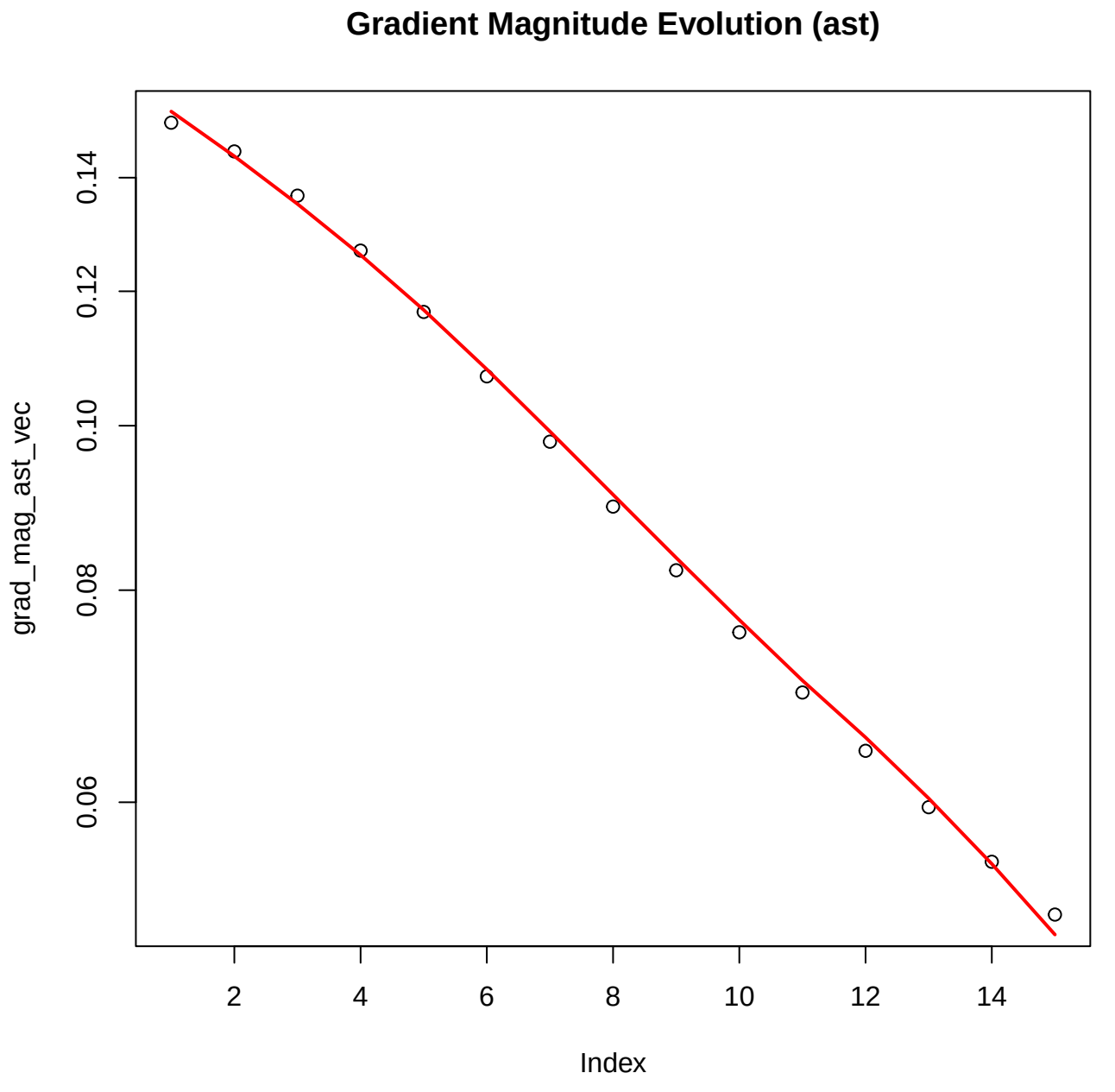
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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14
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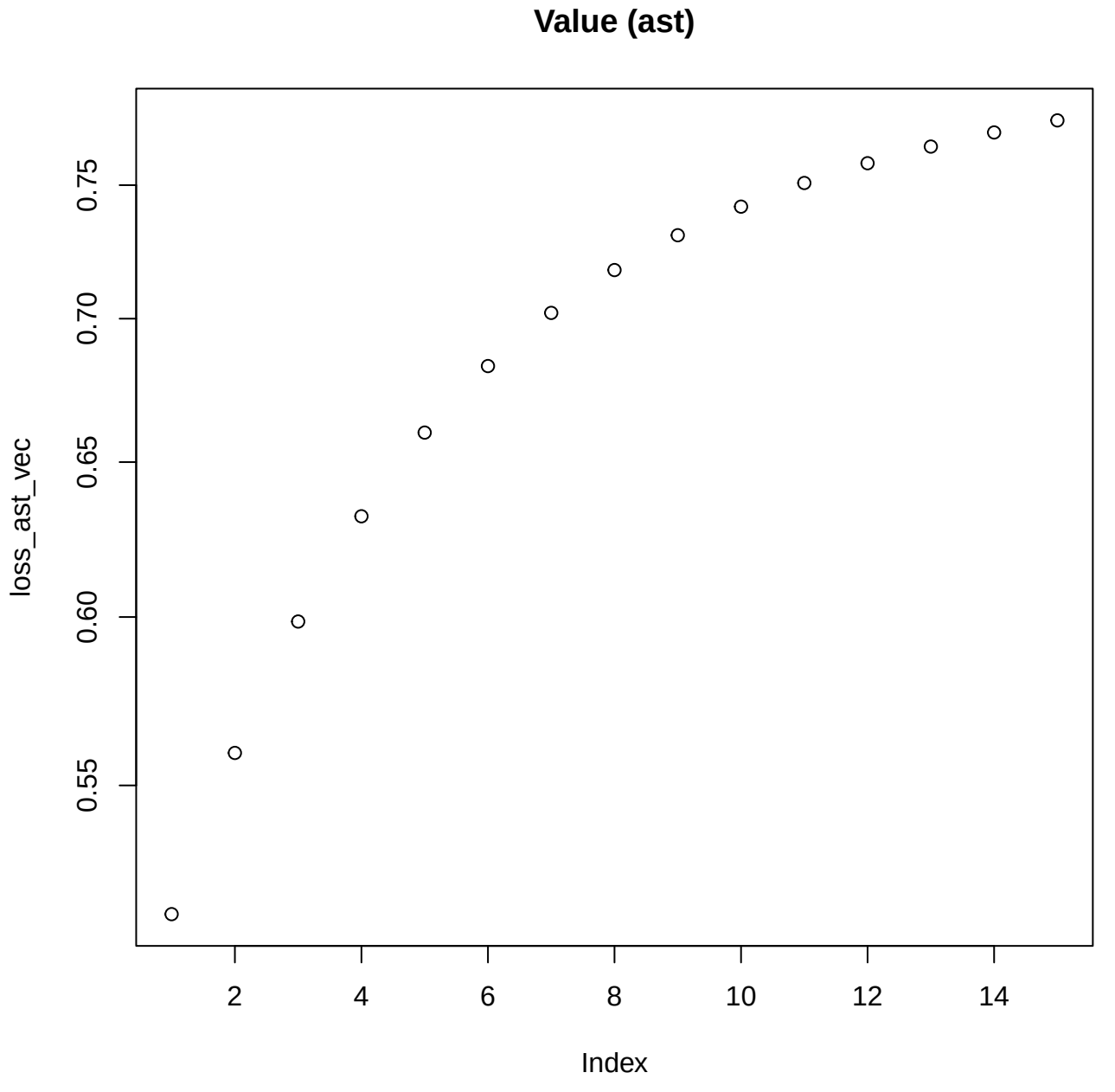
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loss_ast_vec
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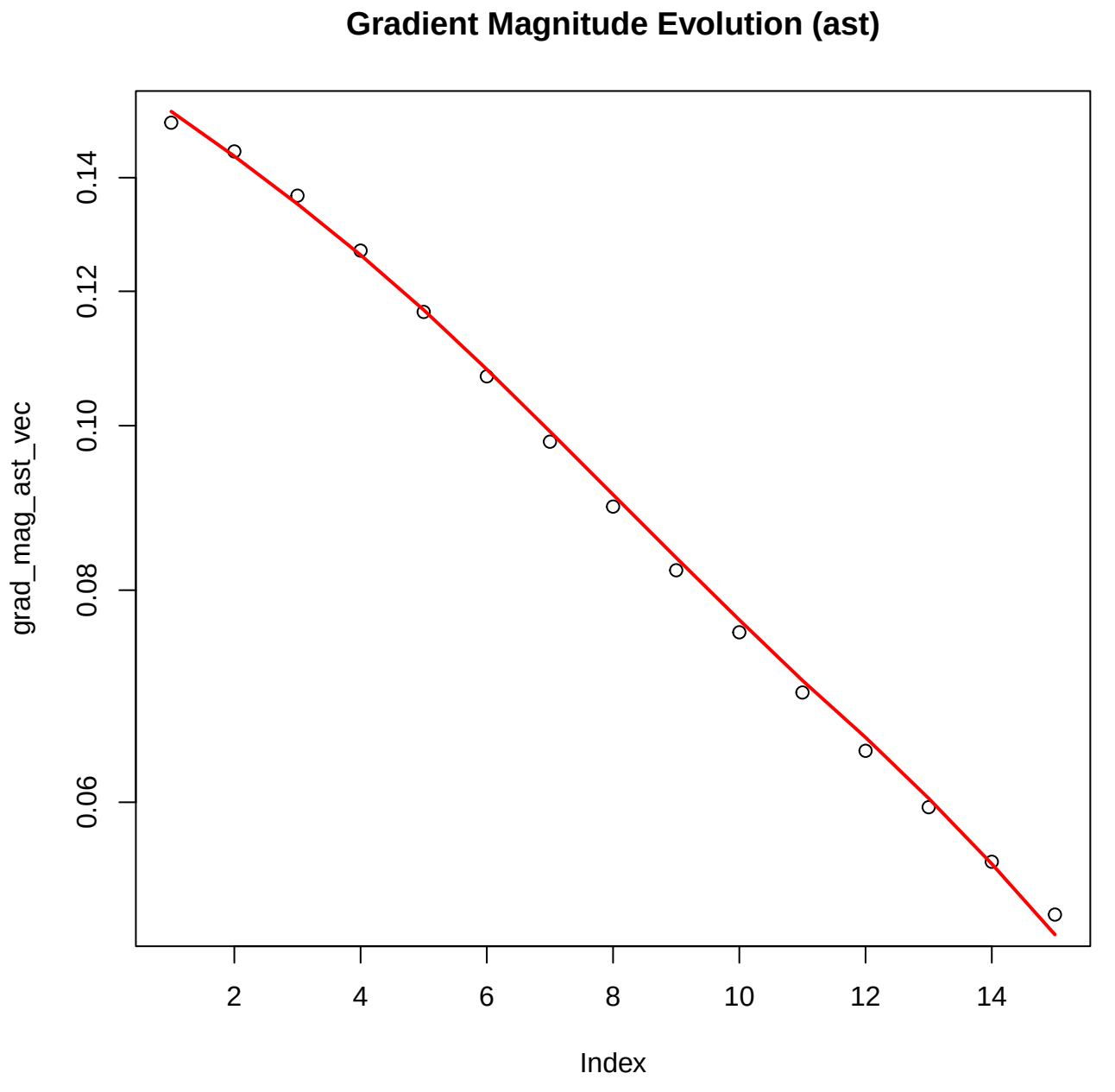
14
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
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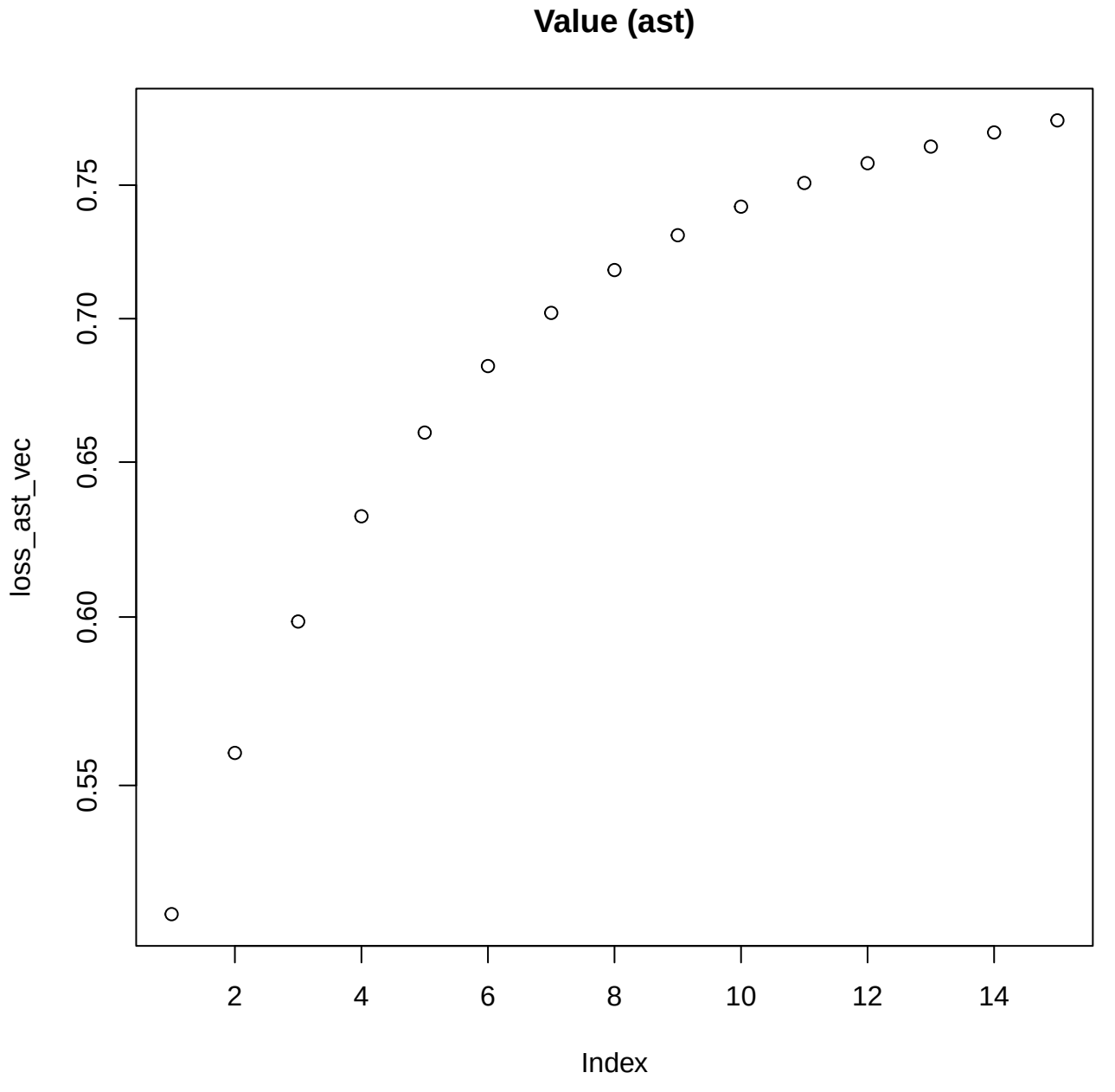
Value (ast)
loss_ast_vec
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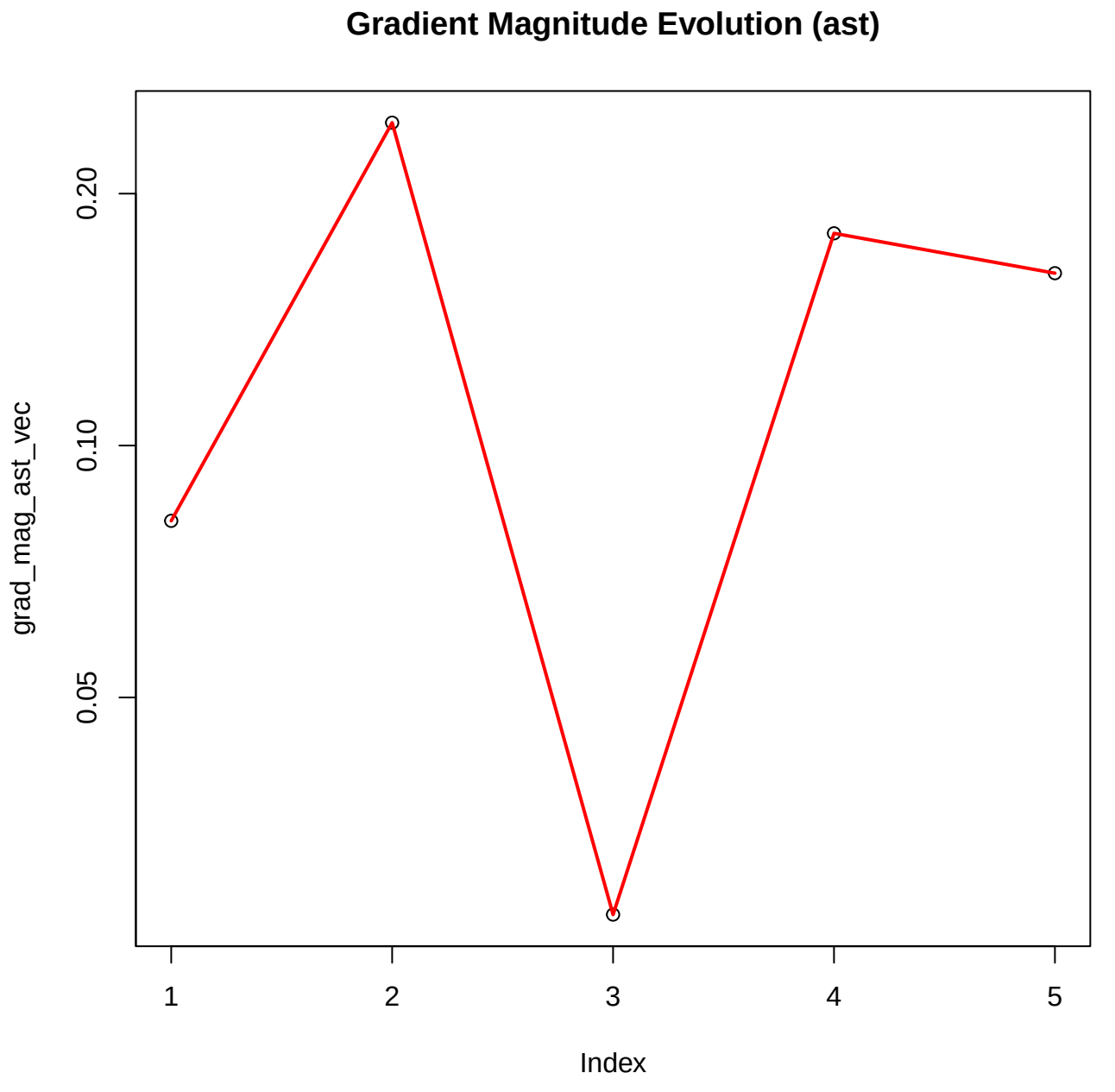
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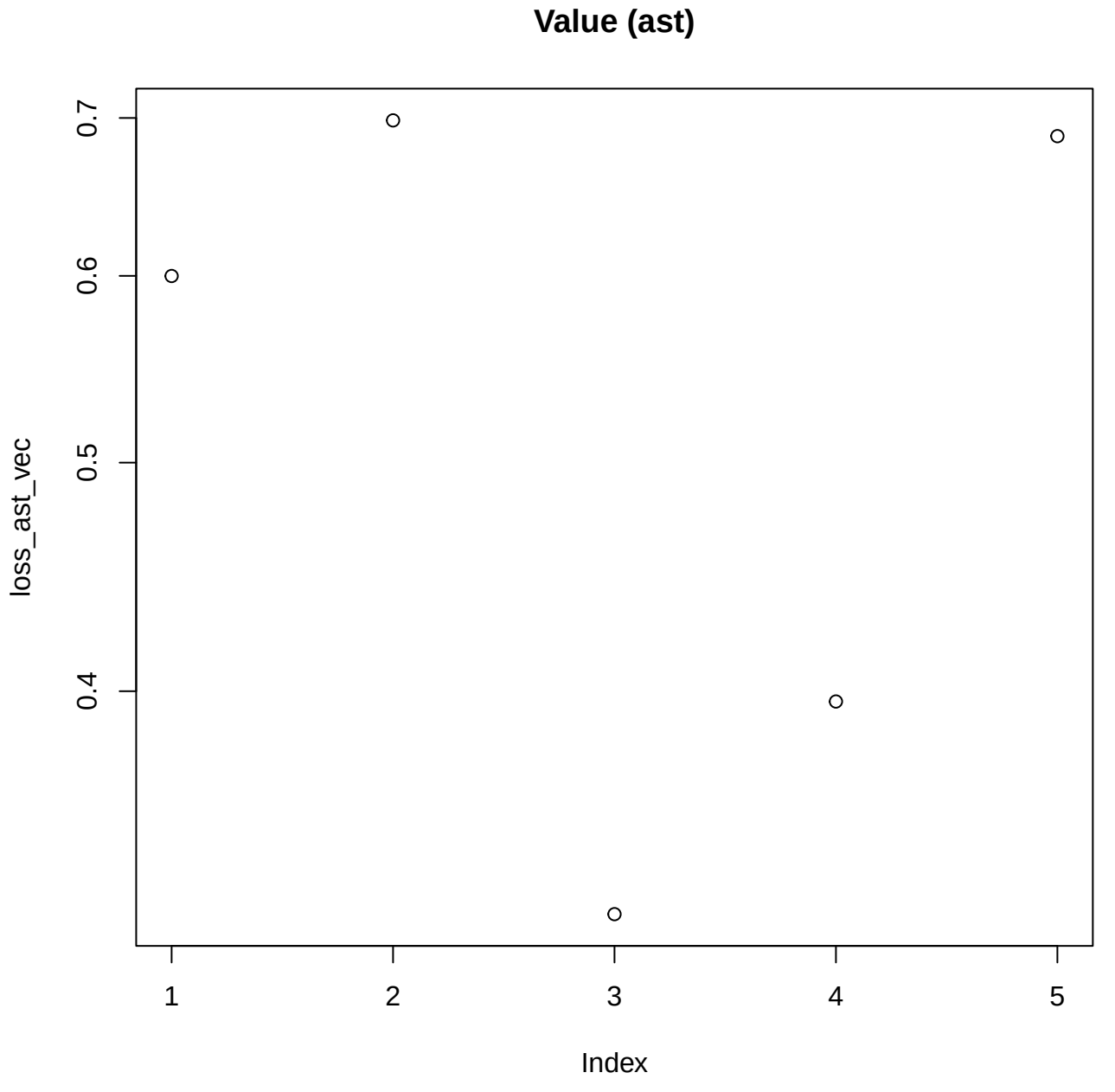
5
Index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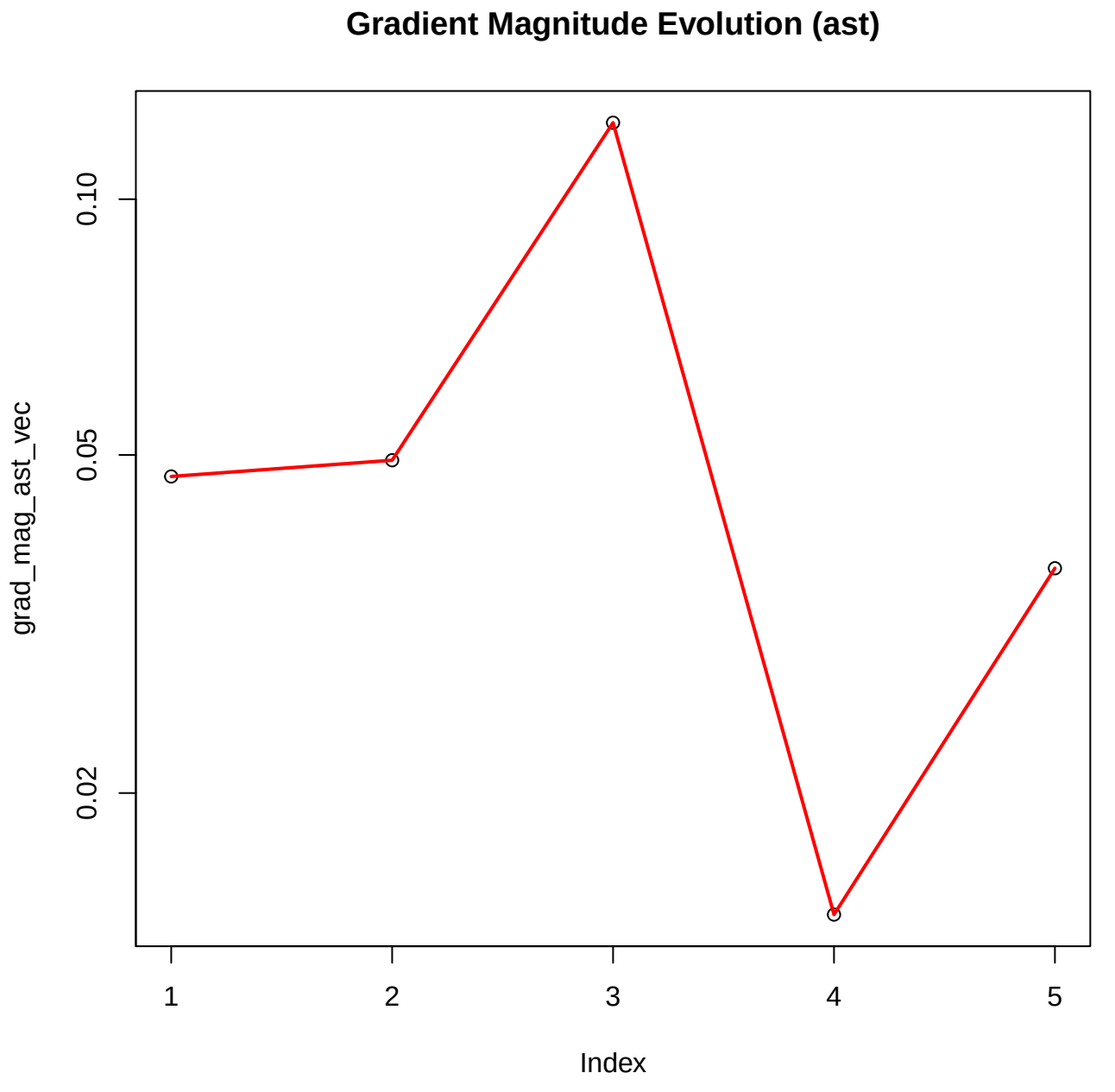
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
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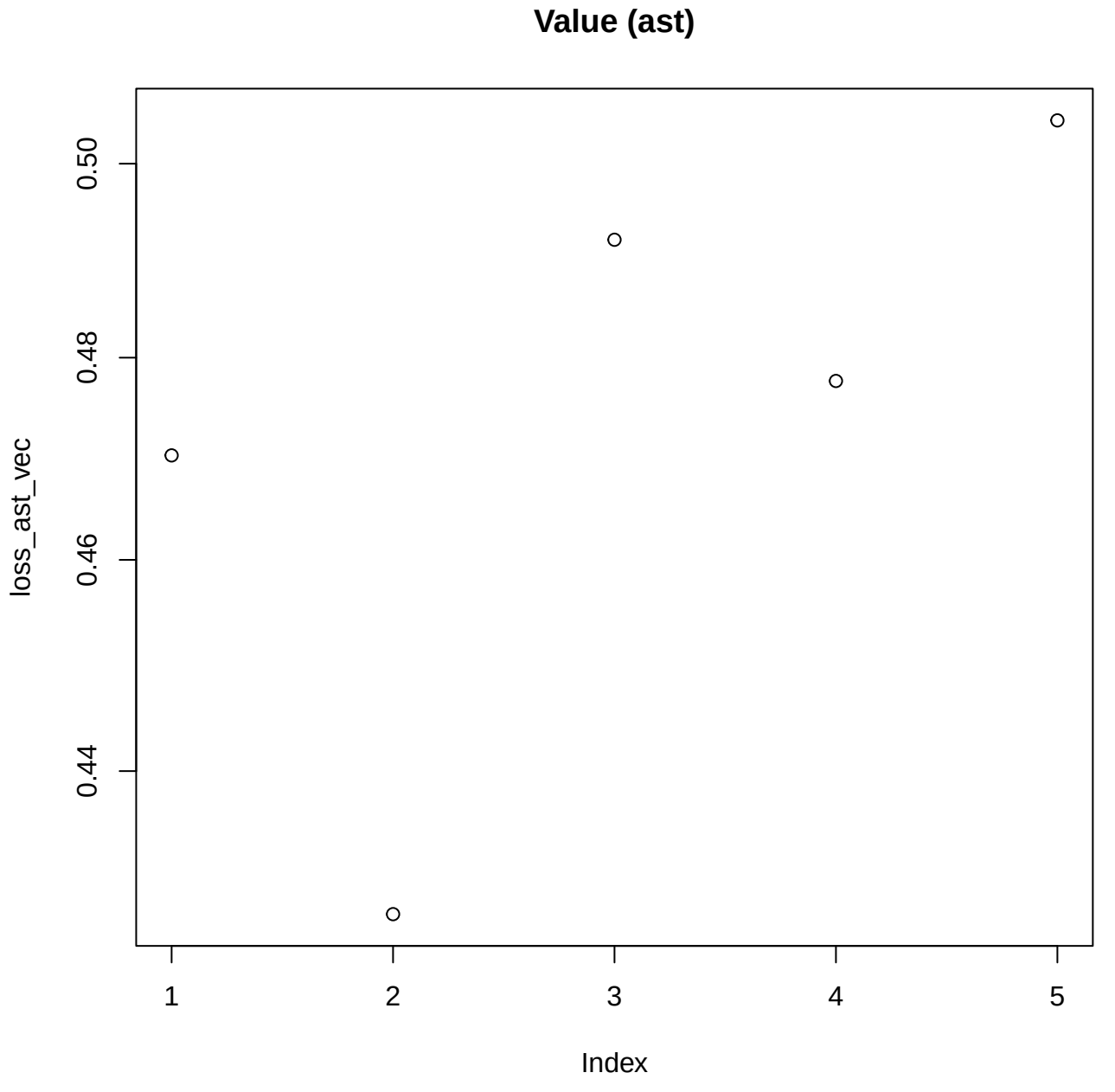
3
4
5
Index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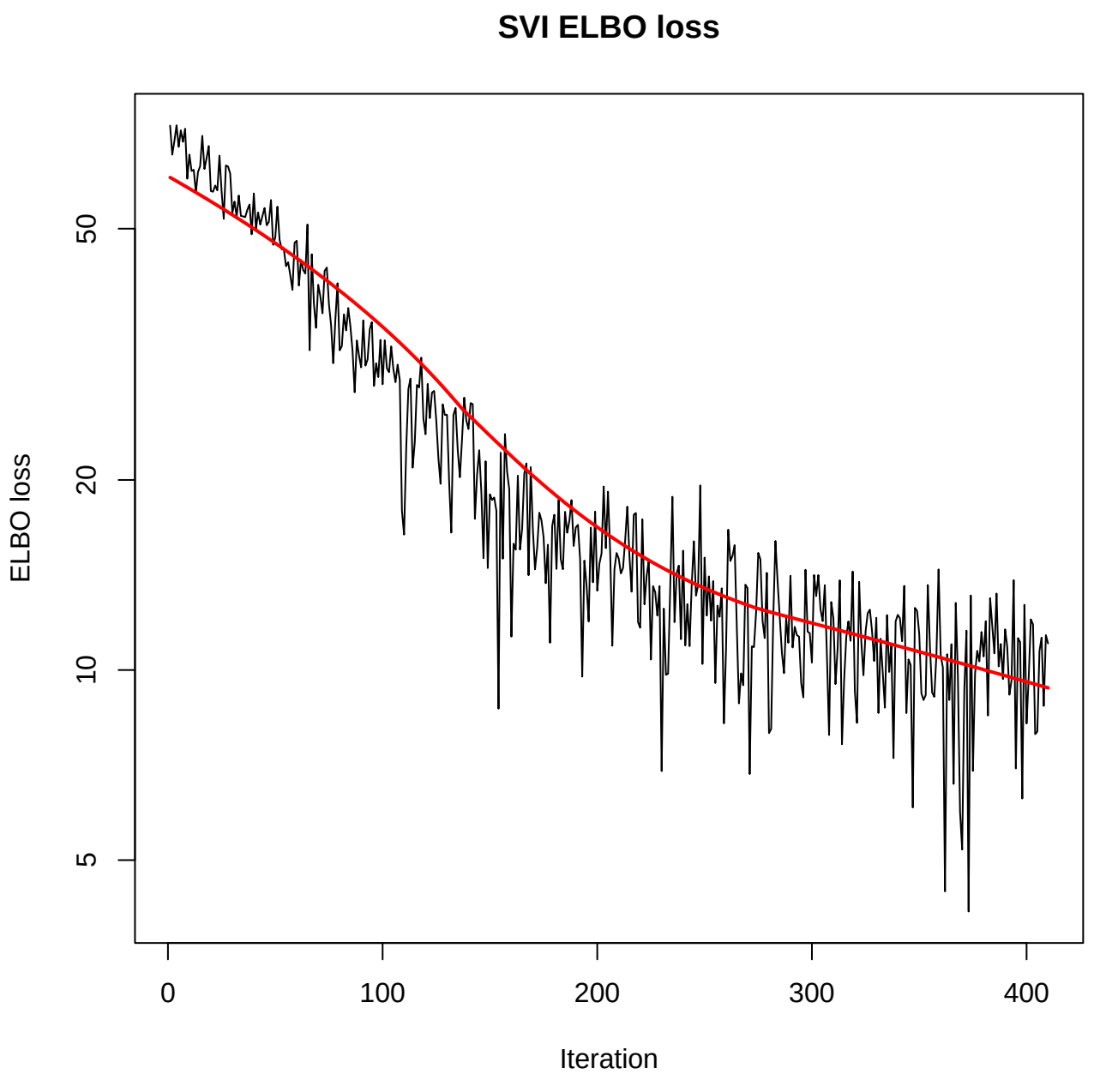
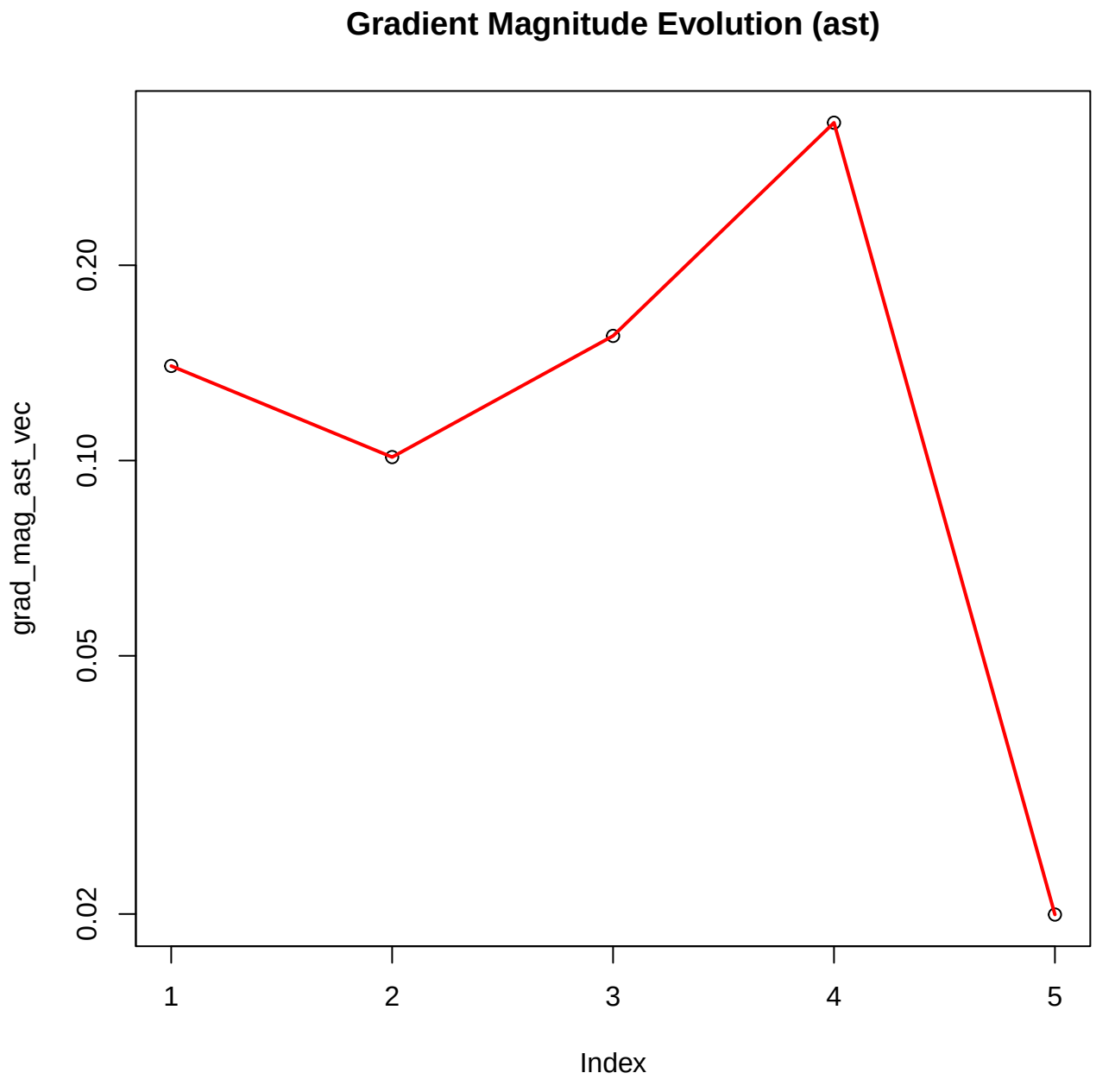
3
4
5
Index


SVI ELBO loss
ELBO loss
50
20
10
5
0
100
200
300
400
Iteration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grad_mag_ast_vec
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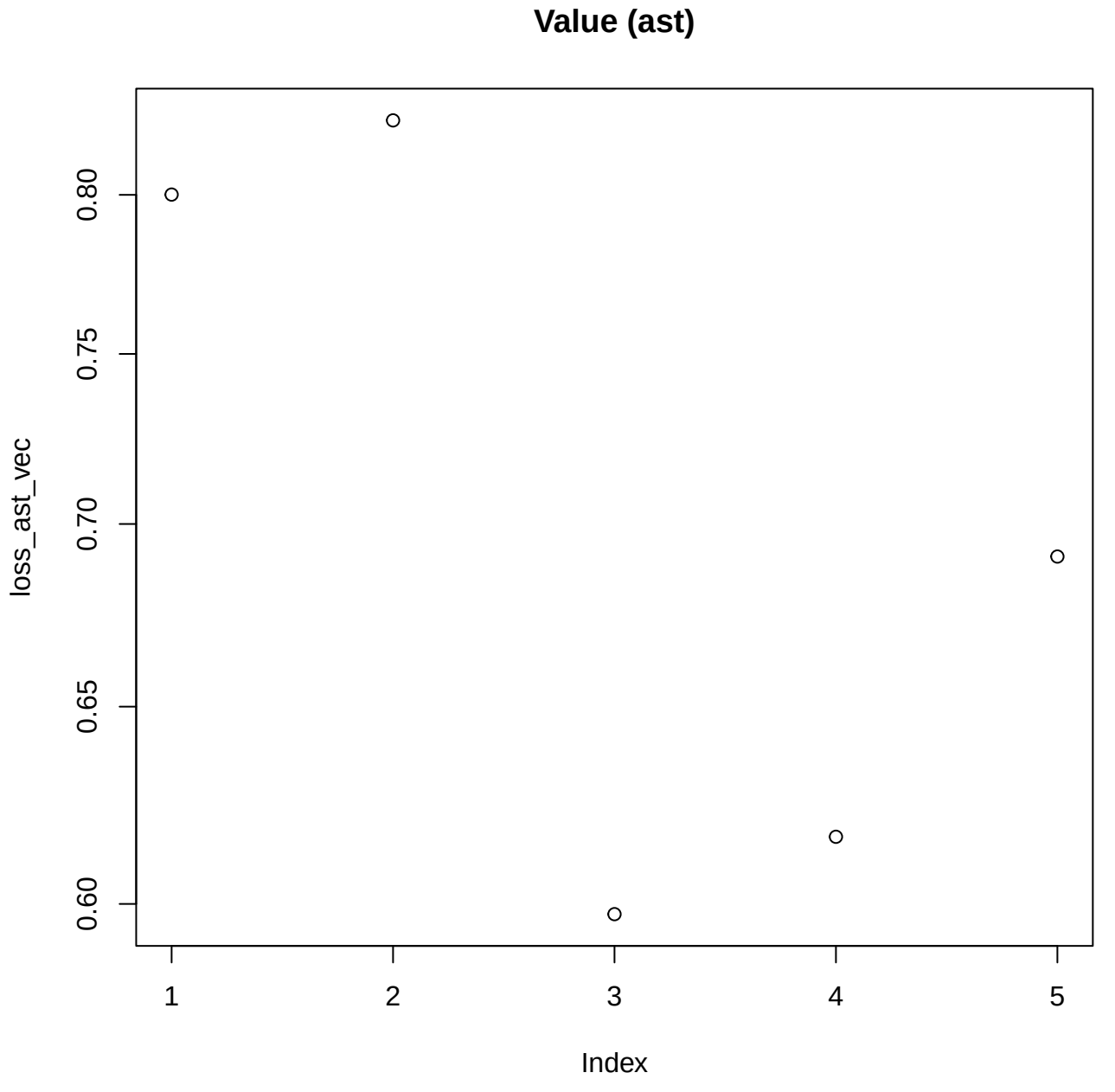
0.05
0.10
0.20
1
2
3
4
5
Index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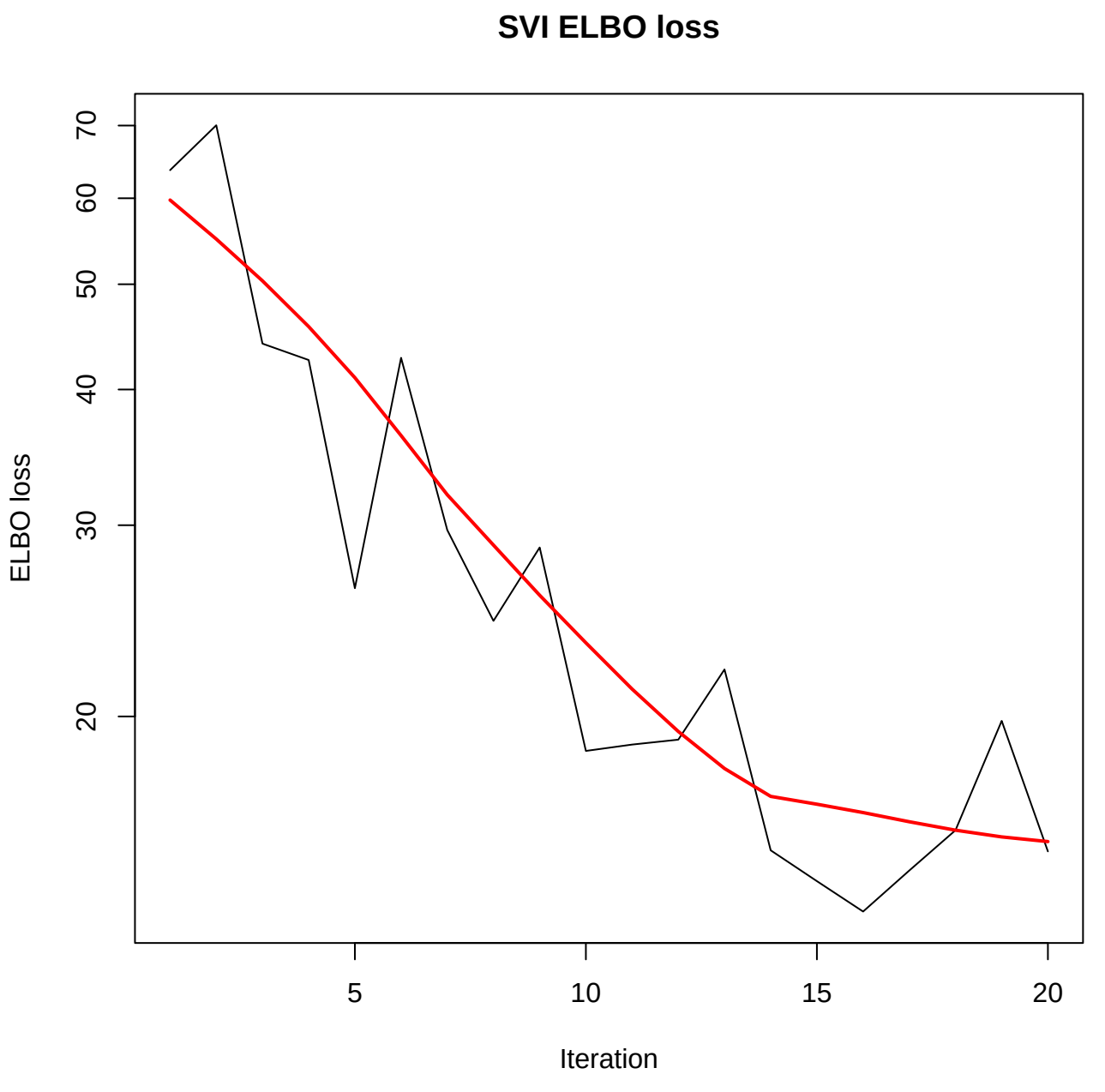
4
5
Index


SVI ELBO loss
ELBO loss
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60
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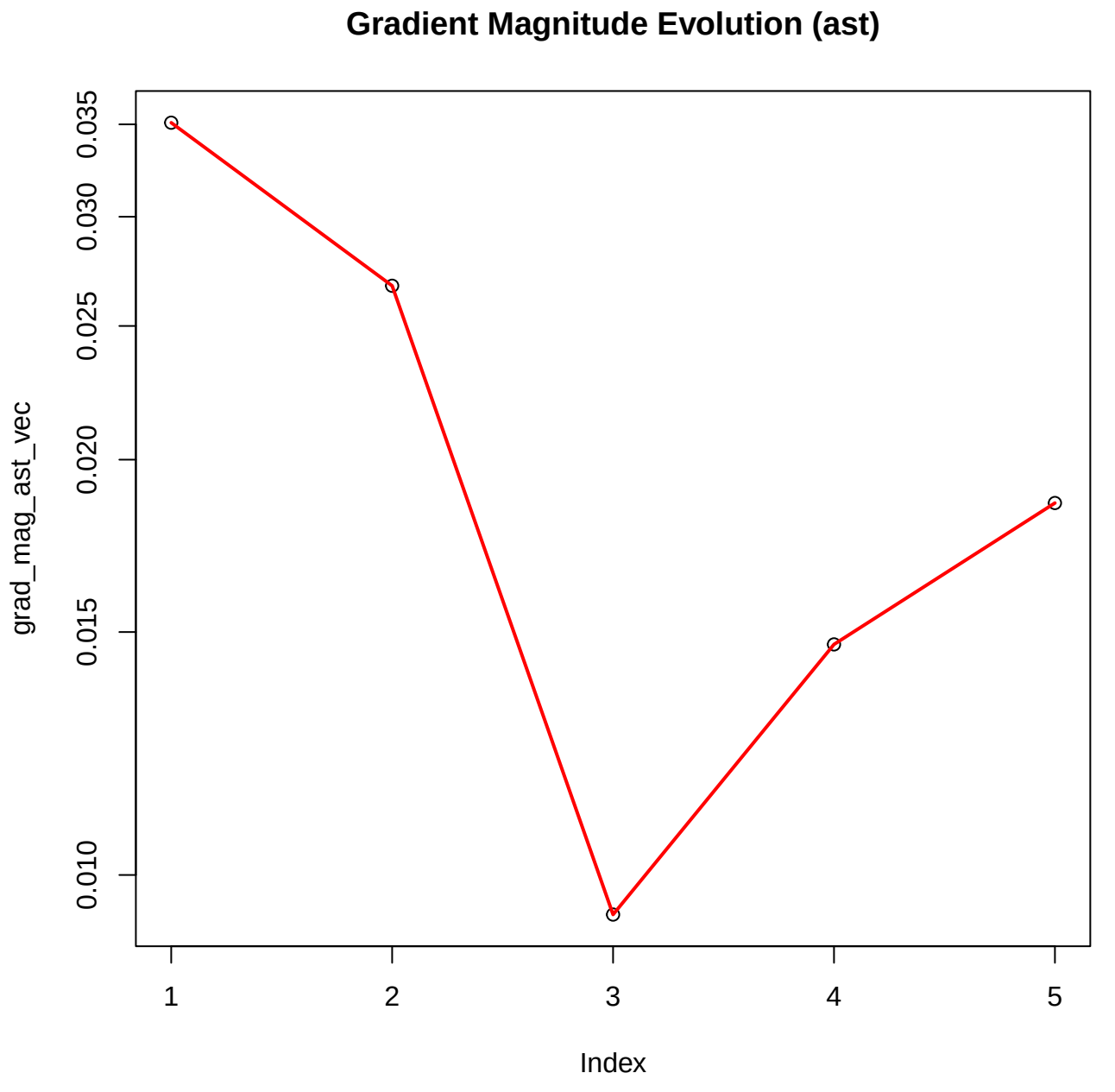
40
30
20
5
10
15
20
Iteration


Gradient Magnitude Evolution (ast)
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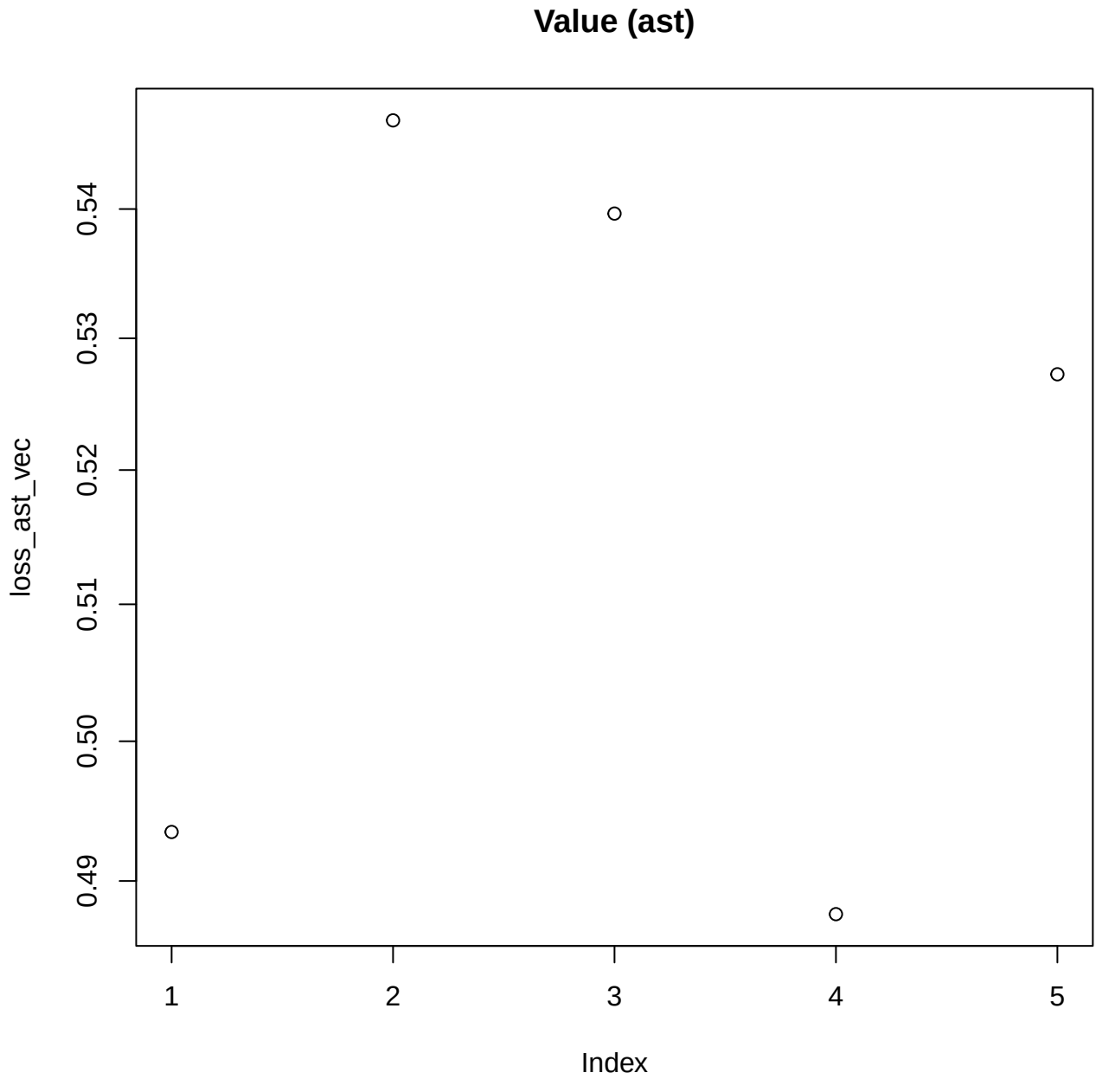
4
5
Index


Value (ast)
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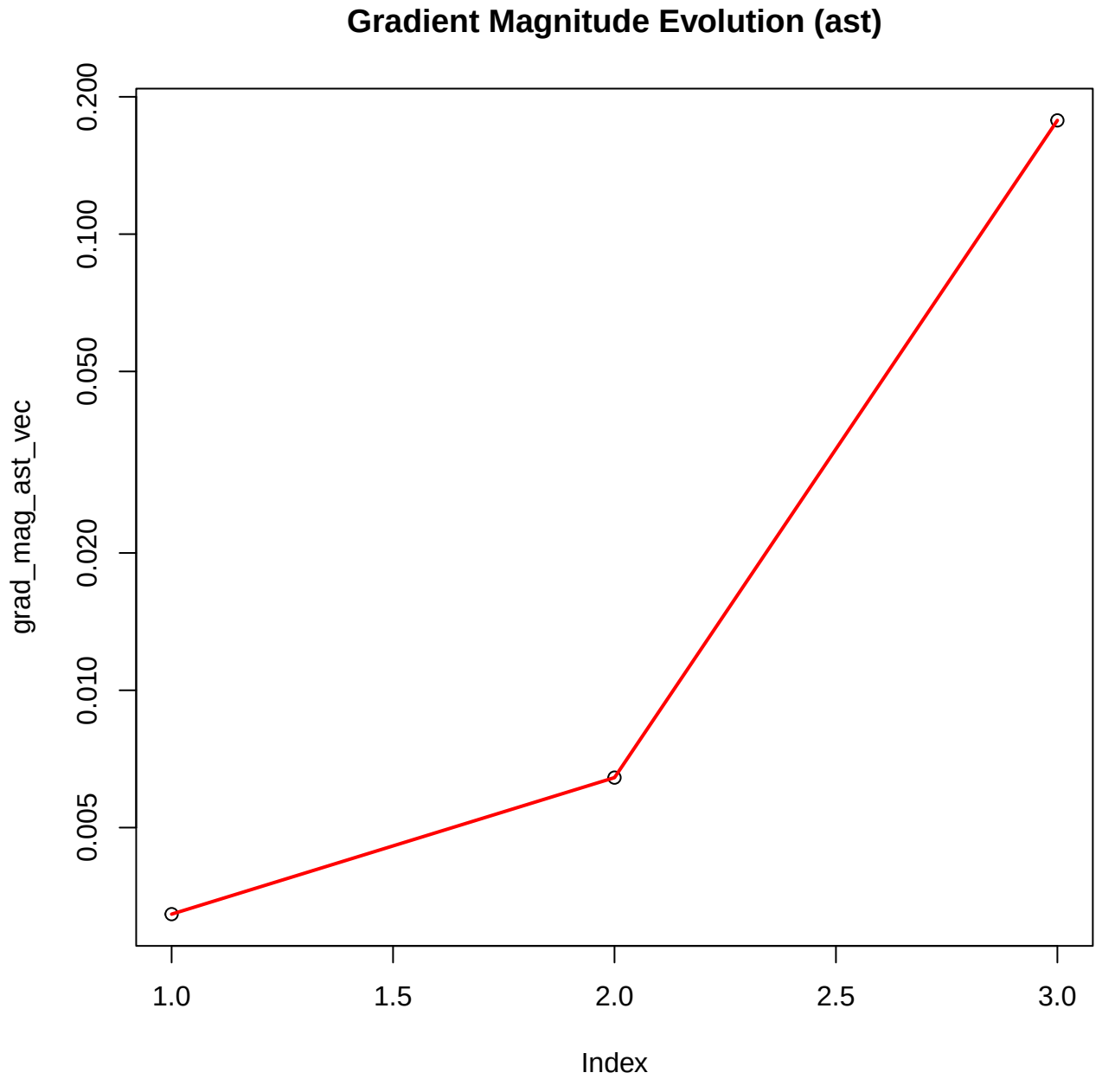
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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2.5
3.0
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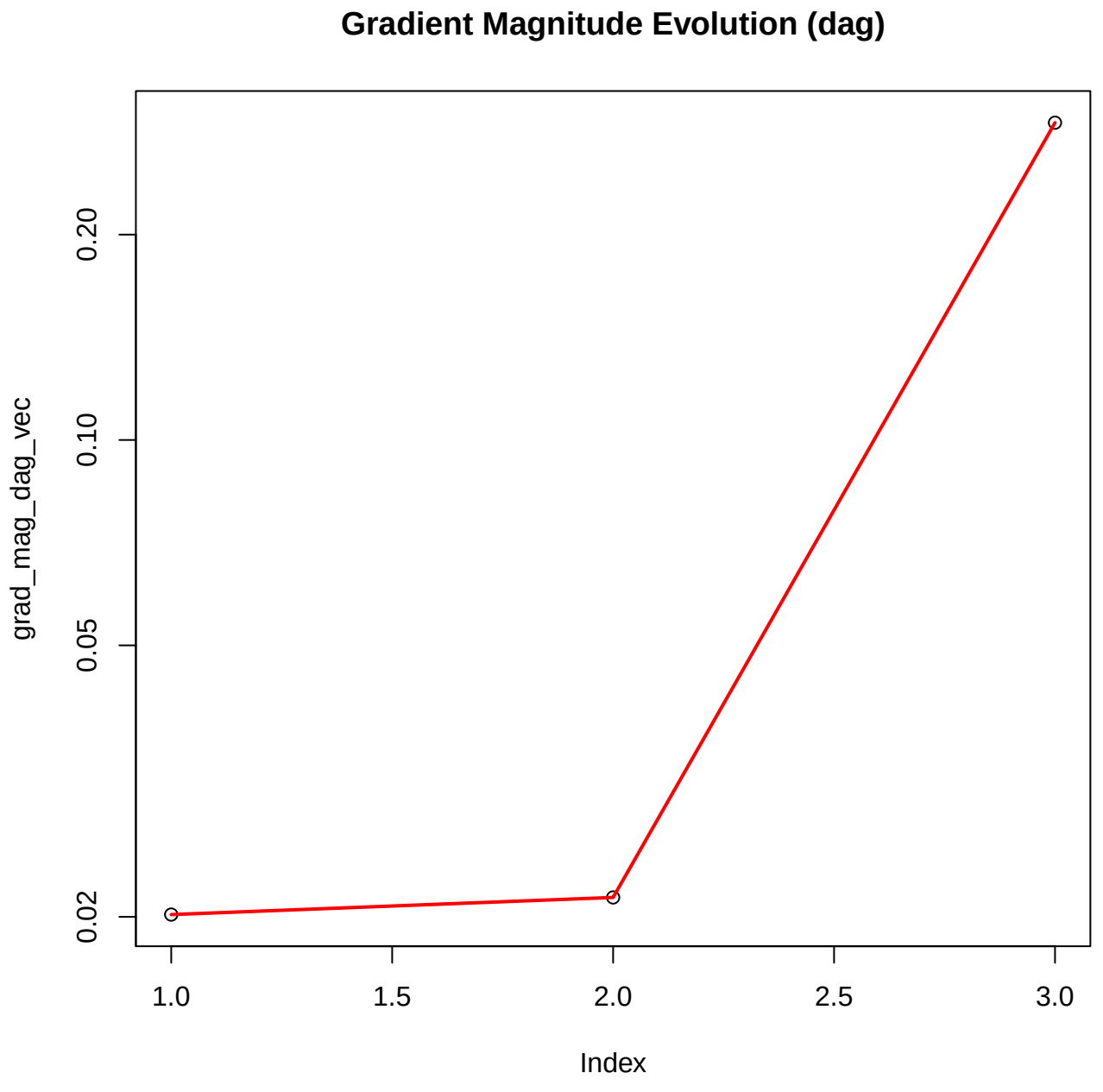
Gradient Magnitude Evolution (dag)
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0.10
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2.0
2.5
3.0
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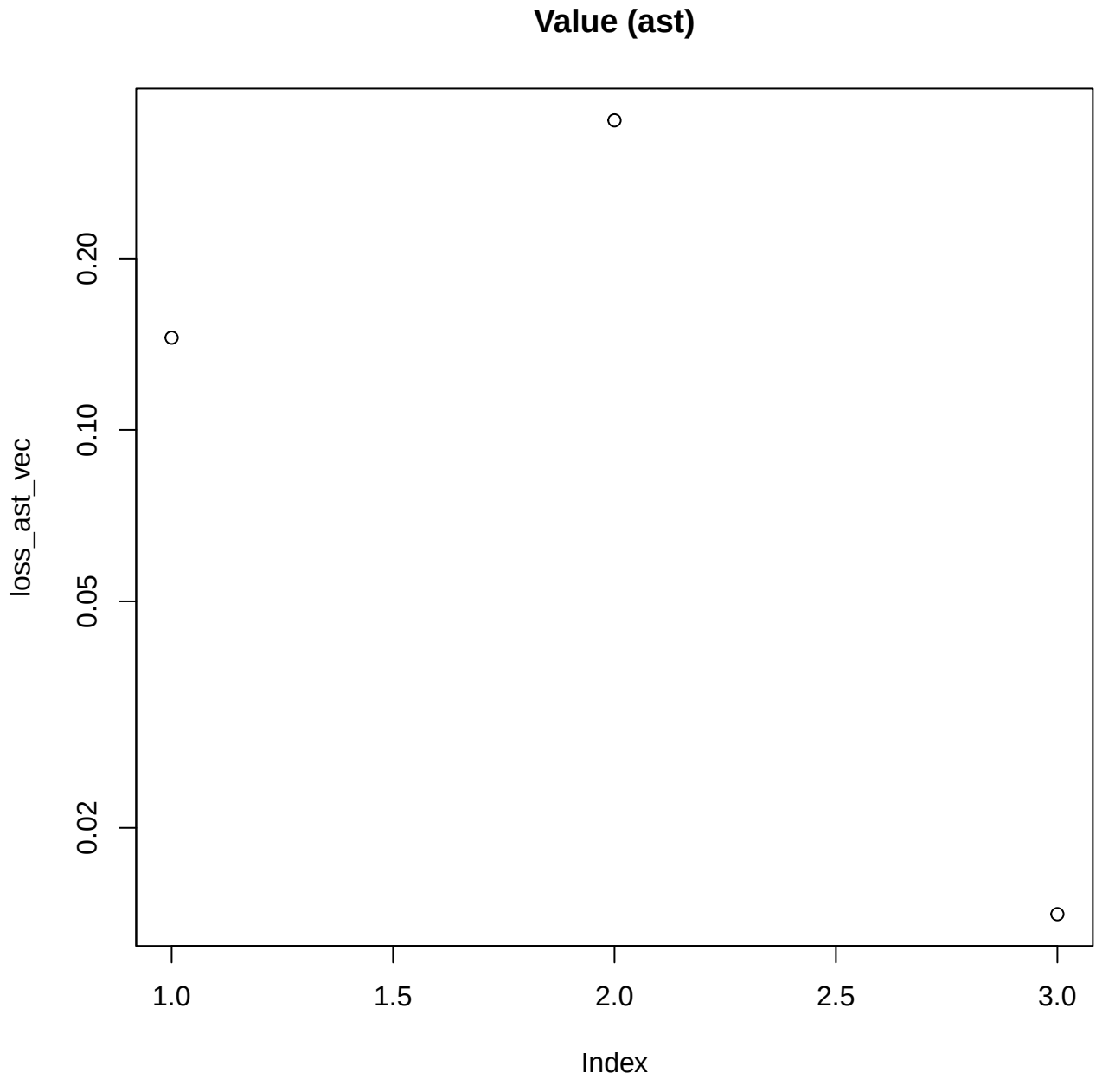
Value (ast)
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0.10
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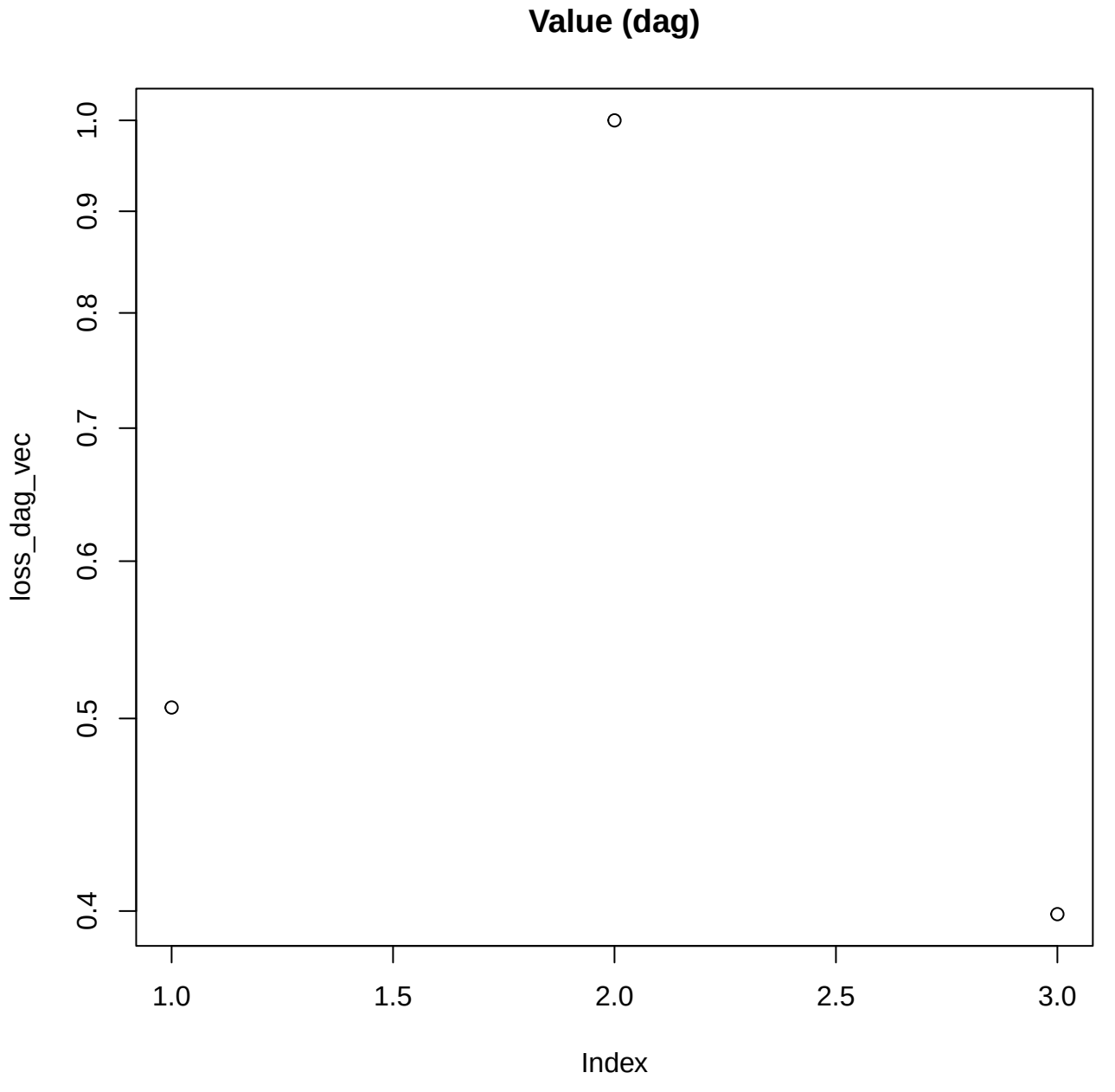
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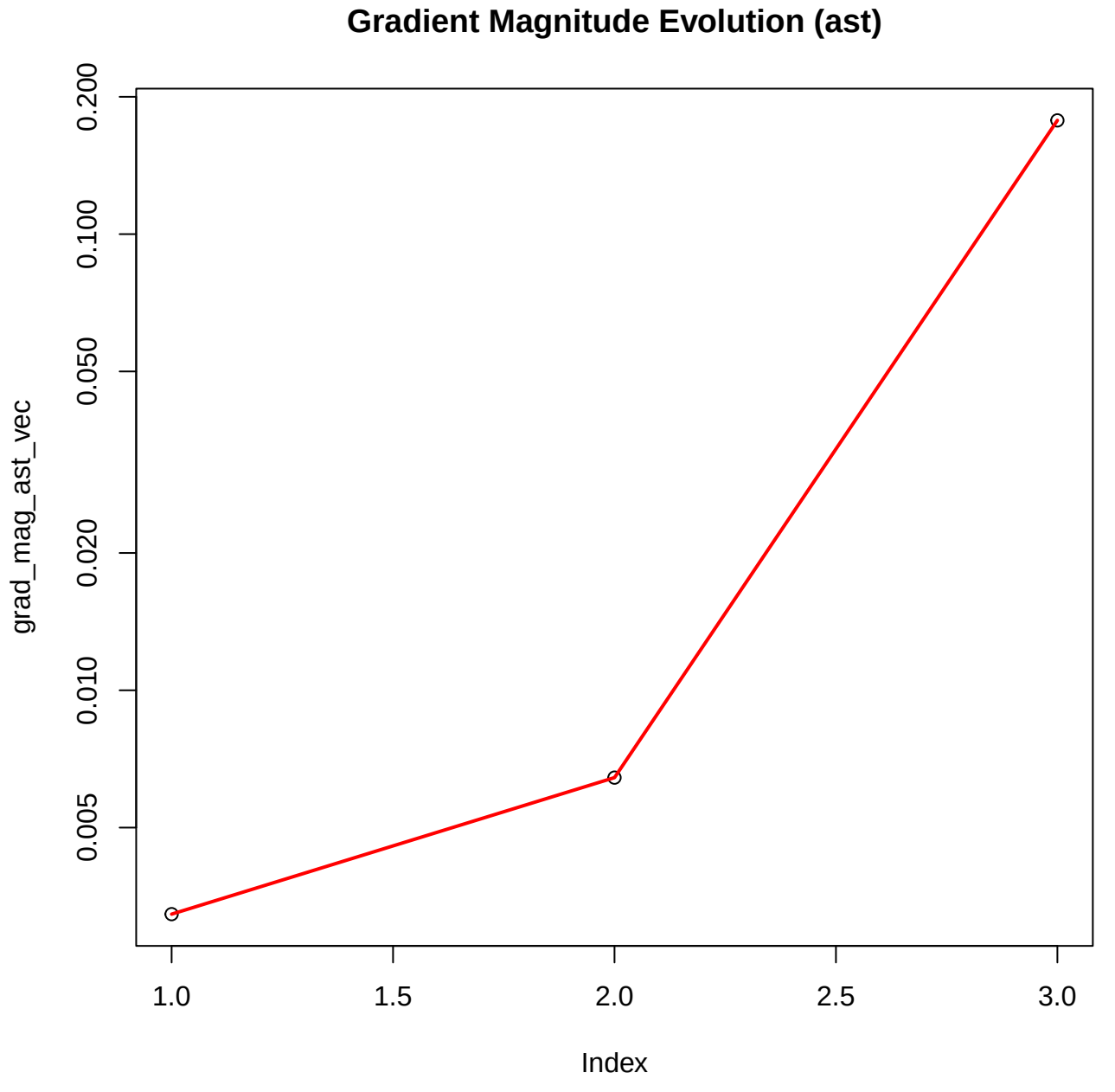
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.010
0.020
0.050
0.100
0.200
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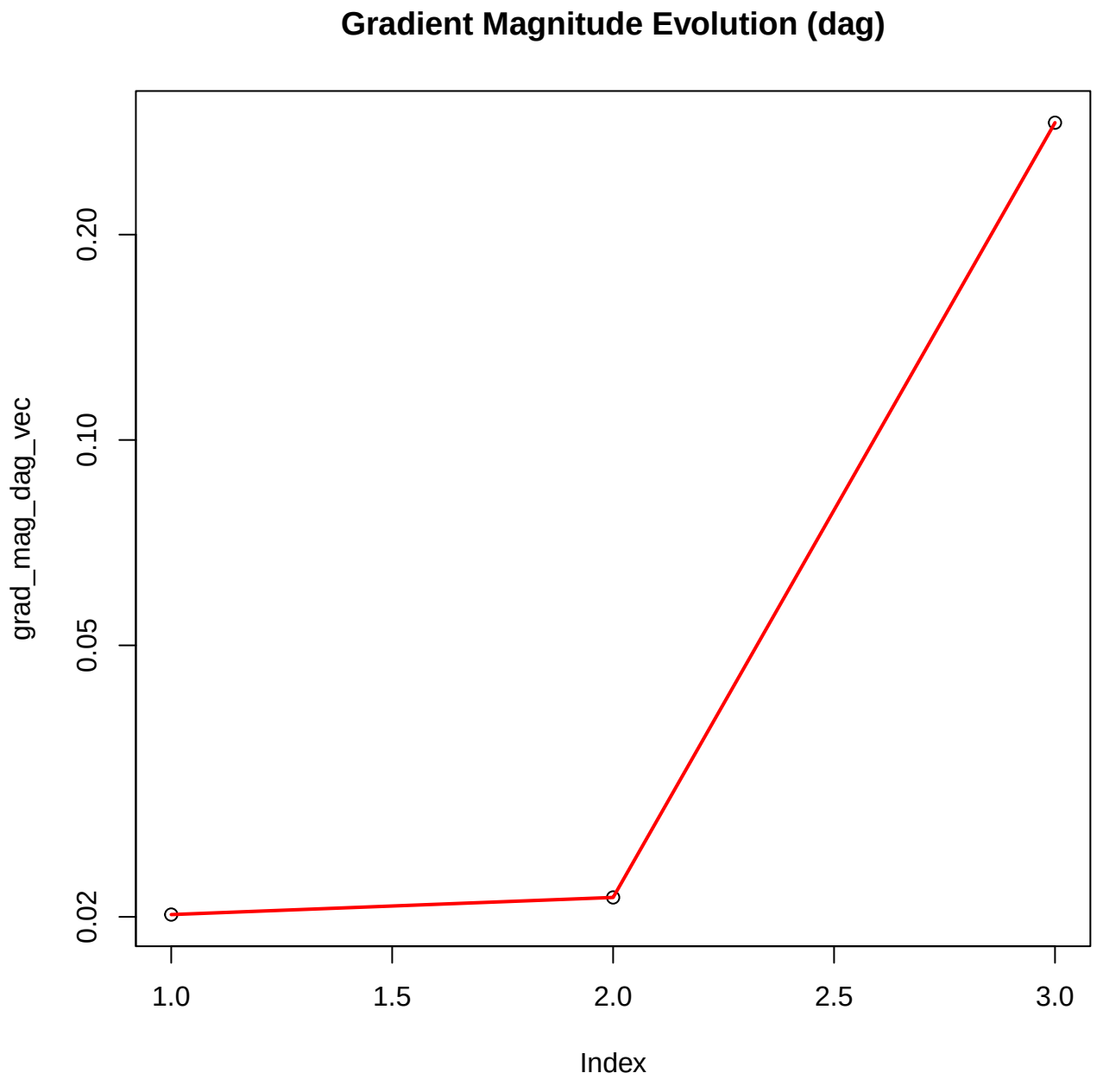
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.05
0.10
0.20
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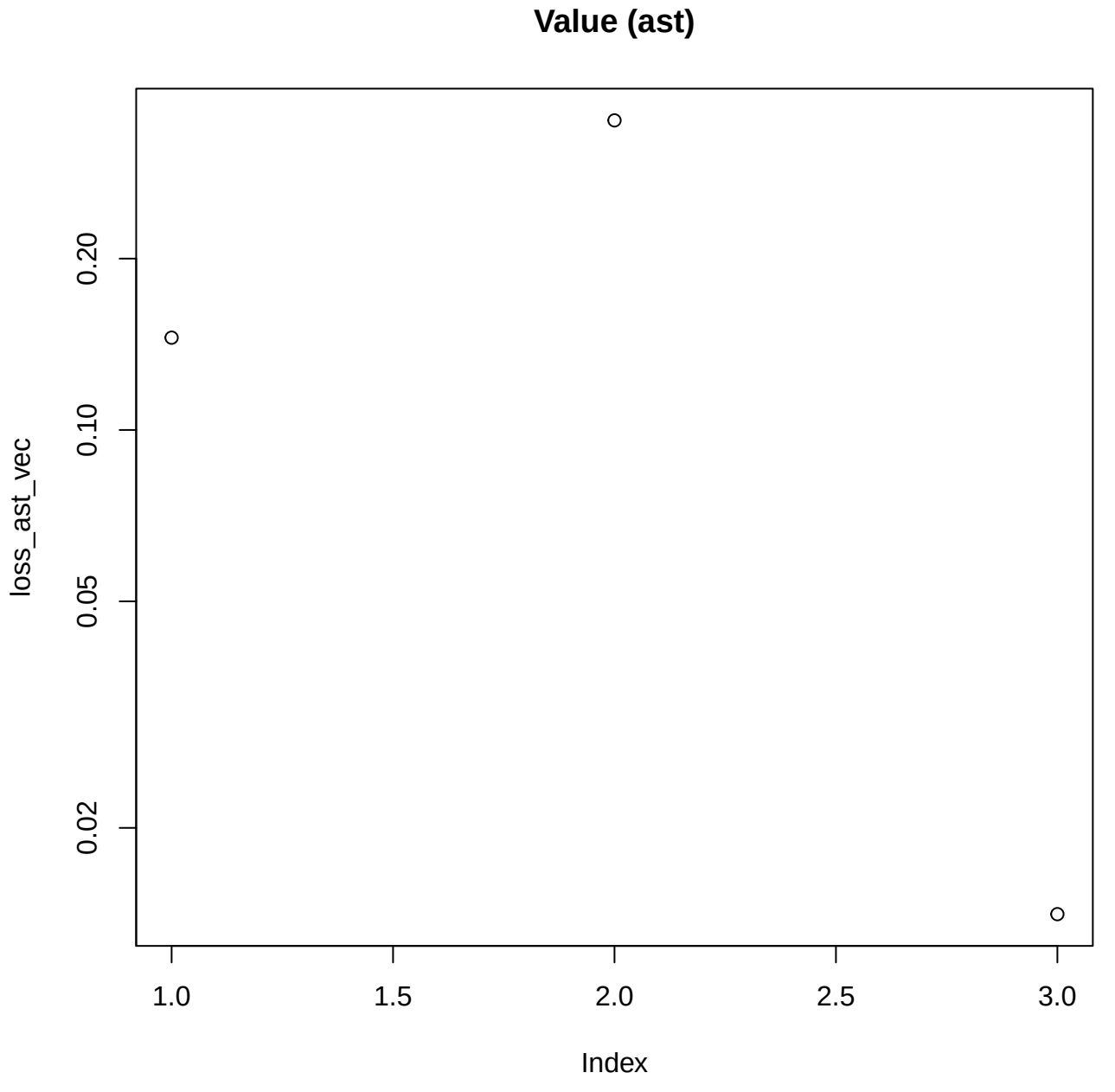
1.5
2.0
2.5
3.0
Index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loss_ast_vec
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0.10
0.20
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1.5
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2.5
3.0
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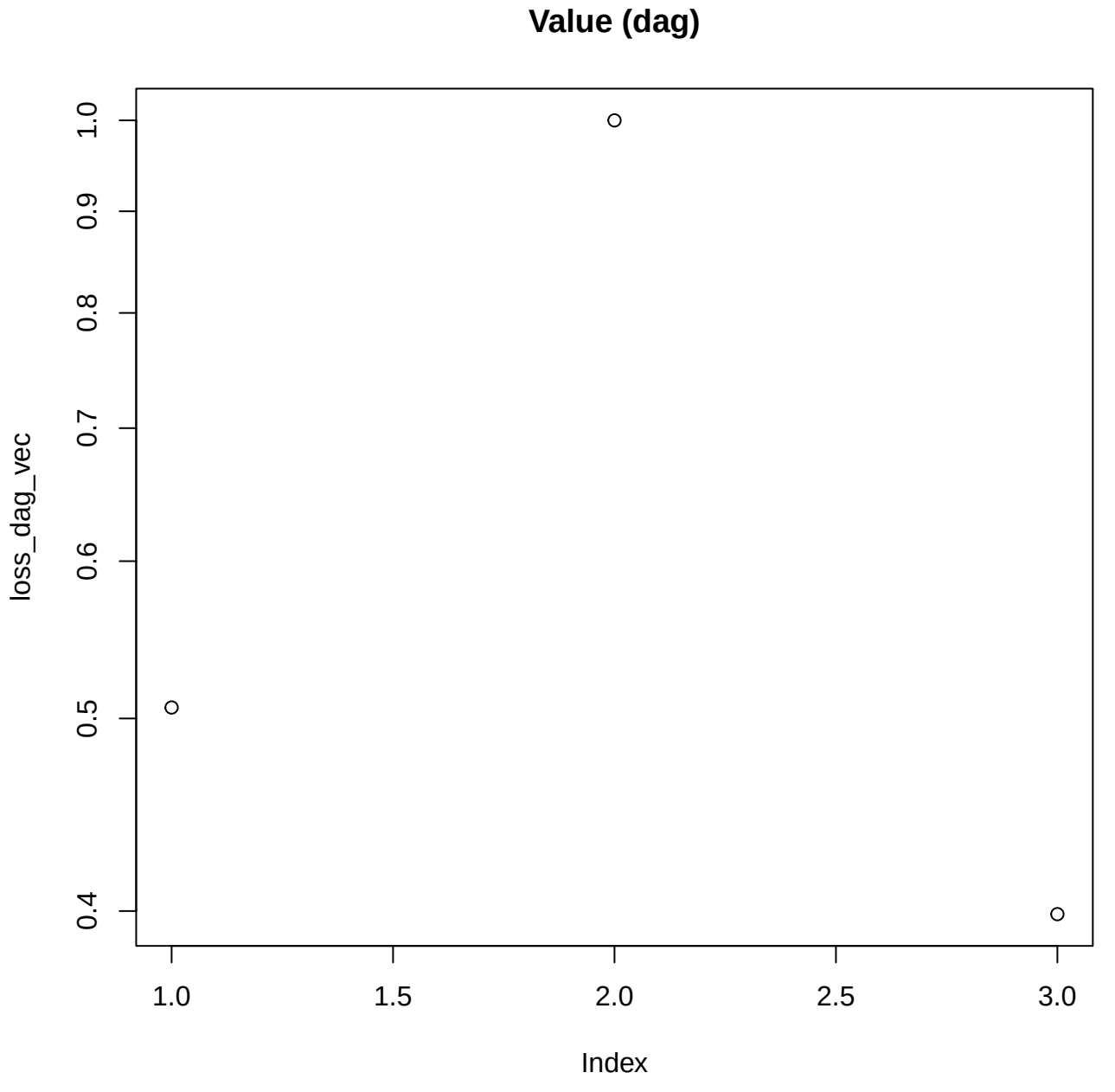
Value (dag)
loss_dag_vec
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2.5
3.0
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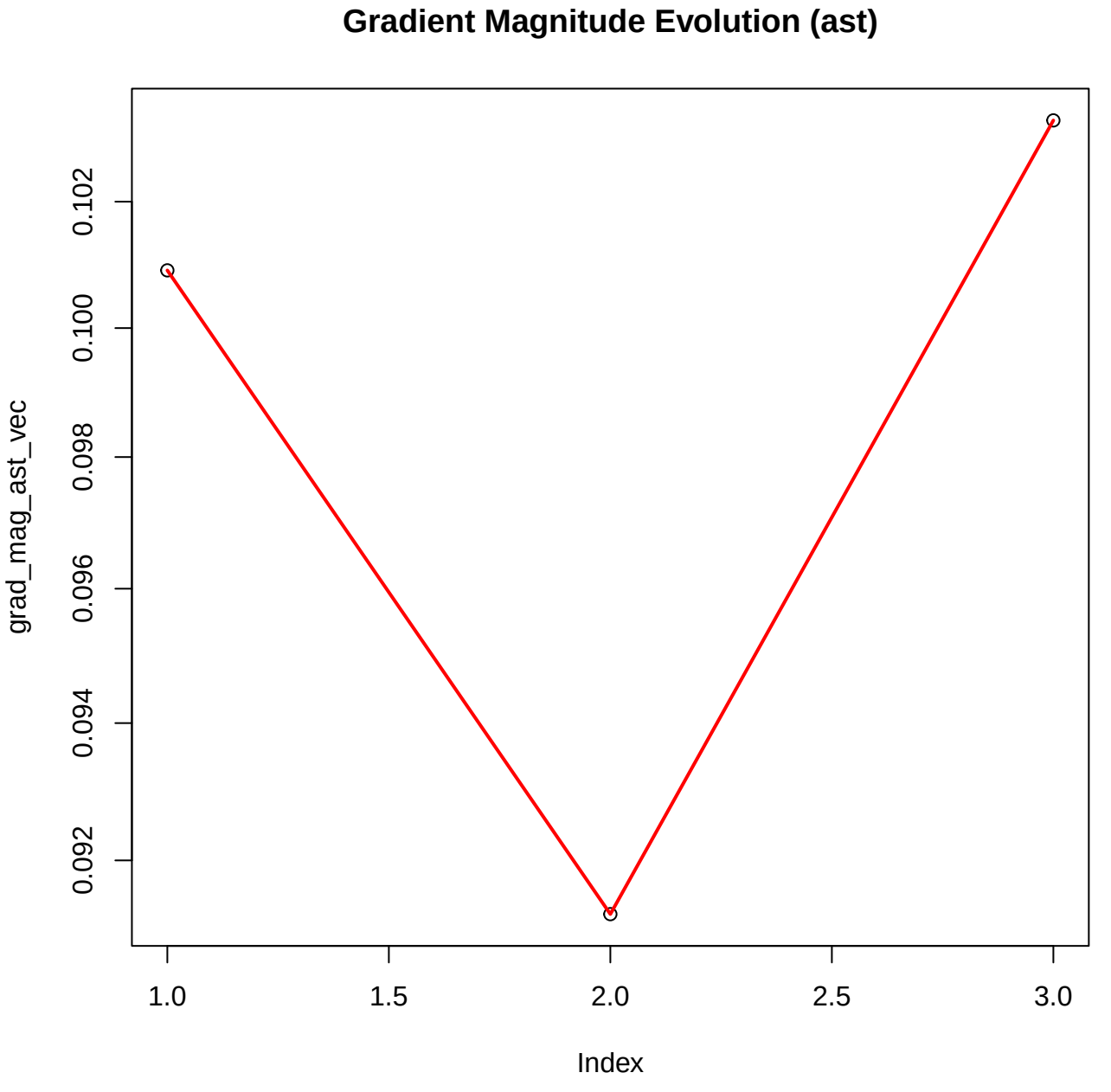
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0.096
0.098
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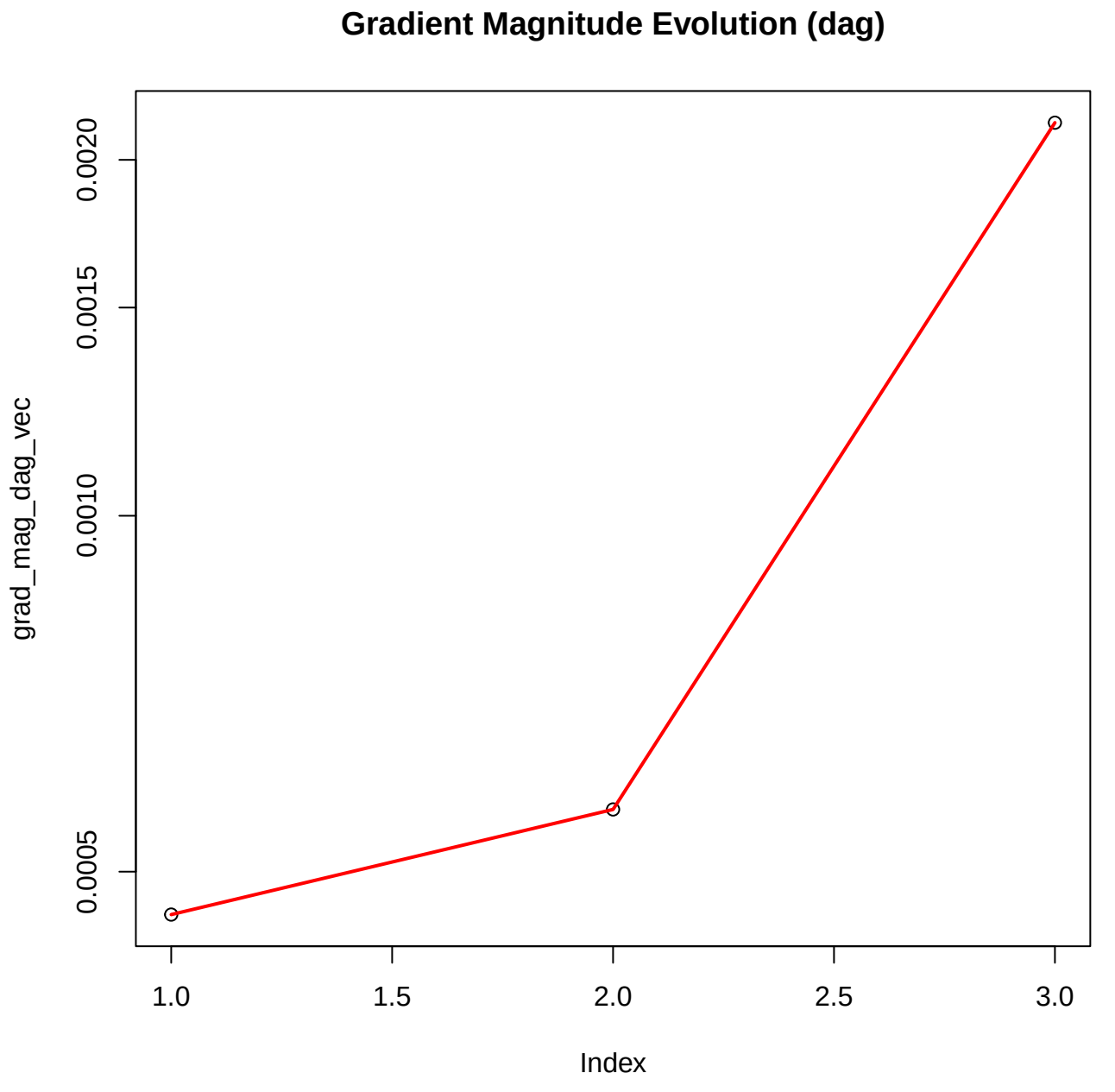
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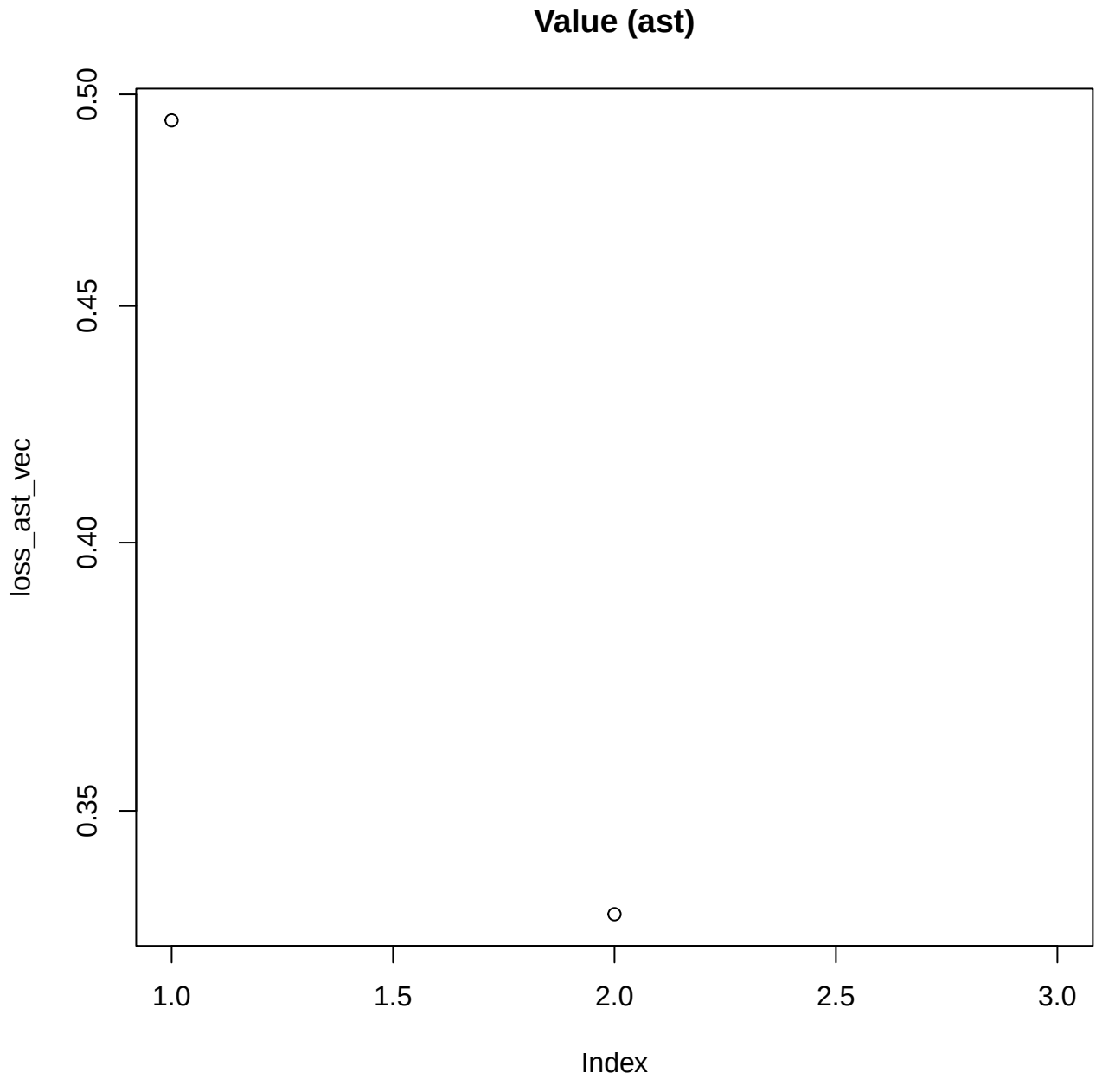
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3.0
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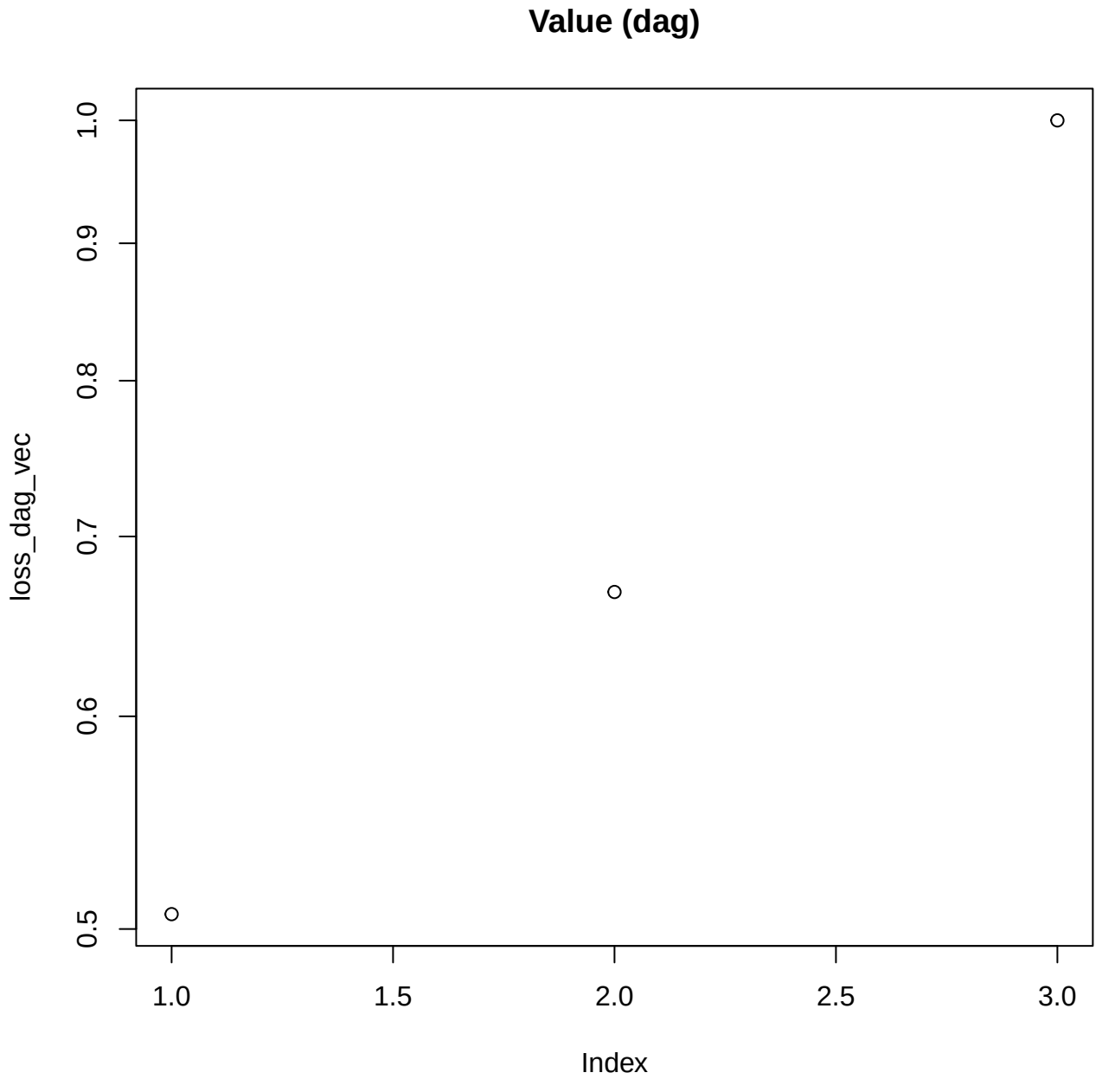
Value (dag)
loss_dag_vec
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2.5
3.0
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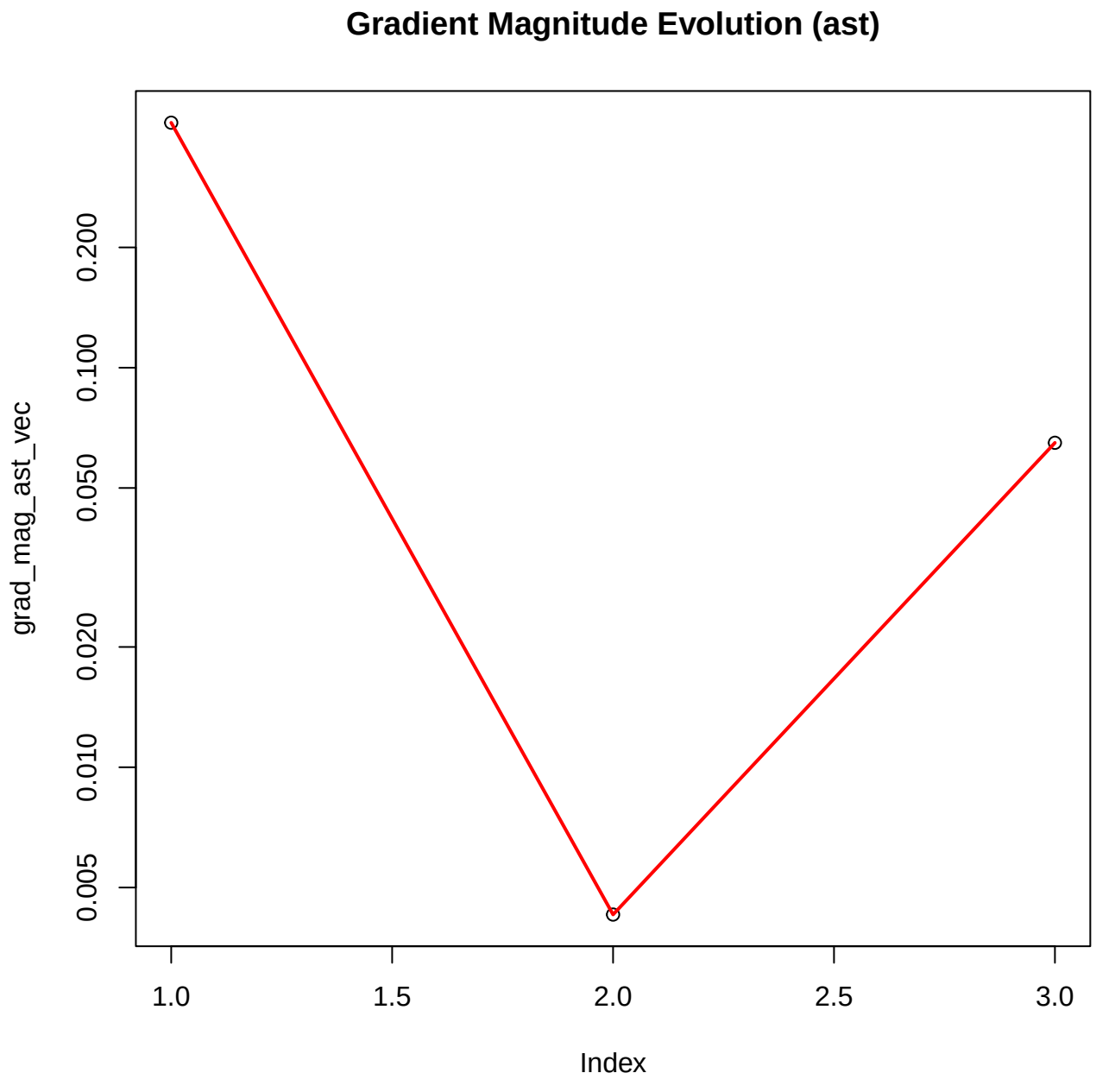
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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2.5
3.0
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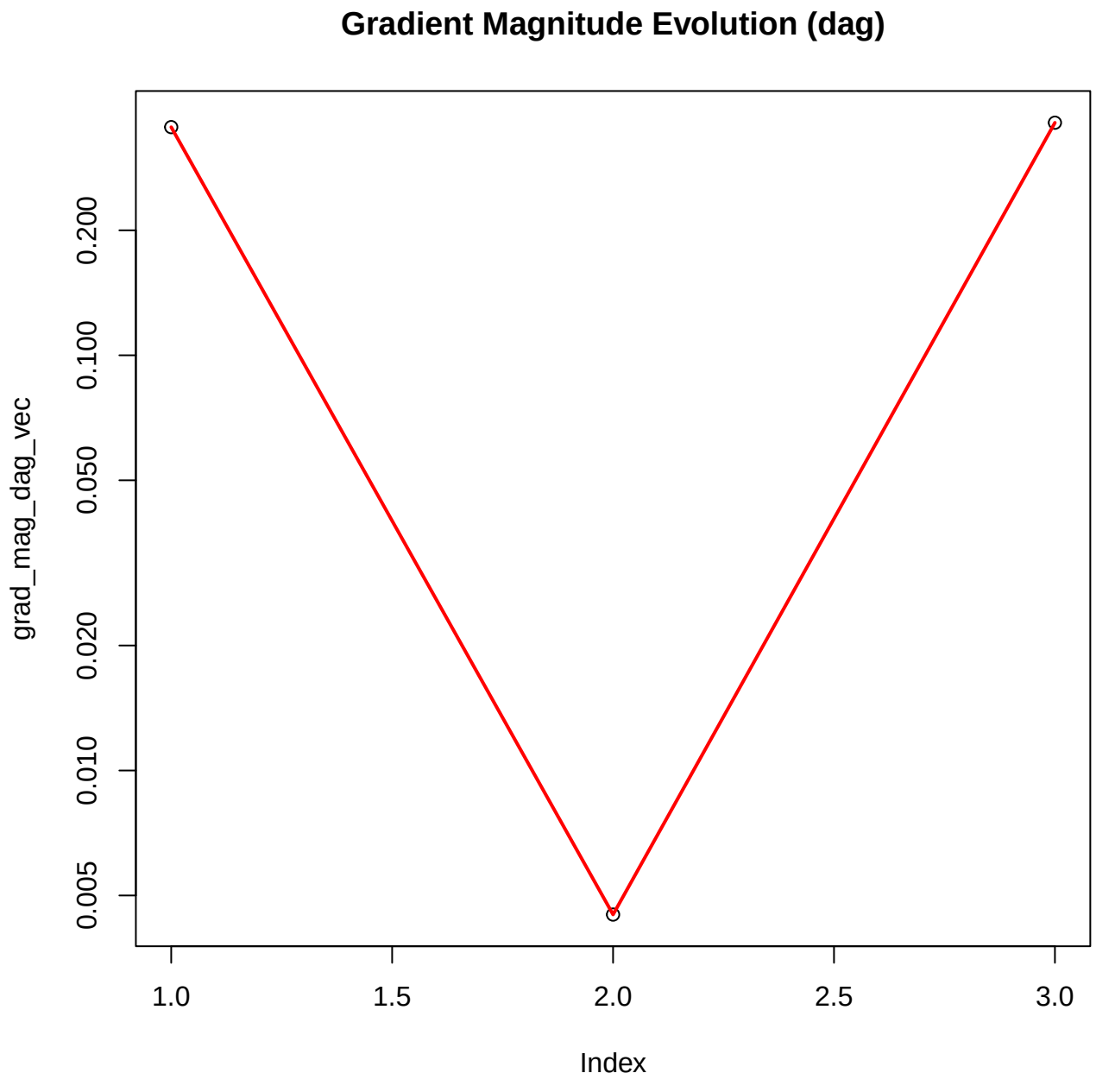
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.200
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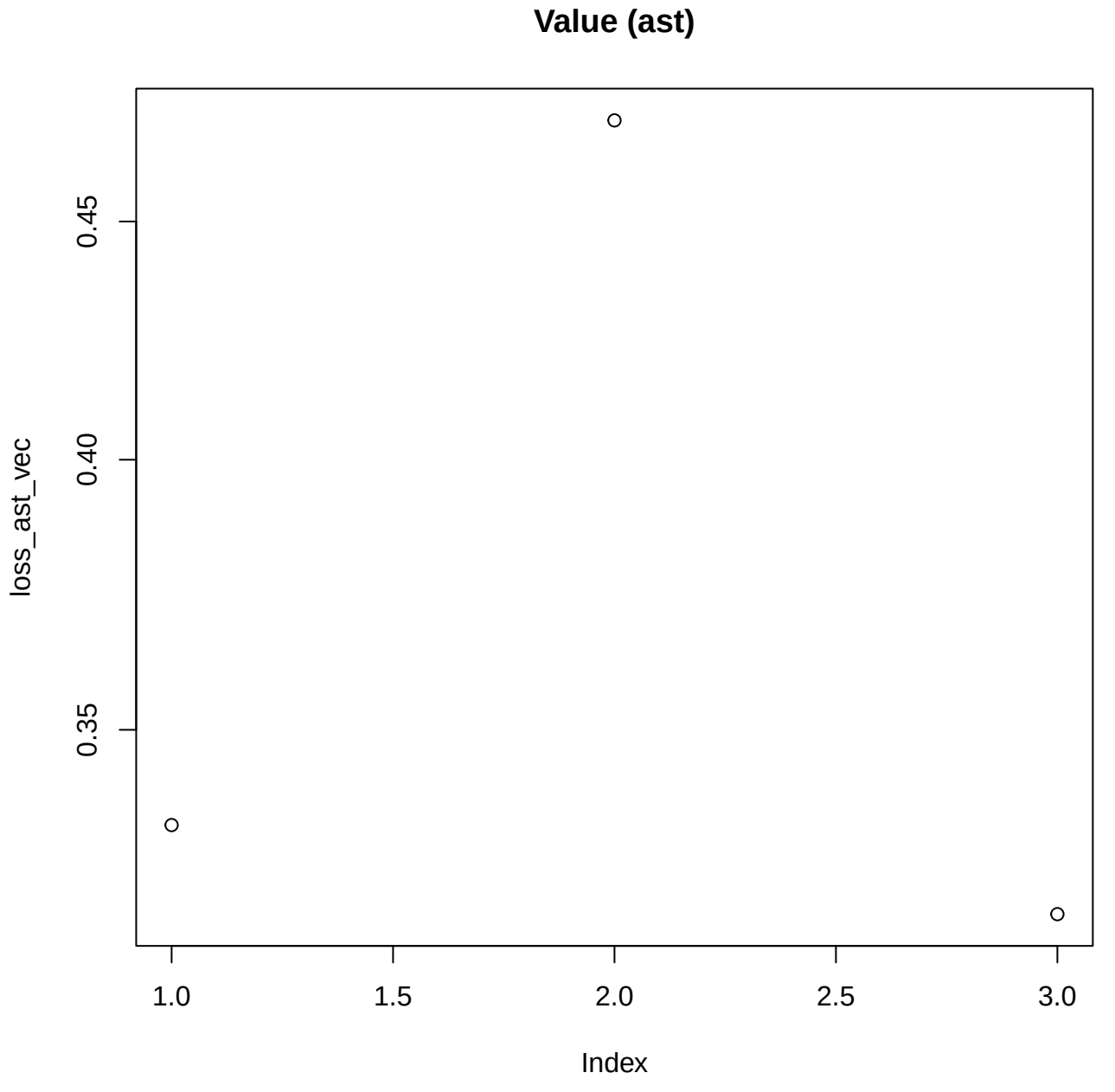
2.5
3.0
Index


Value (ast)
loss_ast_vec
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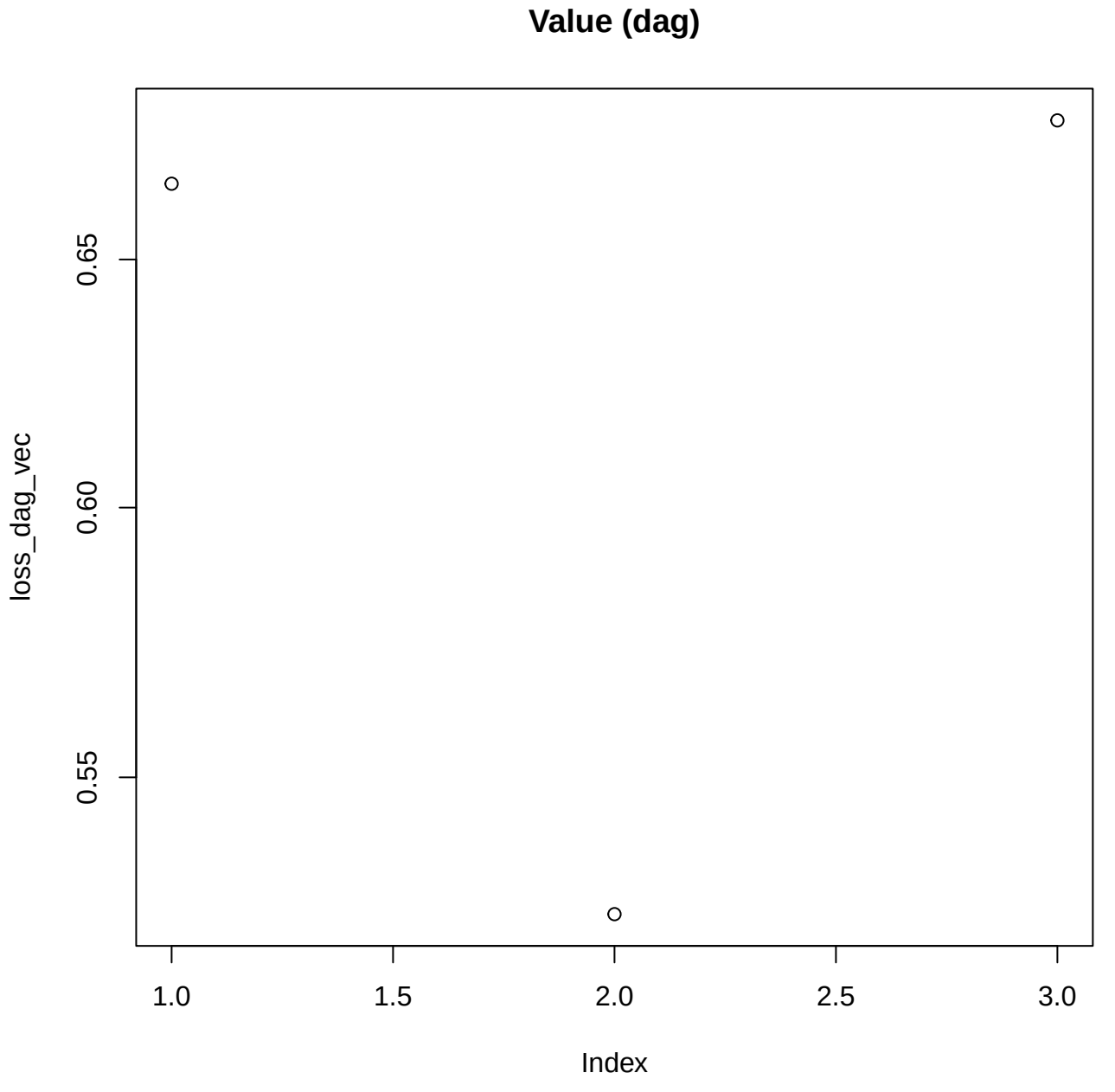
Index


Value (dag)
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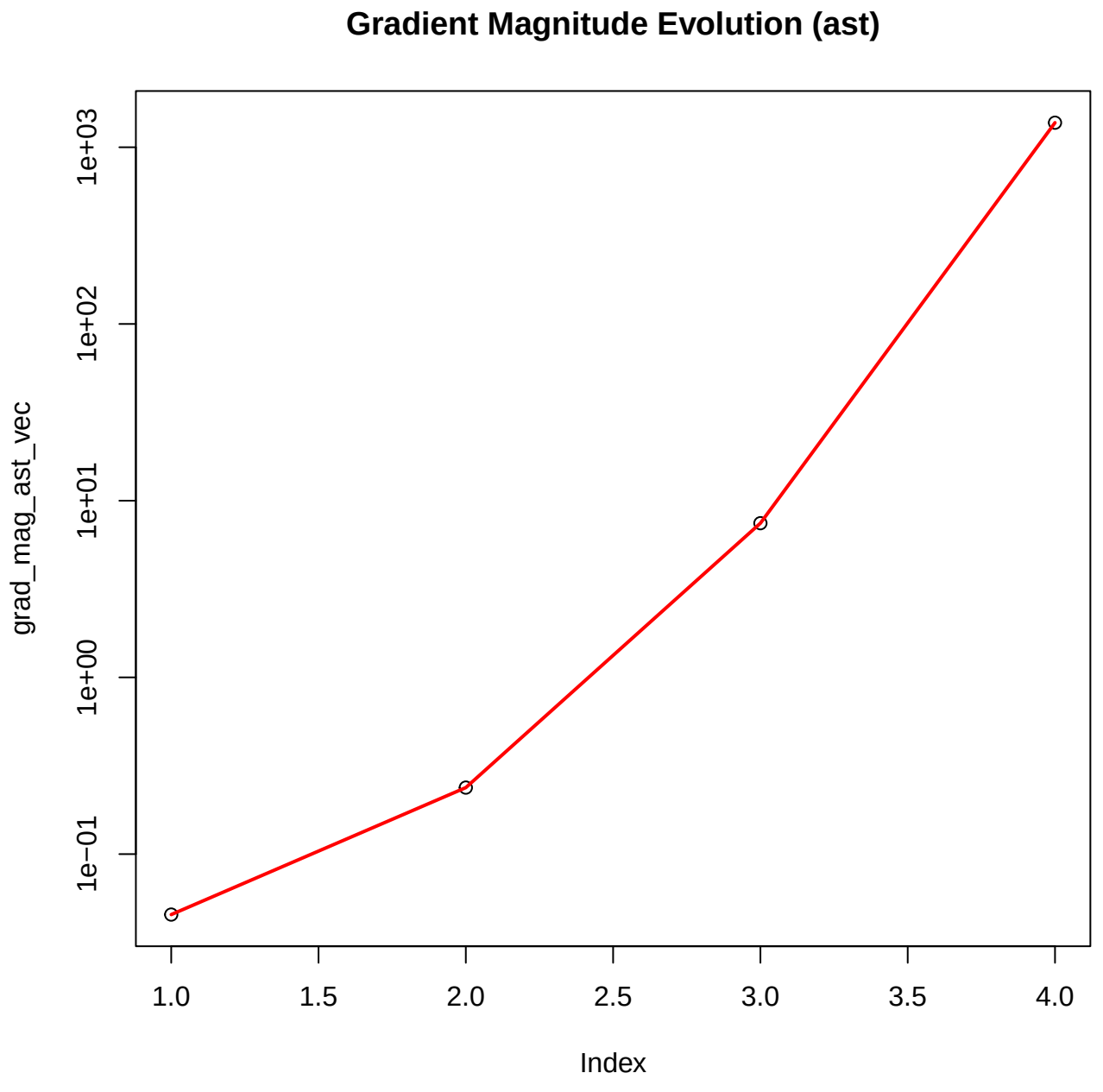
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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1e+00
1e+01
1e+02
1e+03
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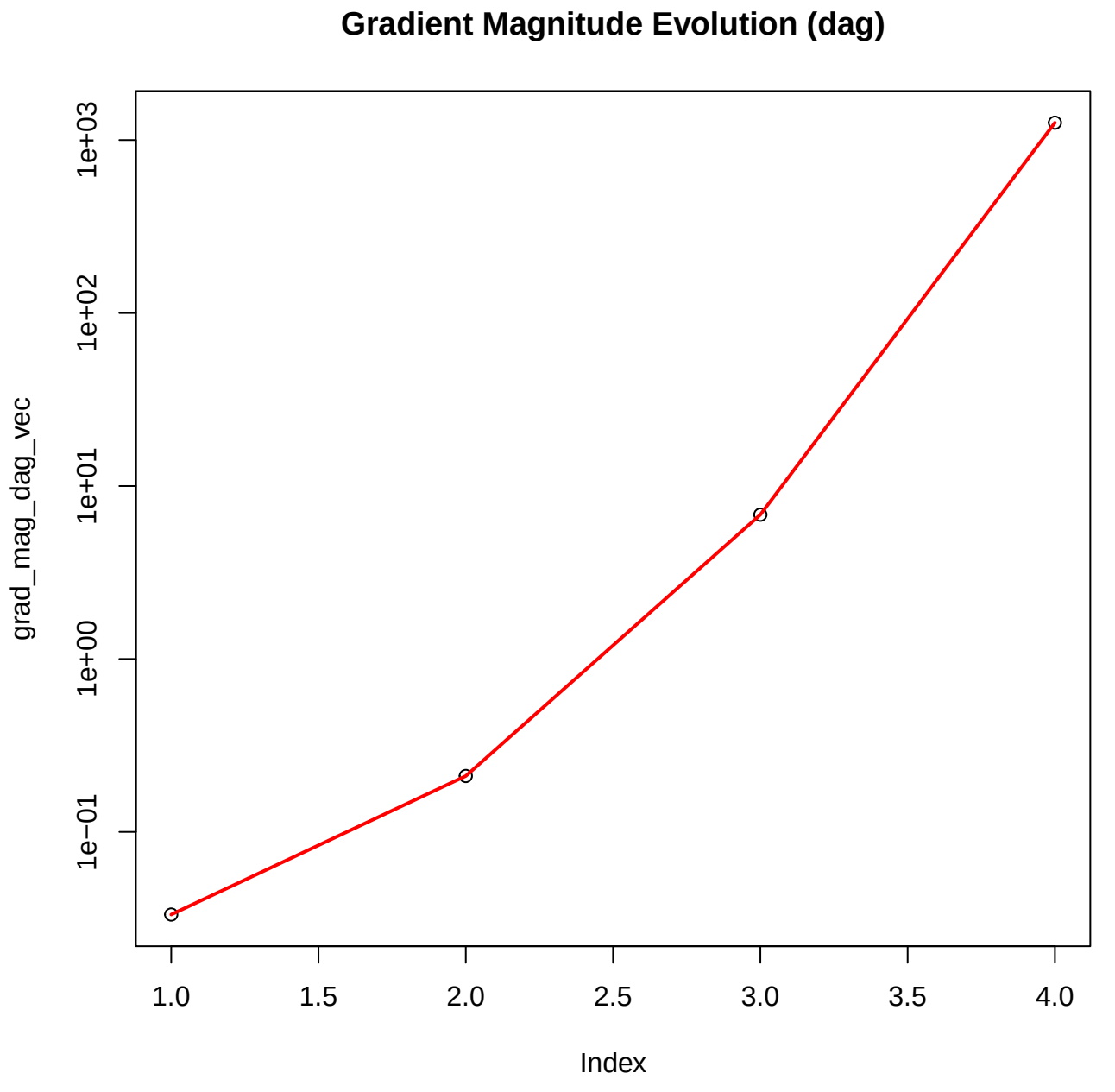
2.5
3.0
3.5
4.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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1e+00
1e+01
1e+02
1e+03
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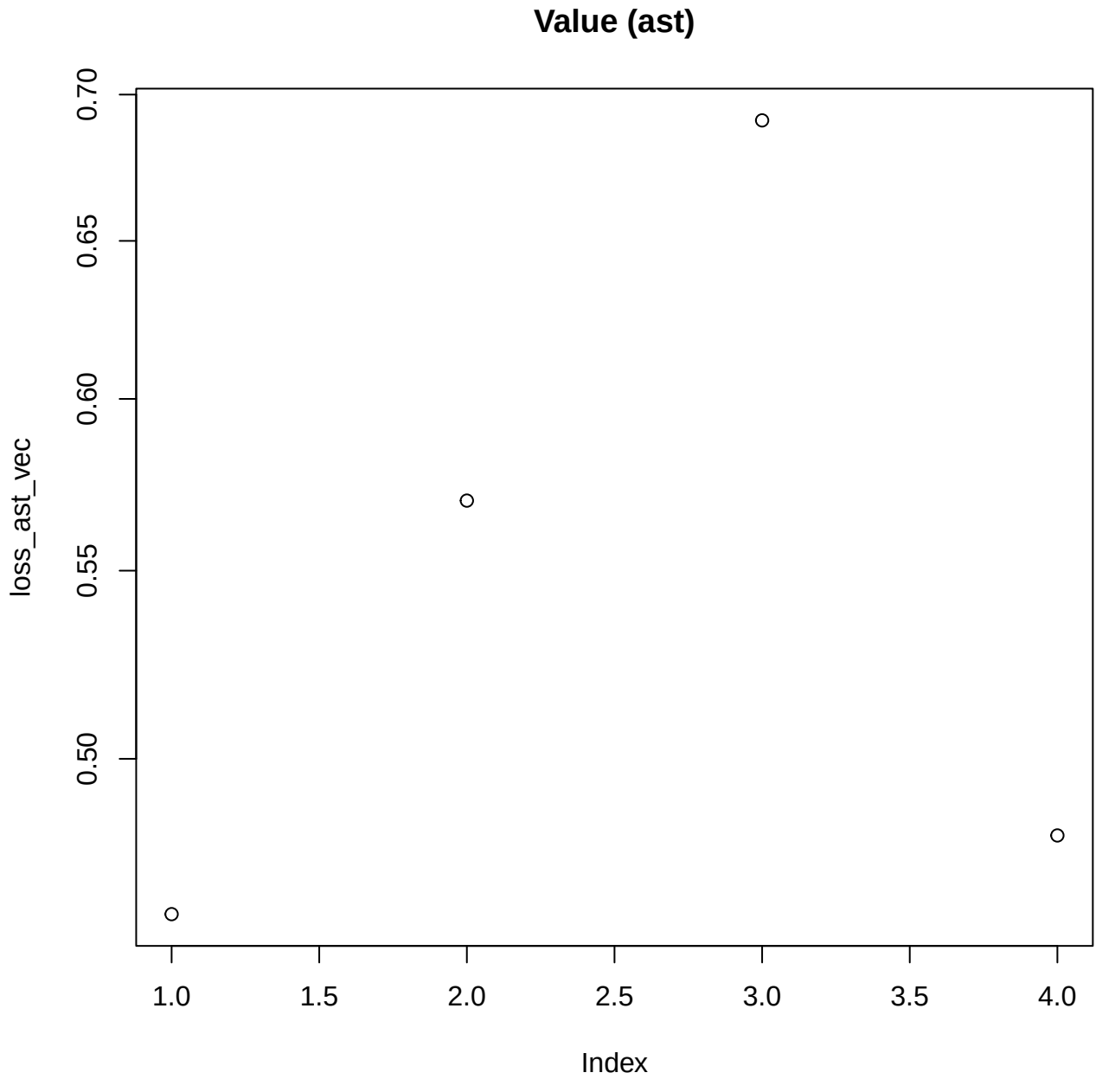
2.0
2.5
3.0
3.5
4.0
Index


Value (ast)
loss_ast_vec
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2.5
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3.5
4.0
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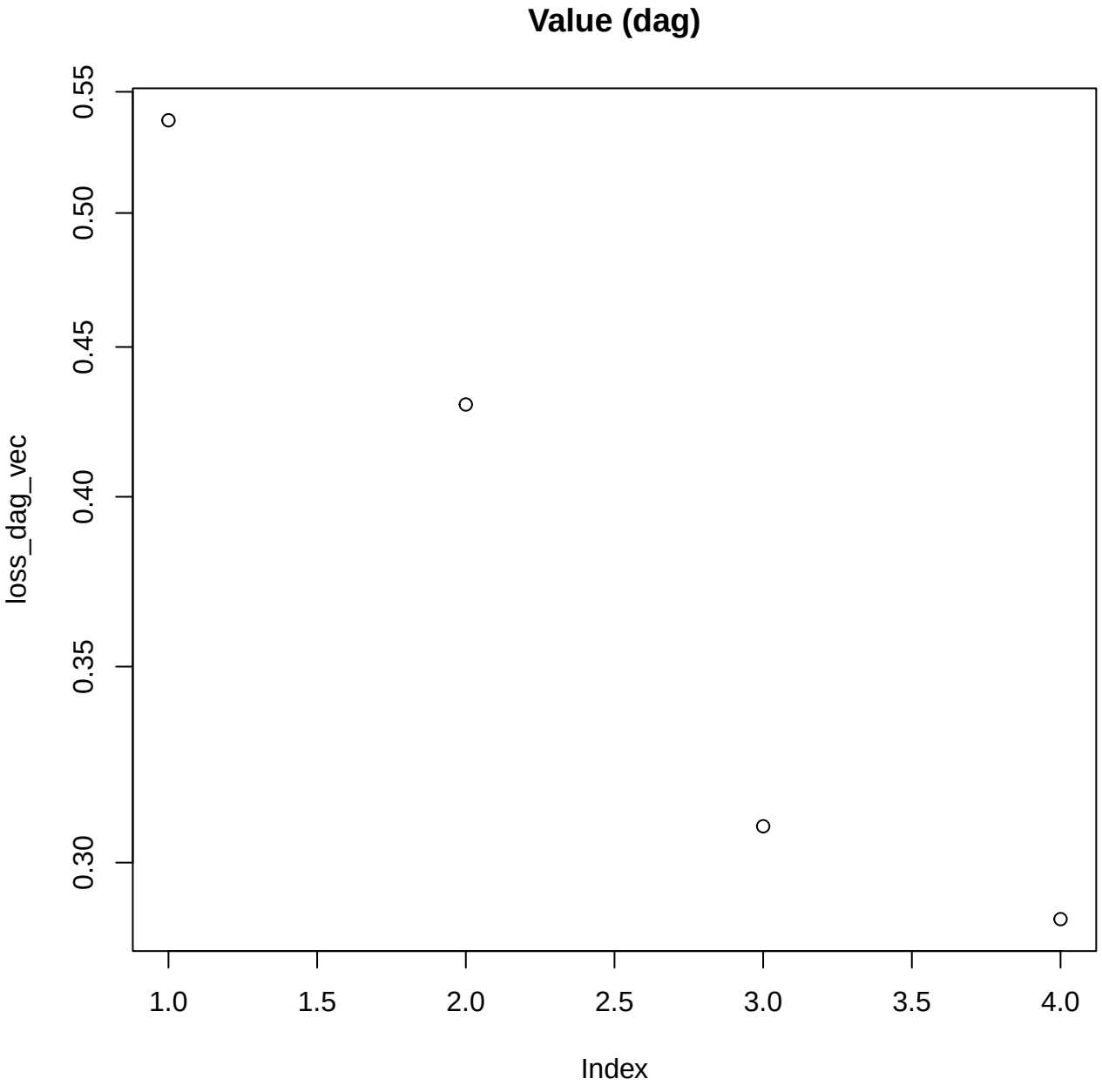
Value (dag)
loss_dag_vec
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2.5
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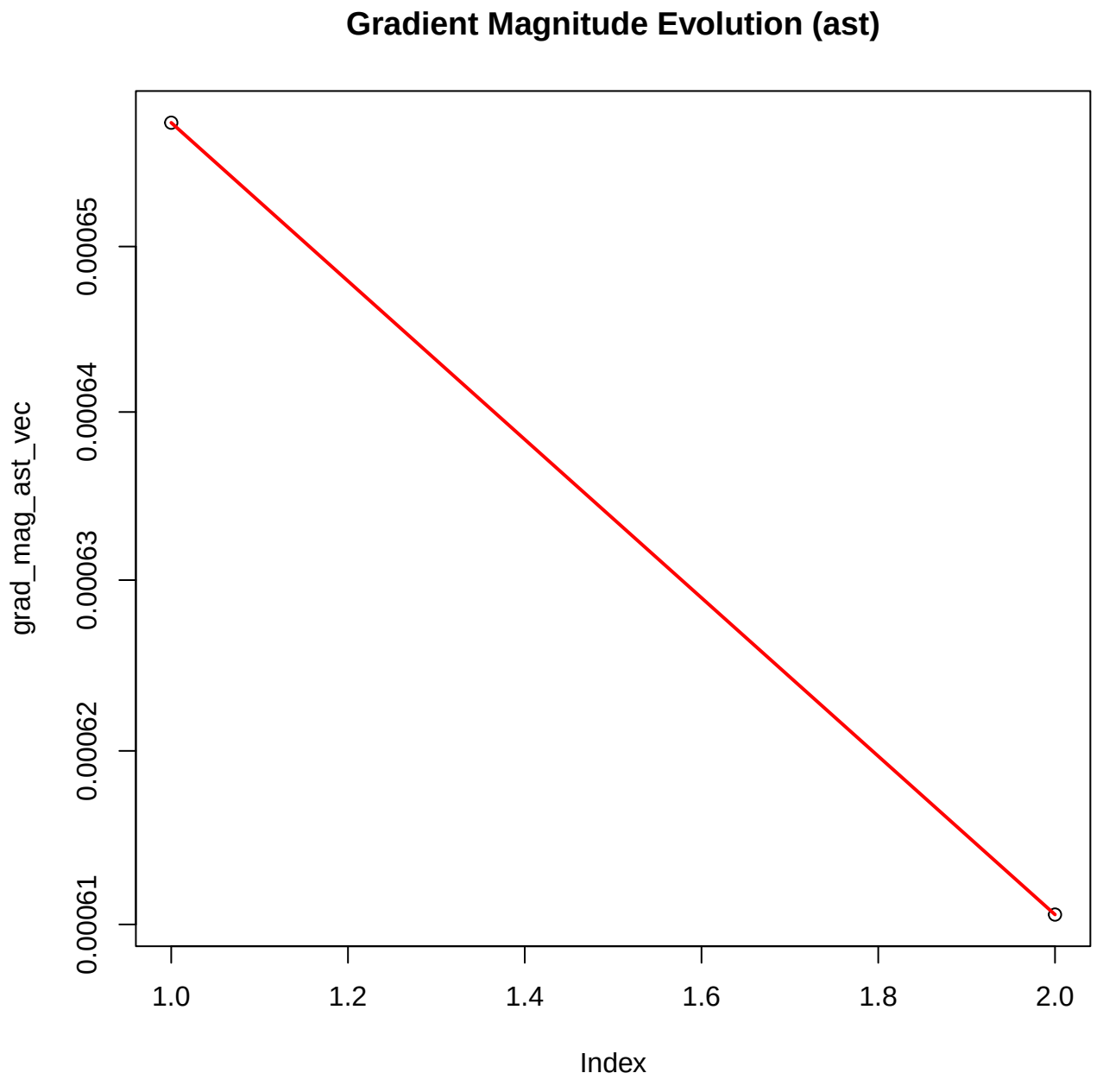
3.5
4.0
Index


Gradient Magnitude Evolution (ast)
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0.00065
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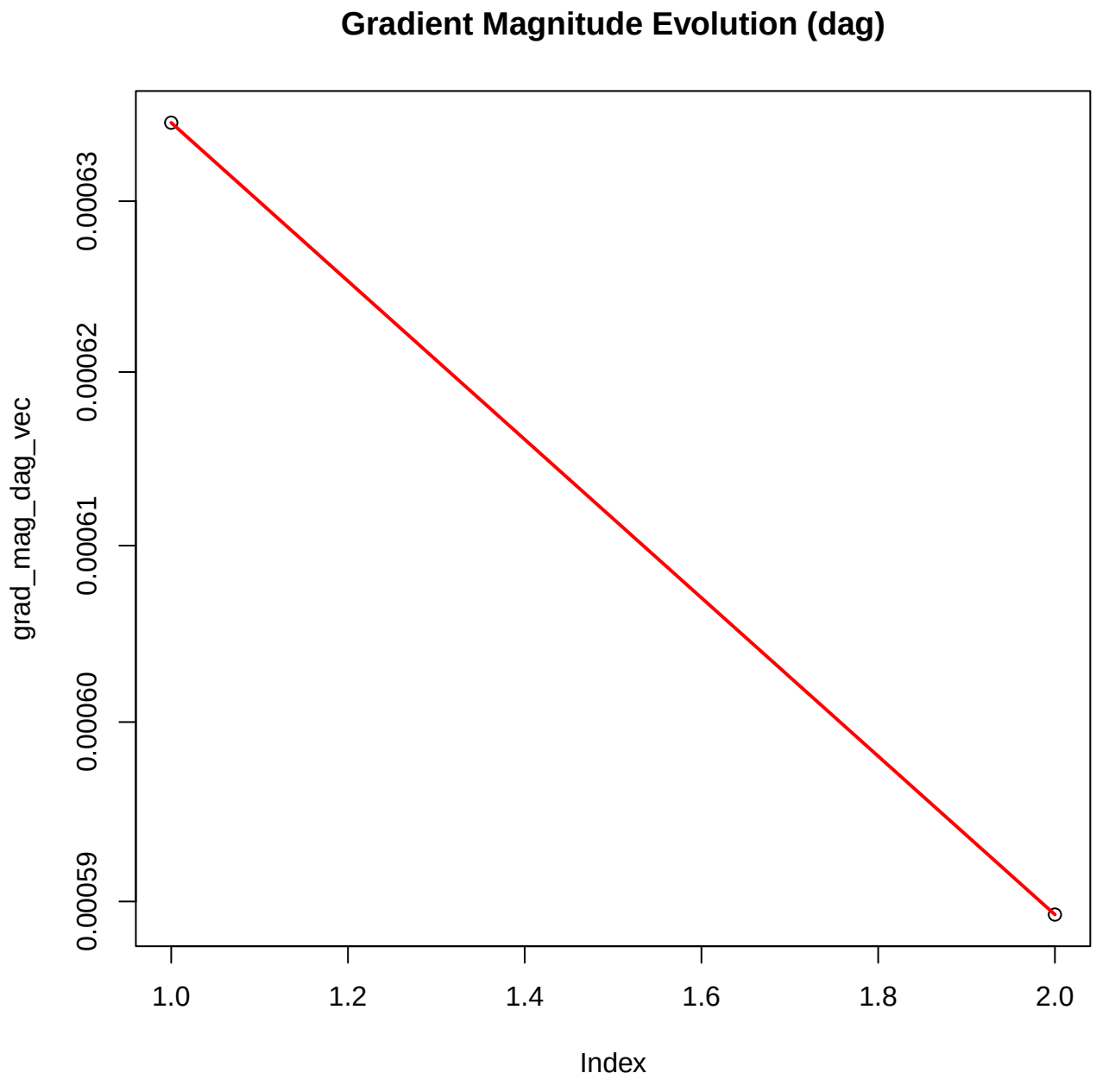
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (dag)
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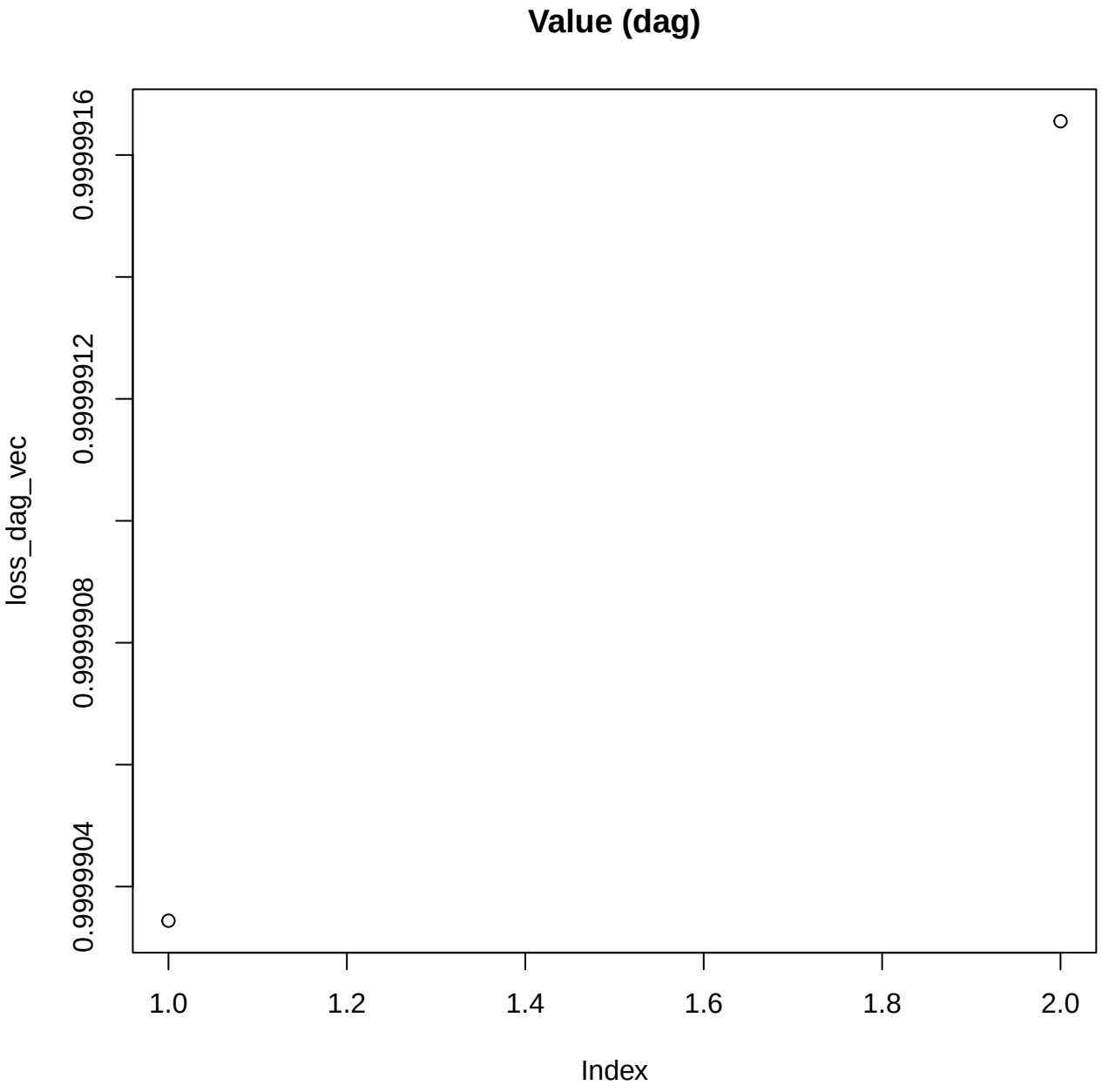
1.4
1.6
1.8
2.0
Index


Value (dag)
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0.9999912
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1.4
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1.8
2.0
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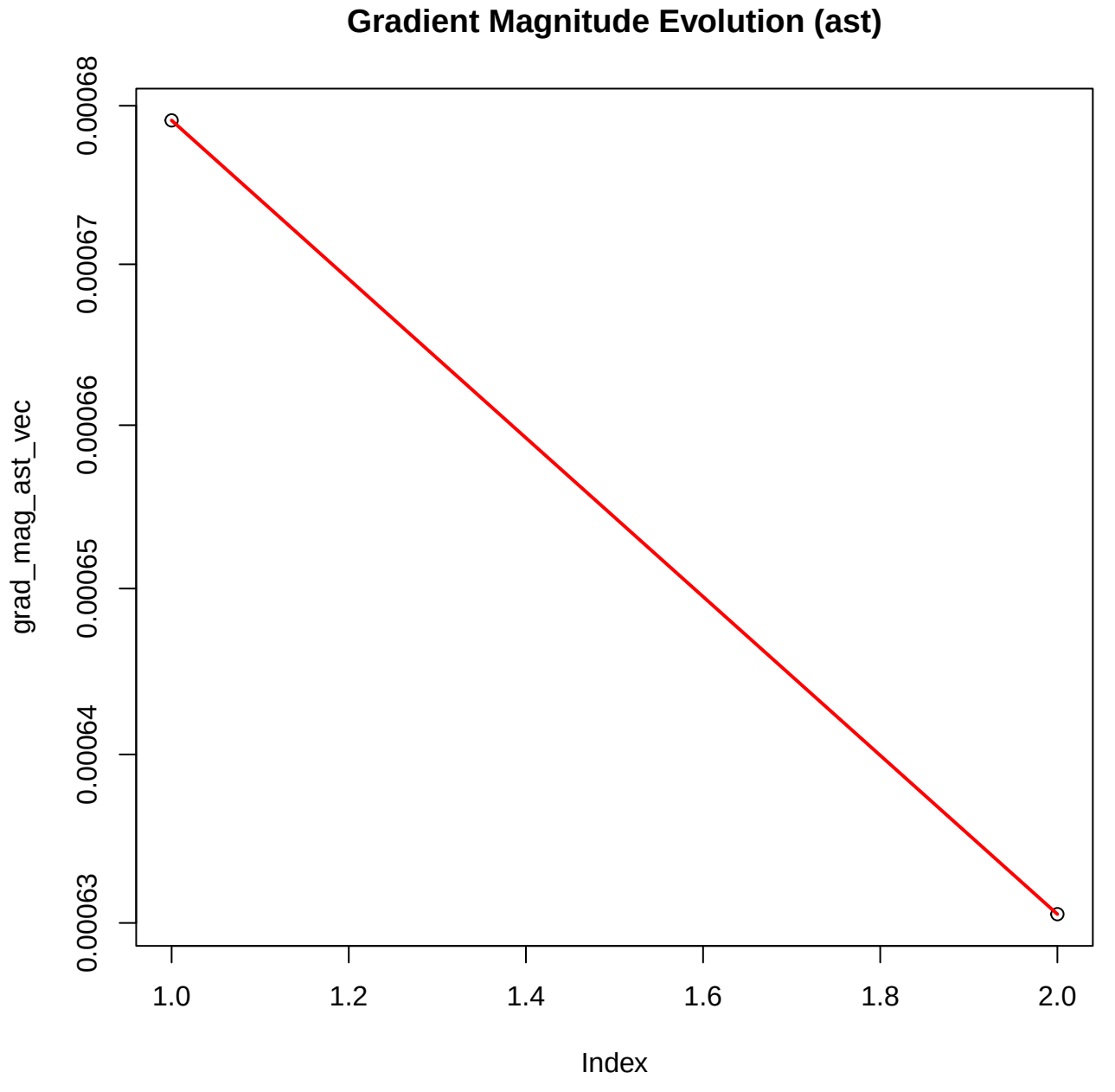
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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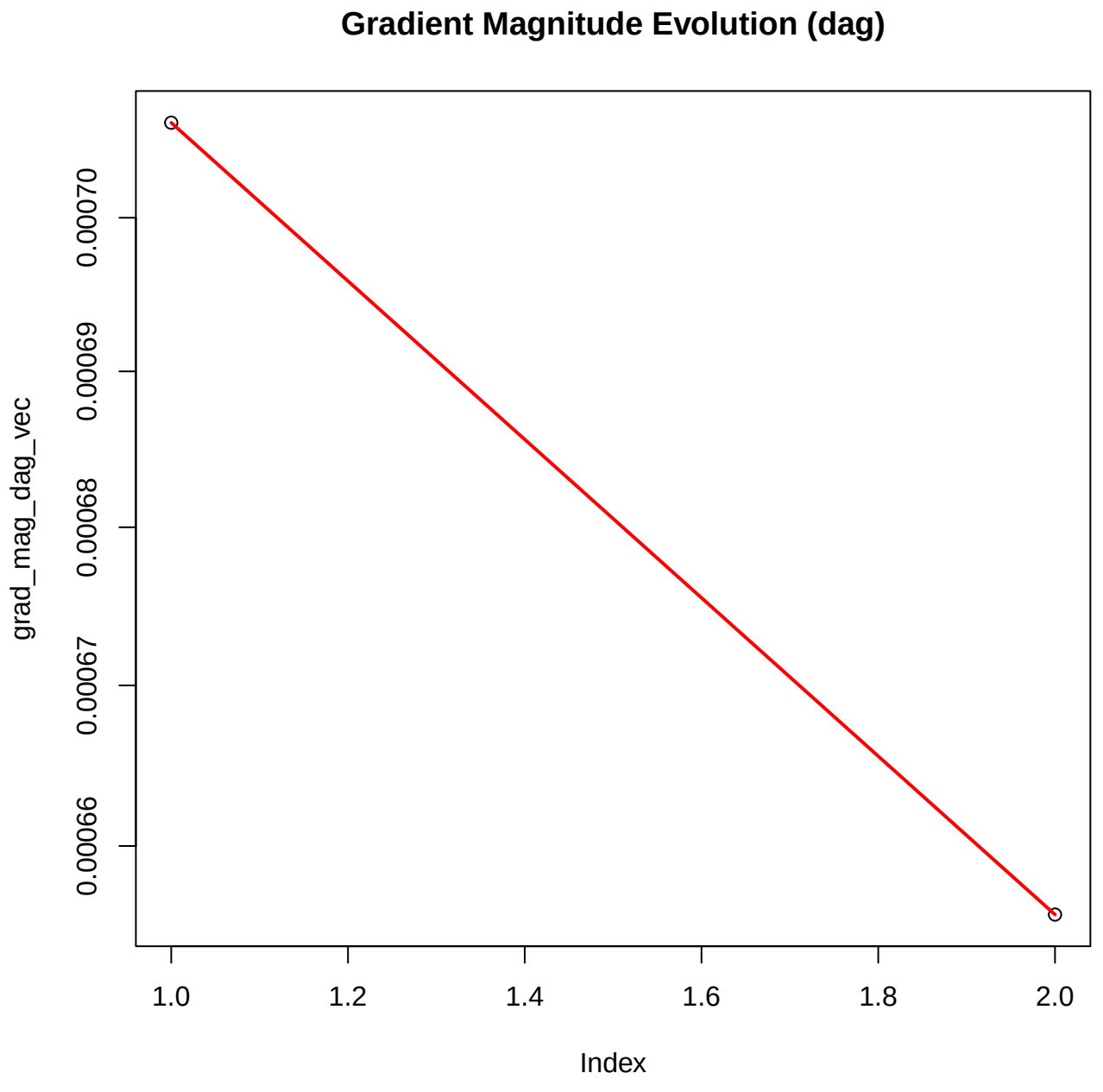
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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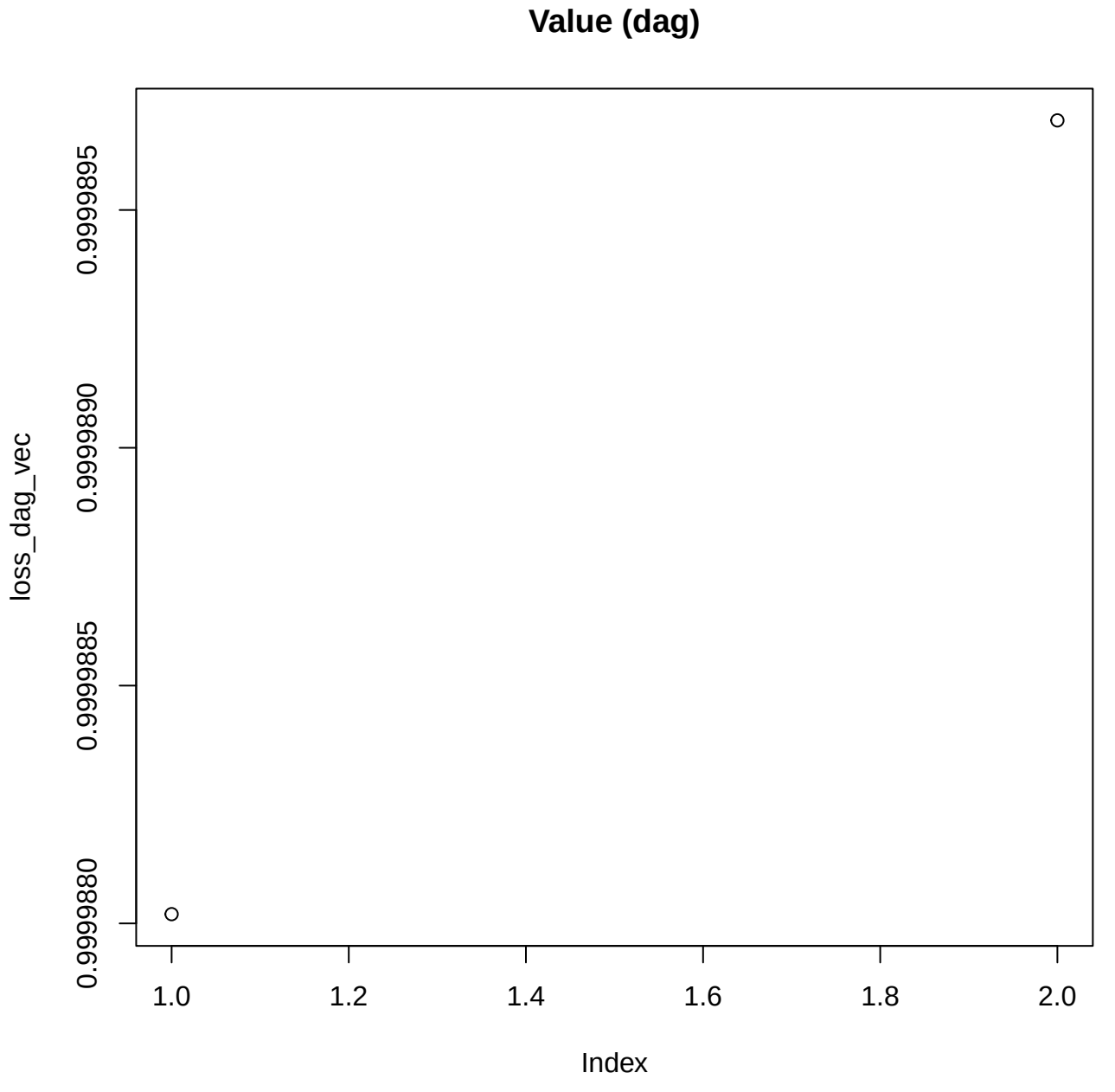
Value (dag)
loss_dag_vec
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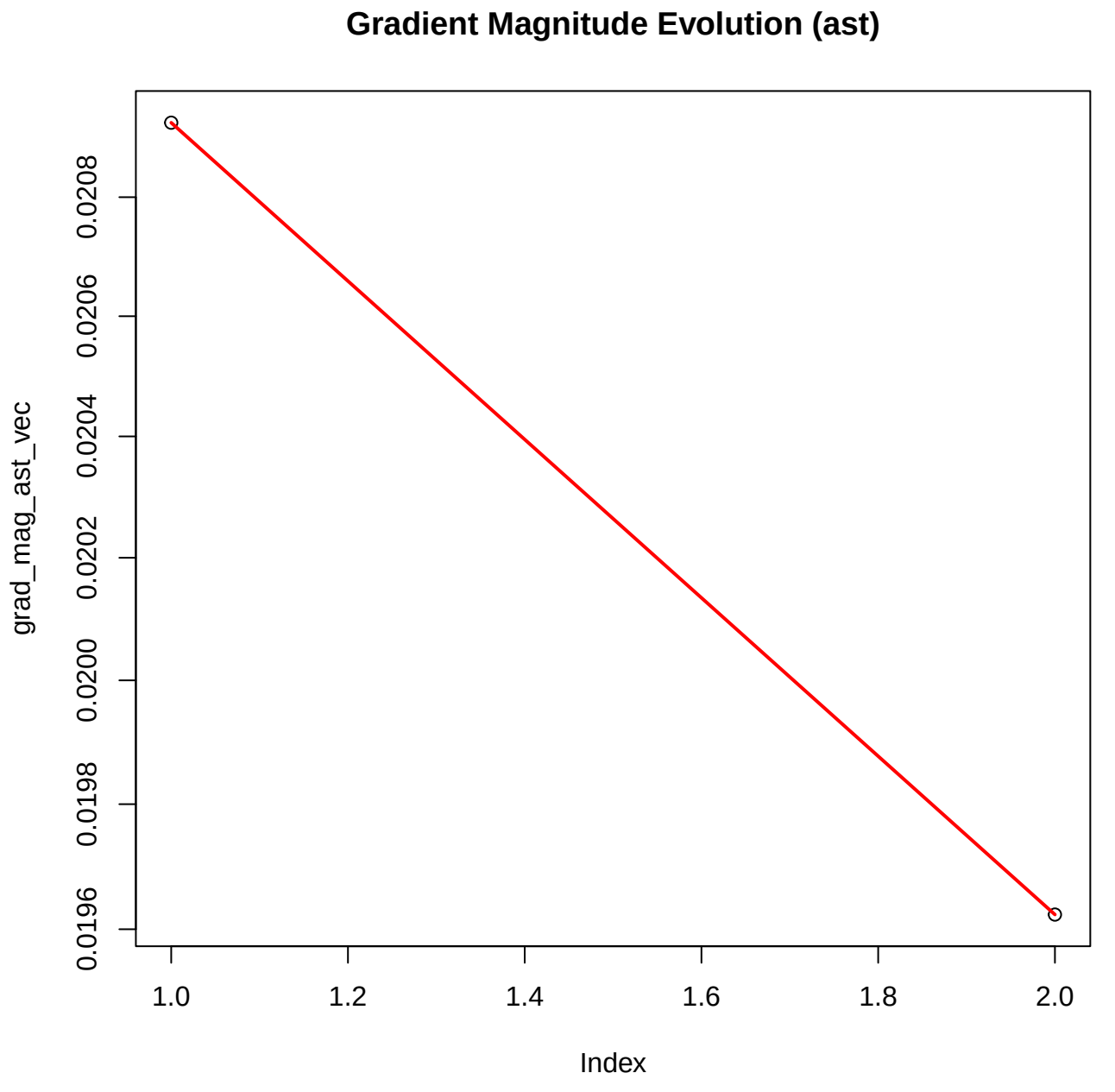
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0.0198
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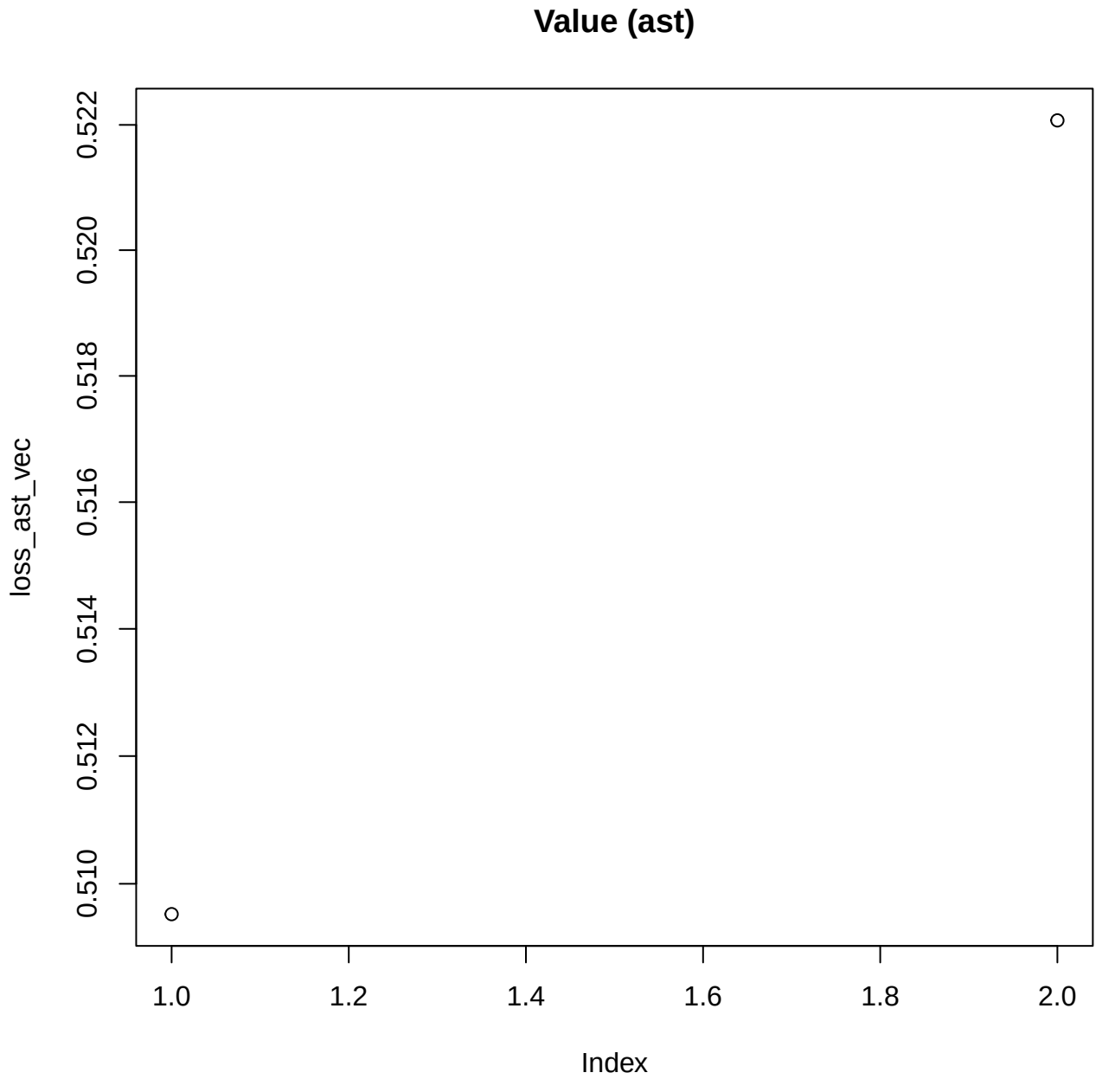
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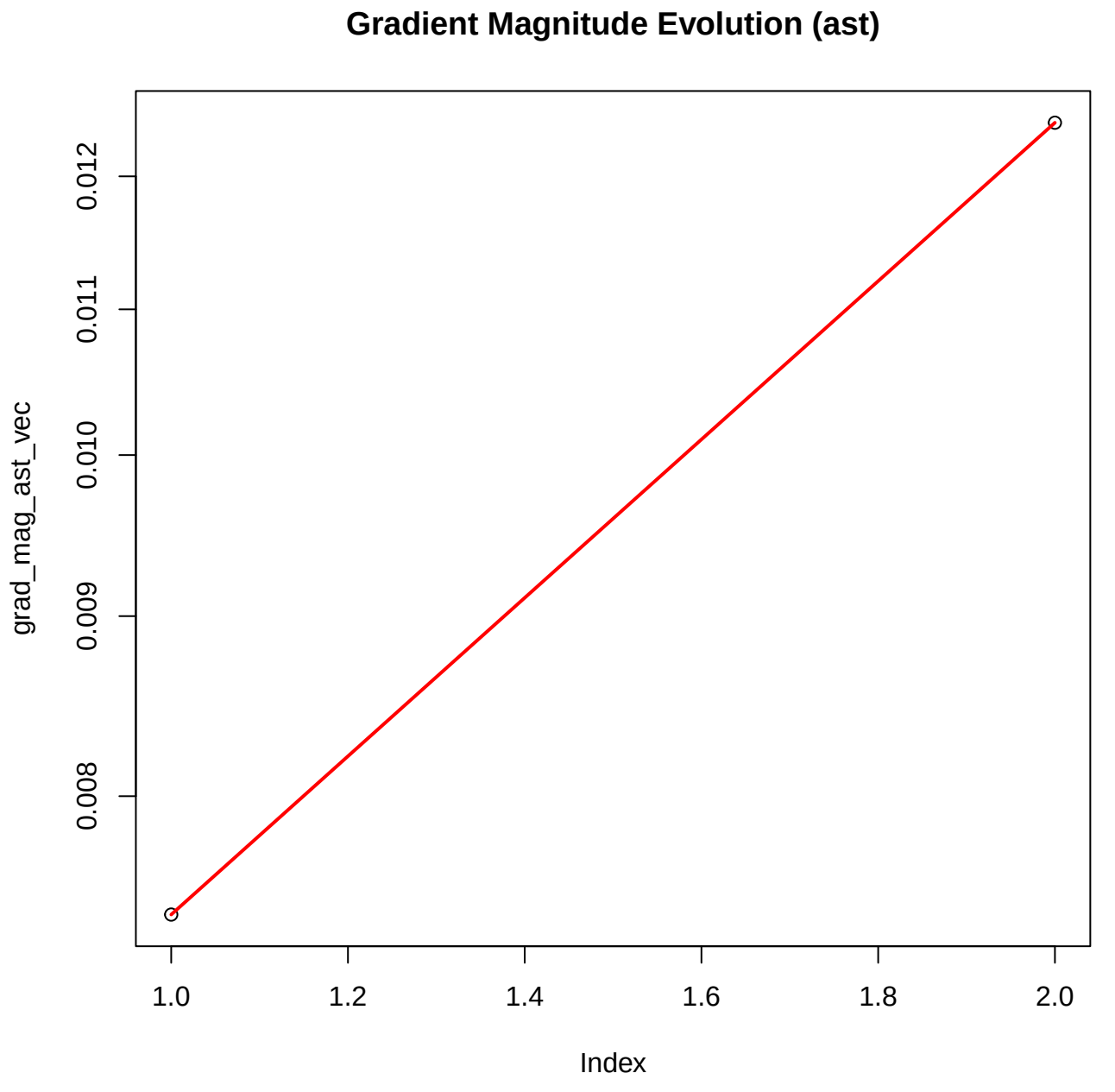
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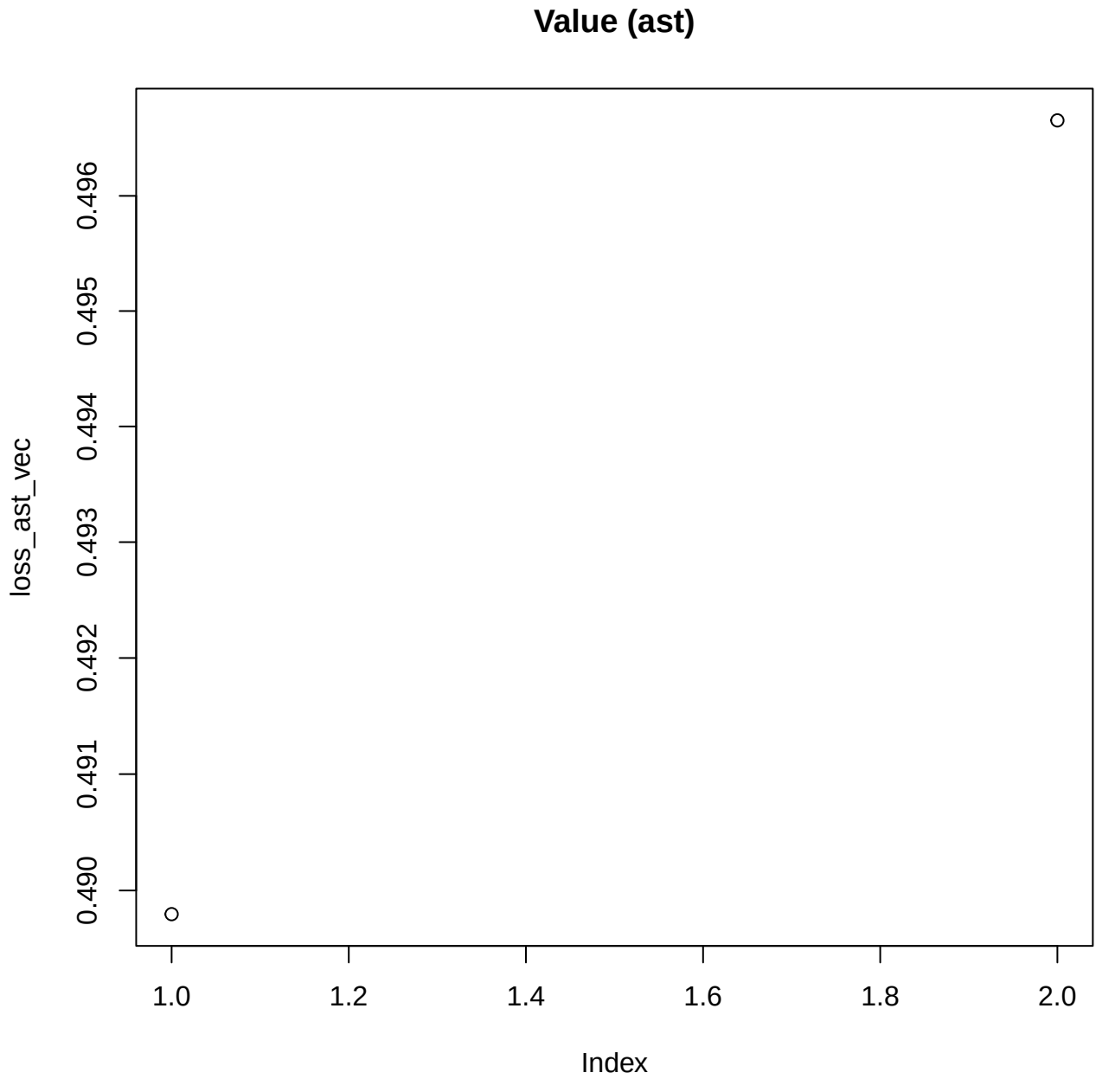
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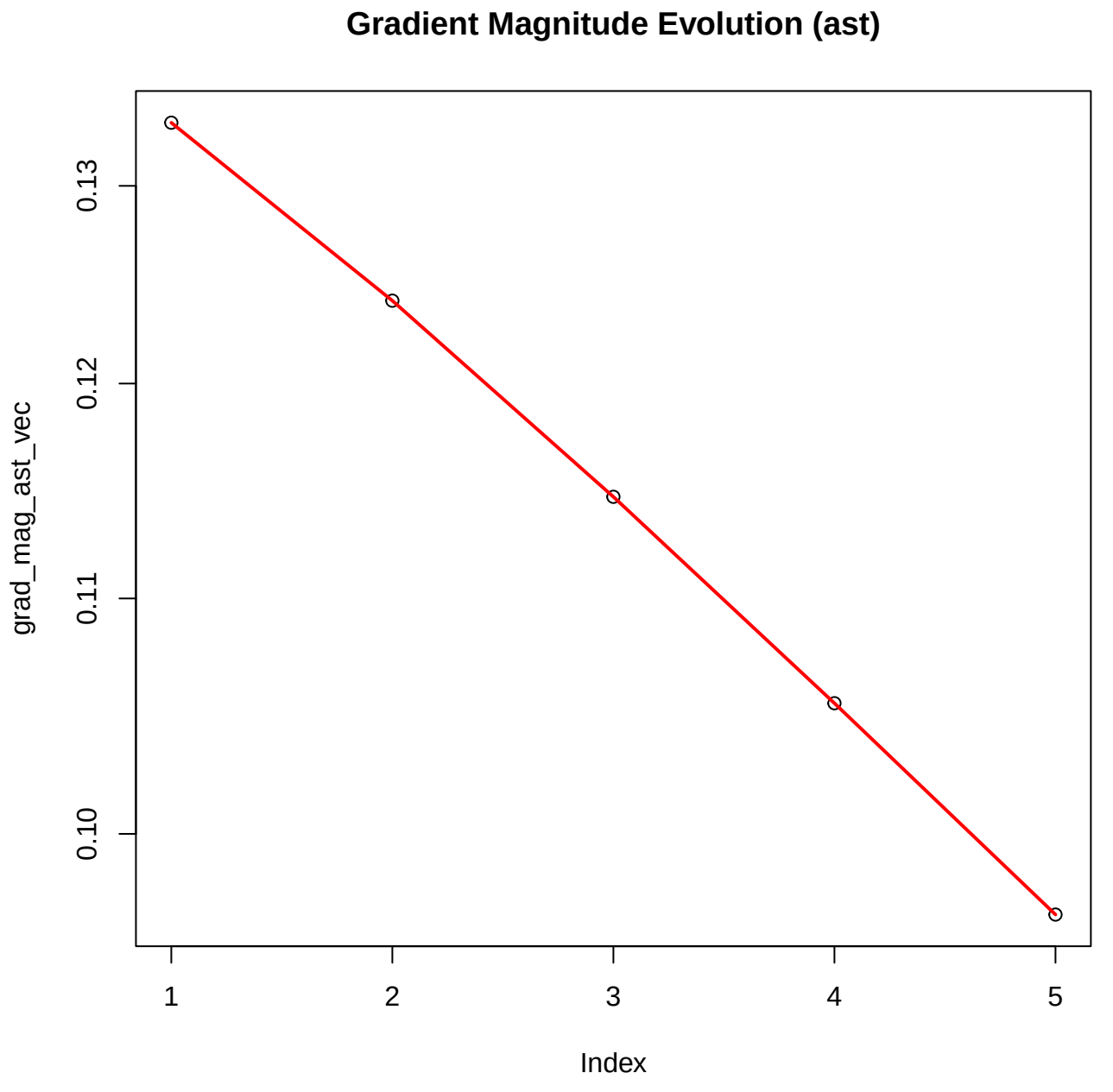
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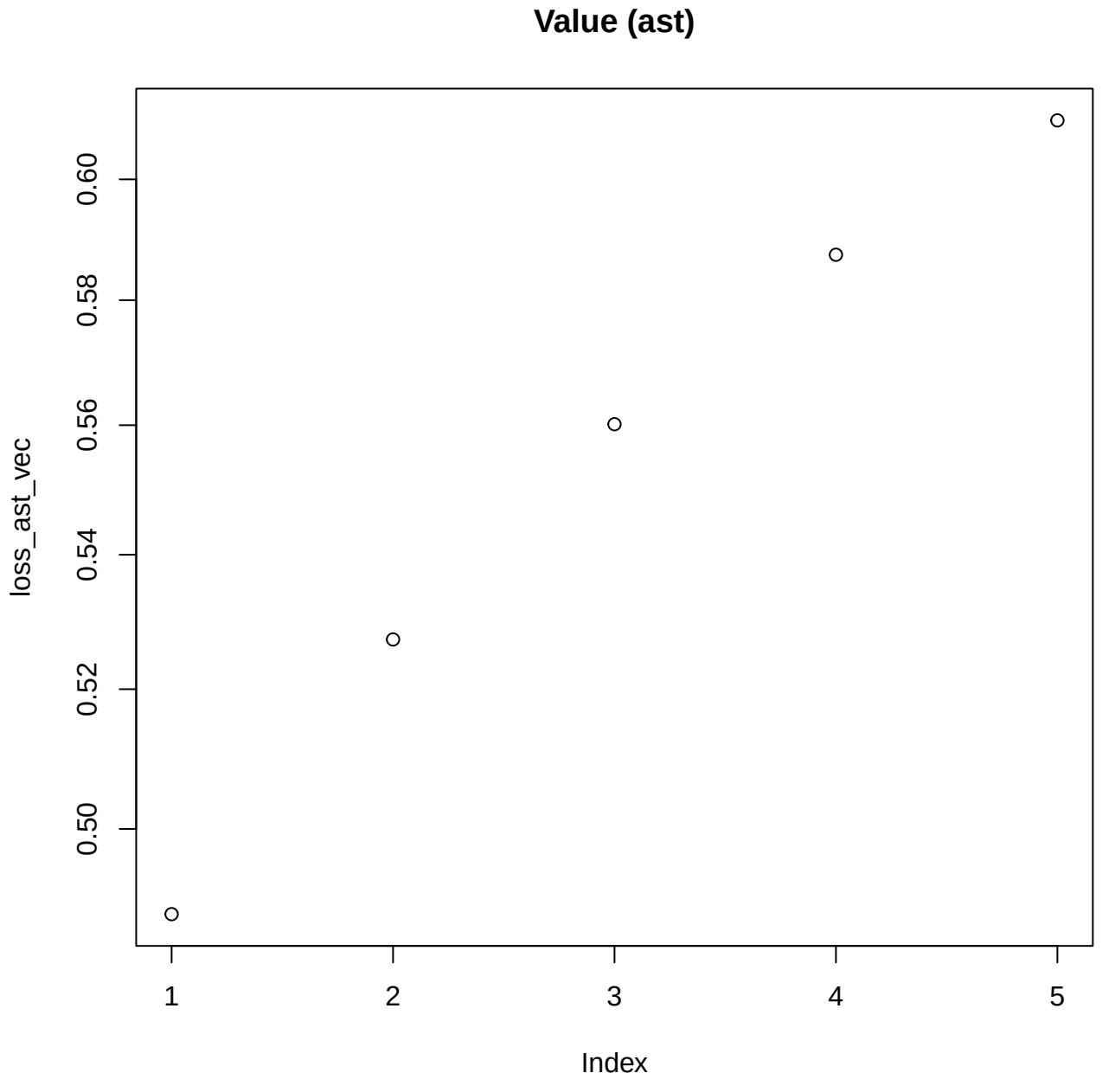
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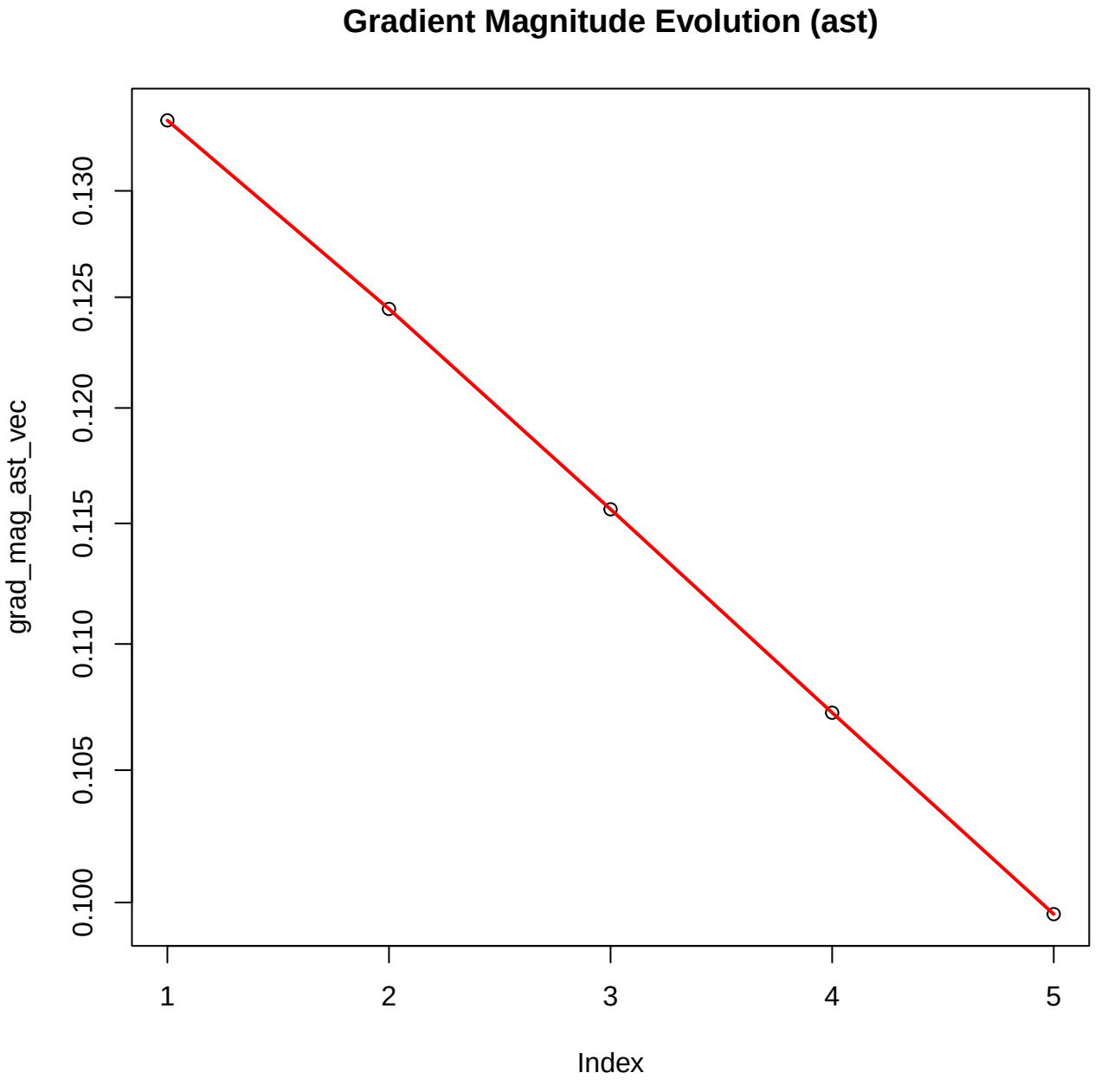
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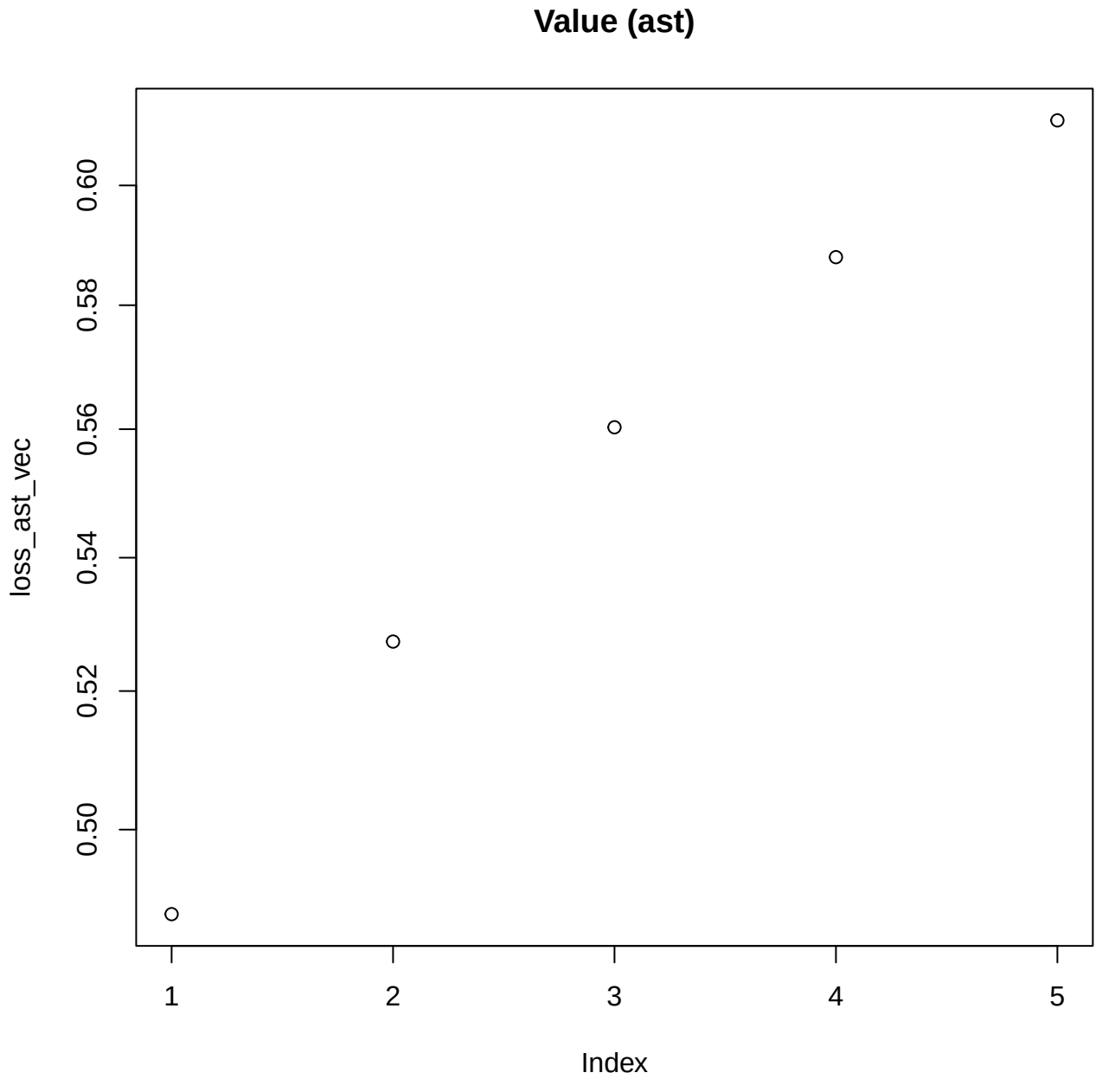
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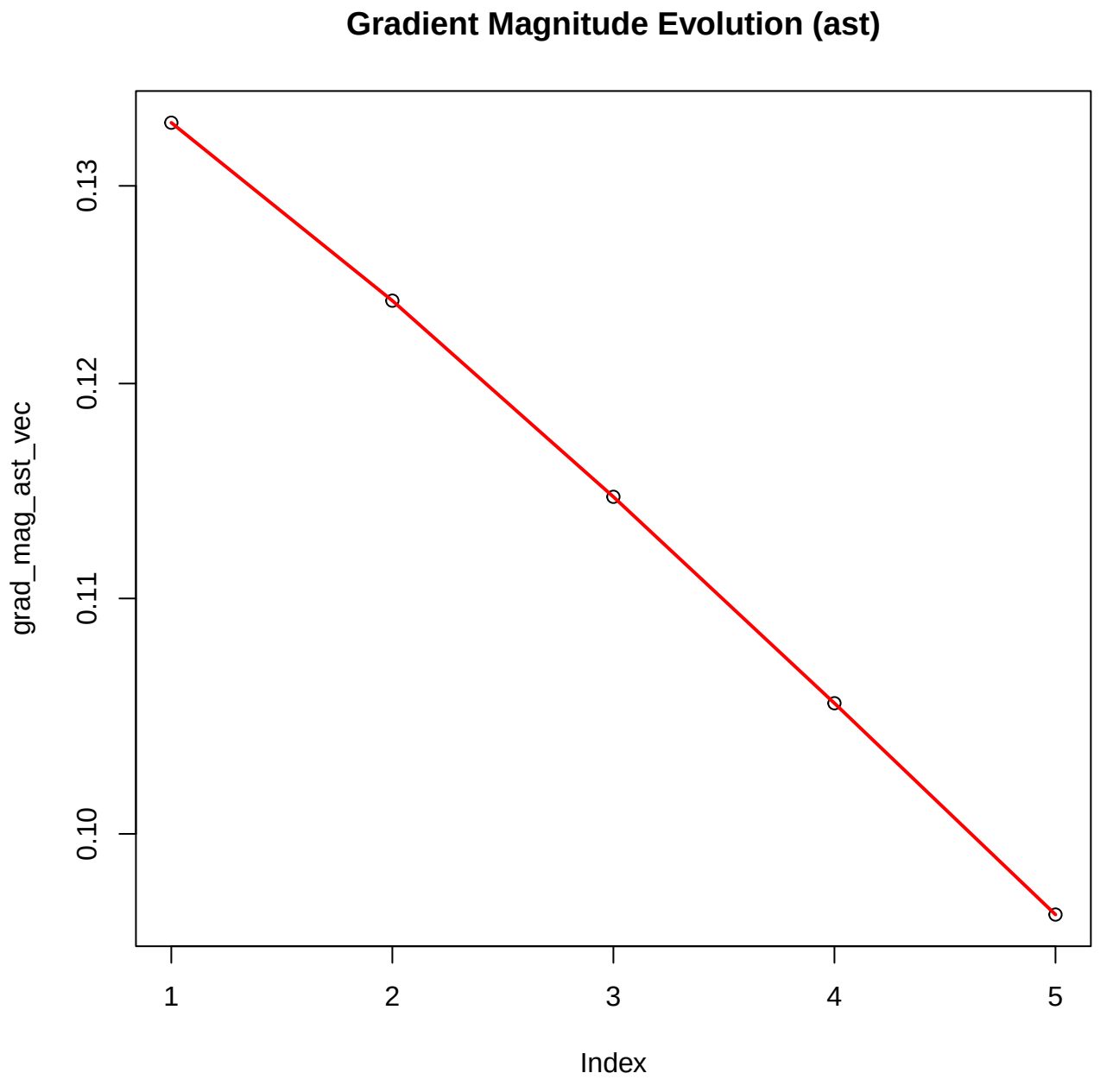
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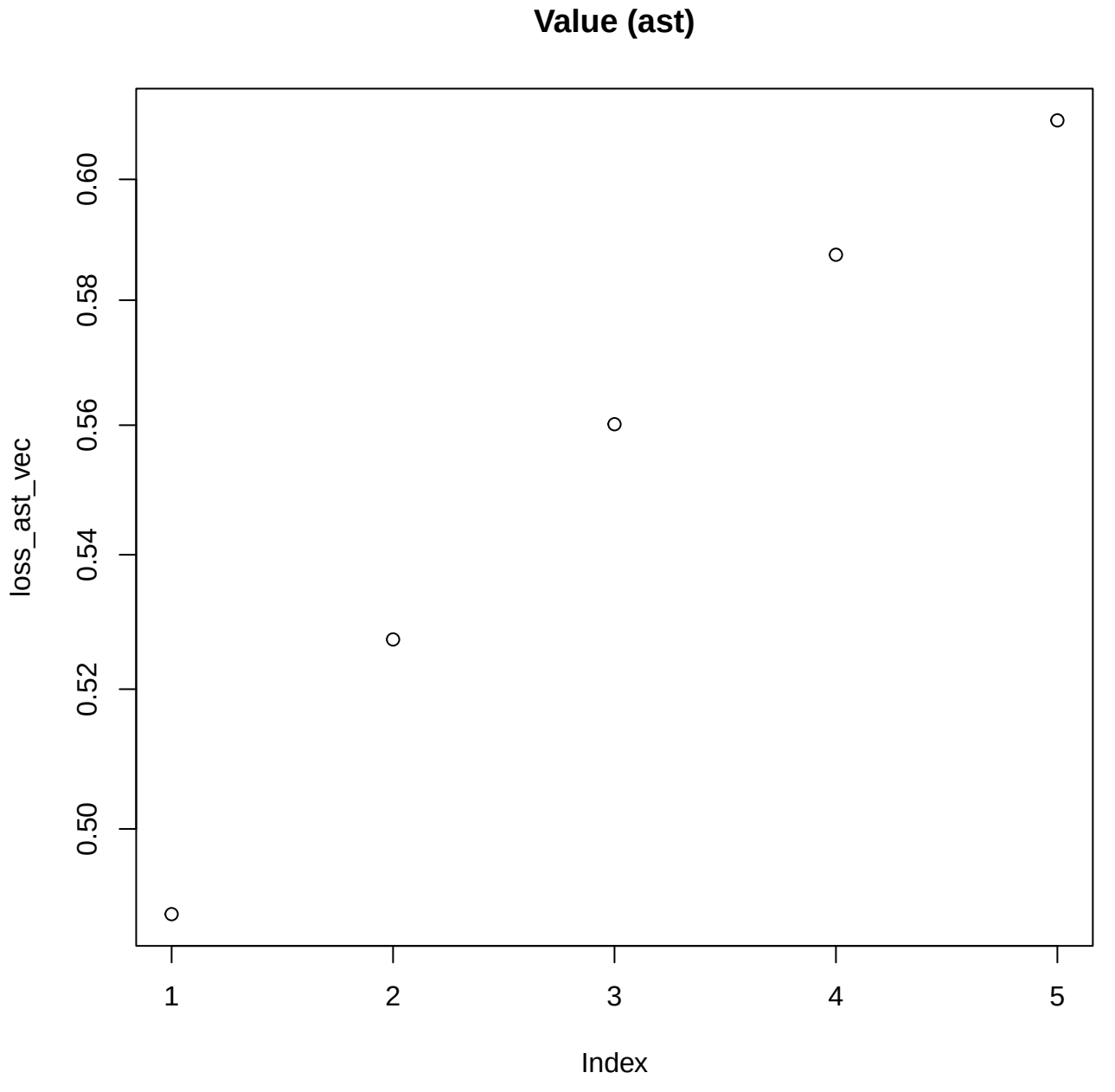
Value (ast)
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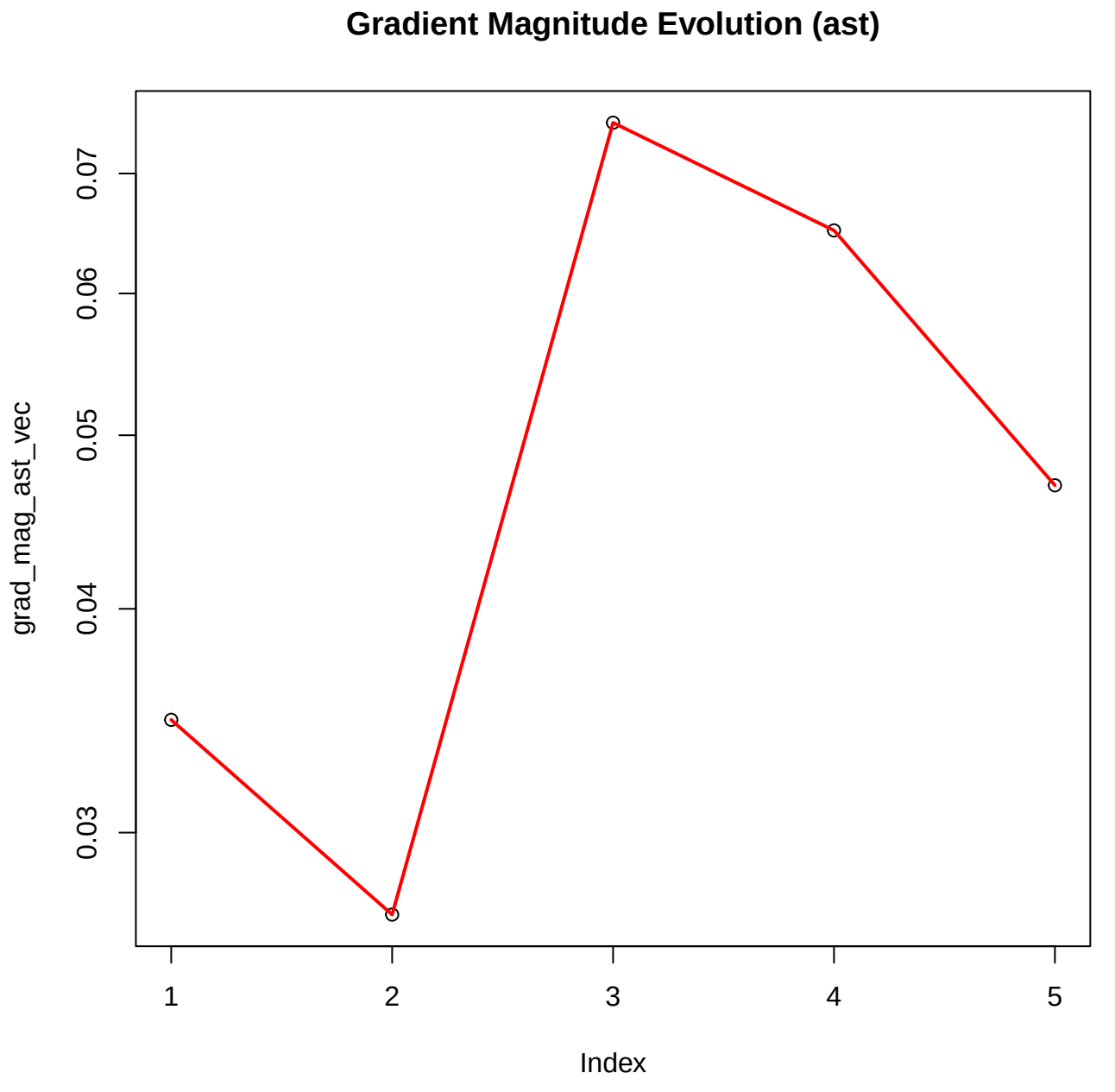
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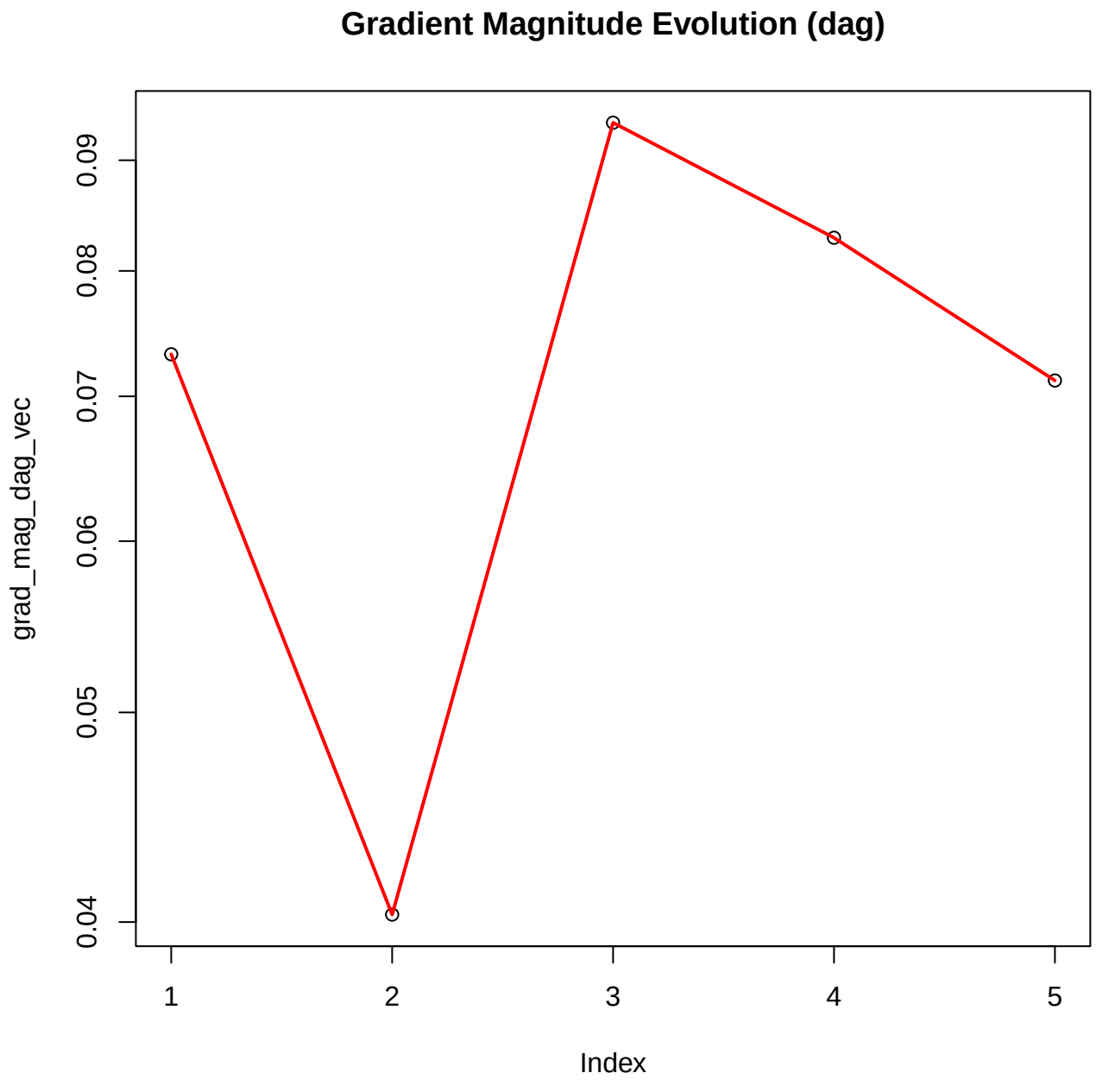
Gradient Magnitude Evolution (dag)
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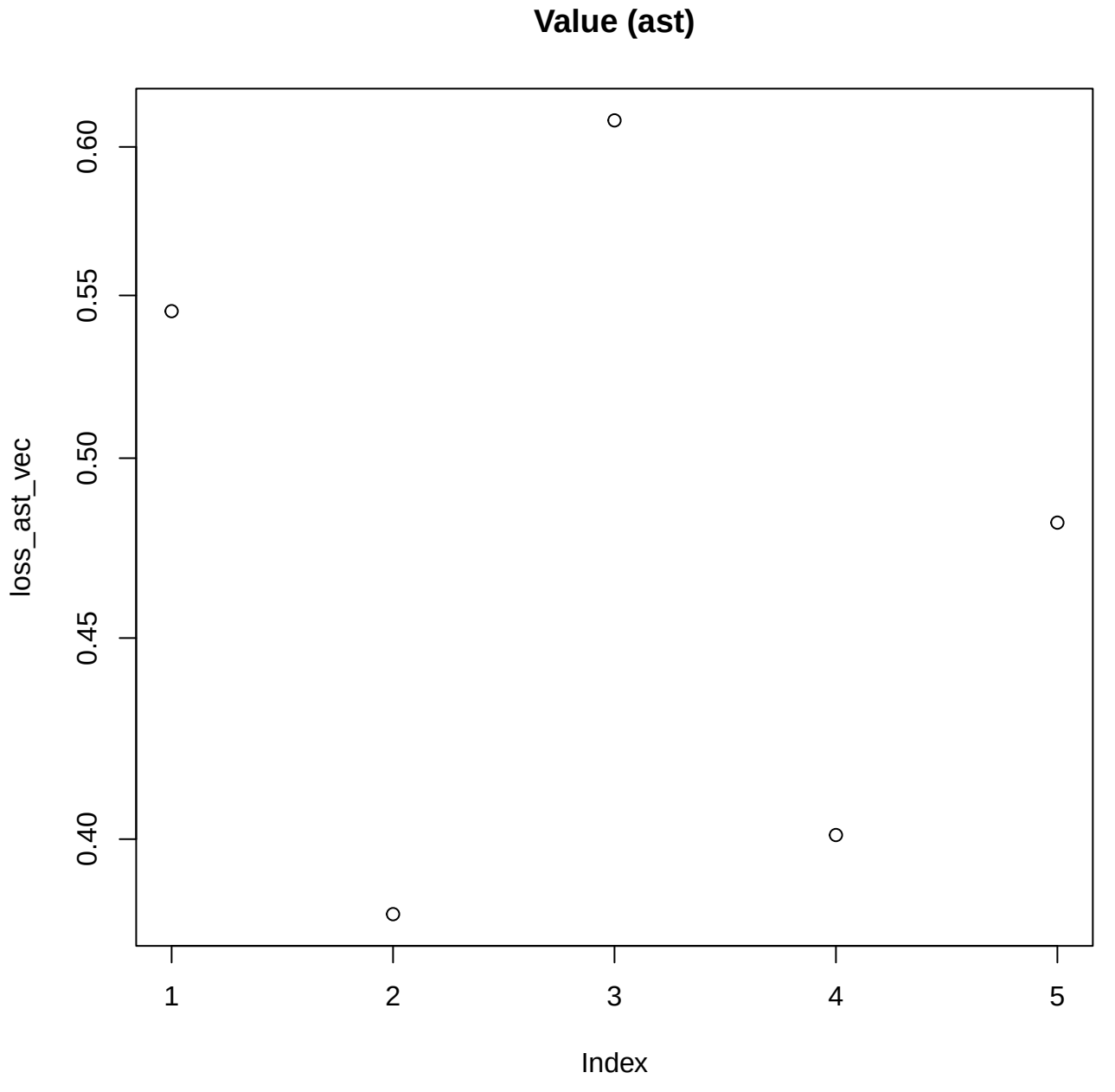
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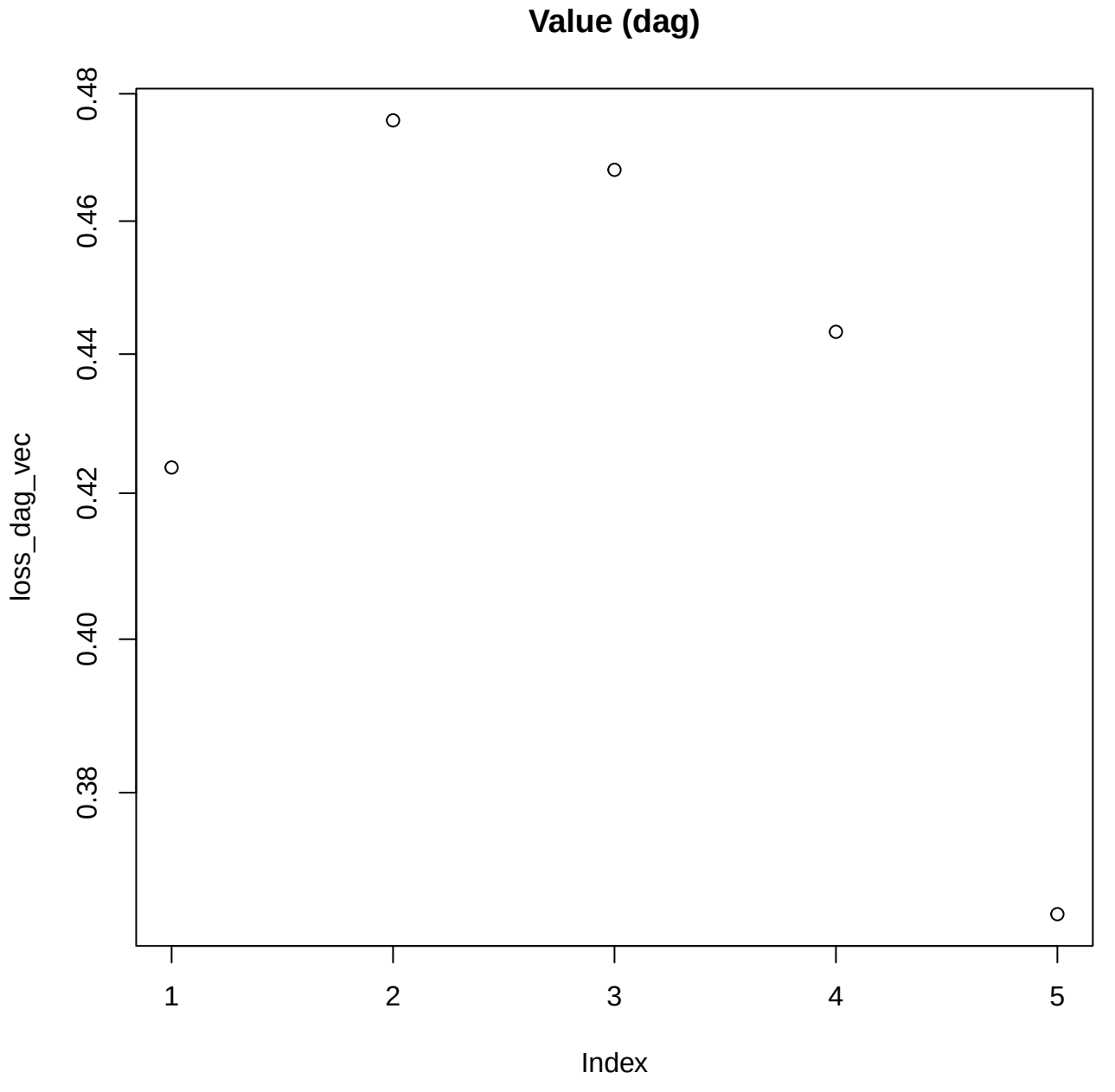
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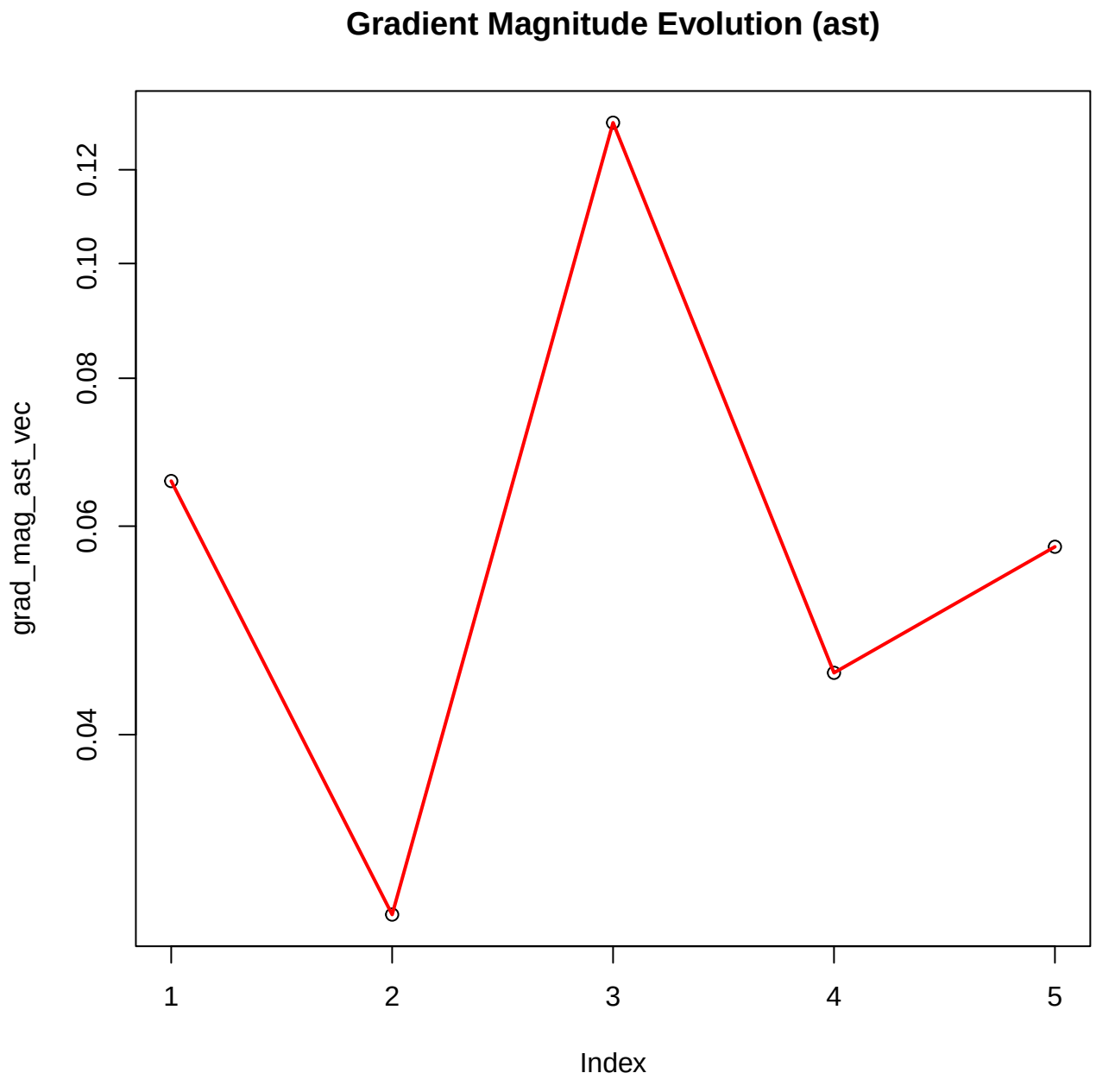
Gradient Magnitude Evolution (ast)
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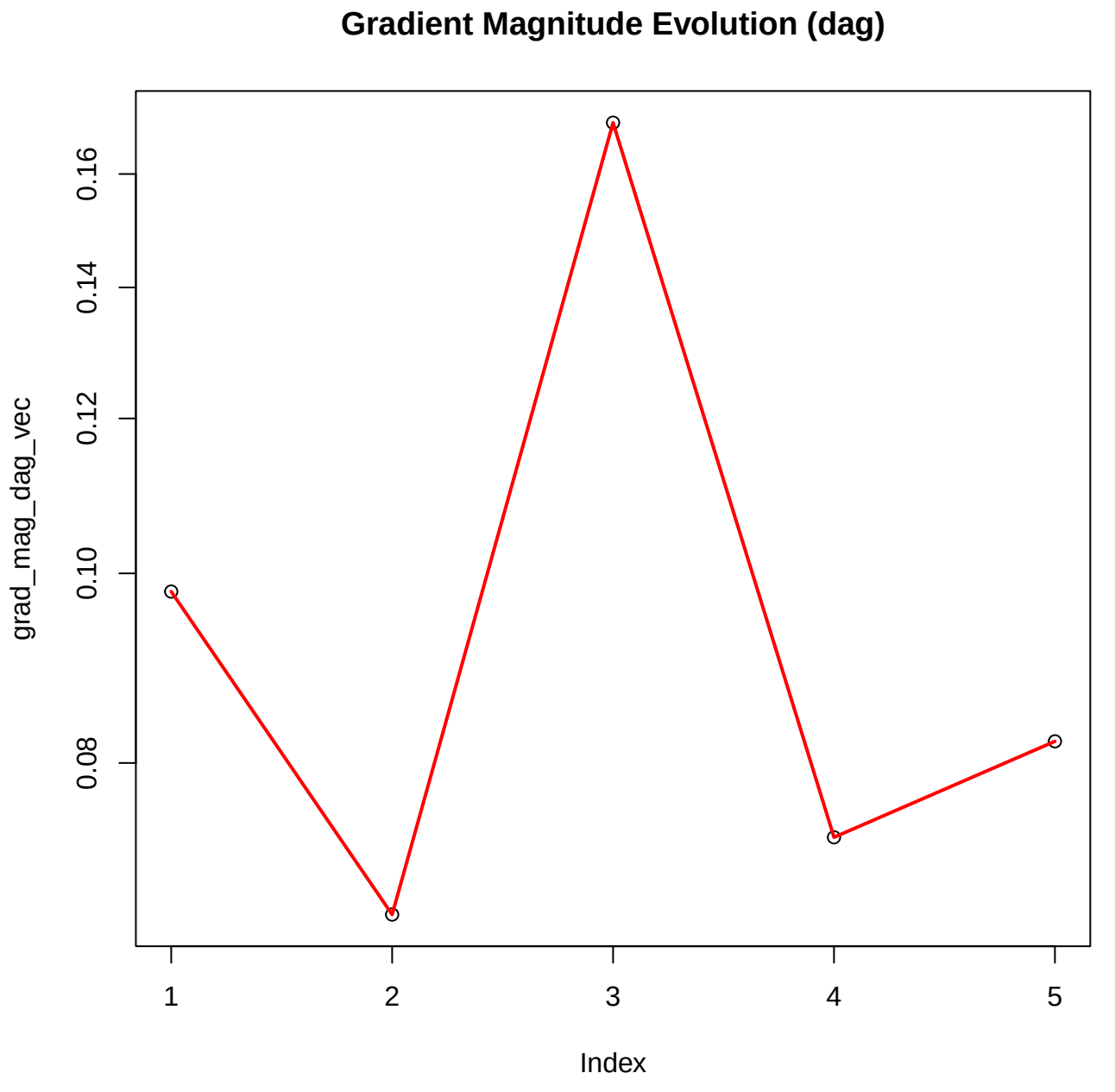
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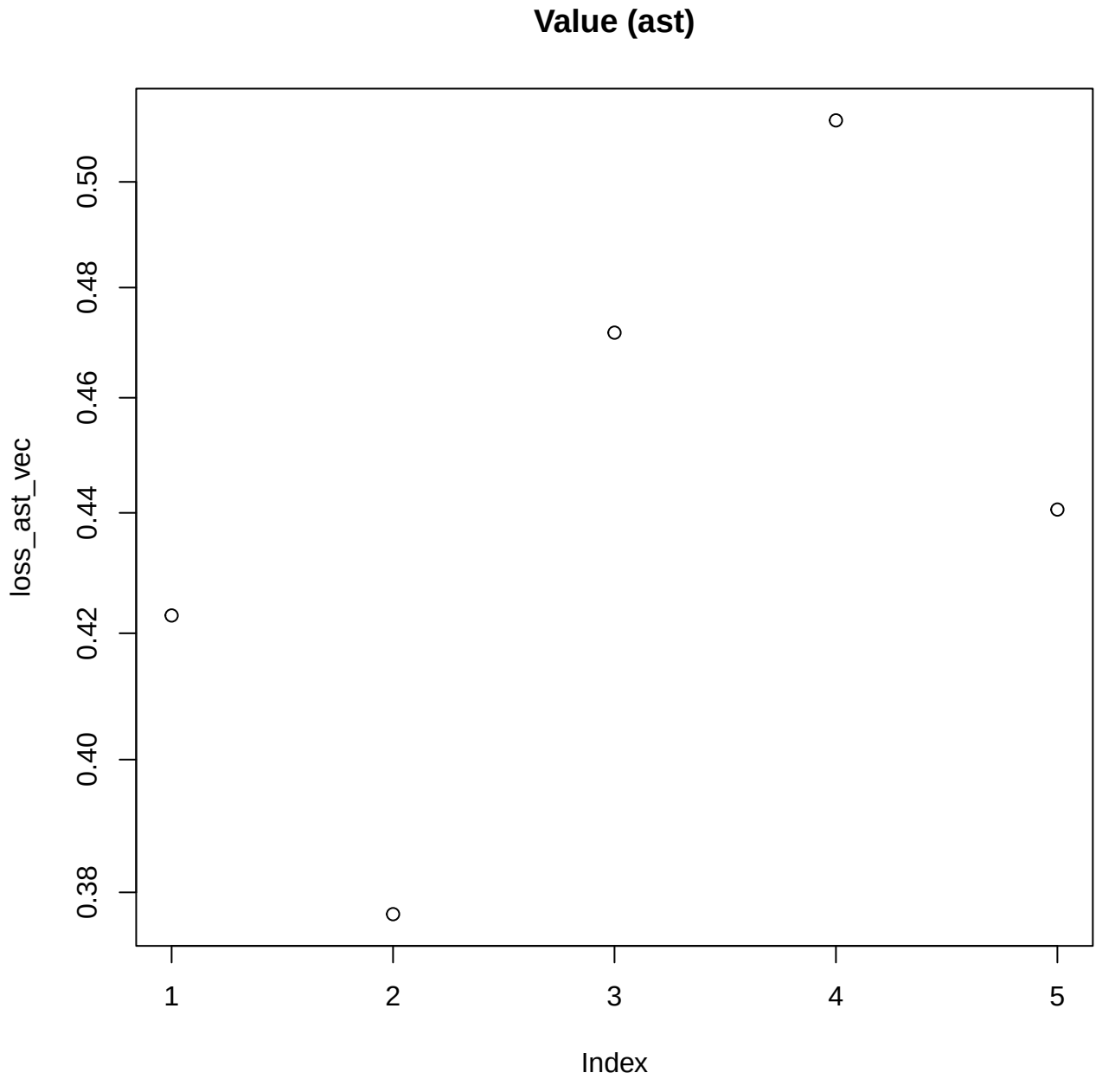
Index


Value (ast)
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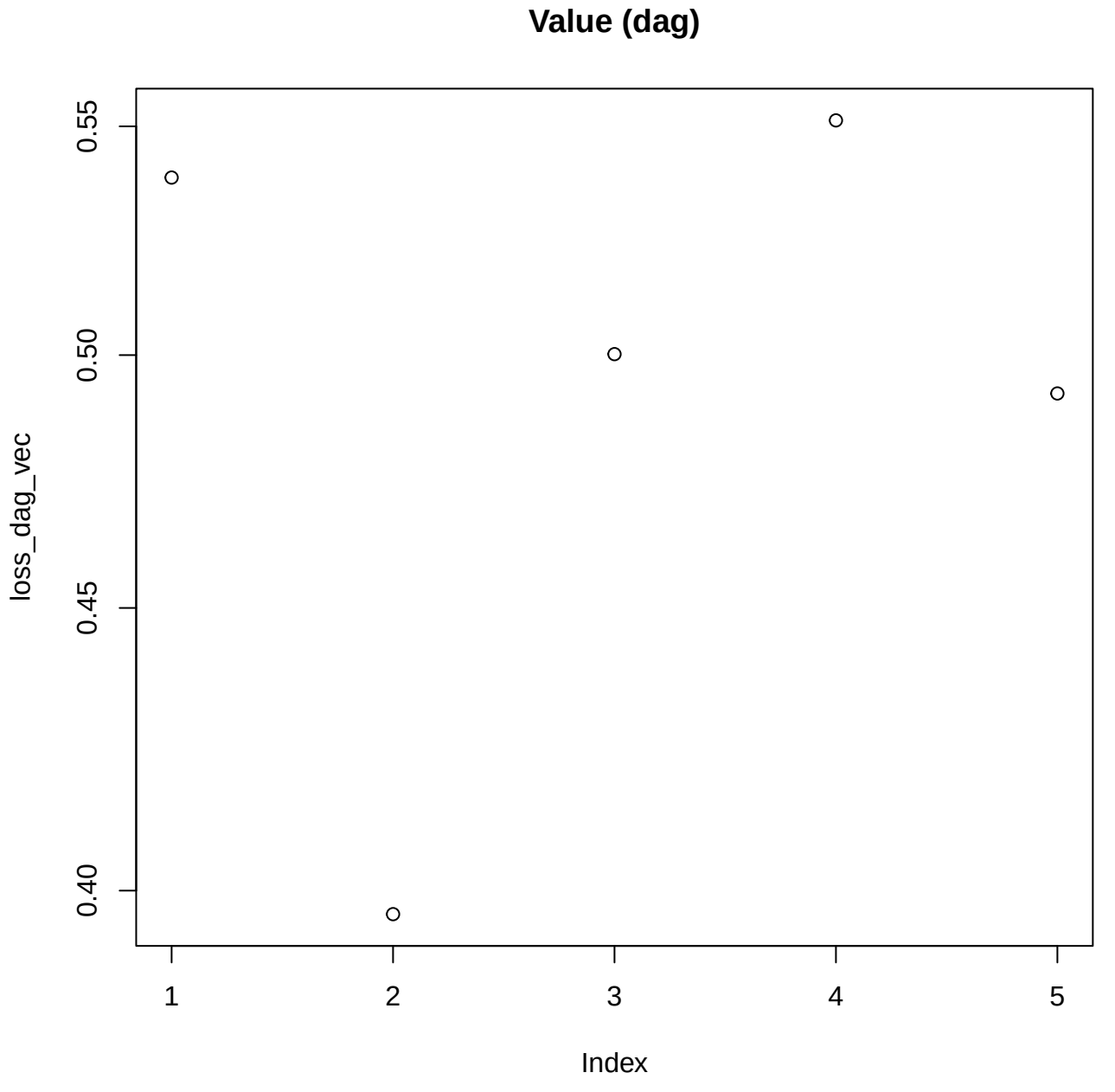
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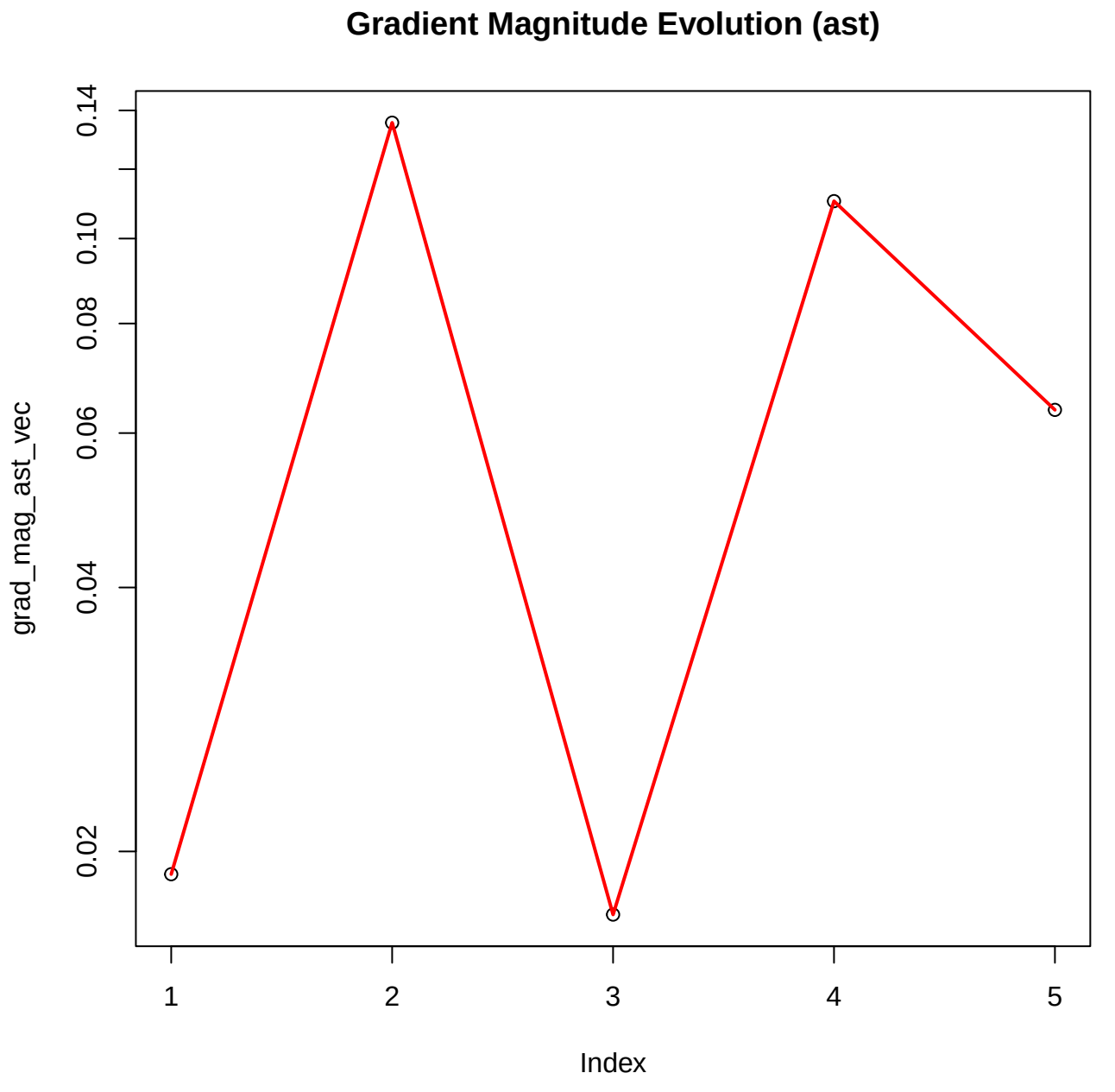
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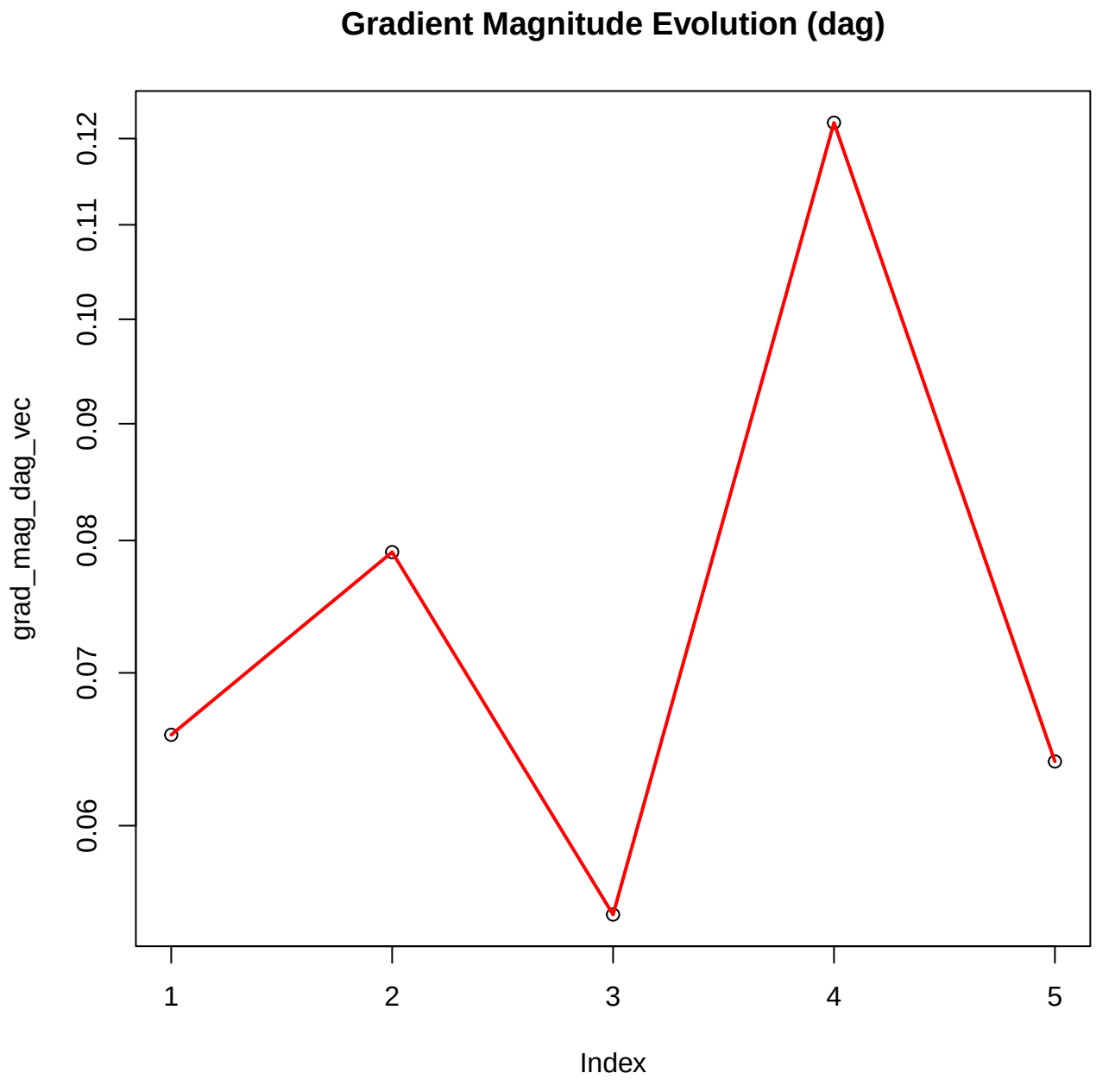
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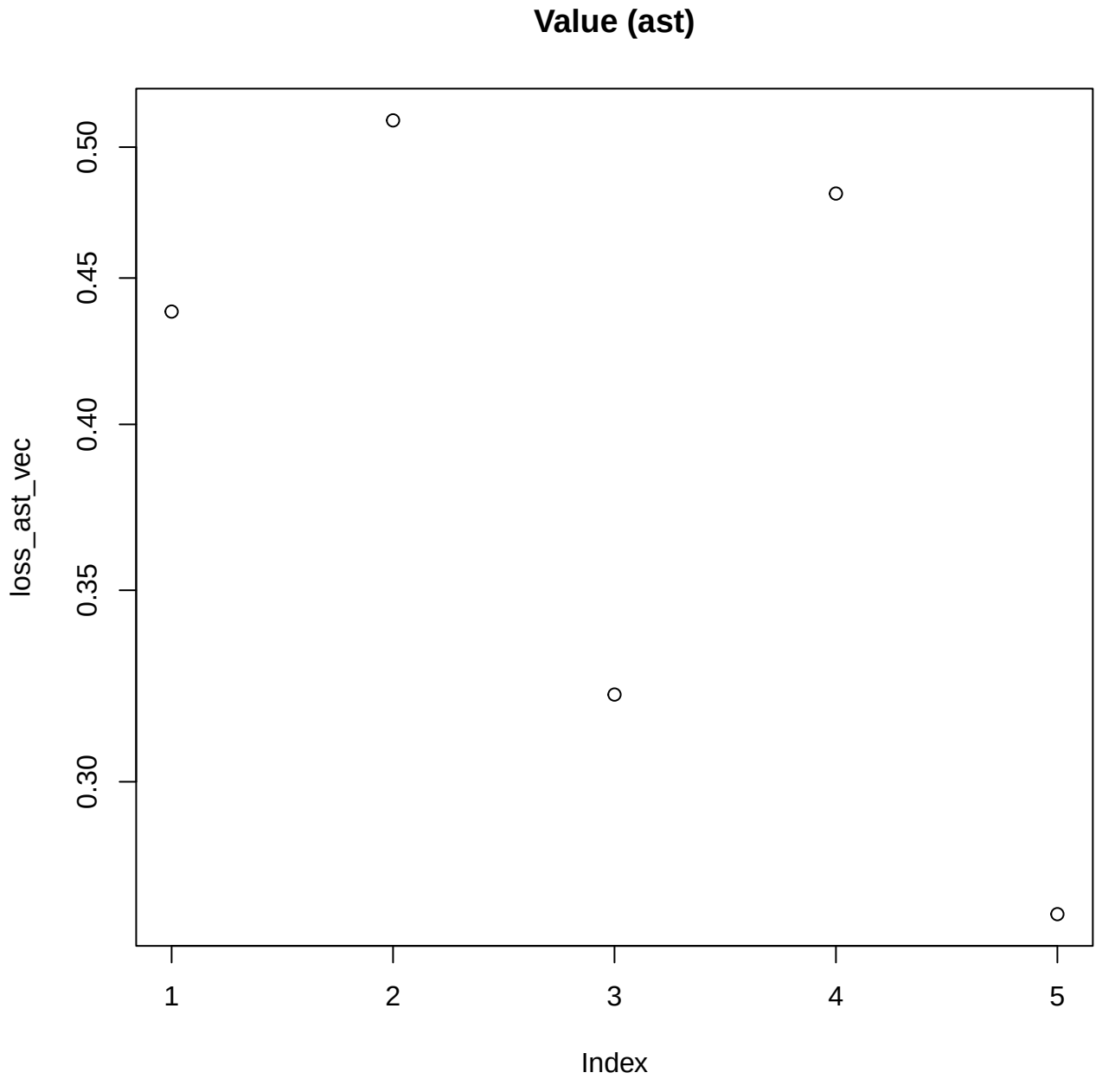
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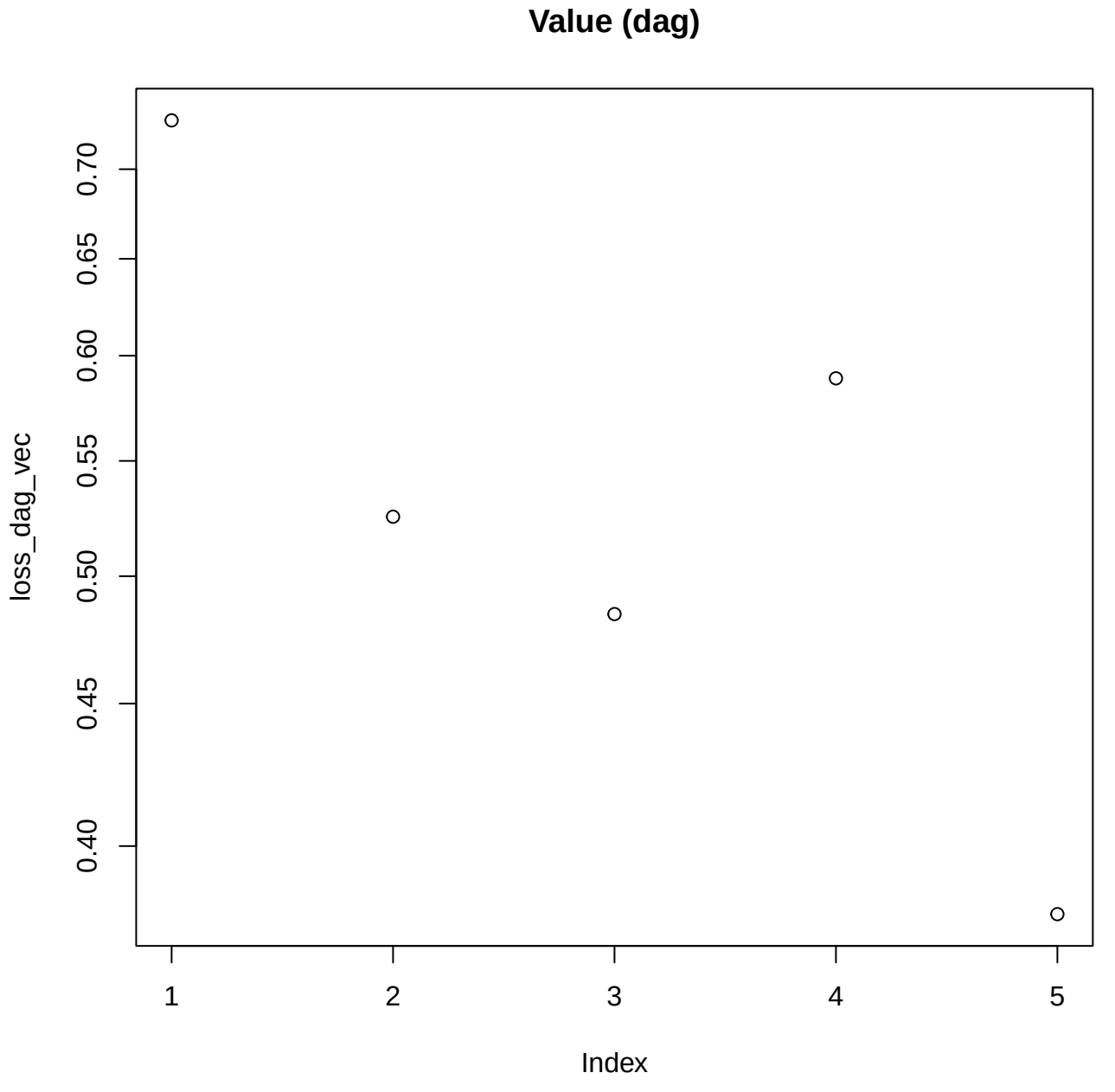
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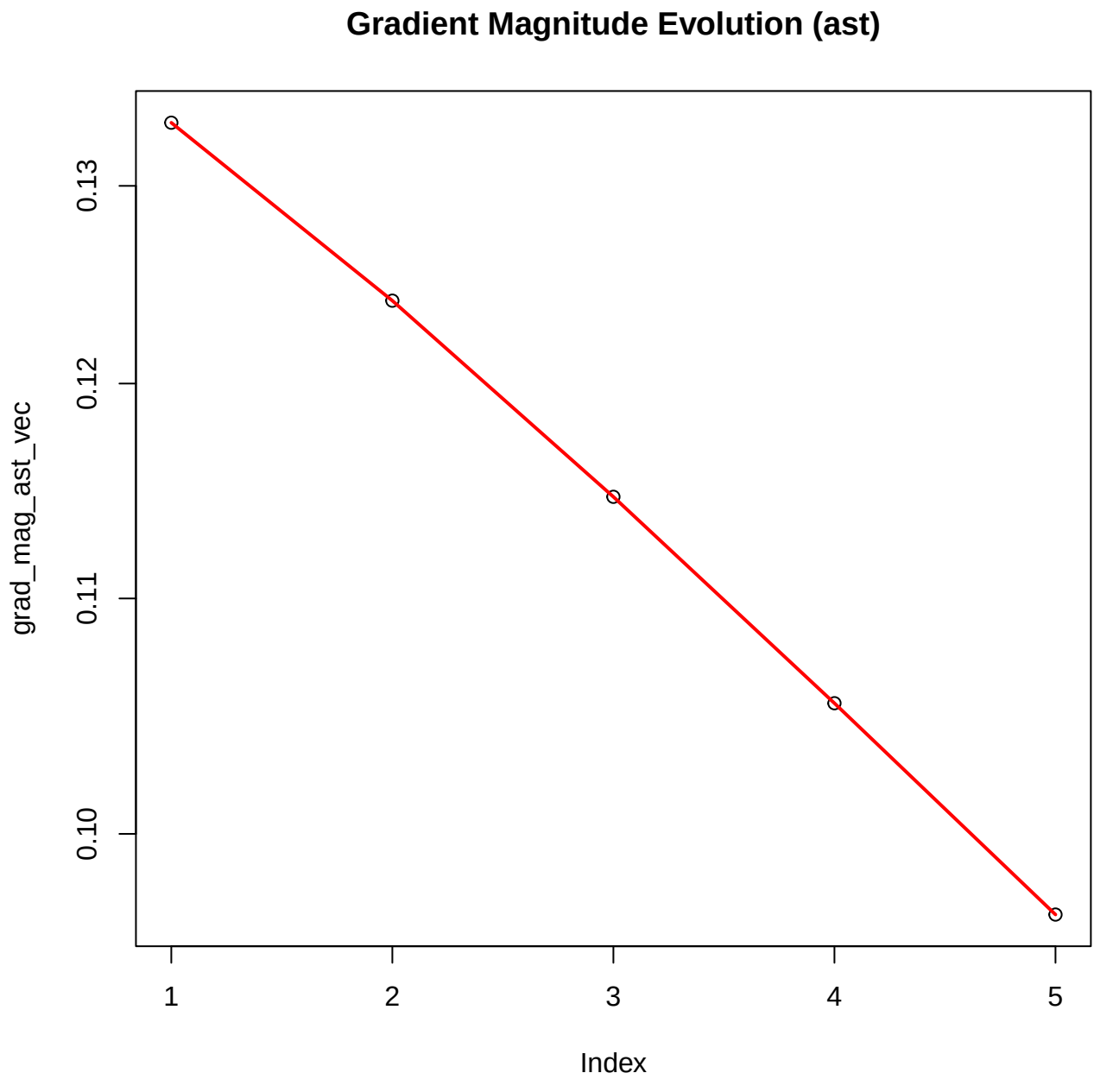
Gradient Magnitude Evolution (ast)
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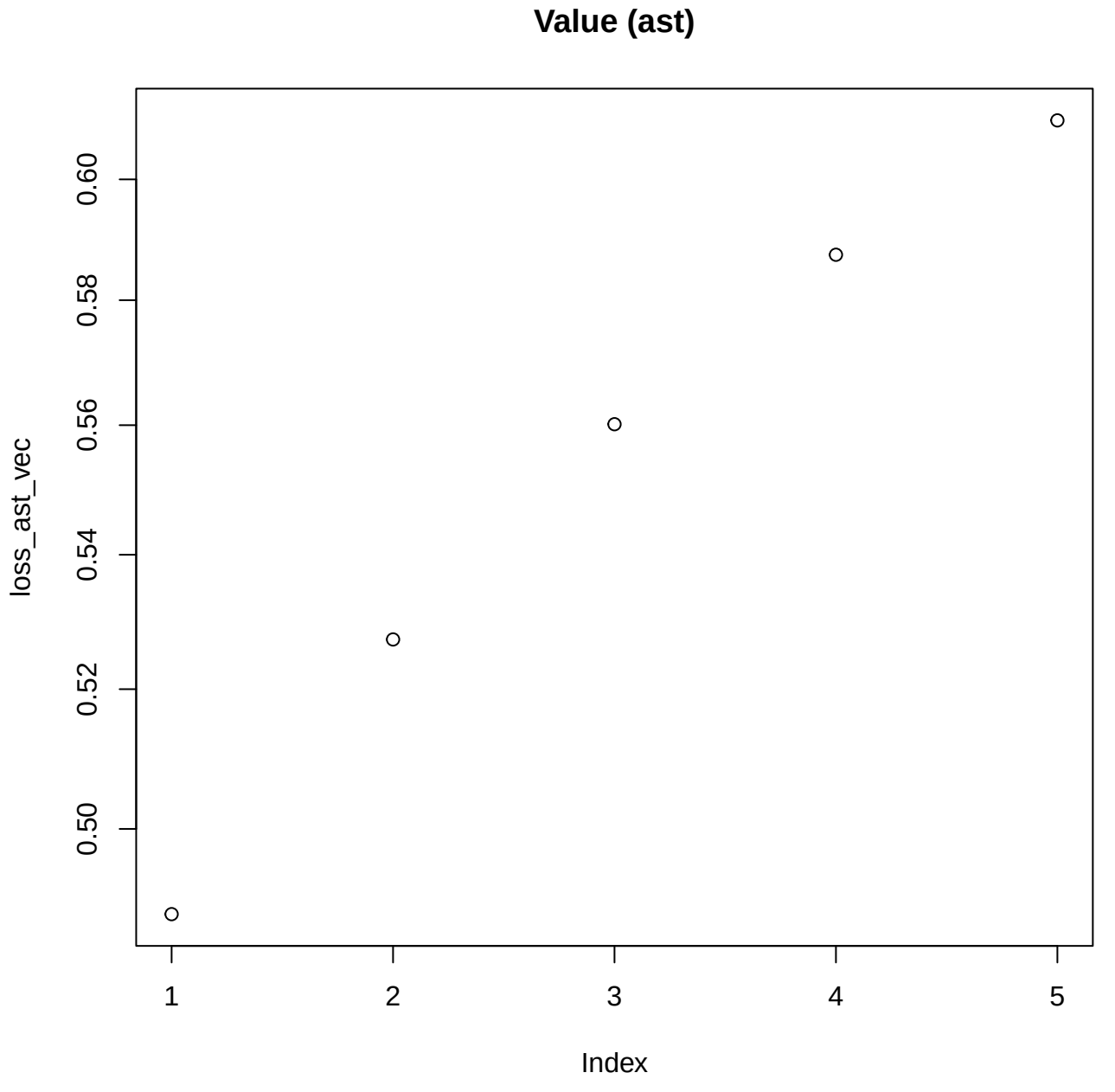
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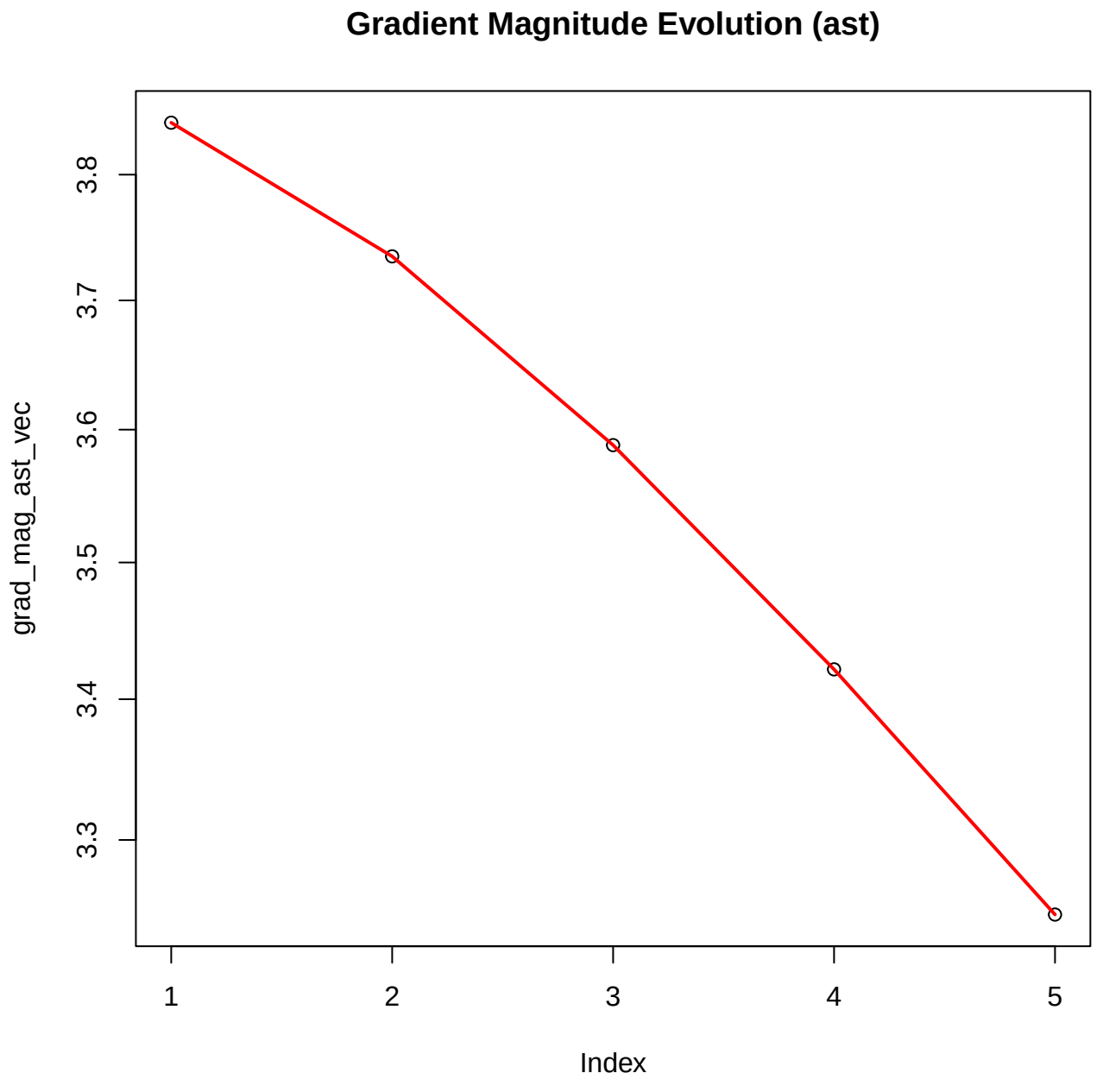
Gradient Magnitude Evolution (ast)
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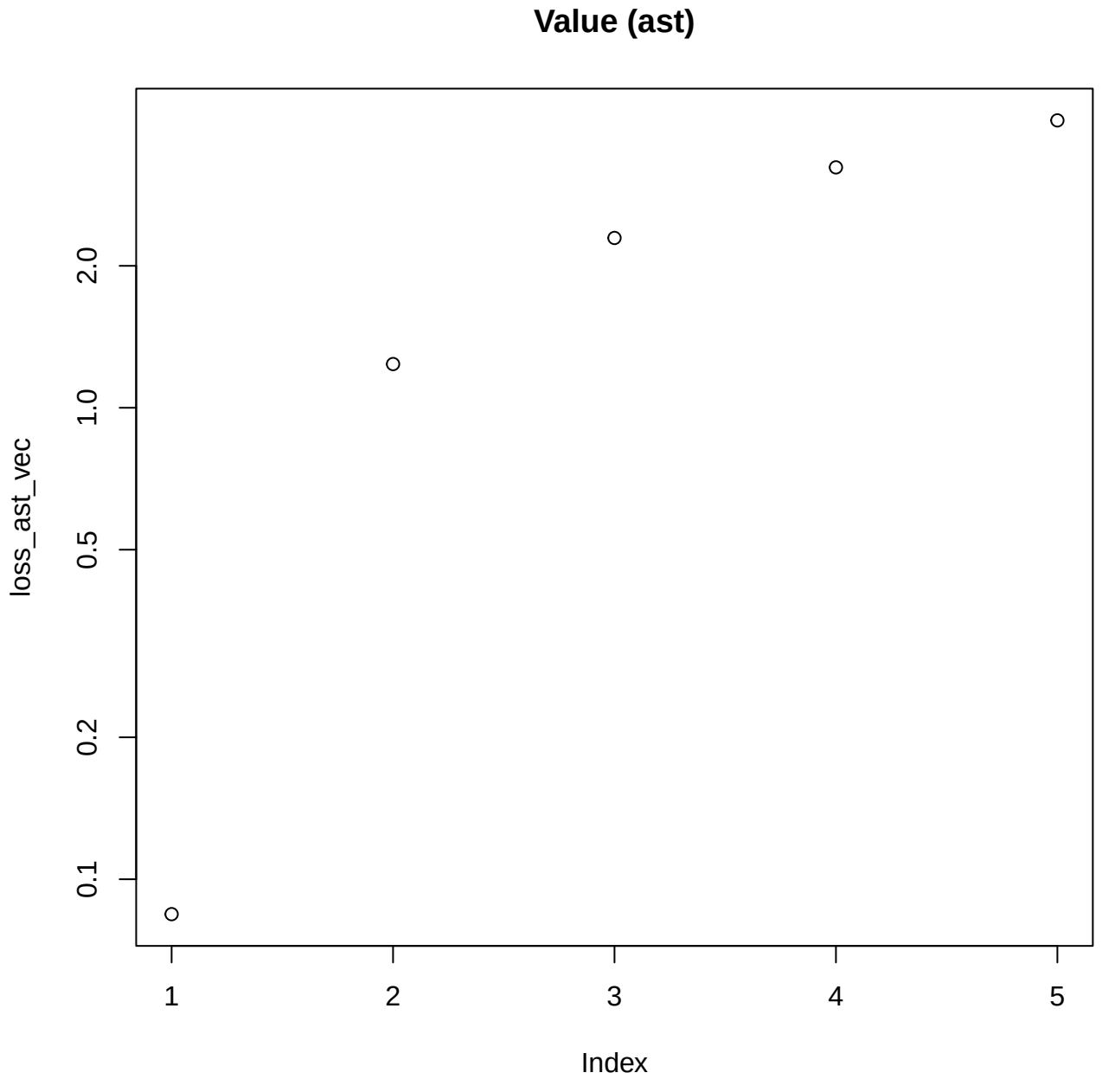
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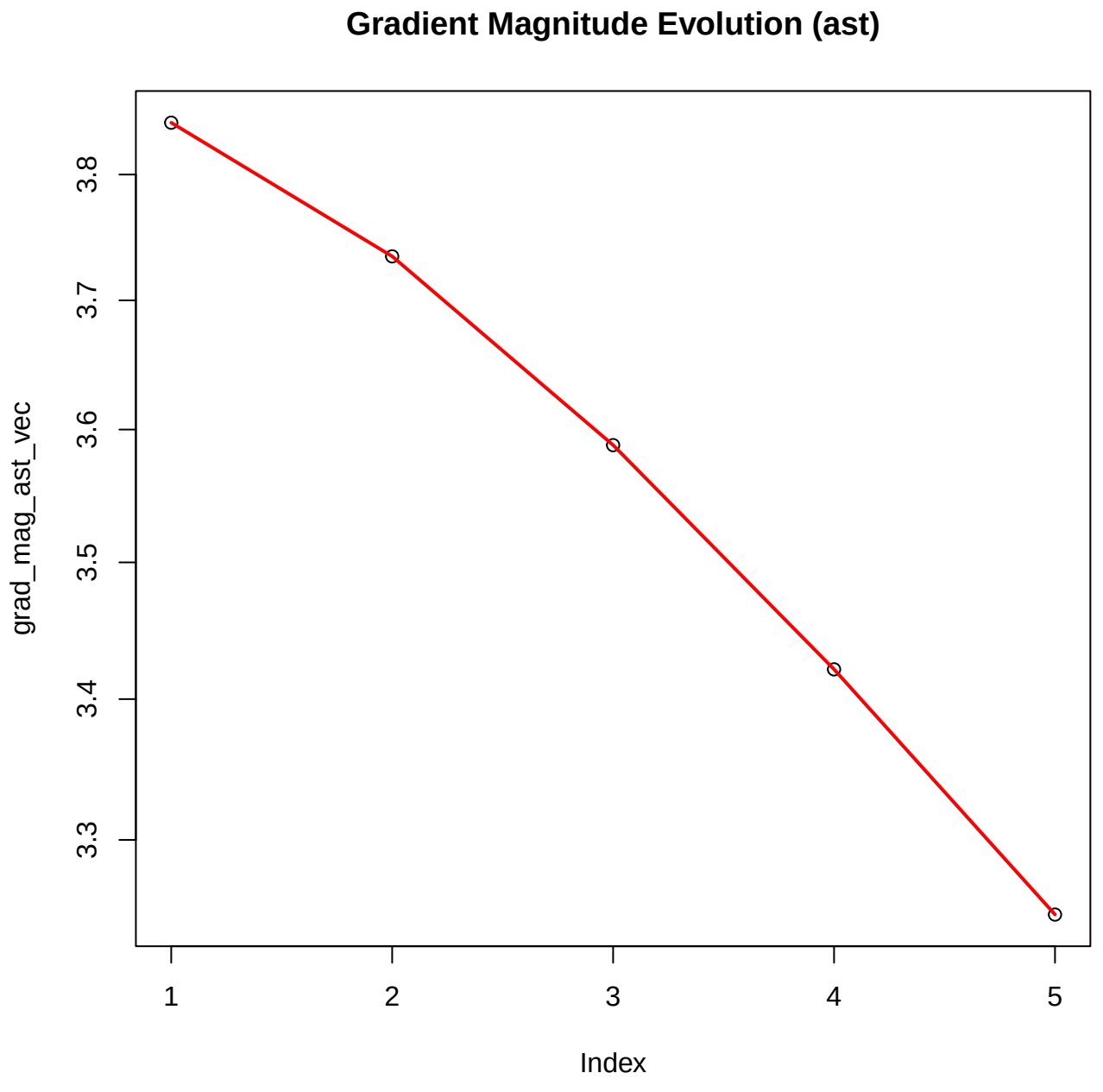
Gradient Magnitude Evolution (ast)
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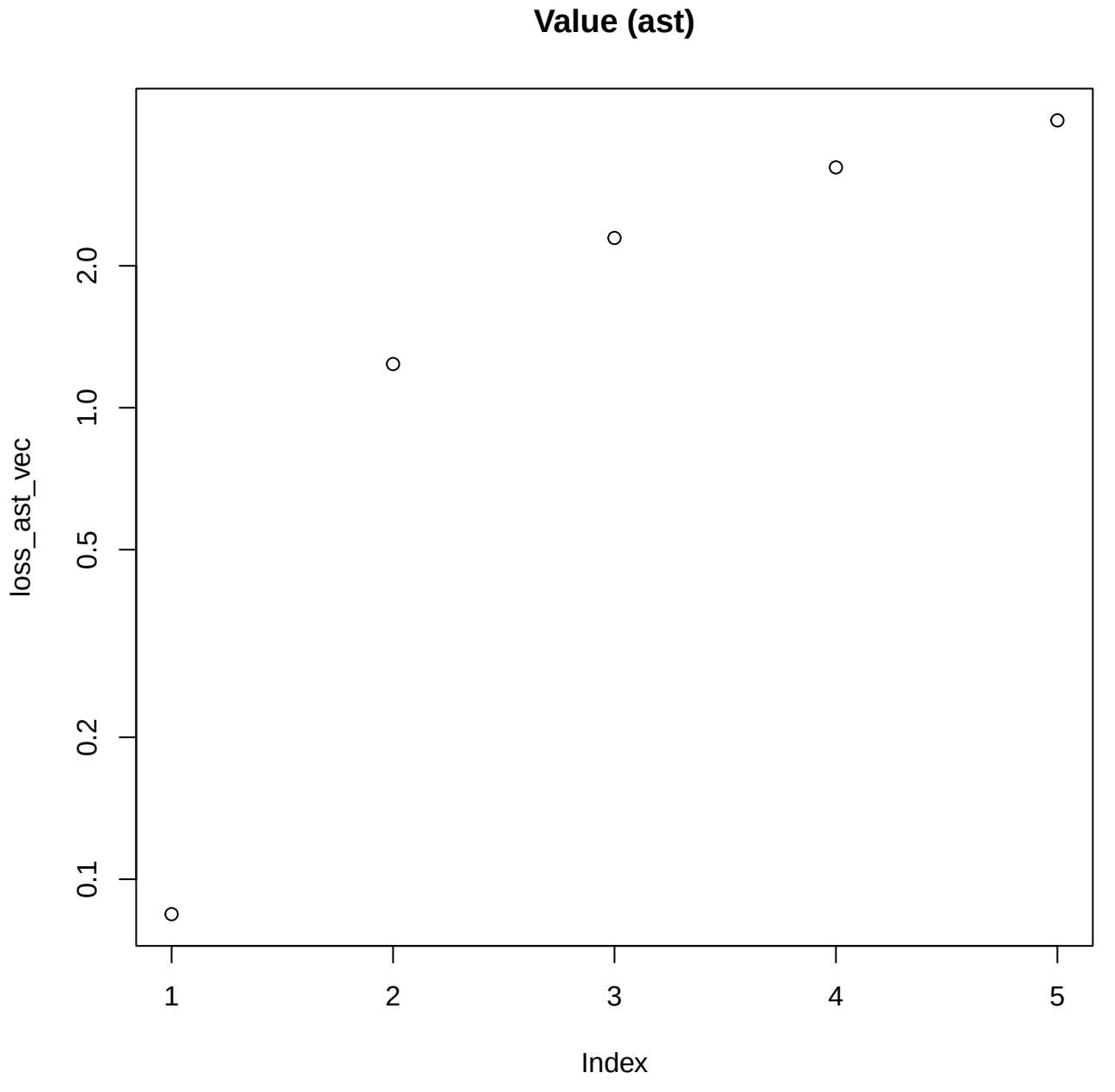
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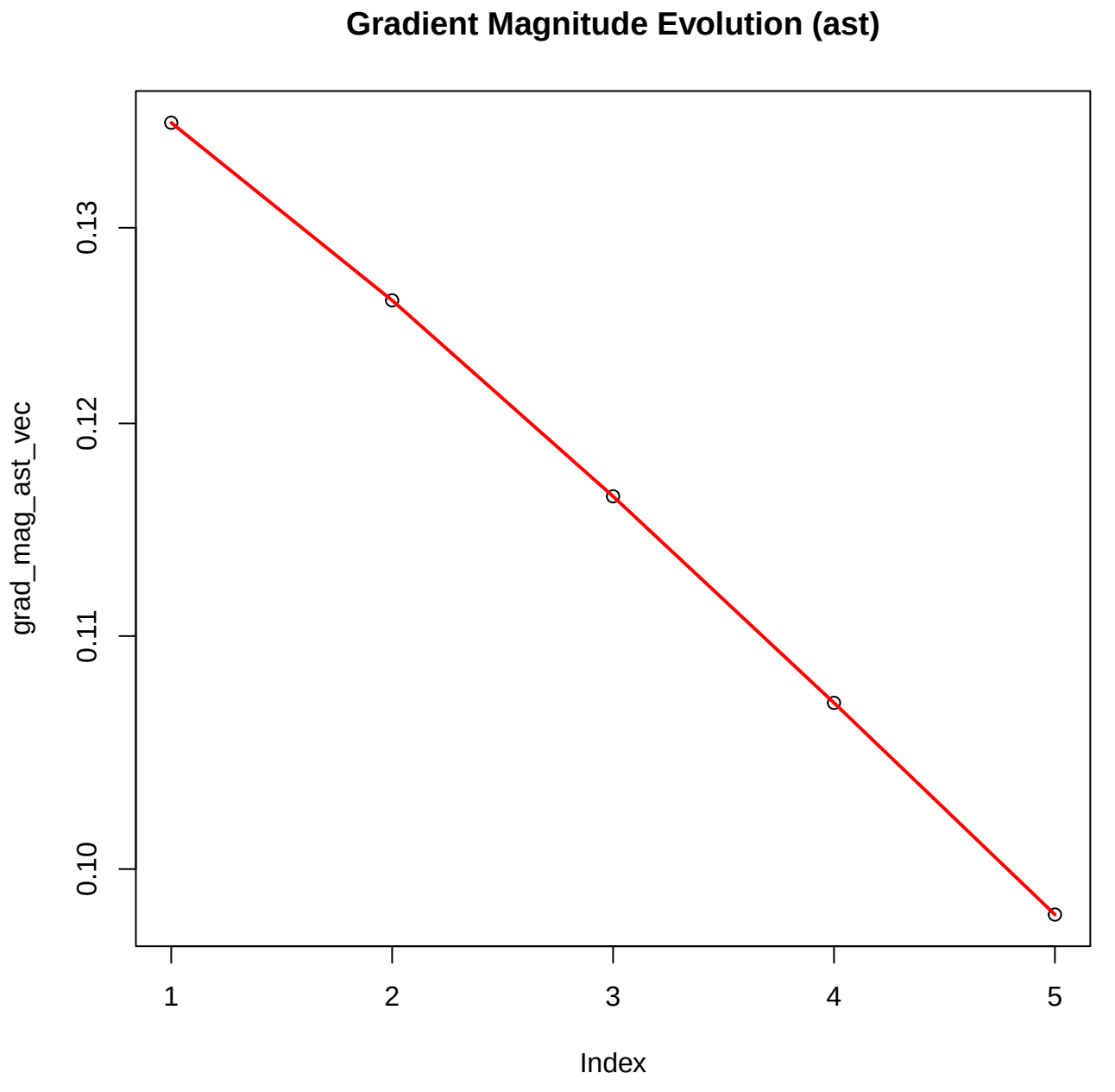
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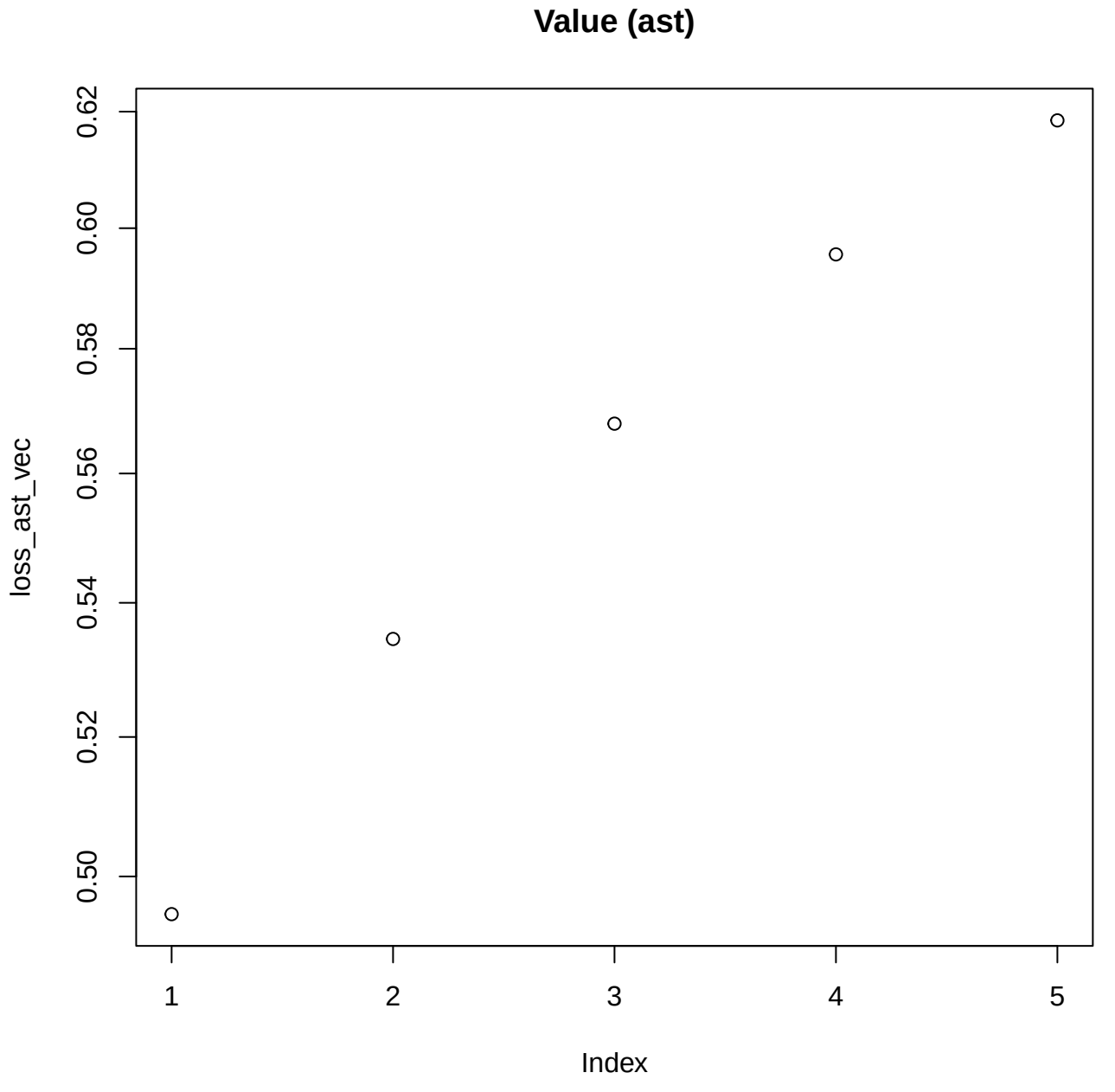
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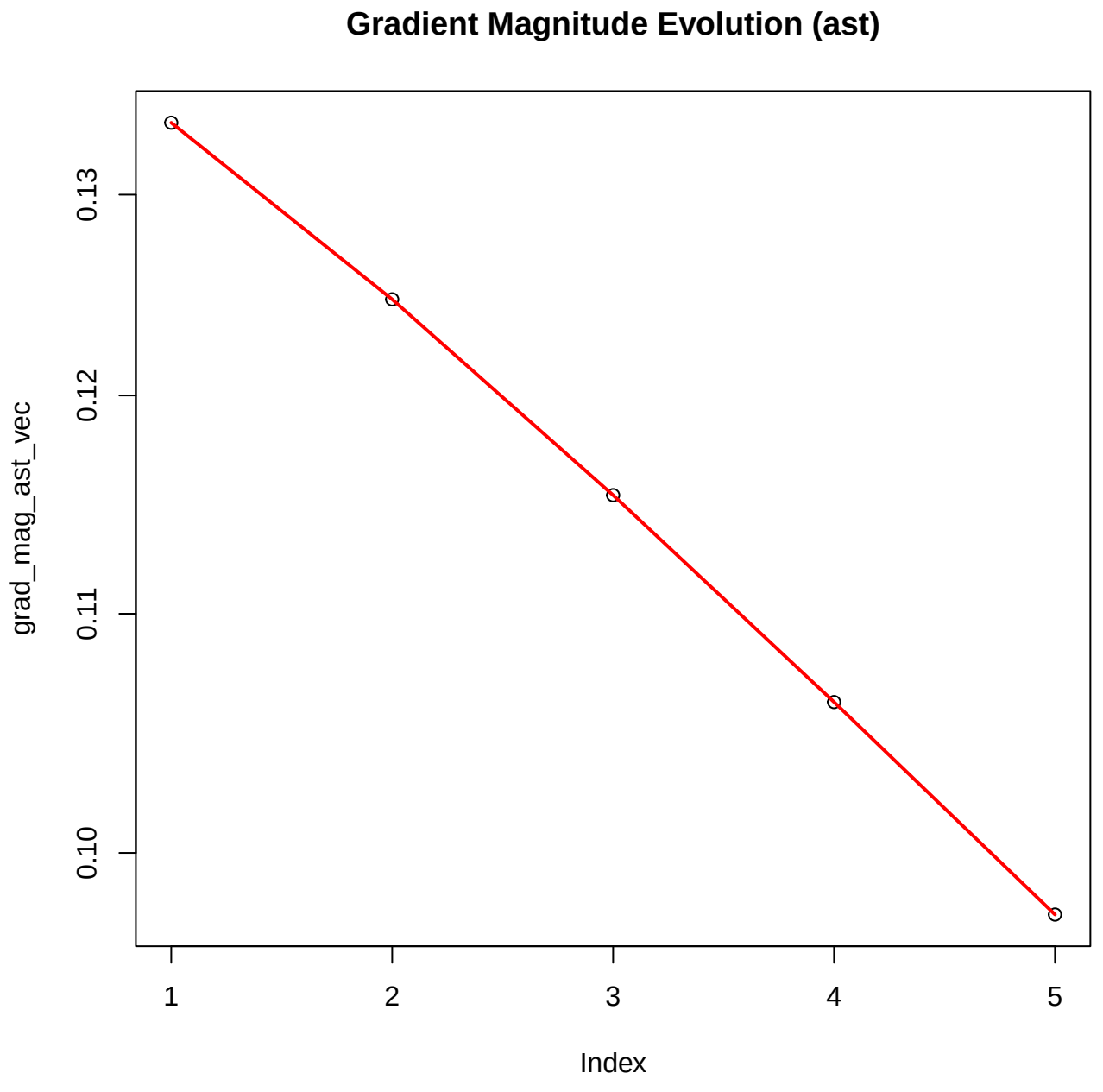
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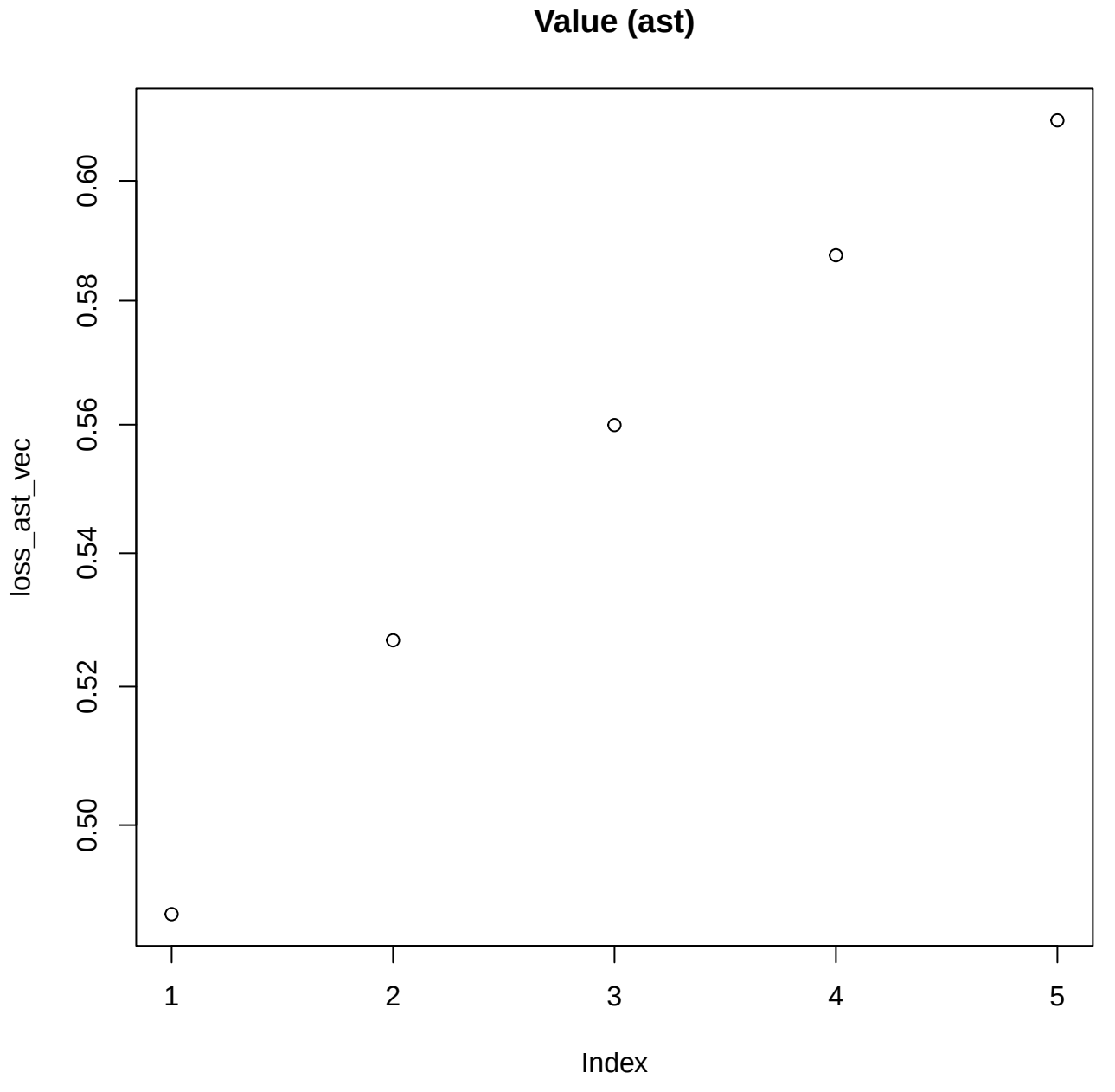
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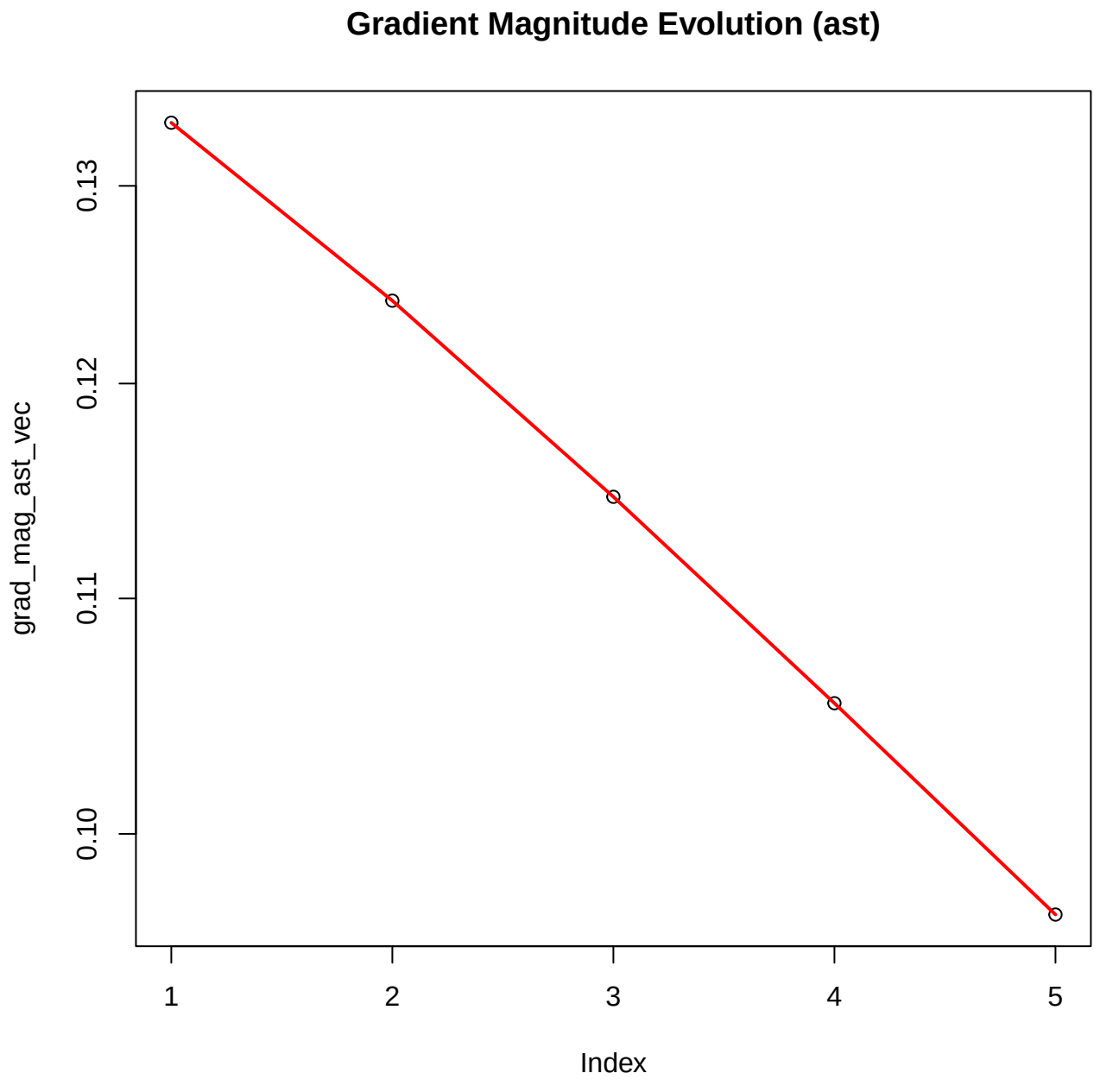
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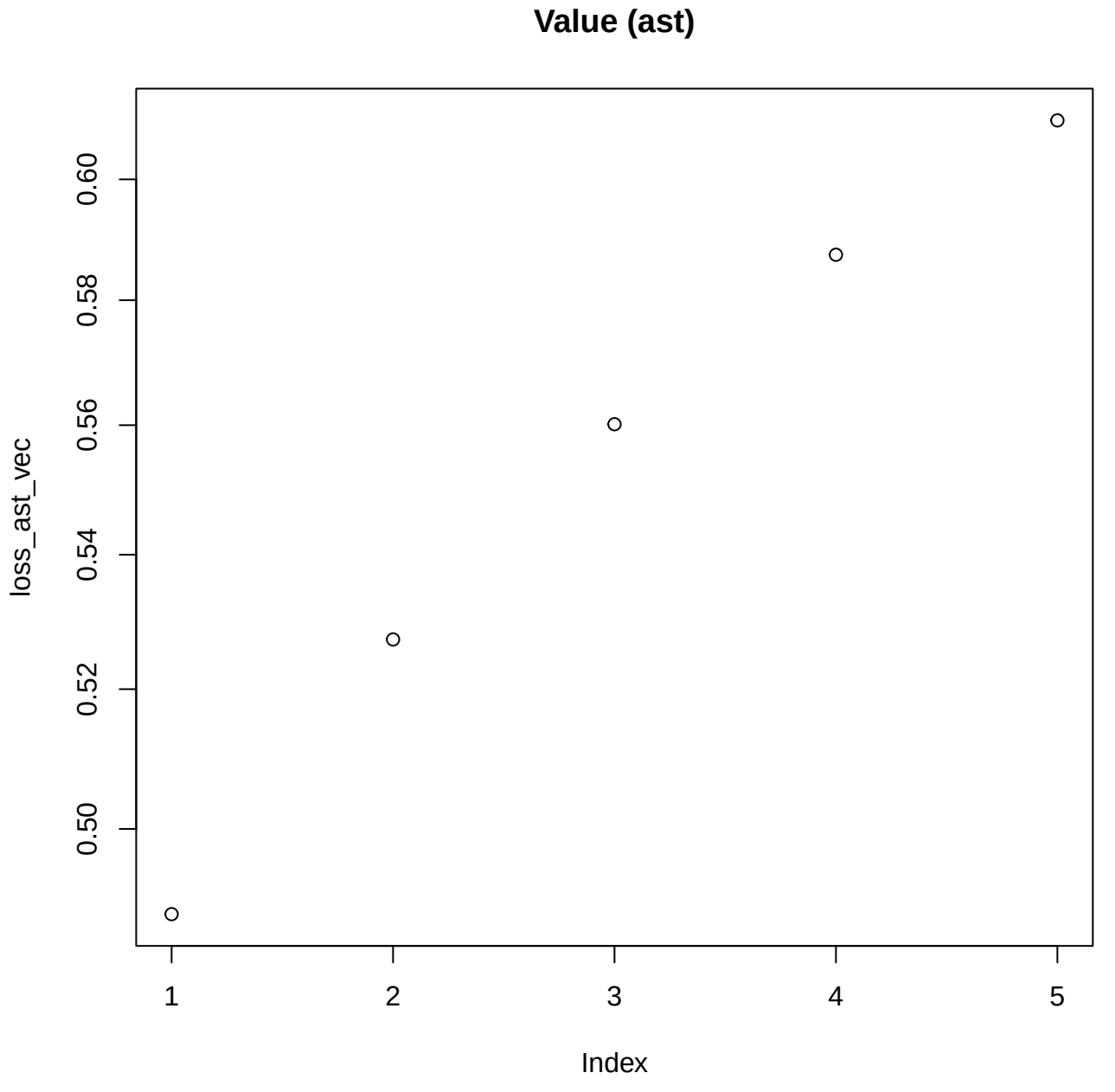
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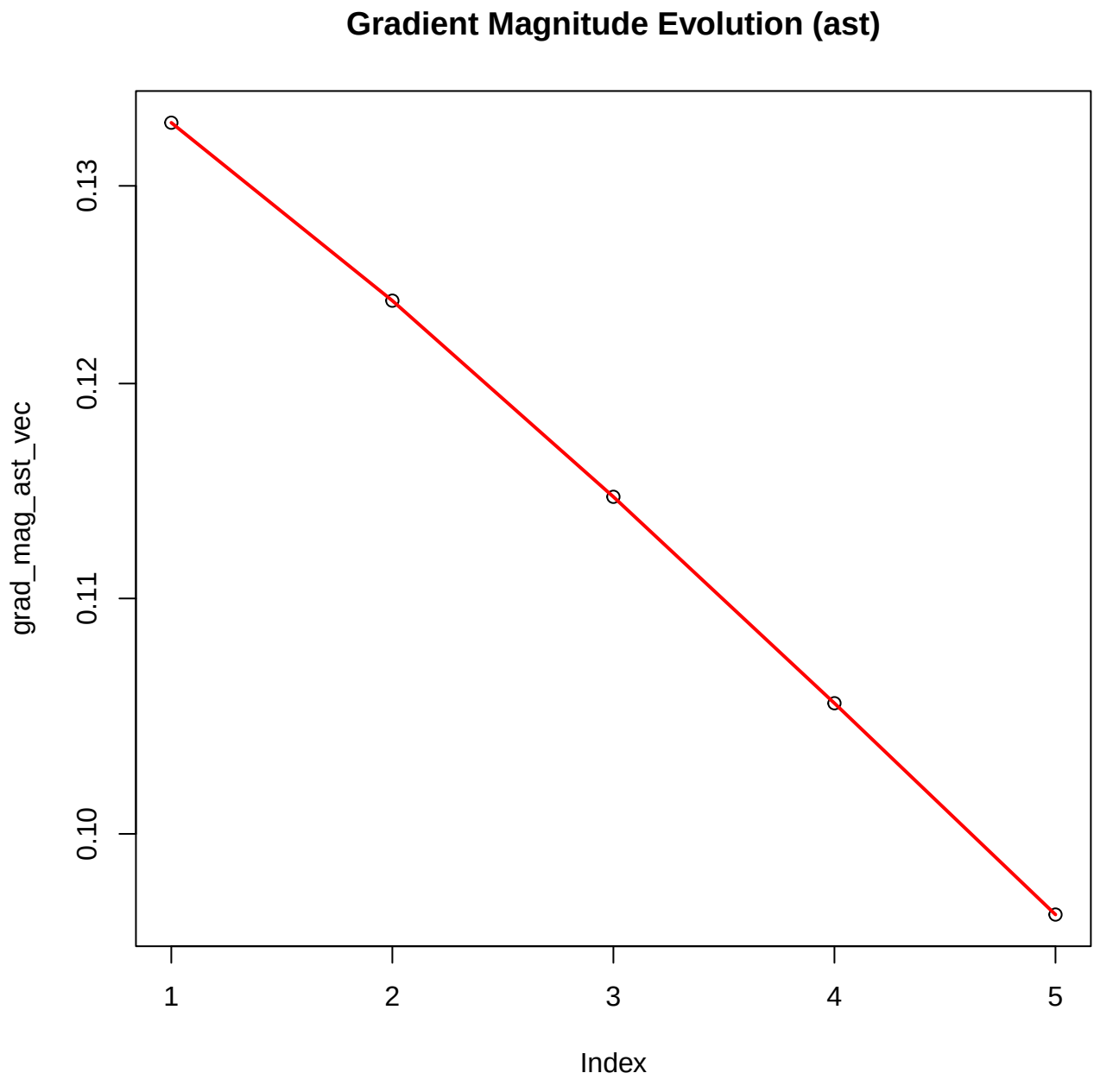
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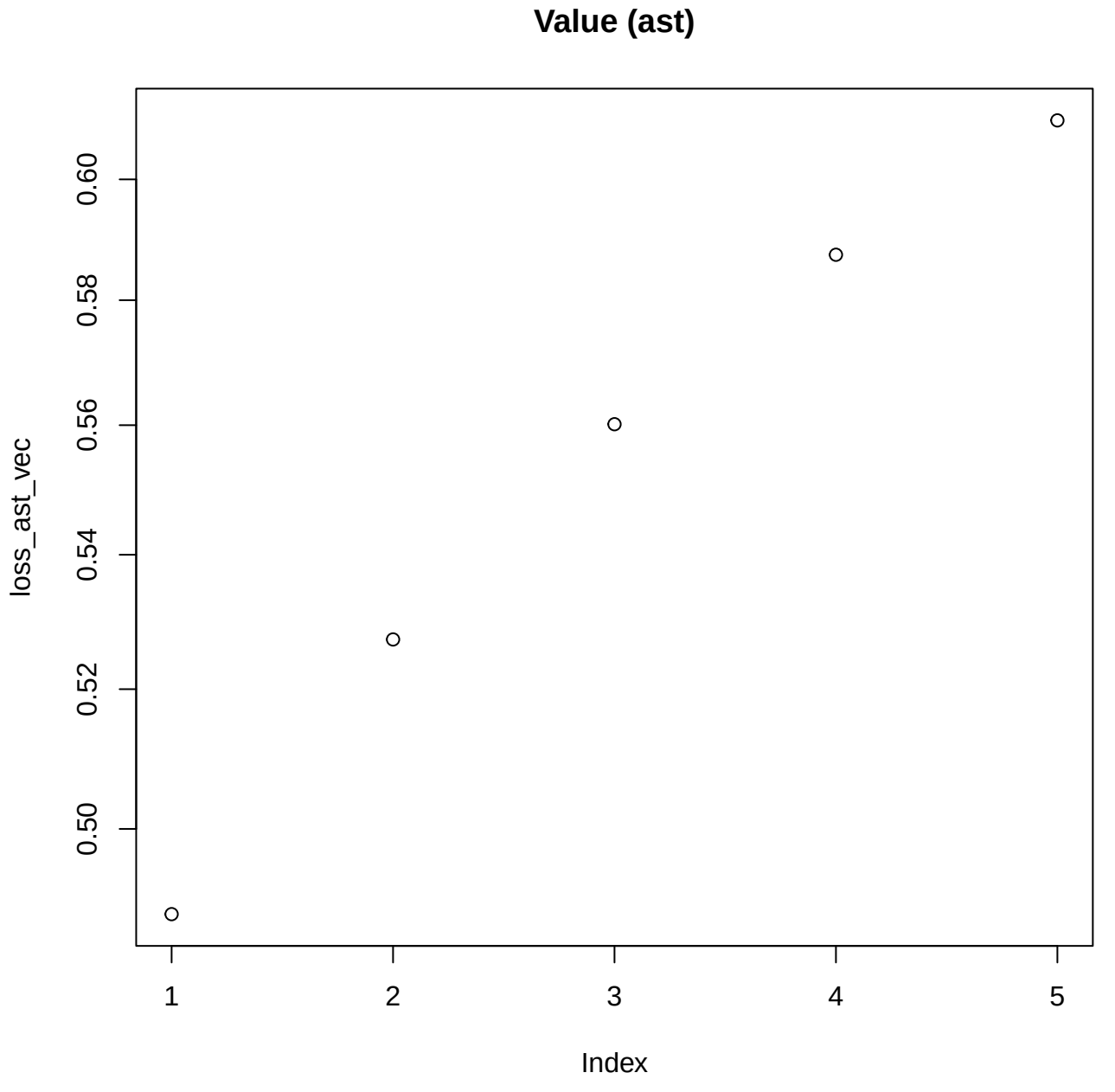
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5
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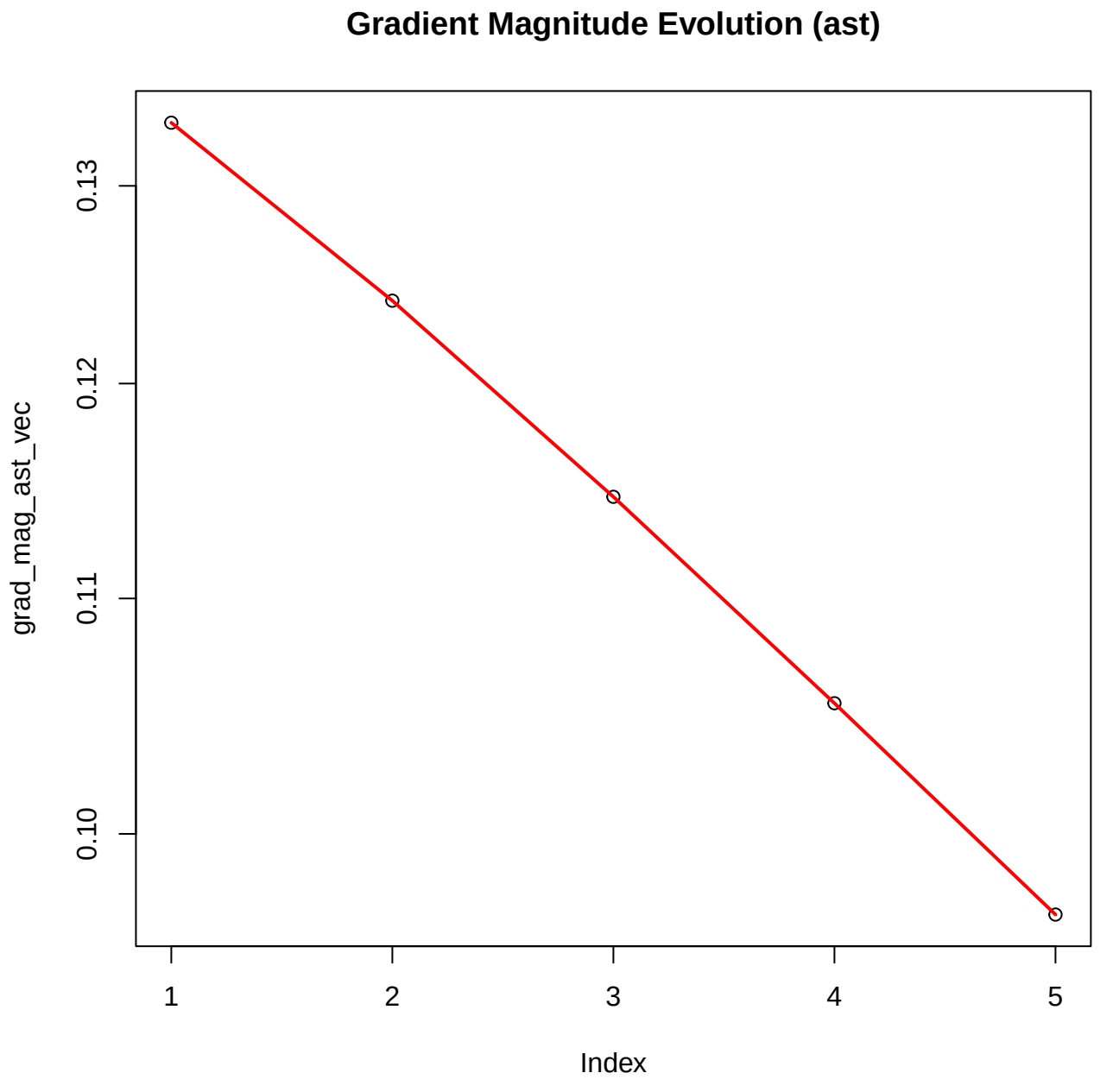
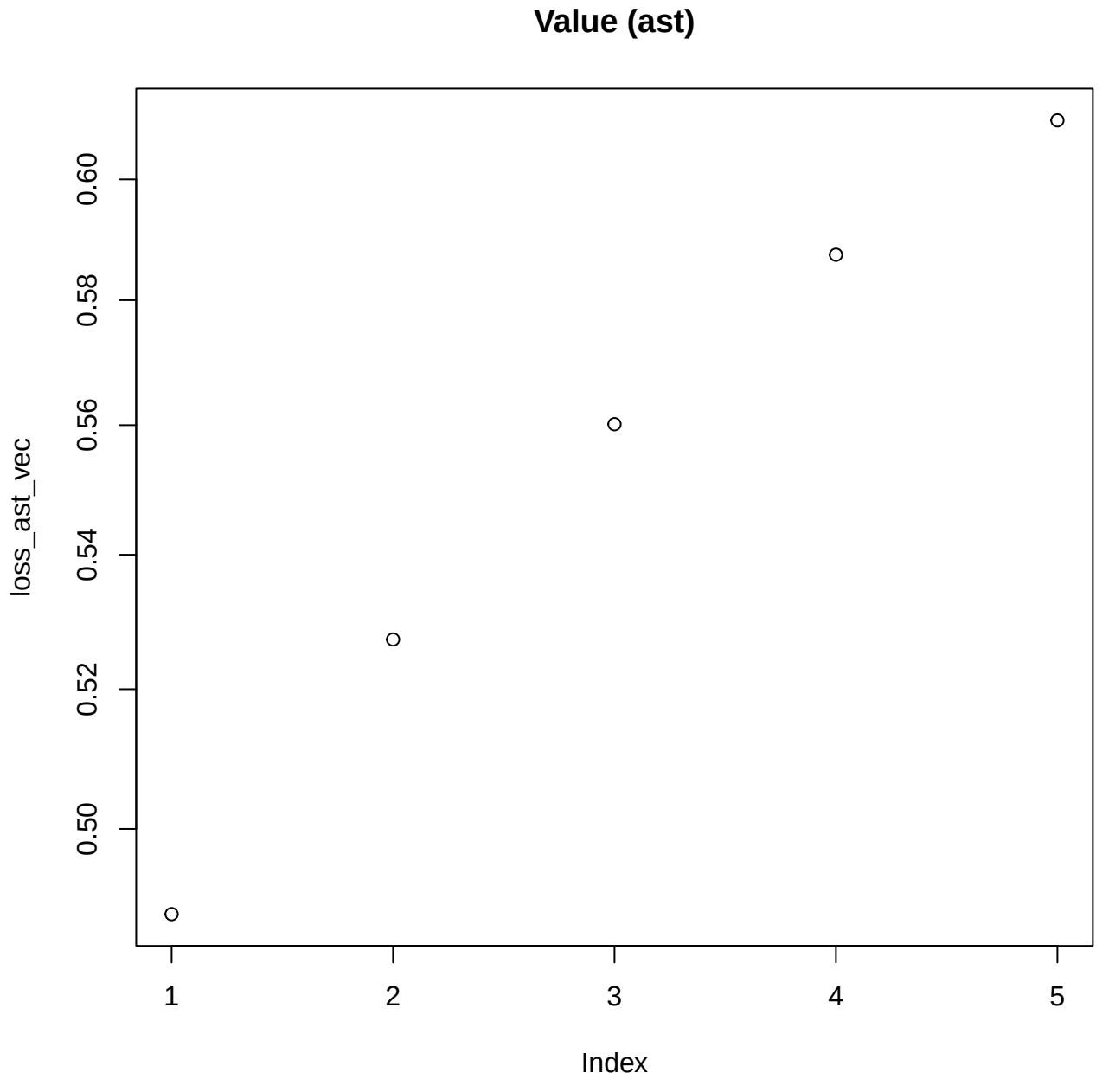
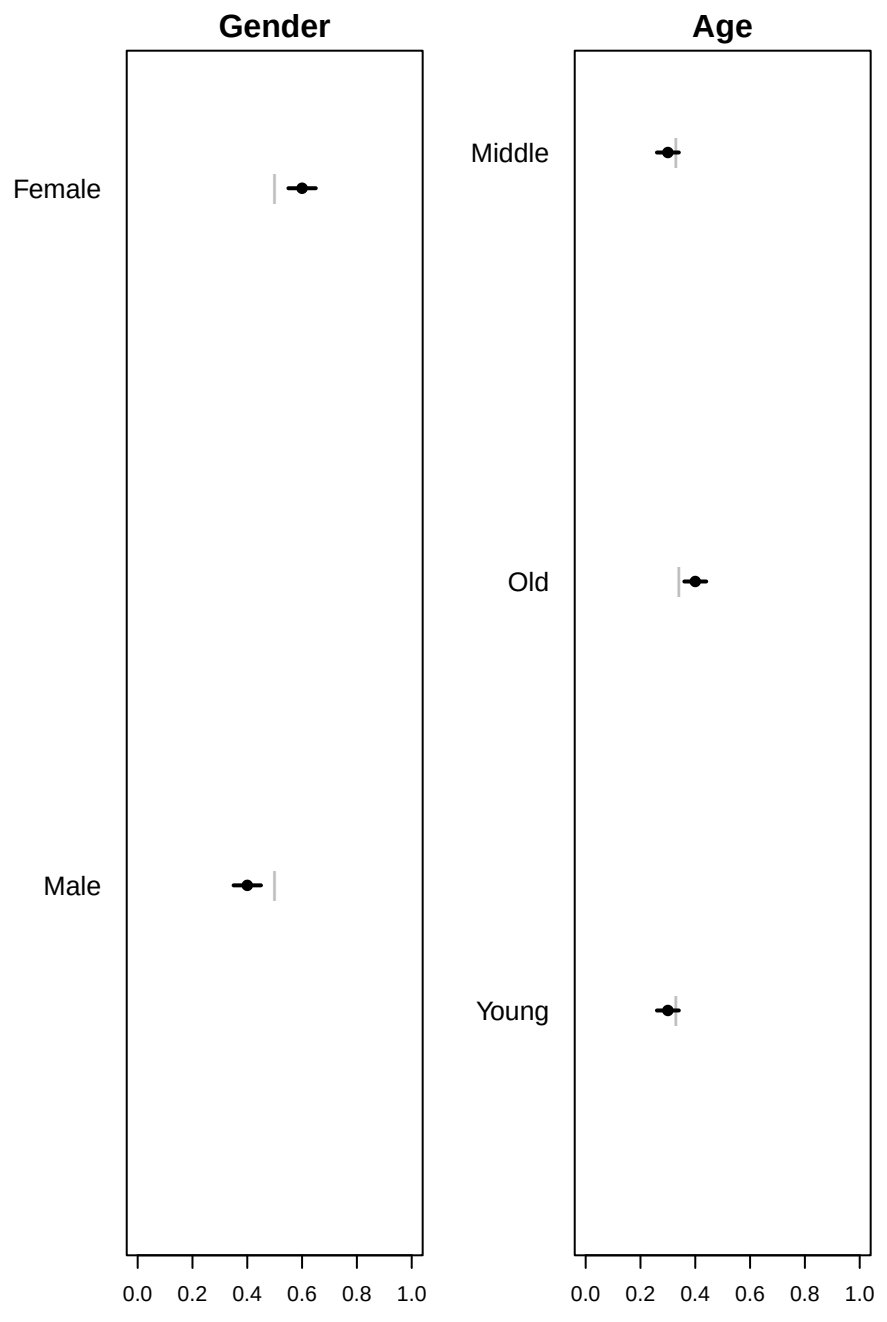
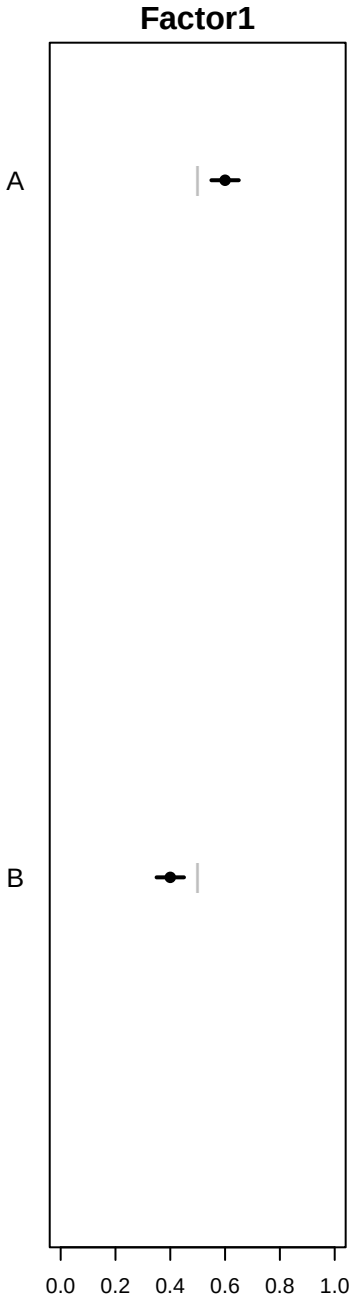
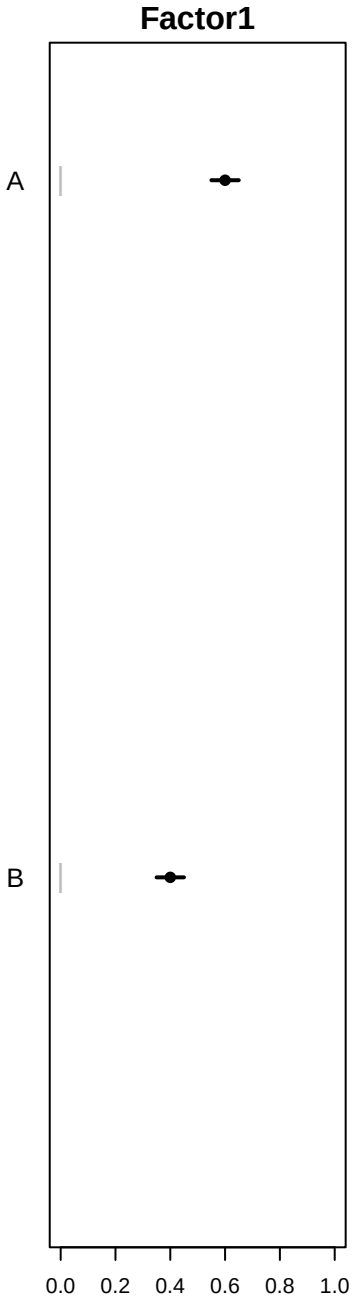
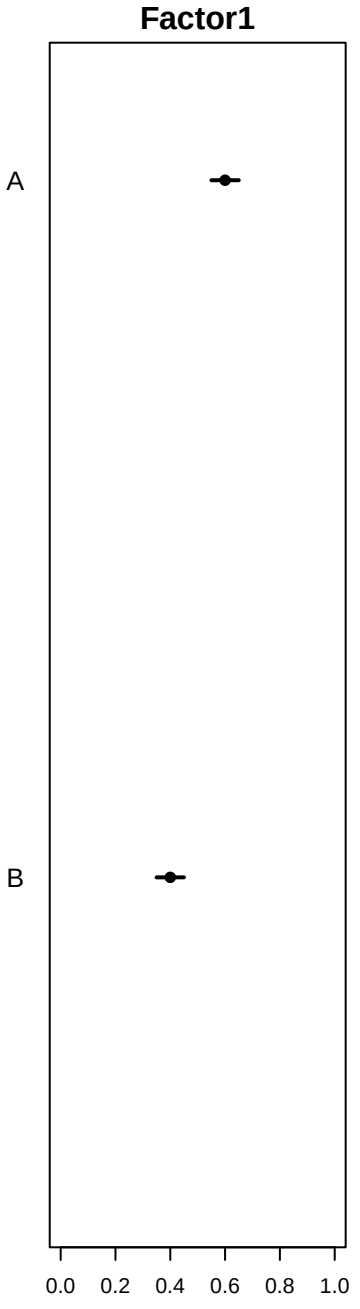

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

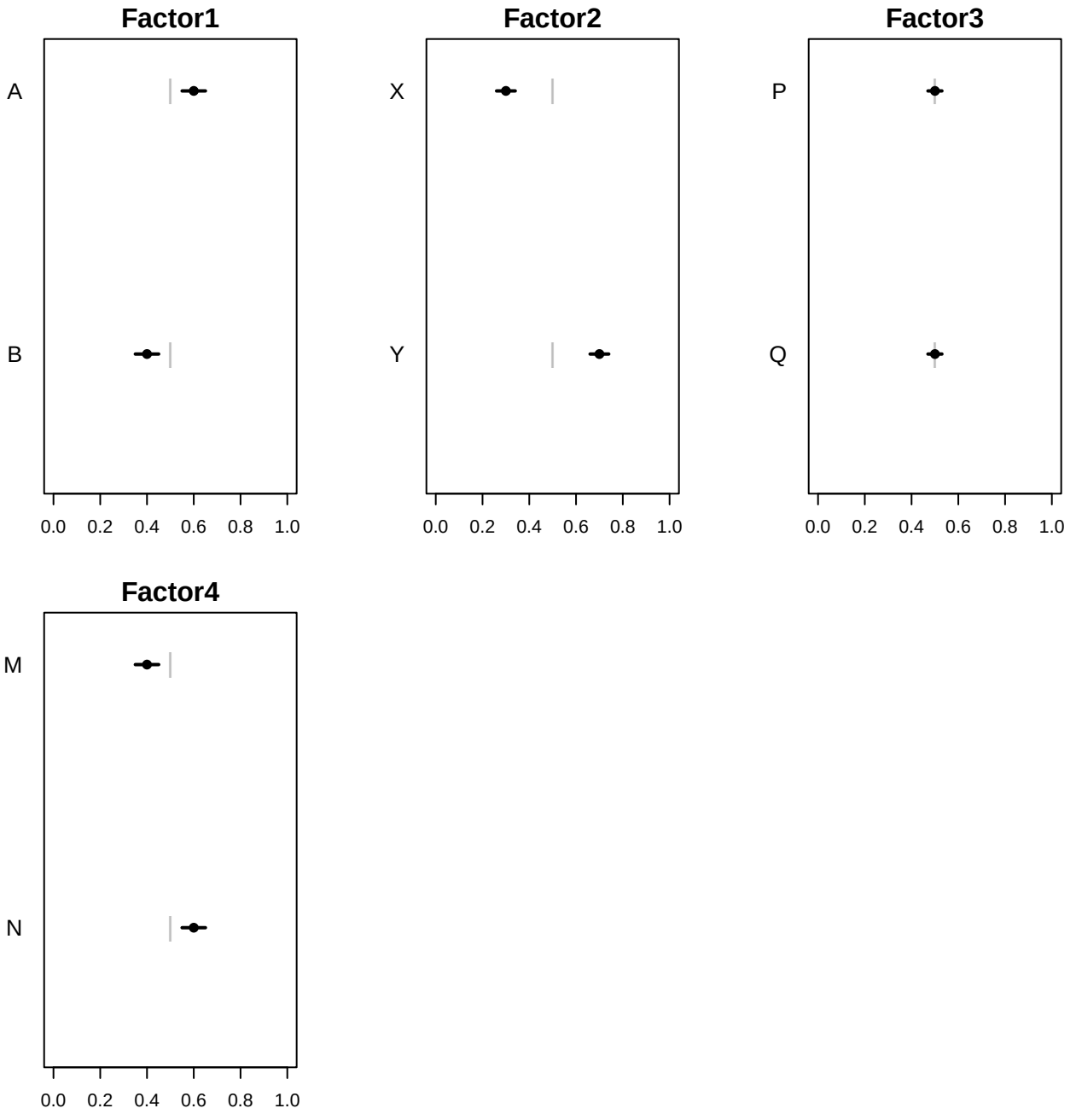

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

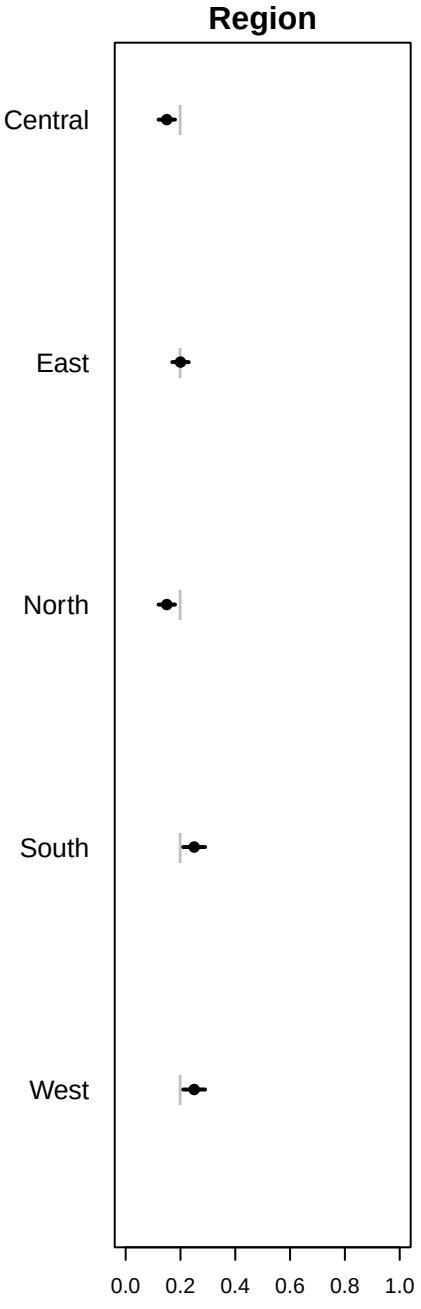

Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0
Age
Middle
Old
Young
0.0
0.2
0.4
0.6
0.8
1.0

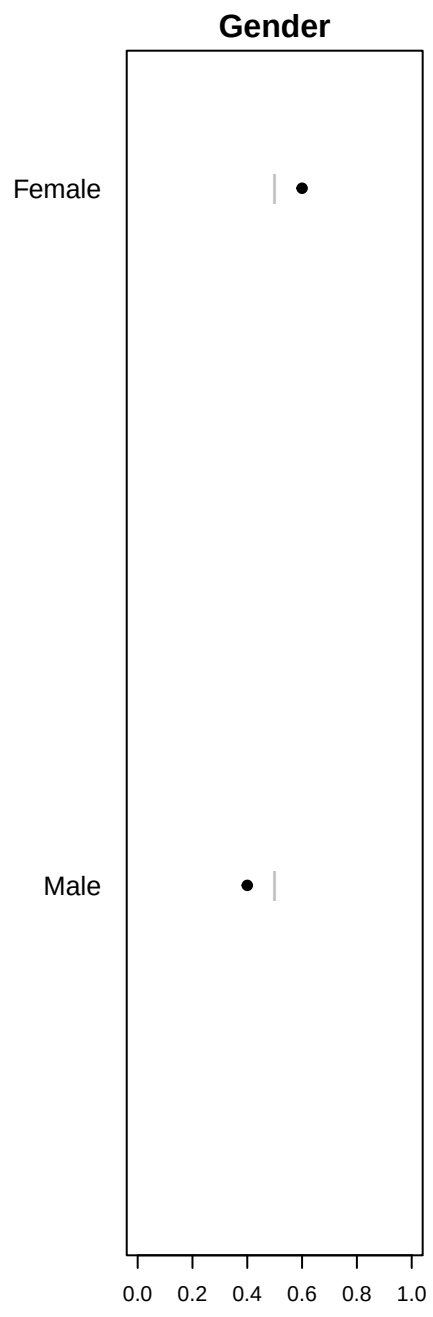

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

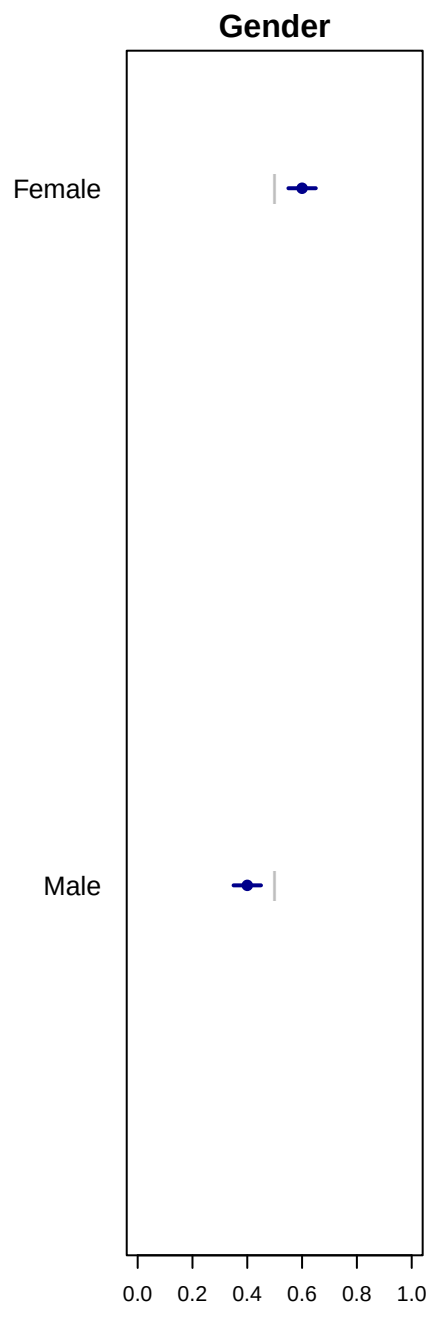

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

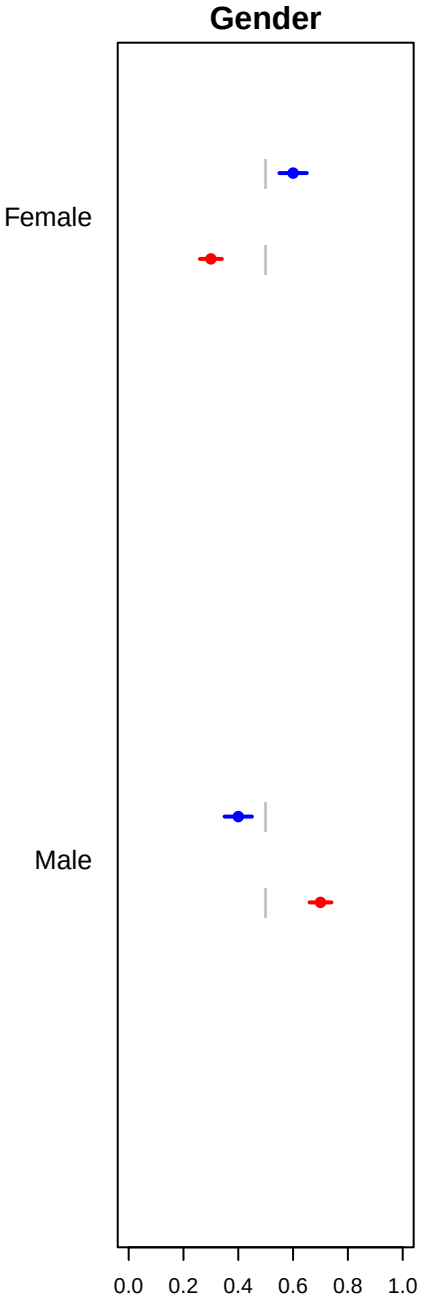

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

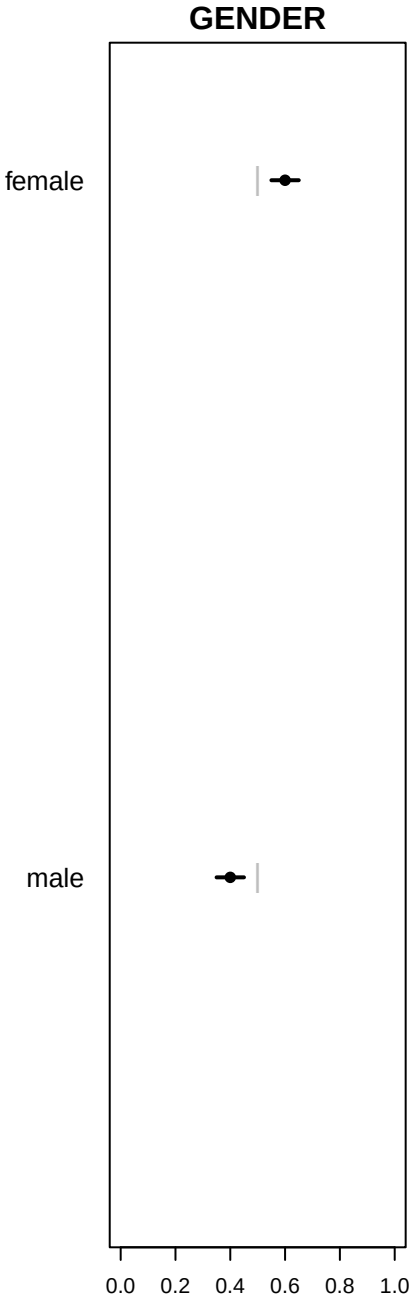

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
0.0
0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
0.0
0.2
0.4
0.6
0.8
1.0

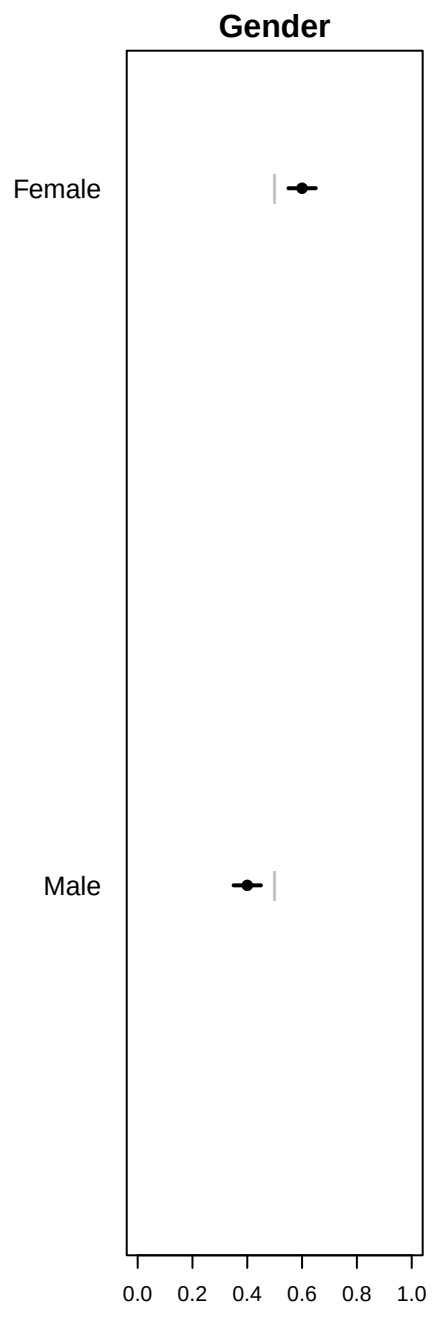

Region
Central
East
North
South
West
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


GENDER
female
male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0